Paper 1 – Basic Sciences in Urology: All Answers by Section

Compiled from: 2015–2025 actual Paper 1 (12 papers), Practice Sets 1–14 Paper 1 and Practice Set 18–22 SAQ sheets — 238 answers.

Repeats merged: questions asked more than once are printed once, with every wording and source listed under "Asked as"; their answers are combined so that no fact from any version is lost. 45 merged topics replace 109 original answers; 174 topics in all.

Layout: the 17 revision-book sections; in each, long answer questions (LAQ) first, then short answer questions (SAQ), then a Rapid Revision Sheet of that section's key points.

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Section 1 · Basic Principles — Long answer questions (LAQ)

1. Iodinated and other contrast agents (use in urology) and their complications *(merged: 3 questions)*
2. Laparoscopic access (Veress, Hasson cannula, extraperitoneal space), gases, and open vs laparoscopic vs robotic radical prostatectomy *(merged: 3 questions)*
3. Pneumoperitoneum physiology, complications of retroperitoneoscopy and laparoscopic donor nephrectomy *(merged: 4 questions)*
4. Describe IVU & its applications in Urology.
5. Principles of tissue engineering and its use in urology
6. Radionuclide (isotope) scans in urology: principles, renal scans, MAG3 and applications *(merged: 4 questions)*

Section 1 · Basic Principles — Short answer questions (SAQ)

1. Newer generation CT scan
2. History of Laparoscopy
3. Foley's (urethral) catheters, guidewires and DJ stents with their complications *(merged: 3 questions)*
4. Andrew Novick.
5. Proteinuria *(merged: 2 questions)*
6. Acetylcysteine and Contrast Induced Nephropathy (CIN) *(merged: 3 questions)*
7. Non iodinated contrast materials
8. Health-related quality of life (HRQOL) instruments in urology, including after urinary diversion *(merged: 2 questions)*
9. PET scans in urology.
10. Bacteriuria in pregnancy.
11. Recist Criteria.
12. Outline algorithm for difficult catheter placement.
13. Indications for outpatient cystourethroscopy.
14. MR urography
15. Urine cytology
16. Patient positioning in lap urology
17. Clinical trial methodology
18. Evidence-based medicine and critical appraisal of research papers *(merged: 2 questions)*
19. Doppler in urology
20. Christian Chaussey
21. Flow cytometry
22. VCUG

23. Pioneers of urology: Hopkins, Huggins, Calne, Gaur and Asopa *(merged: 2 questions)*
24. Interventional radiology in urology and therapeutic angioembolisation *(merged: 2 questions)*
25. Tumour markers in urology
26. Haemostatic agents in urology
27. Sterilisation of urological instruments
28. Ethics and informed consent in urology
29. Operating room set-up
30. Amyloidosis in urology

Section 2 · Oncology, DSD, Gender & Ageing — Short answer questions (SAQ)

1. Oncogenes *(merged: 2 questions)*
2. Premalignant conditions in urology (including premalignant lesions of penis) *(merged: 2 questions)*
3. 46-XX Male
4. Female circumcision
5. Familial malignancies in urology *(merged: 2 questions)*
6. Secondary malignancies in urology *(merged: 2 questions)*
7. Genetic counselling
8. Congenital Penile curvature..
9. Circumcision undone
10. Noonan syndrome

Section 3 · Infections & Inflammation — Long answer questions (LAQ)

1. Genitourinary infections, common organisms, bacteraemia and urosepsis: pathophysiology and management *(merged: 3 questions)*
2. Describe clinical implication of AIDS on the Genito-Urinary Tract?

Section 3 · Infections & Inflammation — Short answer questions (SAQ)

1. Principles of antimicrobial therapy in UTI
2. Renal and perinephric abscess *(merged: 2 questions)*
3. Ciprofloxacin
4. Catheter Associated Urinary Tract Infections (CAUTI)
5. Complicated urinary tract infections.
6. ESBL producing enterobacterial UTI *(merged: 2 questions)*
7. Lichen sclerosus (BXO, balanitis xerotica obliterans) *(merged: 2 questions)*
8. Drug Fosfomycin
9. Chronic Prostatitis : diagnosis, classification and management.
10. Lymphogranuloma venereum
11. Xanthogranulomatous pyelonephritis.
12. EPN. Prognostic factors & conserv. mgt.
13. Papillary necrosis.
14. Cephalsporins
15. Fungal UTI
16. Antimicrobial resistance
17. Biofilm
18. Renal mucormycosis
19. Hydatid disease of kidney

Section 4 · Paediatric Urology 1 — Long answer questions (LAQ)

1. Primary (congenital) VUR: VUJ embryology, pathophysiology, natural history and management *(merged: 2 questions)*

2. Urological evaluation and management of pediatric kidney transplant patient.
3. Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis *(merged: 2 questions)*
4. IVC anomalous development and retrocaval ureter *(merged: 2 questions)*

Section 4 · Paediatric Urology 1 — Short answer questions (SAQ)

1. Glandular hypospadias
2. TIP repair in hypospadias
3. Vanishing testes
4. Pelvicalyceal anomalies other than PUJ obstruction
5. Methods of orchidopexy
6. Hypospadias repairs – single-stage versus staged techniques

Section 5 · Paediatric Urology 2 — Long answer questions (LAQ)

1. Describe embryology of bladder. What are the proposed theories for extrophy epispadias complex? Describe skeletal abnormalities.
2. Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management.
3. Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children

Section 5 · Paediatric Urology 2 — Short answer questions (SAQ)

1. Embryology of Posterior Urethral Valves.
2. Anomalies detected on Prenatal scan between 16th and 20th weeks of gestation
3. USG in neonates
4. Nocturnal enuresis
5. Neurogenic bladder in myelodysplasia

Section 6 · Female Urology — Long answer questions (LAQ)

1. Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia *(merged: 4 questions)*

Section 6 · Female Urology — Short answer questions (SAQ)

1. Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications *(merged: 2 questions)*
2. Burch colposuspension
3. Management of perforated mid urethral sling
4. TOT in SUI
5. Describe clinical presentation of pelic floor prolapse. diagnostic evaluation and brief discussion of various surgical options.
6. Female urethral diverticula
7. Endoscopic procedure of SUI
8. Eponymous techniques of VVF repair
9. Endometriosis in urology

Section 7 · Voiding Dysfunction — Long answer questions (LAQ)

1. Describe pathophysiology of OAB. Briefly outline management &describe nonmedical non-pharmacological management

Section 7 · Voiding Dysfunction — Short answer questions (SAQ)

1. Detrusor–external sphincter dyssynergia (DESD / DSD) *(merged: 2 questions)*
2. Compare and contrast various detrusor relaxants in management of OAB
3. EMG
4. Botulinum toxin in urology: mechanism, indications and doses *(merged: 2 questions)*

Section 9 · Andrology & Sexual Medicine — Long answer questions (LAQ)

1. Spermatogenesis, azoospermia, non-obstructive azoospermia and maturation arrest *(merged: 3 questions)*
2. Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED *(merged: 2 questions)*

Section 9 · Andrology & Sexual Medicine — Short answer questions (SAQ)

1. Male reproductive immunology
2. Management of Peyronie's disease
3. Haematospermia *(merged: 2 questions)*
4. Testicular biopsy and TESA (testicular sperm aspiration) *(merged: 2 questions)*
5. Premature Ejaculation
6. Avanafil

Section 10 · Testis, Penis & Urethra — Long answer questions (LAQ)

1. Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias *(merged: 2 questions)*
2. Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications *(merged: 3 questions)*
3. Enlist various grafts used in urology and role of BMG in urology
4. Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy.

Section 10 · Testis, Penis & Urethra — Short answer questions (SAQ)

1. Surgical management of primary scrotal cancer
2. Varicocele – Pathophysiology
3. Treatment of submeatal stricture urethra in males
4. Perineal urethrostomy.
5. Testicular microlithiasis
6. Growing teratoma syndrome

Section 11 · Stones, Obstruction & Trauma — Long answer questions (LAQ)

1. Describe Pathogenesis of Congenital Pelvi Ureteric Junction Obstruction its diagnosis and Management.
2. Discuss pathophysiology of renal stones and also describe how to prevent its recurrence

Section 11 · Stones, Obstruction & Trauma — Short answer questions (SAQ)

1. Flexible Ureterorenoscopy
2. Distal Renal tubular acidosis.
3. Radionuclide studies and diuretic renography in PUJ obstruction *(merged: 3 questions)*
4. Prone vs Supine PCNL: Indications, positioning, technique, specific complications, advantages and disadvantages of each.
5. Metabolic Evaluation in Stone Disease
6. What are nanobacteria? Describe their role in urolithiasis.
7. Tear drop bladder
8. Retrograde endopyelotomy
9. Autotransplantation
10. Retroperitoneal fibrosis
11. Management of failed pyeloplasty
12. Post-obstructive diuresis
13. Stone complexity scores
14. Calyceal diverticulum – classification and management

15. Pelvic lipomatosis
16. Cystinuria

Section 12 · Bladder Cancer & Diversion — Short answer questions (SAQ)

1. Instestinal segments in urology *(merged: 2 questions)*
2. Diagnosis and management of hemorrhagic cystitis
3. BCG therapy in Bladder Cancer
4. Cancer in augmented bladder

Section 13 · Kidney & UTUC — Long answer questions (LAQ)

1. Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal malignancy *(merged: 4 questions)*

Section 13 · Kidney & UTUC — Short answer questions (SAQ)

1. Role of renal nephrometry in management of RCC.

Section 14 · Adrenal & Retroperitoneum — Long answer questions (LAQ)

1. Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma

Section 15 · Renal, Nephrology & Transplant — Long answer questions (LAQ)

1. Describe in details basic patho-physiology of urine production and transport
2. Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL *(merged: 3 questions)*
3. Lymphatic drainage of the genitourinary system and chyluria (etiopathogenesis and management) *(merged: 3 questions)*
4. Pathophysiology of Renovascular hypertension

Section 15 · Renal, Nephrology & Transplant — Short answer questions (SAQ)

1. Renin-angiotensin system
2. Patho-physiology of Brain stem death
3. Drugs associated with Acute Renal Failure.
4. GFR estimation in a prospective renal donor and creatinine clearance *(merged: 2 questions)*
5. Berger's Disease.
6. Renal allograft rejection.
7. Immunosuppressants in renal transplantation (classification, protocols, ciclosporin) *(merged: 3 questions)*
8. Kidney retrieval techniques for organ donation
9. Extended Criteria for Selection of Live Renal Donor
10. Vascular complications after renal transplantation, including transplant renal artery stenosis *(merged: 2 questions)*
11. Management of acute hyperkalaemia *(merged: 2 questions)*
12. Vascular grafts in urology
13. Hypokalemia
14. Lymphocele and postoperative lymphatic leak (chylous ascites) *(merged: 2 questions)*
15. Hyperfiltration injury in AKI and CKD
16. Management of CAPD peritonitis

Section 16 · BPH — Long answer questions (LAQ)

1. Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer *(merged: 2 questions)*

Section 16 · BPH — Short answer questions (SAQ)

1. HoLEP in BPH
2. Millin's prostatectomy

3. TUR Syndrome
4. TURP instruments and techniques *(merged: 2 questions)*
5. Methods to estimate prostate size

Section 17 · Prostate Cancer — Long answer questions (LAQ)

1. Describe staging of CA prostate and management of nonorgan confined CA prostate

Section 17 · Prostate Cancer — Short answer questions (SAQ)

1. Transrectal ultrasound study of prostate
2. Robotic radical prostatectomy
3. Prostate multiparametric MRI and endorectal coil MRI *(merged: 2 questions)*
4. Docetaxel

Section 1 · Basic Principles — Long answer questions (LAQ)

Iodinated and other contrast agents (use in urology) and their complications

Asked as:

- Discuss Iodinated contrast radiological agents and their complications. — *Summer 2016 Paper I Q1 (LAQ)*
- Complications of intravenous contrast media. — *Summer 2017 Paper I Q3a (SAQ)*
- Use of contrast agents in urology and their complications — *Summer 2022 Paper I Q3d (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2016-Summer Paper 1, Q1 and 2018-Summer Paper 1, Q3a)

Introduction

- Iodine (atomic number 53, K-edge 33 keV) absorbs X-rays strongly. All agents are built on a **tri-iodinated benzene ring** (tri-iodinated benzene rings).
- They are filtered by the glomerulus, not secreted or reabsorbed, so they concentrate in urine. Half-life is about 2 hours with normal function; biliary excretion rises in renal failure.
- Modern practice uses **non-ionic low-osmolar** (iohexol, about 600–850 mOsm/kg) or **iso-osmolar** (iodixanol, 290) agents. High-osmolar ionic agents caused far more reactions.
- Uses: CT (computed tomography) urography, IVU (intravenous urography), pyelography, cystourethrography, angiography.

Classification

Class	Example	Osmolality (mOsm/kg)	Iodine : particles
High-osmolar ionic monomer	Diatrizoate	~1500–2000 (5–8 × plasma)	3 : 2
Low-osmolar non-ionic monomer	Iohexol, iopamidol	~600–850	3 : 1
Low-osmolar ionic dimer	Ioxaglate	~600	6 : 2
Iso-osmolar non-ionic dimer	Iodixanol	~290	6 : 1

- **Why it matters:** a higher ratio gives more opacity per particle. Fewer particles mean less fluid shift, pain and reactions. So non-ionic low- or iso-osmolar agents are standard intravascularly.

Types and uses (contrast agents in urology)

Agent	Principle	Urological use
Iodinated: ionic high-osmolar (diatrizoate); non-ionic low-osmolar (iohexol); iso-osmolar dimer (iodixanol, 290 mOsm/kg)	Iodine absorbs X-rays; glomerular filtration	CT (computed tomography) urography, angiography; retrograde pyelogram, urethrogram, cystogram
Gadolinium (GBCA, gadolinium-based contrast agent)	Shortens T1 relaxation; chelated	MRI (magnetic resonance imaging) of renal mass, prostate, MR urography
Microbubbles	Gas in shell, intravascular	CEUS (contrast-enhanced ultrasound): cysts; voiding urosonography
Carbon dioxide	Negative contrast	Angiography in CKD (chronic kidney disease)

Complications

A. Hypersensitivity (allergic-like) reactions

- Mostly non-IgE (immunoglobulin E) direct mast-cell release: idiosyncratic, not dose-related. Severe reactions: 0.22% with ionic vs 0.04% with non-ionic agents (Katayama, 1990).

Type	Timing	Features
Immediate	Within about 1 hour (most within minutes)	Urticaria, angio-oedema, bronchospasm, hypotension; anaphylaxis if severe
Non-immediate	Hours to days	Maculopapular rash; T-cell mediated, milder

- Risk factors: previous reaction (strongest), asthma, atopy, severe cardiac disease, drug allergy, female sex. **Shellfish ("iodine") allergy is not a cross-reaction.**
- **Physiological reactions:** warmth, metallic taste, nausea, vasovagal hypotension with bradycardia, arrhythmia, seizures.

B. CA-AKI (contrast-associated acute kidney injury)

- **CA-AKI** = AKI within 48 hours of contrast, whatever the cause. **CI-AKI (contrast-induced AKI)** = the subset caused by contrast.
- KDIGO (Kidney Disease: Improving Global Outcomes) AKI: creatinine rise ≥0.3 mg/dL in 48 hours, ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for ≥6 hours.
- **Mechanism:** afferent vasoconstriction causes medullary hypoxia (ischaemic ATN (acute tubular necrosis)), plus direct tubular toxicity.
- **Risk factors:** CKD (chronic kidney disease), diabetes, age, anaemia, contrast volume, intra-arterial route.
- ACR (American College of Radiology)/NKF (National Kidney Foundation) 2020: risk was overstated; it is rare but real, mainly at eGFR (estimated glomerular filtration rate) <30 mL/min/1.73 m^2.

C. Others

- **Metformin:** lactic acidosis only if AKI develops. ACR: eGFR ≥30 and no AKI, continue. eGFR <30 or AKI, withhold 48 hours and restart when function is stable.
- **Thyroid:** iodine load can trigger hyperthyroidism/thyrotoxicosis (Jod-Basedow) in Graves' disease or nodular goitre. It blocks radio-iodine scans or therapy for several weeks.
- **Extravasation:** usually minor, but large power-injected volumes can rarely cause compartment syndrome or skin necrosis. Elevate the limb, apply compresses and observe. Get a surgical review if swelling increases, blisters form, or perfusion or sensation changes.
- **Also know:** iodide "mumps"; myasthenia flare; pregnancy (crosses placenta, not teratogenic); breastfeeding can continue; phaeochromocytoma safe with IV (intravenous) non-ionic agents; dialysis patients need no urgent dialysis.
- 4. **Also:** physiological (warmth, vasovagal), thyrotoxicosis, extravasation. Retrograde studies are safe in allergy.

Complications of gadolinium

- **NSF (nephrogenic systemic fibrosis)** at GFR <30; group II agents carry almost no risk. Brain deposition with linear agents.

Prevention, treatment and management

Premedication for high-risk patients / prior reaction (ACR 2023)

Regimen	Drugs
1	Prednisone 50 mg orally 13, 7 and 1 hour before + diphenhydramine 50 mg 1 hour before

Regimen	Drugs
2	Methylprednisolone 32 mg orally 12 and 2 hours before + diphenhydramine 50 mg 1 hour before

- First consider ultrasound, MRI (magnetic resonance imaging) or non-contrast CT. Breakthrough reactions still occur, so keep resuscitation drugs ready. Retrograde pyelography is safe even after anaphylaxis, because little contrast is absorbed.

Treating reactions (treatment of reactions)

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Stop injection → airway, oxygen, call for help
  Urticaria → observe ± H1-antihistamine (may progress)
  Bronchospasm → salbutamol inhaler 2 puffs (90 µg each), up to 3 times
  Anaphylaxis → intramuscular (IM) adrenaline 0.3–0.5 mg (1 mg/mL) every 5–15 min (repeat),
                max 1 mg; IV fluids
  Vasovagal → legs up, IV fluids, IV atropine
Observe ≥30 min (minutes); later allergy testing; record the reaction
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Preventing CA-AKI

- Lowest dose of a low- or iso-osmolar agent; weight-based dosing; a low tube voltage (80 kVp, kilovolt peak) needs less contrast. Avoid repeat doses within 48 hours; stop NSAIDs (non-steroidal anti-inflammatory drugs).
- **IV isotonic saline** for eGFR <30 not on dialysis: the only proven measure.
- **PRESERVE (2018):** bicarbonate and N-acetylcysteine (NAC) gave no benefit. **AMACING (2017):** no prophylaxis was non-inferior to hydration at eGFR 30–59. Removing contrast by dialysis does not reduce risk.

Diagram to draw: tri-iodinated benzene ring, monomer versus dimer, on an osmolality scale.

Key points to remember:

- Ratio 1.5 (ionic), 3 (non-ionic monomer), 6 (iodixanol, 290 mOsm/kg).
- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg, max 1 mg; shellfish allergy is not a risk.
- Premedication: prednisone 50 mg at 13-7-1 h or methylprednisolone 32 mg at 12-2 h, plus diphenhydramine.
- CA-AKI: saline if eGFR <30; bicarbonate and NAC useless; hold metformin only if eGFR <30 or AKI.
- NSF risk at GFR <30; group II agents almost none.

Laparoscopic access (Veress, Hasson cannula, extraperitoneal space), gases, and open vs laparoscopic vs robotic radical prostatectomy

Asked as:

- Discuss types of access, gases used in Laparoscopic urology and describe comparative analysis of open, Laparoscopic and Robotic Radical prostatectomy. — *Summer 2018 Paper I Q2 (LAQ)*
- Hassan's canula. — *Summer 2017 Paper I Q3e (SAQ)*
- Describe instrumentation for development of extraperitoneal space in laparoscopy? — *Summer 2020 Paper I Q3b (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Access in laparoscopy

A. Transperitoneal: closed (Veress)

- Veress needle: 14-gauge, spring-loaded blunt stylet. Check flow and tip recoil first. Two "pops": fascia, then peritoneum.
- **Sites:** umbilicus (thinnest wall; aim towards pelvis in non-obese); lower quadrants; **Palmer point** (left subcostal, midclavicular line) with suspected adhesions; avoid scars.
- **Confirmation tests:** aspiration–injection–aspiration of saline; **drop test**; advancement test (1 cm free). Then insufflate: opening pressure <8–10 mmHg, set limit 15 mmHg.
- First trocar inserted blind, or under vision with an **optical trocar** (0° lens inside a clear tip).

B. Open (Hasson)

- 2-cm infra-umbilical incision; fascia and peritoneum opened under vision; stay sutures; blunt cannula.
- Safer with adhesions or previous surgery and early in training; more gas leak.

Hasson cannula (Hassan's canula)

Assumed to mean the Hasson cannula, used for open laparoscopic access.

Introduction

- Described by **Harrith Hasson (1971)** for open ("direct vision") laparoscopy. The first port is placed under vision instead of a blind Veress needle and trocar.
- **Parts:** outer sleeve with a cone-shaped (olive) adapter, **blunt obturator**, struts to anchor fascial stay sutures, and a gas inlet with tap.

Indications

- **Previous abdominal surgery or expected adhesions** (main indication).
- Organomegaly, pregnancy, aortic aneurysm, children, thin patients.
- Retroperitoneoscopy and extraperitoneal access (open start, then balloon dilatation).

Technique (transperitoneal)

1. 2-cm semicircular infra-umbilical incision.
2. Open fascia and then peritoneum separately, under vision, finger-sized.
3. Confirm entry by sight and finger sweep (free adhesions).
4. Place two 0 silk fascial stay sutures.
5. Insert the cannula with its **blunt obturator**.
6. Seat the cone in the defect; wrap stay sutures round the struts to seal.
7. Remove obturator; insufflate at high flow (to 12–15 mmHg).
8. At closure, tie the stay sutures to close the fascia.

- **Retroperitoneal:** 2–2.5 cm incision below the 12th rib tip, finger dissection onto psoas, balloon dilatation, then port.

Advantages over Veress

- No blind puncture: fewer major vessel and bowel injuries; no preperitoneal insufflation; no gas embolism from intravascular insufflation.
- Fast insufflation; easy fascial closure (less port-site hernia).

Disadvantages

- Larger incision; **gas leak** around the port; slightly longer in obese patients.
- Early retroperitoneal use caused subcutaneous emphysema and hypercarbia, now solved by balloon-cuffed blunt ports.

Complications

- Bowel injury if adherent under the wound (repair through the same incision); wound infection; port-site hernia; vascular injury is still possible.

Evidence

- Reviews show no single entry method is clearly safest for major injury; open entry has fewer failed entries.

C. Retroperitoneal

- 2-cm incision below the tip of the 12th rib; finger dissection behind Gerota's fascia, in front of psoas → **balloon dilatation** (commercial or **Gaur** glove-on-catheter balloon) → balloon-tip port to prevent gas leak and emphysema.

D. Extraperitoneal (pelvic): infra-umbilical, behind rectus into the Retzius space with a balloon (used for extraperitoneal prostatectomy).

Instrumentation for development of the extraperitoneal space

- Extraperitoneal laparoscopy works outside the peritoneum: the **flank retroperitoneum** (kidney, adrenal, ureter) or the **pelvic prevesical (Retzius) space** (extraperitoneal prostatectomy, lymphadenectomy).
- There is no natural cavity, so a working space must be **dissected before insufflation**.

Instrument	Description and use
Open (Hasson) access set	Scalpel, S-retractors, artery forceps
Index finger	Blunt first pocket; feels psoas behind, Gerota's fascia in front
Commercial balloon dissector	Transparent silicone balloon on a stiff hollow shaft; inflated with room air by bulb (maximum 800 mL, 40 pumps). Oblong for the retroperitoneum, round for the pelvis. Laparoscope inside confirms position
Gaur balloon (1992)	Size 7 glove (or glove fingers, condom) tied with 0 silk on an 8-Fr red rubber catheter; filled with saline (600–800 mL). Cheap, but no rigid shaft or view; latex can rupture into fragments
Optical (visual) obturator trocar with 0° scope	Enters under vision; yellow retroperitoneal fat confirms the plane; scope sweeps the space
Blunt-tip balloon trocar	Internal balloon plus external foam cuff gives an airtight seal; prevents gas leak, emphysema and hypercarbia

Technique: Retroperitoneal

1. Full flank, table flexed; 1.5–2.5-cm incision below the 12th rib tip.
2. Split muscles; open the lumbodorsal fascia; finger dissection between psoas and Gerota's fascia.
3. Balloon inflated cephalad then caudad, in front of psoas, behind Gerota's fascia.

4. Blunt-tip balloon trocar; insufflate; secondary ports under vision.

Technique: Pelvic extraperitoneal

- Infraumbilical incision → open anterior rectus sheath, separate recti → finger tunnels behind the rectus to the pubis → pierce transversalis fascia → balloon midline, then each side.

Pitfalls

- Balloon in muscle → flank hernia. Too anterior → peritoneal tear or colon injury; too posterior → psoas or quadratus bleeding.
- Peritoneal tear shrinks the space; convert if needed. Extraperitoneal CO_2 needs more ventilation.

E. Others: hand-assisted (6.5–7.5 cm port); single-port/LESS (laparoendoscopic single-site); direct trocar entry without pneumoperitoneum.

- **Trocars: blunt dilating trocars** have replaced bladed ones (about fivefold less epigastric injury, fewer hernias). Close fascia of midline ports >10 mm or stretched sites (all ≥5 mm in children).
- **Access complications:** preperitoneal insufflation, vessel or bowel puncture, gas embolism, subcutaneous emphysema.

Gases used

Gas	Advantages	Disadvantages
CO_2 (carbon dioxide; standard)	Non-combustible, cheap, **highly blood-soluble** (embolism less lethal), quickly absorbed	Absorbed → hypercarbia, acidosis, arrhythmia; dangerous in COPD (chronic obstructive pulmonary disease)
Nitrous oxide	Less peritoneal pain, fewer acid–base effects	**Supports combustion**: no electrosurgery
Helium	Inert, no hypercarbia; use in lung disease	Poorly soluble → embolism; costly
Argon	Inert	Embolism risk
Air, oxygen	–	Embolism, combustion: abandoned

- **Physiology:** keep pressure 10–15 mmHg (12 mmHg more physiological). Pressure reduces venous return, raises afterload, causes **oliguria** (renal vein and parenchymal compression) and splints the diaphragm.
- CO_2 absorption plateaus at 30–60 minutes; higher with extraperitoneal insufflation. Hypercarbia peaks after extubation.
- **Gas embolism:** sudden collapse, millwheel murmur, end-tidal CO_2 rise then fall. Treat: stop insufflation, **left lateral, head-down**, 100% oxygen, aspirate via central line.

Comparison of open (RRP), laparoscopic (LRP) and robotic (RALP) radical prostatectomy

- **Background:** anatomical nerve-sparing open RRP (Walsh, 1982) was the standard. First LRP (Schuessler, 1997) was too slow; French centres standardised it in 2000, but laparoscopic suturing of the anastomosis remained a barrier. The robot (da Vinci, 2000) removed that barrier, and RALP is now dominant in the USA.

Parameter	Open RRP	LRP	RALP
Vision	Direct, loupes	2D, unstable	**3D, 10× magnified**, stable
Operating time	Shortest	Longest early	Longer than open; falls with experience

Parameter	Open RRP	LRP	RALP
Instruments	Hands, haptic feedback	Rigid, fulcrum effect	**Wristed** (7 degrees of freedom); no haptics
Learning curve	Moderate	**Steepest** (suturing)	Shorter
Blood loss	300–1000 mL	Less (pneumoperitoneum tamponade)	Least; **transfusion 1–2%**
Hospital stay, catheter	Longer; catheter ~2 weeks	Shorter	Usually 1 day; catheter ~1 week
Pain, return to work	More	Less	Less; normal activity 3–4 weeks
Positive margins	Similar	Similar	Similar (~15%; ~9% if pT2)
Continence (1 year)	~93% (expert)	Similar	Similar; possibly earlier
Potency	Surgeon-dependent	Similar	Similar
Bladder neck stricture	Up to 16% (older series)	Lower	**<2%** (mucosa-to-mucosa)
Unique risks	Wound infection, hernia	Conversion, CO_2 effects	Steep Trendelenburg (eye, nerve), robot failure (<0.5%)
Cost	Lowest	Moderate	**Highest**

Evidence

- **Yaxley 2016 (randomised, one open vs one robotic surgeon):** equal continence, potency and early margins at 3 months; less blood loss and shorter stay with robot.
- **Coughlin 2018:** no difference in urinary or sexual function at 24 months.
- **LAP-01 (robotic vs laparoscopic):** earlier continence with robot.
- **Concept:** the **surgeon, not the platform**, decides outcome; margins depend mainly on tumour (Gleason ≥7, stage) and surgeon volume. The robot's advantages are recovery and wider reach (obese patients, prior surgery).
- **Also know:** transperitoneal vs extraperitoneal approach (no consistent difference); Retzius-sparing RALP (earlier continence, more margins); single-port (SP) robot; new robotic platforms; telesurgery.

Diagram to draw: Veress needle sites (umbilicus, Palmer point) and trocar layout for RALP; Hasson cannula with cone, stay sutures and blunt obturator in the abdominal wall; flank section with balloon between psoas and Gerota's fascia.

Key points to remember:

- Veress tests: aspiration, drop, advancement; opening pressure <10 mmHg.
- CO_2 standard (soluble, non-flammable); helium in COPD; N_2O supports combustion.
- Gas embolism: left lateral head-down, oxygen, aspirate.
- Yaxley 2016: equal function and margins; robot less blood loss and stay.
- Hasson 1971: open entry under vision.
- Main indication: previous surgery or adhesions.
- Avoids blind Veress injuries; drawback is gas leak.
- Space is created first: finger, then balloon.
- Gaur balloon: glove on red rubber catheter.
- Balloon trocar seals and prevents emphysema.
- Balloon in muscle causes hernia.

Pneumoperitoneum physiology, complications of retroperitoneoscopy and laparoscopic donor nephrectomy

Asked as:

- Describe physiological and hemodynamic changes due to creation of pneumoperitoneum during laparoscopy. Describe complications of Retroperitoneoscopy. — *Summer 2019 Paper I Q2 (LAQ)*
- Describe hemodynamic & physiologic changes d/t pneumoperitoneum during lap. — *Practice Set 3 Paper I Q1 (LAQ)* (also answered, with Q2, in 2019-Summer Paper 1, Q2)
- Complications of retroperitoneoscopy. — *Practice Set 3 Paper I Q2 (LAQ)* (also answered, with Q1, in 2019-Summer Paper 1, Q2)
- Physiological and hemodynamic effects of laparoscopy. Write in detail about lap donor nephrectomy — *Practice Set 10 Paper I Q2 (LAQ)* (also answered in Practice Set 3 Paper 1, Q1 and Practice Set 2 Paper 4, Q3a)

Type: LAQ (25 marks)

"d/t" is read as "due to" and "lap" as laparoscopy.

Part A: Physiological and haemodynamic effects of pneumoperitoneum (laparoscopy)

Introduction

- **Pneumoperitoneum** is insufflation of gas, usually CO_2 (carbon dioxide), into the peritoneal cavity at 10–15 mmHg to create a working space.
- Its effects have **two drivers**: **pressure** (compression of the cava, renal veins, kidneys and diaphragm) and **absorbed CO_2** (hypercarbia, acidosis, sympathetic drive). Patient position and volume status modify both (position modifies both).
- **CO_2** is standard: non-combustible, cheap, highly blood-soluble, so embolism is less lethal (less dangerous). Helium or argon in severe lung disease; N_2O (nitrous oxide) supports combustion.

Choice of insufflant

Gas	Advantages	Disadvantages
CO_2 (standard)	Non-combustible; highly blood-soluble, so embolism is less lethal; cheap	Hypercarbia, acidosis, arrhythmia; COPD (chronic obstructive pulmonary disease) patients cannot clear it
Helium, argon	Inert; no hypercarbia	Poorly soluble, so gas embolism risk; costly. Helium used in lung disease or to salvage hypercarbia
N_2O (nitrous oxide)	Less peritoneal irritation	Supports combustion; no electrosurgery
Room air, oxygen	—	Embolism, combustion; abandoned

Effects by pressure (supine, Campbell Table 18.1)

Effect	5 mmHg	10 mmHg	20 mmHg	40 mmHg
Heart rate (HR)	↑	↑	↑	↓
MAP (mean arterial pressure), SVR (systemic vascular resistance)	↑	↑	↑	↑
Cardiac output	↔/↓	↔/↑	↔/↓	↓

Effect	5 mmHg	10 mmHg	20 mmHg	40 mmHg
Venous return, cardiac output (Summer 2019 source)	—	Variable	Variable	↓ sharply
GFR (glomerular filtration rate), urine output	↔	↓	↓↓	↓↓
Arterial PCO_2	↔	↑	↑	↑
Arterial pH	↔	↔/↓ (another source: ↔)	↓	↓

- Renal function is the first casualty (from 10 mmHg). Afterload rises at every pressure. Output collapses at 40 mmHg. Acidosis builds from 20 mmHg.

Cardiovascular (haemodynamic) changes

- **Afterload rises** at every pressure: aortic and splanchnic compression plus catecholamine and **vasopressin** release raise MAP and SVR.
- **Preload depends on volume status.** In normovolaemia or hypovolaemia, pressure up to 20 mmHg compresses the IVC (inferior vena cava; the cava collapses) and venous return falls (IVC compression lowers venous return). In hypervolaemia the cava resists and venous return may rise. At ≥40 mmHg venous return falls sharply in everyone; above 40 mmHg output collapses.
- **Arrhythmias**: hypercapnia causes tachycardia and ventricular ectopics; peritoneal stretch causes **vagal bradycardia** or asystole. A new arrhythmia may warn of (herald) pneumothorax, hypoxia or gas embolism.
- **CVP (central venous pressure) reads falsely high** because intrathoracic pressure rises; do not guide fluids by it.
- Leg venous return falls (leg venous stasis): theoretical DVT (deep vein thrombosis) risk.

Respiratory changes

- **Diaphragm splinted** (splinted diaphragm lowers FRC and compliance, raises airway pressure): FRC (functional residual capacity) and compliance fall, peak airway pressure rises; atelectasis and V/Q (ventilation–perfusion) mismatch. Dead space is unchanged. PEEP (positive end-expiratory pressure) helps in lung disease.
- **Head-down** (steep Trendelenburg) adds a further fall in vital capacity, endobronchial tube slip and facial oedema.
- **Barotrauma** (pressure >15 mmHg for long; at high pressure): pneumothorax, pneumomediastinum.

Acid–base changes (CO_2 absorption)

- CO_2 is absorbed across (through) the peritoneum and, extraperitoneally, through fat and disrupted lymphatics. Absorption rises over 30–60 minutes, then plateaus.
- **Hypercarbia and respiratory acidosis** and rising end-tidal CO_2. Treat (managed) by raising minute ventilation and lowering pressure.
- **COPD** patients cannot blow it off: arterial gases if insufflation exceeds 1 hour (>1 hour), continued in recovery; consider helium. **Hypercarbia peaks after extubation**, as stored CO_2 is released.

Renal changes

- **Oliguria** comes from **renal vein and parenchymal compression**, not ureteric compression or hormones. Cortical flow falls; medullary flow rises up to 20 mmHg, then falls. It reverses on desufflation (reversible).
- Keep pressure ≤10 mmHg (another source: ≤10–12 mmHg); avoid fluid overload. If needed give furosemide, mannitol 12.5–25 g or dopamine 2 µg/kg/min to keep urine >100 mL/hour,

rather than fluid boluses (overload, oedema). Important in live donor nephrectomy.

Other systems (others)

- **Splanchnic**: hepatic and mesenteric flow fall (rare mesenteric thrombosis); less ileus than open surgery; no rise in reflux.
- **ICP (intracranial pressure) and IOP (intraocular pressure)** rise with hypercarbia and head-down tilt: caution in head injury, shunts and glaucoma.
- **Hormonal:** catecholamines, cortisol and **vasopressin** rise; renin–angiotensin activation adds to afterload.
- **Stress and immune response**: cortisol, catecholamines and prolactin rise as in any surgery, but hyperglycaemia, insulin resistance, catabolism and IL-6 (interleukin-6) rise less than after open surgery (smaller stress and immune response), which may explain faster recovery; oncological benefit unproven.
- **Also know:** hypothermia from cold dry gas; shoulder-tip pain from diaphragmatic CO_2 irritation.

Effects of position and approach

Position	Heart rate	MAP	Cardiac output	Other effects
Head-up	↑	↓	↓	Output ↓, hypotension
Head-down (Trendelenburg)	↓	↑	↑ (but lungs worse)	Venous return and output ↑; lung volumes ↓, ICP and IOP ↑
Lateral flank, kidney rest	—	—	—	Lung V/Q mismatch; rhabdomyolysis risk with kidney rest

- **Summary:** head-down raises MAP and output but worsens lungs; head-up lowers output.
- **Retroperitoneal** insufflation causes smaller swings in output and venous pressures (smaller haemodynamic swings) than transperitoneal; pelvic extraperitoneal insufflation raised arterial CO_2 more.

Safe practice

- Preoperative cardiopulmonary assessment; correct arrhythmias; optimise volume.
- **Practical rule: work at 10–12 mmHg** (12 mmHg keeps stroke volume unchanged); 20 mmHg only briefly (first trocar, venous bleeding); never exceed 20 mmHg for long.
- Monitor end-tidal CO_2, saturation, airway pressure and urine output.
- **Recent advances:** low-pressure (8 mmHg) surgery with valveless insufflators (AirSeal); helium (or argon) for lung disease (poorly soluble, so embolism risk).

Diagram to draw: abdomen under pressure with arrows to diaphragm (lungs), IVC (preload), aorta (afterload) and renal vein (oliguria), plus CO_2 absorption to blood.

Part B: Complications of retroperitoneoscopy

Introduction

- **Retroperitoneoscopy** is laparoscopic or robotic surgery in the retroperitoneum without entering the peritoneum. It gives direct hilar access and avoids bowel, but the space must be **created**.
- **Why different (why its complications differ):** small space with no natural cavity; space made by finger and **balloon dissection** (commercial or Gaur glove balloon) behind Gerota's fascia and in front of the psoas; fewer landmarks; dissection close to the lens; gas tracks along tissue planes; CO_2 (carbon dioxide) is absorbed through fat and disrupted lymphatics, so hypercarbia appears early if the port leaks.

- Minimally invasive renal surgery: about 15% complications, mostly postoperative. Learning curve about 20 cases.

Classification (by phase of surgery)

Phase	Complications
Access and space creation	Wrong plane, balloon rupture, peritoneal tear, muscle bleeding
Insufflation	Subcutaneous emphysema, hypercarbia, pneumothorax, gas embolism
Dissection	Vascular, visceral, nerve, thermal injury
Exit and recovery	Port bleeding, hernia, rhabdomyolysis, urinoma, lymphocele

1. Access and balloon-related

- **Balloon in muscle** (wrong plane, intramuscular) → postoperative **flank hernia** or bulge.
- **Too anterior** → peritoneal breach or colon injury; **too posterior** → bleeding from quadratus lumborum or psoas.
- **Balloon rupture**: latex glove balloon breaks into many fragments (silicone leaves one) that must be retrieved.
- **Peritoneal tear**: CO_2 enters the peritoneum and collapses the small working space. Manage (fix) by venting with a Veress needle or an extra port (to vent), enlarging the tear, or converting to transperitoneal.
- Disorientation: keep the **psoas horizontal** as the landmark.

2. Gas-related

- **Subcutaneous emphysema** (commonest; commoner than transperitoneal): crepitus over flank, chest and neck; pneumoscrotum. Cause: leak around the port. Prevent with a **balloon-tip (blunt-tip) trocar (port)**, tight incisions and 12 rather than 15 mmHg.
- **Hypercarbia**, especially with emphysema; a sealed port removes the difference from transperitoneal surgery.
- **Pneumothorax (1.6–4%, commoner retroperitoneally)**: pleura is near the upper pole and 12th rib. Signs: neck and chest emphysema; tension gives hypotension, absent breath sounds and high airway pressure. Treat with needle decompression (16-G needle in the second or third intercostal space, midclavicular line), then chest drain. Pneumomediastinum and pneumopericardium (about 0.8%) also occur.
- **Gas embolism** (gas into an opened renal vein [renal vein injury], IVC [inferior vena cava] or lumbar vein): sudden collapse; end-tidal CO_2 rises abruptly, saturation falls, then end-tidal CO_2 falls; "millwheel" (mill-wheel) murmur. Stop insufflation (gas), **left lateral, head down**, 100% oxygen, aspirate through a central line.

3. Vascular injury

- Commonest complication of minimally invasive urology; major injury <1 in 1000 but fatal in 8–17%. Upper-tract (retroperitoneal-organ) work carries about five times the pelvic vascular risk (commoner in upper-tract work).
- Sites: renal artery and vein, **lumbar and gonadal veins** (gonadal vessels), adrenal vein, IVC, aorta. In the small posterior space the IVC or aorta can be mistaken for hilar vessels.
- **Management of venous bleeding**: raise (increase) pressure temporarily to 25 mmHg, suction to see the site, compress / tamponade with a sponge, haemostatic sealant, extra 5-mm port for suturing or a Satinsky clamp, hand-assist, add ports, suture or clip, or convert if uncontrolled. Arterial: clip or staple; aortic injury needs conversion. Never clip blindly.

4. Visceral injury

- **Peritoneum and colon**; duodenum (right); spleen, pancreas (left); liver; pleura; ureter; adrenal; usually by thermal spread.

- **Thermal bowel injury** by four mechanisms: direct activation, direct coupling, capacitive coupling, insulation failure. It may present up to 18 days later with low-grade fever and pain; diagnose by CT (computed tomography) with oral contrast.

5. Positioning and nerve injury

- **Rhabdomyolysis** (about 1% retroperitoneal; 0.4% overall): men, BMI (body mass index) ≥25, kidney rest, full flexion, long cases. Brown urine, hip pain, CK (creatine kinase) >5,000 units. Prevent: limit kidney rest to <1 hour, cases <5 hours, avoid hypotension.
- **Nerves** (neuropraxia): brachial plexus (axillary roll), sciatic; in the field, **genitofemoral and lateral femoral cutaneous** on the psoas, ilioinguinal and iliohypogastric (thigh numbness, flank bulge).

6. Postoperative

- Bleeding or haematoma; port-site hernia or haematoma; wound infection.
- **Urinoma** (partial nephrectomy, pyeloplasty); **lymphocele** or chylous leak after hilar or node dissection.
- Delayed bowel injury; DVT (deep vein thrombosis).
- **Conversion** to transperitoneal or open surgery for bleeding, peritoneal tear or lost orientation.

Retroperitoneal versus transperitoneal

Retroperitoneal more	Retroperitoneal less
Emphysema, pneumothorax, hypercarbia, disorientation	Bowel injury, ileus, adhesions; peritoneum spared

Prevention

- Open (Hasson) access below the 12th rib tip; balloon wholly in the retroperitoneum; balloon-tip port; 12 mmHg; trocars under vision; respect the 12th rib and pleura; check haemostasis at low pressure.

Diagram to draw: flank cross-section showing balloon between psoas and Gerota's fascia, with peritoneum pushed medially and port positions (flank port positions for retroperitoneoscopy).

Part C: Laparoscopic donor nephrectomy

Introduction

- First described by **Ratner (1995)**; now the preferred approach (open under 2% in the United States). **Donor safety comes first** at every stage.

Donor evaluation

- Questionnaire → physical and psychosocial review, blood group, HLA (human leukocyte antigen) typing → **GFR by at least two methods, one measured** → metabolic, serology, cancer screening → **CT (computed tomography) angiography** for vessels → cardiac review if needed.
- Exclusions: kidney disease, malignancy, active infection, cardiovascular disease, diabetes, significant obesity; bilateral stones generally exclude.

Side selection

- **Left preferred:** longer renal vein, simpler dissection. **Right** if the left has multiple arteries or complex veins, or the right kidney is smaller: **the donor keeps the better kidney.**
- **Anatomy matters:** arterial variation in 25–40%; early (prehilar) branching and accessory arteries must be seen on CT, as they may contraindicate the laparoscopic route.

Techniques

Technique	Features
Pure transperitoneal	3–4 ports; colon reflected; Pfannenstiel extraction
Hand-assisted (about two-thirds)	Hand port doubles as extraction site; tactile feedback, quick haemostasis
Retroperitoneoscopic	No bowel mobilisation; direct hilum; less room
Robotic; LESS (laparoendoscopic single-site surgery)	Wristed instruments; single incision, less pain

Operative steps (left)

1. Reflect colon; divide **gonadal, adrenal and lumbar veins**.
2. Preserve periureteric tissue and the **gonadal–ureter triangle** (ureteric blood supply).
3. Keep pressure ≤10 mmHg; hydration and mannitol or furosemide to keep urine >100 mL/hour (counters pneumoperitoneum oliguria).
4. Divide ureter, then **artery first, then vein**. Vein stapled lateral to the adrenal vein; Endo-TA (non-cutting) stapler gives extra vein length.
5. Extract quickly; cold flush. Warm ischaemia a few minutes.

- **Hem-o-lok:** manufacturer contraindicates it on the donor renal artery after stump bleeds.
- **Open donor nephrectomy** is kept for complex anatomy or previous surgery, more often on the right.

Advantages versus open

- Less pain, shorter stay, faster return to work, better cosmesis; **more people volunteer to donate**; equal graft function.

Disadvantages and complications

- Longer operation, learning curve, shorter vessels (right vein), slightly longer warm ischaemia.
- Complications about **16%** (mostly gastrointestinal); Clavien ≥IV 2.5%; **mortality 3 per 10,000**; conversion 0.26–3.34%, mainly for vascular injury.

Long-term donor counselling

- **Also know:** kidney paired donation for ABO-incompatible or crossmatch-positive pairs; living-donor grafts survive longer than deceased-donor grafts.
- Function 60–80% of pre-donation; lifetime ESRD (end-stage renal disease) **90 per 10,000** versus 14 in matched non-donors (Muzaale, 2014); more gestational hypertension.

Diagram to draw: left renal hilum with gonadal, adrenal and lumbar veins, stapled artery and vein, and the gonadal–ureter triangle.

Key points to remember:

- Two drivers: pressure and absorbed CO_2; work at 10–12 mmHg.
- Afterload rises at all pressures; output collapses at 40 mmHg.
- Oliguria from renal vein compression; reversible.
- Hypercarbia peaks after extubation; COPD needs arterial gases.
- Gas embolism: left lateral, head down, oxygen, aspirate.
- Retroperitoneoscopy: space is created, not natural: wrong plane gives hernia, peritoneal tear or bleeding.
- Emphysema commonest; balloon-tip port and 12 mmHg prevent it (prevents leak).
- Pneumothorax 1.6–4%, commoner retroperitoneally.
- Rhabdomyolysis about 1%: limit kidney rest to <1 hour.
- Ratner 1995; left kidney preferred; hand-assisted in about two-thirds.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Donor ESRD 90 versus 14 per 10,000 (matched non-donors).

Describe IVU & its applications in Urology.

Type: LAQ (25 marks)

Source: 2021-Summer Paper 1, Q2

Definition and principle

- IVU (intravenous urography; also IVP, intravenous pyelography or excretory urography) is a series of timed abdominal radiographs after intravenous iodinated contrast.
- Contrast is **freely filtered by the glomerulus and not reabsorbed**. As water is reabsorbed it is concentrated in the tubules (**nephrogram**) and then outlines the calyces, pelvis, ureters and bladder (**pyelogram**).
- So IVU shows **anatomy plus a rough, qualitative idea of function and drainage** of each kidney.
- **Current status:** largely replaced by CTU (CT [computed tomography] urography). It is still used where CT is unavailable (common in India) or when dynamic excretion and drainage must be seen.

Contrast media

Type	Example	Feature
High-osmolar ionic	Diatrizoate	More reactions; obsolete for IV use
Low-osmolar non-ionic (standard)	Iohexol, iopamidol	Fewer reactions
Iso-osmolar non-ionic	Iodixanol	High-risk renal patients

Preparation

- Check creatinine or eGFR (estimated glomerular filtration rate), allergy, asthma, pregnancy, diabetes on metformin, myeloma.
- Laxative the night before (bowel gas hides the kidneys); short fast but **no dehydration**.
- Premedicate high-risk patients, e.g. prednisone 50 mg at 13, 7 and 1 h plus diphenhydramine 50 mg at 1 h.

Technique: film sequence

Film	Timing	Shows
Scout (control) KUB (kidney–ureter–bladder)	Before contrast	Calcification, bones, gas, technique
Contrast	Low-osmolar, about 1 mL/kg (usually 50–100 mL) IV	—
Nephrogram / nephrotomogram	About 1 min (60–90 s)	Renal size, outline, parenchyma
Pyelogram	5 min	Calyces; then apply **ureteric compression**
Compression film	10 min	Well-distended calyces and pelvis
Release (full-length) film	About 15 min	Whole ureters
Bladder and **post-void** films	20–30 min	Bladder outline, residue, lower ureters
Delayed films	Up to 24 h	Level of obstruction; **no excretion at 24 h = non-functioning kidney**

- **Also know:** films in expiration; prone film fills the ureters; oblique views; erect film for nephroptosis.

- Compression is avoided in suspected obstruction or colic, aortic aneurysm, recent abdominal surgery and trauma.

Contraindications

- Previous severe contrast reaction; significant renal impairment; pregnancy (relative); dehydration, myeloma.

Nephrogram patterns

Pattern	Causes
Absent	Renal artery occlusion, agenesis, avulsion
Delayed, increasingly dense	**Acute obstruction**, renal artery stenosis
Striated	Acute pyelonephritis, obstruction
Rim (cortical rim) sign	Renal infarction (capsular collaterals)
Bilateral persistent dense	Hypotension, ATN (acute tubular necrosis)

Applications in urology

Condition	IVU features
Calculi	Confirms an opacity is in the tract; **radiolucent stone = filling defect**; site, obstruction, calyceal anatomy before SWL (shock wave lithotripsy) or PCNL (percutaneous nephrolithotomy)
Acute obstruction	Delayed dense nephrogram, delayed excretion, dilatation down to the block, **standing column** of contrast, forniceal rupture
PUJ (pelviureteric junction) obstruction	Ballooned pelvis, delayed drainage, non-dilated ureter; crossing-vessel indentation
Duplex system	Upper moiety obstructed or non-functioning: lower moiety displaced, "**drooping lily**"
Ureterocele	"**Cobra head**" (spring onion) at the bladder
Horseshoe kidney	Low, medially rotated; lower calyces point medially
Retrocaval ureter	"Reverse J" or **fish-hook** at L3
Medullary sponge kidney	"**Paintbrush**" papillary striations
Papillary necrosis	**Ring sign**, "lobster claw" calyces
Genitourinary TB (tuberculosis)	**Moth-eaten** calyx, **phantom calyx**, Kerr's kink, **pipestem** or beaded ureter, contracted bladder, autonephrectomy
Urothelial tumour	Irregular filling defect, non-visualised calyx
Retroperitoneal fibrosis	Medial deviation and tapering of ureters with hydronephrosis
Pelvic lipomatosis	Pear-shaped bladder
Schistosomiasis	Ureteric striations, calcified bladder

- **Trauma: one-shot intraoperative IVP:** 2 mL/kg IV push, single film at 10 min. It confirms a functioning contralateral kidney before exploration in the unstable patient.
- **Also know:** nephroptosis (descent >5 cm or two vertebral bodies supine to erect); urinary fistulae (ureterovaginal); renal masses (distortion); post-operative assessment; diverticula.

Complications

- **Contrast reactions:** immediate (urticaria to anaphylaxis) or delayed (rash). Severe reaction: stop contrast, **IM (intramuscular) adrenaline 0.3–0.5 mg** every 5–15 min

(maximum 1 mg), oxygen, β2-agonist. Observe 30 min.
- **CA-AKI (contrast-associated acute kidney injury):** prevent with hydration, low or iso-osmolar contrast, lowest dose.
- Radiation; metformin-related lactic acidosis in renal impairment.

Limitations and current role

- Two-dimensional; bowel gas; poor parenchymal detail; misses small masses; poor in renal failure.
- CTU is more sensitive for stones, masses and urothelial lesions, but gives a higher radiation dose than IVU. IVU remains useful in resource-limited settings.

Recent advances

- Split-bolus CTU (fewer phases); dual-energy CT (virtual unenhanced images); MR urography without radiation.

Diagram to draw: normal IVU with minor and major calyces, pelvis, ureter and the three narrowings; beside it a "drooping lily".

> **Key points to remember:**
> - Contrast is filtered, not reabsorbed: nephrogram then pyelogram.
> - Delayed, increasingly dense nephrogram = acute obstruction; no excretion at 24 h = non-functioning.
> - Classic signs: drooping lily, cobra head, fish-hook, paintbrush, ring sign, phantom calyx.
> - Severe reaction: IM adrenaline 0.3–0.5 mg; CTU has replaced IVU.

Principles of tissue engineering and its use in urology

Type: LAQ (25 marks)

Source: 2023-Winter Paper 1, Q1 (also answered in 2022-Summer Paper 4, Q2; this answer stresses the principles)

Definition and rationale

- **Tissue engineering** applies engineering and life sciences to build biological substitutes that restore, maintain or improve tissue function (Langer and Vacanti, 1993).
- **Why in urology:** bowel in the urinary tract causes mucus, stones, metabolic acidosis, infection and cancer; grafts for long urethral defects are scarce. The aim is to replace "wrong tissue" with "right tissue".

Principles: the triad plus bioreactor and blood supply

```
Biopsy -> cells expanded in culture -> seeded on scaffold (+ signals)
       -> conditioned in bioreactor -> implanted -> host vascularises and remodels
```

1. Cells

Source	Comment
Autologous differentiated cells (urothelial and smooth muscle cells from a bladder biopsy)	No rejection; expand well; not usable if the organ is malignant
Adult stem cells: bone marrow or adipose mesenchymal, muscle-derived, urine-derived	Easy harvest; act largely by paracrine (growth factor) effects
Embryonic stem cells	Pluripotent; ethical issues, teratoma risk
iPSCs (induced pluripotent stem cells)	Adult cells reprogrammed; patient-specific; tumour risk

2. Scaffolds (biomaterials)

- **Requirements:** biocompatible; degrade at the rate new tissue forms; porous for cell entry and vessel ingrowth; mechanically suited (a compliant bladder, a non-collapsing tube).

Type	Examples	Feature
Natural polymers	Collagen, alginate, silk	Good cell recognition; weak
Acellular tissue matrices	Bladder acellular matrix, SIS (small intestinal submucosa)	Native architecture and stored growth factors
Synthetic polymers	PGA (polyglycolic acid), PLA (polylactic acid), PLGA (their copolymer)	Controlled shape and degradation; no bioactivity
Newer	Electrospun nanofibres, hydrogels, bio-ink	Mimic extracellular matrix

- **Seeding:** urothelial cells on the luminal side and smooth muscle cells on the outer side, copying native layers.
- **Unseeded** matrices rely on host cells migrating in, so they work only for short onlay defects (about 0.5 cm for tubes in animal work). Longer or tubular gaps need **seeded** scaffolds, or they fibrose and contract.

3. Signals

- Growth factors (for example VEGF, vascular endothelial growth factor, for vessels) released from the scaffold, extracellular-matrix cues and mechanical stimuli direct cell growth and differentiation.

4. Bioreactors

- Culture chambers that perfuse nutrients and oxygen through thick constructs and apply **physiological conditioning**, such as cyclic stretch and filling for bladder smooth muscle, so the tissue is functional before implantation.
- Types: static culture, spinner flask, rotating wall, perfusion and mechanical-stretch bioreactors. They also allow monitoring and sterile, standardised production.

5. Vascularisation and innervation

- Oxygen diffuses only about 100–200 µm, so thick constructs die centrally without vessels. This is the main barrier.
- Strategies: **omental wrap** (used by Atala), angiogenic factors, co-seeding endothelial cells, built-in channels by bioprinting.
- Innervation and coordinated contraction cannot yet be recreated.

Uses in urology

Organ	Status
Bladder	**Atala et al. (2006):** seeded collagen–PGA constructs with omental wrap in 7 myelomeningocele patients; compliance improved. Industry phase II trial: **no significant gain in compliance or capacity**. Experimental (Chapter 48).
Urinary conduit	Neo-Urinary Conduit trial (NCT01087687): survived about 250 days on average; **half developed stenosis**; stiff material, no blood supply (Chapter 135)
Urethra	Acellular matrix onlay for short strictures and hypospadias; **Raya-Rivera et al. (2011):** seeded tubular PGA–PLGA urethras in **5 boys**, functional up to 6 years; tissue-engineered oral mucosa
Sphincter	Autologous muscle-derived cells for SUI (stress urinary incontinence); biodegradable scaffold slings
Vagina	Engineered vaginas in MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome (Raya-Rivera, 2014)
Penis	Corporal tissue and tunica grafts (stem cell-seeded SIS in Peyronie disease): animal or experimental; stem cells for ED (erectile dysfunction) investigational
Kidney	Decellularised–recellularised scaffolds (animal), kidney organoids, renal assist devices
Testis	Spermatogonial stem cell banking in prepubertal cancer patients

Recent advances

- **3D bioprinting:** cells in hydrogel "bio-ink" laid layer by layer from CT (computed tomography) or MRI (magnetic resonance imaging) data; vascular channels built in. Printed (non-living) models already aid planning.
- **Organoids:** kidney organoids from iPSCs; bladder and prostate tumour organoids for drug testing.
- Robotic vascularised composite bladder allograft (Chapter 135): a transplant route that avoids bowel but needs immunosuppression.

Challenges

- Vascularity, innervation, fibrosis and contracture; scale-up, cost, culture time, regulation; stem cell tumour risk; promising pilots followed by disappointing controlled trials.
- **Also know:** gene-modified cells; growth factor-releasing "smart" scaffolds; cell sheet engineering.

Diagram to draw: biopsy → culture → seeding on scaffold → bioreactor → implant with omental wrap.

Key points to remember:

- Triad: cells, scaffold, signals; plus bioreactor and blood supply.
- Diffusion limit about 100–200 µm: vascularity is the main barrier.
- Atala 2006 promising; phase II trial showed no gain; conduit half stenosed.
- Raya-Rivera 2011: seeded urethras in 5 boys.

Radionuclide (isotope) scans in urology: principles, renal scans, MAG3 and applications

Asked as:

- Radionuclide scans and its applications in uro — *Practice Set 13 Paper I Q1 (LAQ)*
- Isotope studies of urology — *Practice Set 2 Paper I Q3d (SAQ)*
- Renal isotope scans — *Practice Set 2 Paper I Q3e (SAQ)*
- MAG 3 — *Practice Set 7 Paper I Q3b (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(Practice Set 13: also answered in Practice Set 2 Paper 1, Q3d–Q3f. Practice Set 2 Q3d: renal scans in depth: see Practice Set 2 Paper 1, Q3e; diuretic renography: see Practice Set 2 Paper 1, Q3f. Practice Set 2 Q3e: also answered in 2017-Winter Paper 2, Q3f and 2021-Summer Paper 1, Q3e; diuretic protocol: see Practice Set 2 Paper 1, Q3f)

"Uro" is read as urology.

MAG 3 — assumed meaning: Tc-99m MAG3 (mercaptoacetyltriglycine) renal scan: principle, technique, interpretation and uses.

Principles

- Nuclear medicine shows **function, not structure**. A tracer, given in a tiny (subpharmacological) dose, follows a physiological or molecular process; the patient is the radiation source.
- **Gamma camera** (sodium iodide crystal, photomultipliers, collimator): photon, collimator (admits photons from one direction), sodium iodide crystal (light), photomultiplier tubes, image. Planar (about 20 mm resolution) or **SPECT (single-photon emission computed tomography)** with CT (about 10 mm).
- **PET (positron emission tomography):** the positron annihilates with an electron (positron annihilation), giving two 511 keV photons 180° apart, detected in coincidence. This rejects noise (less noise) and allows true quantitation (quantitative SUV, standardised uptake value); sharper (about 5 mm).

	Planar	SPECT (single-photon emission computed tomography)	PET
Image	2D	3D, with CT (computed tomography)	3D, with CT or MRI (magnetic resonance imaging)
Resolution	About 20 mm	About 10 mm	About 5 mm
Quantitation	Semiquantitative	Semiquantitative	SUV
Main use	Renal function	Problem-solving	Cancer

- **^{99m}Tc (technetium-99m):** 6-hour half-life, single 140 keV photon: the workhorse. **^{18}F:** about 110 minutes, lowest positron energy, so the sharpest PET image. ^{68}Ga comes from a generator; ^{11}C (20 minutes) needs an on-site cyclotron.

Applications (overview)

Area	Tracer	Use
Kidney	^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid), -MAG3 (mercaptoacetyltriglycine), -DMSA (dimercaptosuccinic acid)	GFR (glomerular filtration rate), split function, obstruction, scars, transplant

Area	Tracer	Use
Bone	^{99m}Tc-MDP (methylene diphosphonate)	High-risk prostate cancer staging; low specificity
Prostate	**PSMA (prostate-specific membrane antigen) PET**; choline; fluciclovine	PSMA preferred (NCCN [National Comprehensive Cancer Network]) for staging and recurrence; specificity >90%
Therapy	^{177}Lu-PSMA-617; radium-223	mCRPC (metastatic castration-resistant prostate cancer); VISION trial (2021)
FDG (fluorodeoxyglucose) PET	Glucose analogue trapped in cells	**Residual seminoma >3 cm**; penile cancer with palpable nodes; distant urothelial disease. Poor in prostate and clear cell RCC (renal cell carcinoma)
Renal mass	Sestamibi SPECT; girentuximab PET	Oncocytoma "hot" on sestamibi; clear cell RCC "hot" on girentuximab
Adrenal	^{123}I-MIBG (metaiodobenzylguanidine), ^{68}Ga-DOTATATE, FDG	Metastatic or extra-adrenal phaeochromocytoma; MIBG best in MEN-2 (multiple endocrine neoplasia type 2)
Sentinel node	^{99m}Tc-nanocolloid plus blue dye	Penile cancer, impalpable nodes
Scrotum	^{99m}Tc-pertechnetate	Torsion cold, epididymitis hot
Reflux	Direct radionuclide cystography	VUR (vesicoureteral reflux) follow-up: more sensitive, lower dose than VCUG (voiding cystourethrogram)
Infection	^{111}In-labelled leucocytes; ^{67}Ga	Occult abscess; not in transplant

Renal tracers

- Each tracer images the part of the nephron that handles it. Plasma clearance is about 20% filtration and 80% tubular secretion.

Feature	^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid)	^{99m}Tc-MAG3 (mercaptoacetyltriglycine)	^{99m}Tc-DMSA (dimercaptosuccinic acid)
Handling	Glomerular filtration only	Protein-bound; tubular secretion	Filtered, then retained by (in) proximal tubules (megalin, cubilin); tubular cortical binding
Study	Dynamic	Dynamic	**Static cortical**, at 2 hours (2 h)
Main use	**GFR (glomerular filtration rate)**; captopril study; renovascular	**Split function, obstruction (first choice)**, drainage; transplant	**Pyelonephritis and scars**
Renal failure	Poor images in renal failure	Good images in renal failure	Delay imaging
Dose	5–10 mCi	5–10 mCi	5 mCi adult; 50 µCi/kg child

MAG3: principle

- **^{99m}Tc-MAG3:** protein-bound tracer cleared mainly by **proximal tubular secretion**. Extraction is nearly **60%** per pass, versus about 20% for DTPA (diethylenetriaminepentaacetic acid).
- So it gives **good images even with poor GFR (glomerular filtration rate)**, lower background and lower cost. **Agent of choice for split function and obstruction.**

Dynamic renography (MAG3): method and interpretation

- **Preparation:** hydrate (**hydration**); (**bladder**) catheter for outlet obstruction or neurogenic bladder, and in children; cap nephrostomy. Dose 185–370 MBq (5–10 mCi) IV (intravenous) bolus (intravenously) in adults.
- **Perfusion phase** (first minute): 1–2 second frames (frames every 1–2 seconds); the kidney appears within seconds of the aorta (renal curve follows aortic upstroke).
- **Function (uptake) phase:** regions of interest with background subtraction. **Split function (DRF, differential renal function) = area under each curve at 1–3 minutes**; normal up to 45:55 (**normal 45–55%**). Normal (MAG3) peak at 3–5 minutes (MAG3 peaks at 3–5 minutes); bladder by 10–15 minutes.
- **Drainage phase:** a rising curve needs furosemide (see Q3f). Upright or post-void images test gravity drainage.
- **GFR:** DTPA camera method (Gates) or plasma-sample clearance.
- **Diuretic renography ("well-tempered" diuretic renogram):** dilatation is not obstruction. Furosemide 0.5 mg/kg (adult maximum 40 mg) in adults; 1 mg/kg in children, given when the system is full; separates a patulous system from obstruction. Timing varies (F+20, F+15, F0, F−15) and must be stated, so the report must state it.

T½ (half-clearance time) after furosemide	Meaning
<10 minutes (min)	Not obstructed
10–20 minutes (min)	Indeterminate
>20 minutes (min)	Obstructed

- **Curve types (O'Reilly):** I normal; II rising (obstructed); IIIa dilated, washes out with diuretic; IIIb equivocal.

MAG3 uses

- PUJ (pelviureteric junction) obstruction and megaureter: **surgery if DRF <40% with poor drainage or falling DRF**; follow-up after pyeloplasty.
- Transplant: acute rejection (perfusion falls) versus delayed graft function (perfusion kept); urine leak.
- **Also know:** indirect cystography for VUR (vesicoureteric reflux); split function before nephrectomy.

MAG3 pitfalls

- Dehydration, full bladder, wrong ROI (region of interest), mistimed diuretic, massive dilatation and neonatal immaturity mimic obstruction. **Trend on serial scans** matters.

Other renal applications (special studies)

- **Captopril renography (DTPA):** the stenotic kidney keeps GFR by angiotensin II efferent constriction. Captopril removes this (it), so uptake falls. Repeat off ACE (angiotensin-converting enzyme) inhibitors: improvement **≥10%** off ACE inhibitors means renovascular hypertension is likely. Largely replaced by CT (computed tomography) angiography.

- **Transplant (MAG3):** acute rejection: perfusion falls and worsens; delayed graft function: perfusion kept, improves. Detects (shows) urine leak without iodinated contrast.
- **DMSA (cortical scan):** pyelonephritis gives an ill-defined defect that resolves by about 6 months; a scar is sharp, with cortical thinning, and persists.

	Acute pyelonephritis	Scar
Defect	Ill-defined	Sharp, cortical thinning
Kidney	Normal size	Often small
At 6 months	Resolves	Persists

- **Radionuclide cystography:** direct (tracer instilled into the bladder) or indirect (after MAG3). More sensitive and lower dose than VCUG (voiding cystourethrogram), but little anatomy. So VCUG first; nuclear cystography for follow-up.

Infection imaging

- **^{111}In-oxine leucocytes** (image at 24 hours): best for occult abdominal or genitourinary infection. ^{99m}Tc-HMPAO (hexamethylpropyleneamine oxime) leucocytes give better images, but urinary excretion obscures the tract. Neither in a transplant (leucocytes home to the graft).
- **Also know:** ^{67}Ga-citrate (non-specific, slow).

Oncology

- **FDG (^{18}F-fluorodeoxyglucose):** a glucose analogue phosphorylated and trapped (Warburg effect). Urinary excretion limits it to distant disease.
  - **Seminoma residual mass >3 cm** after chemotherapy (EAU [European Association of Urology]). Not for NSGCT (non-seminomatous germ cell tumour), because teratoma is not FDG-avid.
  - Penile cancer with palpable nodes; equivocal bladder or upper-tract metastases.
  - Poor in prostate cancer and clear cell RCC (renal cell carcinoma).
- **Prostate: PSMA (prostate-specific membrane antigen) PET** is preferred (NCCN [National Comprehensive Cancer Network]) for high-risk staging and biochemical recurrence. Specificity >90%; nodal sensitivity only 30–40%. Calais trial (2019): detection 56% (PSMA) versus 26% (fluciclovine). Bone scan (^{99m}Tc-MDP, methylene diphosphonate) remains conventional.
- **Renal mass: sestamibi SPECT** "hot" in oncocytoma; **girentuximab** (anti-CAIX, carbonic anhydrase IX) PET "hot" in clear cell RCC (ZIRCON trial).
- **Also know:** ^{123}I-MIBG (metaiodobenzylguanidine) and ^{68}Ga-DOTATATE for phaeochromocytoma; ^{99m}Tc-nanocolloid sentinel node biopsy in penile cancer; ^{99m}Tc-pertechnetate scrotal scan (torsion cold, epididymitis hot) when ultrasound is equivocal.

Therapy (theranostics)

- **^{177}Lu-PSMA-617:** beta emitter for mCRPC (metastatic castration-resistant prostate cancer) after ARPI (androgen receptor pathway inhibitor) and taxane: **VISION (2021)**. PSMA PET confirms the target first.
- **Radium-223:** alpha emitter for bone-only mCRPC.

Advantages and limitations

- Quantitative function, low radiation, safe in contrast allergy and pregnancy (MAG3 fetal dose <5 mGy).
- Poor anatomy; PET is costly; results depend on technique.

Diagram to draw: MAG3 renogram: perfusion spike, peak at 3–5 minutes, washout, 1–3 minute split window; obstructed curve rising; renogram curves types I, II, IIIa and IIIb. Gamma camera chain: patient, collimator, crystal, photomultiplier, image.

Key points to remember:

- ^{99m}Tc: 6 hours, 140 keV; PET: two 511 keV photons.
- DTPA filtered = GFR; MAG3 secreted = obstruction and transplant; DMSA cortical = scars.
- MAG3: tubular secretion, about 60% extraction; best for split function and drainage.
- Split function (DRF) from 1–3 minute counts; normal 45–55%.
- T½ >20 minutes (min) obstructed, <10 min not.
- Hydration, catheter, correct diuretic timing: the well-tempered renogram.
- Captopril renogram: ≥10% improvement off ACE inhibitors.
- Scar persists at 6 months; pyelonephritis resolves.
- PSMA PET preferred; FDG for residual seminoma >3 cm.
- MIBG best in MEN-2; sentinel node with ^{99m}Tc-nanocolloid.
- Sestamibi hot = oncocytoma; ^{177}Lu-PSMA-617 (VISION).

Section 1 · Basic Principles — Short answer questions (SAQ)

Newer generation CT scan

Type: SAQ (10 marks)

Source: 2015-Summer Paper 1, Q3a

Introduction

- Computed tomography (CT) is the reference imaging for stones, masses and trauma. Attenuation is expressed in Hounsfield units (HU): water 0, air −1000.

Evolution

Generation	Feature
Helical (~1989)	Continuous table movement, volumetric data
Multidetector (1998 → 320-row)	Sub-millimetre slices, 3D reconstruction
Dual-source/dual-energy (2005)	Two energies: material analysis
Photon-counting (US approval 2021)	Energy-sorts each photon: less noise, dose and metal artefact

Dual-energy CT (DECT)

- Two energies (e.g. 80 and 140 kVp) compared pixel by pixel: **uric acid vs non-uric-acid stones**, **virtual unenhanced** images (one phase fewer) and **iodine maps** (hyperdense cyst vs enhancing mass).

Low-dose CT

Protocol	Dose
Conventional abdomen/pelvis	10–20 mSv
Low-dose	<3–4 mSv
Ultra-low-dose	1–2 mSv

- Stone sensitivity stays ~95%; use standard dose if body mass index >30. EAU (European Association of Urology): ultrasound first; AUA (American Urological Association): CT first.

Reconstruction

- Filtered back projection is noisy; iterative reconstruction halves dose; **deep learning reconstruction** gives good images at ~1 mSv.

Contrast phases and protocols

Phase	Timing	Use
Arterial	20–30 s	Renal arteries, donor anatomy, active bleeding
Corticomedullary	30–70 s	Hypervascular (clear cell) tumours, renal vein
Nephrographic	90–120 s	Most sensitive for renal masses
Excretory	7–10 min	Collecting system, ureter, urine leak

- **CT urography:** non-contrast, nephrographic and excretory phases (~20–30 mSv). **Split-bolus** merges phases, lowering dose but may miss subtle urothelial lesions.

- **Renal mass:** enhancement ≥20 HU = solid, 10–20 equivocal, <10 cyst; Bosniak 2019 for cysts.
- **Adrenal:** <10 HU unenhanced = lipid-rich adenoma; otherwise 15-minute washout (absolute >60% or relative >40% = adenoma).
- **Also know:** stone HU (uric acid 200–400, calcium oxalate 600–1200; indinavir invisible); bone window for fragments; partial volume causes pseudoenhancement of small cysts; ALARA (as low as reasonably achievable) dosing, since radiation risk has no threshold.

Diagram to draw: Timeline: axial → helical → multidetector → dual-source → photon-counting.

Key points to remember:

- Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
- DECT: uric acid stones, virtual unenhanced, iodine maps.
- Enhancement ≥20 HU = solid; deep learning enables ~1 mSv CT.

History of Laparoscopy

Type: SAQ (10 marks)

Source: 2015-Summer Paper 1, Q3b

Introduction

- Laparoscopy is endoscopic viewing of the peritoneal cavity through a pneumoperitoneum; it grew from optics to robotic surgery.

Optics and access

Year	Pioneer	Milestone
1806	**Bozzini**	**Lichtleiter**, candle-lit light conductor
1877	Nitze	Distal electric light and lenses
1901	**Kelling**	Insufflated abdomen of a **dog**
1910	**Jacobaeus**	First human series; "laparoscopy"
1938	**Veress**	Spring-loaded needle
1966	**Hopkins**	**Rod-lens system**
1970s–80s	**Semm**	Automatic insufflator; operative laparoscopy
1970s	Hasson	Open access
1987	Mouret	Video cholecystectomy

- **Also know:** Desormeaux 1853, first therapeutic endoscopy; fibreoptics (Tyndall's principle) for flexible scopes; CO_2 became the standard insufflant.

Urological milestones

Year	Pioneer	Milestone
1976	**Cortesi**	Impalpable testis, first urological use
1990–91	**Clayman**	**First laparoscopic nephrectomy**; nephroureterectomy 1991
1992	**Gaur (India)**	**Balloon retroperitoneal access**
1995	**Ratner**	Live-donor nephrectomy
1997	Schuessler	Laparoscopic radical prostatectomy (slow, not adopted)
2000	Guillonneau & Vallancien	Standardised prostatectomy
2000	**da Vinci**; **Binder**	Robot cleared in US; first robotic prostatectomy
2000s	**Menon**	Vattikuti Institute Prostatectomy
2007	Raman; Rane	Single-port nephrectomy

Why the robot spread

- Laparoscopic prostatectomy stalled because intracorporeal suturing was hard. The robot gave wristed instruments and a 3D magnified view.
- da Vinci generations: S 2006 (fourth arm), Si 2009 (dual console), Xi 2014 (flexible docking), SP 2018 (single port).

Indian contribution and recent advances

- Gaur's balloon made retroperitoneoscopy practical; indigenous robot SSI Mantra with telesurgery.
- New robots (Versius, Hugo, Hinotori), telesurgery, haptic feedback.

Diagram to draw: Timeline 1806 → 1901 → 1910 → 1938 → 1990 → 2000.

Key points to remember:

- Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
- Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
- da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.

Foley's (urethral) catheters, guidewires and DJ stents with their complications

Asked as:

- Foley's catheter. — *Summer 2016 Paper I Q3e (SAQ)*
- Urethral catheters, guidewires and DJ stents — *Practice Set 18 SAQ Q3f (SAQ)* (also answered in part in 2016-Summer Paper 1, Q3e and 2015-Summer Paper 3, Q1)
- Complications if DJS — *Summer 2021 Paper I Q3c (SAQ) — Taken as complications of a DJ (double-J) ureteric stent.*

Type: SAQ (10 marks)

Foley's / urethral catheters

History

- **Frederic Foley**, a Boston urologist, developed the modern balloon catheter (Campbell: 1929; published in the 1930s) for **haemostasis after prostatectomy**. Reybard (1855) had made the first self-retaining one.

Design

- **Size: Fr (French) = 3 × diameter in mm.** Adults (adult) usually 14–16 Fr; clot retention 20–24 Fr three-way. Newborn 6 Fr.
- Normal calibre: (male) meatus 24 Fr (narrowest point), bladder neck 28 Fr, prostatic urethra 32 Fr, female urethra 22 Fr.
- **Balloon:** usually 10 mL; larger (30 mL) for traction. Use (inflate balloon with) **sterile water**, only after urine flows at the hub; volume is printed on the port.
- **Lumens:** one (intermittent, specimen); two-way (drainage + balloon) is routine; three-way adds irrigation for haematuria (stiff wall resists collapse). Each lumen reduces (shrinks) drainage size.
- **Tips:** straight; **coudé** (curved, for prostatic enlargement, tip up); **Councill** (open tip, over a wire).
- **Materials:** latex (cheap, allergy); silicone (long term, less urethritis and encrustation; better urethral tolerance); PVC (polyvinyl chloride); coatings (silver, antibiotic) reduce bacteriuria but not clinical infection.
- **Special:** Malecot (winged, no balloon); Lapides diagnostic (rings 1 cm apart); urodynamic.

Indications and contraindications

- **Indications:** retention, clot retention, output monitoring, long surgery, immobility, after urological surgery.
- **Absolute contraindication:** suspected urethral injury (blood at meatus, pelvic fracture). Do an RUG (retrograde urethrogram) first (**suspected urethral injury → RUG first**).
- **Relative:** AUS (artificial urinary sphincter): deactivate, then 12–14 Fr; stricture.

Technique

- Aseptic; 10–15 mL lignocaine gel. In men, advance **to the hub** and see urine before inflating. Replace the foreskin.
- **Difficult:** 18 Fr coudé → guidewire → cystoscopy, Councill over wire → suprapubic catheter.

Complications

- **CAUTI (catheter-associated urinary tract infection):** 75% of hospital UTIs; bacteriuria 5–10% per catheter-day; **biofilm** forms within minutes.
- Trauma (about 0.3% of inpatients): false passage, balloon inflated in the urethra. Also stricture, traumatic hypospadias, encrustation, paraphimosis, non-deflating balloon

(puncture with a wire).

CAUTI prevention

- Review daily and remove early; closed dependent system; clean meatus with water; culture only if symptomatic. Treat only symptomatic infection and **change the catheter**.

Guidewires

Feature	Details
Uses	Access, safety wire, straightening ureter, guiding dilation and stents
Construction	Stainless-steel core with outer wrap; **nitinol** core is kink-resistant
Coating	PTFE (polytetrafluoroethylene): standard safety wire. **Hydrophilic** "glide": passes impacted stones, but too slippery to be a safety wire. Hybrid: hydrophilic tip, PTFE shaft
Size	0.018–0.038 inch (commonly 0.035/0.038); 80–260 cm, 145 cm most useful; floppy tip 1–3 cm
Tip and stiffness	Straight, angled, J; standard to super-stiff (straightens tortuous ureter)

DJ (double-J) stents: introduction

- The **double-J (double-pigtail)** stent (Finney 1978) is a self-retaining polymer tube with coils in the pelvis and bladder; adult usually 6 Fr, length by height. Consent should cover its symptoms and complications.
- **Materials:** polyurethane, silicone (hydrocoated silicone less painful); **metallic (Resonance)** for extrinsic malignant compression.
- **Indications:** obstruction (infected obstructed kidney, malignancy, pregnancy), after URS (ureteroscopy) with injury, pyeloplasty, transplant, leak.
- **Placement:** retrograde pyelogram → wire → stent over wire under fluoroscopy or vision.
- **Problems:** stent symptoms (bladder irritation and reflux) eased by α-blockers and antimuscarinics; migration; encrustation (forgotten stent); failure with extrinsic compression.

Classification of DJ stent complications

Timing	Complications
At insertion	**Misplacement**: proximal end below the obstruction; **submucosal** passage or perforation (suspect when force is needed)
Early	**Stent-related symptoms**; haematuria; infection and urosepsis; **vesicoureteric reflux** up the stent (flank pain on voiding)
Late	**Migration** (proximal or distal; short stent); **encrustation**; **stent failure** (obstruction); biofilm and colonisation; **fracture**; **forgotten stent**; rarely **ureteroarterial fistula**

Stent-related symptoms

- About **4 in 5 patients**: urgency, frequency, dysuria, haematuria, flank and groin pain; sexual and work problems.
- **Mechanism:** the **intravesical coil irritates the trigone**, and **reflux up the stent** distends the kidney on voiding.
- Measured by **USSQ** (Ureteral Stent Symptom Questionnaire).
- **Reduced by: α-blockers and antimuscarinics**; intraureteral placement; hydrocoated silicone; shortest dwell. Diameter does not matter.

Encrustation and forgotten stent

- Minerals and biofilm deposit on the stent. Risk rises with **dwell time**, infection, stone-forming urine and pregnancy. Leads to obstruction, sepsis, fracture, **ureteric avulsion on**

removal, renal loss.

- **FECal grading (Forgotten, Encrusted, Calcified):** I minimal linear; II circular; III partial on one or both coils; IV complete on one coil; V both coils.
- **Management:** non-contrast CT (computed tomography); never pull by force; clear **bladder coil** (cystolithotripsy), then **ureter** (ureteroscopy), then **renal coil** (flexible ureteroscopy or PCNL [percutaneous nephrolithotomy]).

Stent failure

- Extrinsic compression (malignancy, radiation), debris, encrustation. Obstructive symptoms → image; change stent, **PCN** (percutaneous nephrostomy) or **metallic stent**.

Prevention

- Stent only when indicated; correct length; **stent register**; long-term stents changed every 3–6 months (local practice); **pregnancy: about every 4 weeks**.

Diagram to draw: cross-section of a three-way Foley; DJ stent in situ with bladder coil on the trigone and arrows showing reflux.

Key points to remember:

- Foley balloon catheter: designed for post-prostatectomy haemostasis.
- Fr = 3 × diameter (mm); inflate with sterile water; coudé for prostatic enlargement, Councill over a wire.
- Suspected urethral injury → RUG first; AUS → deactivate, 12–14 Fr.
- CAUTI 5–10% per day; remove early; change the catheter when treating.
- Hydrophilic wire to pass, PTFE wire to stay (safety).
- DJ symptoms in about 80%: trigonal irritation and reflux; α-blockers and antimuscarinics help.
- Long-dwelling stents: monitor for encrustation; encrustation rises with dwell time; FECal I–V.
- Encrusted stent: bladder, then ureter, then kidney; keep a stent register.

Andrew Novick.

Type: SAQ (10 marks)

Source: 2016-Summer Paper 1, Q3f

Assumed meaning: Andrew C. Novick and the urological work linked to his name, mainly nephron-sparing surgery.

Who he was

- **Andrew C. Novick** was a **Cleveland Clinic** urologist and long-time chairman of its urology department. He was a world authority on renal surgery, an editor of *Campbell-Walsh Urology* (9th edition), and died in 2008.

Contributions

1. **Nephron-sparing surgery (NSS) / partial nephrectomy (PN):** he established its technique, indications and long-term cancer control. His group showed tumours ≤4 cm did as well with NSS as with radical nephrectomy, supporting the T1a/T1b split.
2. **Renovascular surgery:** aortorenal, splenorenal and hepatorenal bypass; **bench (ex vivo) surgery with autotransplantation** for complex lesions.
3. **Renal transplantation** surgery.

Indications for PN

Category	Situations
Imperative	Bilateral RCC (renal cell carcinoma); solitary functioning kidney
Relative	Contralateral kidney threatened (stones, stenosis, diabetes, hypertension); hereditary RCC
Elective	cT1a (≤4 cm) with a normal other kidney; now larger selected tumours

- **Relative contraindications:** <20% of nephron mass left; pedicle encasement; venous thrombus.

Principles of NSS

1. Three-dimensional CT (computed tomography) maps tumour, vessels and collecting system.
2. **Early hilar control.** Keep warm ischaemia under about 25–30 minutes; if longer, use **ice-slush hypothermia** and mannitol.
3. Excise with a **negative margin** (width immaterial); intraoperative ultrasound for endophytic tumours.
4. **Watertight collecting-system closure**, vessel ligation, capsular renorrhaphy.
5. Drain for urine leak.

- Function afterwards depends on parenchymal **quality and quantity preserved**; ischaemia matters less. At least 20–30% of one kidney is needed to avoid dialysis.

Legacy

- PN is now the AUA (American Urological Association) and EAU (European Association of Urology) standard for cT1a tumours.

Diagram to draw: PN steps: clamp, ice slush, excision, closure, renorrhaphy.

Key points to remember:

- Novick (Cleveland Clinic): nephron-sparing, renovascular and bench surgery.
- PN indications: imperative, relative, elective (cT1a ≤4 cm).
- Warm ischaemia under about 25–30 minutes; ice slush if longer.
- Negative margin is enough; keep ≥20–30% of one kidney.

Proteinuria

Asked as:

- Proteinuria — *Summer 2017 Paper I Q3f (SAQ)*
- Proteinuria — *Summer 2021 Paper I Q3d (SAQ)*

Type: SAQ (10 marks)
(also answered in 2017-Summer Paper 1, Q3f)

Definition

- Normal protein excretion is **<150 mg/day** (another source: **80–150 mg/day**) (albumin <30 mg/day). The filter (glomerulus) blocks (holds back) albumin and the proximal tubule reabsorbs filtered small proteins.
- Normal mix: about 30% albumin, 30% globulins, 40% tissue proteins (Tamm–Horsfall).
- **Nephrotic range:** >3.5 g/day, with hypoalbuminaemia, oedema and hyperlipidaemia.

Detection and quantification (KDIGO, Kidney Disease: Improving Global Outcomes, categories by ACR)

Category	ACR (albumin-to-creatinine ratio)	PCR (protein-to-creatinine ratio)	24-h protein
A1 normal–mild	<30 mg/g (<3 mg/mmol)	<150 mg/g (<15 mg/mmol)	<150 mg
A2 moderate	30–300 mg/g	150–500 mg/g	150–500 mg
A3 severe	>300 mg/g (>30 mg/mmol)	>500 mg/g	>500 mg

- **Spot early-morning ACR is first line**; PCR if non-albumin protein suspected; 24-h urine with creatinine check. Alternatives: PCR, 24-h urine.
- **Dipstick** (tetrabromophenol blue) detects albumin only (mainly albumin), from 20–30 mg/dL; misses dilute urine and **light chains**. **Sulfosalicylic acid** detects (precipitates) all proteins; dipstick negative but sulfosalicylic acid positive → suspect myeloma.

Types

Type	Mechanism	Amount	Example
Glomerular (commonest)	Leaky filter	Often >2 g; albumin ≈70% (about 70%)	Diabetes, glomerulonephritis, FSGS (focal segmental glomerulosclerosis), amyloid
Tubular	Failed reabsorption of small proteins	<2 g (seldom >2–3 g); mainly globulins	Fanconi syndrome, heavy metals, drugs
Overflow	Excess plasma protein	Variable (0.3–2 g)	**Myeloma** (Bence Jones), myoglobin, haemoglobin

By timing	Features
Transient	Fever, exercise, heart failure; commonest in children
Orthostatic	Young men, upright only, <1 g; benign (normal first-morning sample). Test with a split (supine overnight vs upright day) collection
Persistent	Mostly glomerular; needs full work-up

Evaluation

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Positive dipstick → repeat; exclude UTI (urinary tract infection), fever, exercise
→ persistent → quantify (ACR, PCR or 24-h protein)
→ creatinine/eGFR (estimated glomerular filtration rate), sediment (dysmorphic red cells /
RBCs, casts), ultrasound
→ glomerular pattern (>2 g, albumin) → nephrology ± renal biopsy; C3, C4, ANA if systemic
→ globulins → electrophoresis, immunofixation (myeloma)
```

- UTI: urinary tract infection; eGFR: estimated glomerular filtration rate; RBC: red blood cell; ANA: antinuclear antibody.

Urological relevance

- **Haematuria work-up:** proteinuria, casts or dysmorphic cells suggest (suggests) medical renal disease; urological bleeding rarely raises protein much (adds little protein).
- **Reflux nephropathy:** proteinuria marks progression to renal failure.
- **After nephrectomy or partial nephrectomy:** >50% loss of mass causes hyperfiltration, proteinuria and FSGS. AUA (American Urological Association) 2021 counts proteinuria as a reason to prefer (favour) nephron sparing and refer to nephrology.
- **Living donors:** significant proteinuria excludes donation. **Also know:** mTOR (mammalian target of rapamycin) inhibitors cause it.

Management

- Treat the cause; blood pressure control; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)**; SGLT2 (sodium–glucose cotransporter-2) inhibitors.

Diagram to draw: glomerulus and proximal tubule marking glomerular, tubular and overflow sites.

Key points to remember:

- Normal <150 mg/day; nephrotic >3.5 g/day.
- ACR A2 30–300, A3 >300 mg/g.
- Glomerular, tubular, overflow; orthostatic proteinuria is benign.
- Dipstick misses light chains; use sulfosalicylic acid or immunofixation.

Acetylcysteine and Contrast Induced Nephropathy (CIN)

Asked as:

- Acetylcysteine. — *Summer 2018 Paper I Q3b (SAQ)*
- Acetylcysteine — *Summer 2019 Paper I Q3e (SAQ)*
- Contrast Induced Nephropathy (CIN) — *Winter 2017 Paper I Q3d (SAQ)*

Type: SAQ (10 marks)

Answered in 2018-Summer Paper 1, Q3b; core answer repeated here. (Summer 2019 Paper I Q3e)

Acetylcysteine

Introduction

- **NAC (N-acetylcysteine):** acetylated form of the amino acid cysteine (acetylated cysteine); a **thiol (sulfhydryl, –SH) compound**. Given orally, intravenously or by nebuliser (oral, intravenous or nebulised).

Mechanisms of action

- **Glutathione precursor:** supplies cysteine, the rate-limiting substrate for **glutathione** synthesis.
- **Antioxidant:** the free –SH group scavenges reactive oxygen species.
- **Mucolytic:** –SH breaks **disulfide bonds** in mucoproteins, lowering mucus viscosity.
- **Thiol–disulfide exchange:** theoretically (in theory) forms a soluble mixed disulfide with cystine.

Clinical uses

1. Paracetamol (acetaminophen) overdose: main indication

- Overdose saturates normal pathways; the toxic metabolite **NAPQI** (N-acetyl-p-benzoquinone imine) accumulates (builds up) once glutathione is used up (exhausted), causing hepatic (liver) necrosis.
- NAC restores glutathione. It is most effective **within 8 hours**, but still worth giving later (still given later).
- **IV (intravenous) 21-hour regimen:** 150 mg/kg, then 50 mg/kg over 4 h, then 100 mg/kg over 16 h. Shorter two-bag regimens are now used in some countries.

2. Mucolytic: COPD (chronic obstructive pulmonary disease), cystic fibrosis.

Urological relevance

- **Contrast-induced AKI (acute kidney injury):** once widely used (early small trials, KDIGO (Kidney Disease: Improving Global Outcomes) 2012 suggestion). **PRESERVE (2018):** NAC gave no benefit over saline. It is **no longer recommended**; isotonic saline hydration is the only proven measure.
- **Cystinuria:** thiol action is theoretical; standard thiols are **tiopronin** and D-penicillamine, not NAC.
- **Cyclophosphamide cystitis: mesna** (also a thiol; another thiol) is the agent used.
- **Mucus after enterocystoplasty or neobladder:** sometimes used orally or in irrigation; evidence weak.
- **Male infertility:** antioxidant for oxidative stress (high ROS (reactive oxygen species), sperm DNA fragmentation); limited evidence.
- **Ischaemia–reperfusion** (renal surgery, transplant): antioxidant protection studied; not established.

Adverse effects

- **Anaphylactoid reactions** with IV loading (flushing, rash, bronchospasm): slow the infusion, give antihistamine.
- Nausea, vomiting; unpleasant sulphur smell; bronchospasm when nebulised in asthma.
- Safe in pregnancy and renal impairment for paracetamol poisoning; benefit outweighs risk.

Diagram to draw: Paracetamol → NAPQI → glutathione conjugation, with NAC replenishing glutathione.

Contrast Induced Nephropathy (CIN)

Definitions

- **CIN (contrast-induced nephropathy)** (older term): creatinine rise ≥25% or ≥0.5 mg/dL within 48–72 hours of iodinated contrast, with no other cause.
- Current terms (ACR (American College of Radiology)):
  - **CA-AKI (contrast-associated acute kidney injury):** AKI within 48 hours of contrast, whatever the cause.
  - **CI-AKI (contrast-induced AKI):** the subset actually caused by contrast.
- **KDIGO (Kidney Disease: Improving Global Outcomes) AKI:** creatinine rise ≥0.3 mg/dL in 48 hours, ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for 6 hours.

Pathophysiology

- **Renal vasoconstriction** causes **medullary hypoxia** and ischaemic tubular injury, plus **direct tubular toxicity**. Usually transient; dialysis rarely needed.

Risk factors

- **Patient:** CKD (chronic kidney disease, strongest), diabetes, age, dehydration, heart failure, anaemia, NSAIDs (non-steroidal anti-inflammatory drugs).
- **Procedure:** high volume, repeated doses within 48–72 hours, intra-arterial route, high-osmolar agents.
- **eGFR (estimated glomerular filtration rate) thresholds:** real risk mainly at **eGFR <30 mL/min/1.73 m²** (IV (intravenous) contrast); ESUR (European Society of Urogenital Radiology) uses **<45** for intra-arterial first-pass exposure.
- Older uncontrolled studies overstated the risk; ACR: rare but real.

Prevention

- Consider ultrasound or non-contrast CT (computed tomography) instead.
- Lowest dose of **low- or iso-osmolar** agent; low kVp (80); avoid repeat doses within 48 hours; stop NSAIDs.
- **IV isotonic saline** (e.g. 1 mL/kg/h before and after) for high-risk patients (eGFR <30 not on dialysis): the only proven measure.
- **PRESERVE (2018):** sodium bicarbonate and **NAC (N-acetylcysteine)** gave no benefit over saline.
- **AMACING (2017):** no prophylaxis was non-inferior to hydration at eGFR 30–59.
- Prophylactic dialysis does not help.

Metformin

- Risk is lactic acidosis if AKI develops. **eGFR ≥30 and no AKI: continue.** eGFR <30 or AKI: **withhold for 48 hours**, restart when creatinine is stable.

Diagram to draw: contrast → vasoconstriction + tubular toxicity → medullary hypoxia → AKI.

Key points to remember:

- Thiol; glutathione precursor, antioxidant, mucolytic.
- Antidote for paracetamol overdose; best within 8 h.
- PRESERVE 2018: no benefit in contrast AKI.
- Cystinuria uses tiopronin, not NAC.
- CIN = rise ≥25% or 0.5 mg/dL in 48–72 h; now CA-AKI vs CI-AKI.
- Main risk: eGFR <30 (IV); <45 intra-arterial.
- Isotonic saline only proven; PRESERVE: bicarbonate and NAC useless.
- Metformin: withhold 48 h only if eGFR <30 or AKI.

Non iodinated contrast materials

Type: SAQ (10 marks)

Source: 2018-Summer Paper 1, Q3a

Introduction

- Used when iodinated contrast is risky (allergy, renal impairment) or with MRI (magnetic resonance imaging) and ultrasound.

1. Gadolinium-based contrast agents (GBCAs)

- **Mechanism:** paramagnetic gadolinium **shortens T1 relaxation** of nearby water → bright enhancement. Free gadolinium is toxic, so it is **chelated** (linear or macrocyclic; macrocyclic is more stable). Usual dose 0.1 mmol/kg; renal excretion.
- **NSF (nephrogenic systemic fibrosis):** skin and organ fibrosis, joint contractures, no proven treatment. Risk: eGFR (estimated glomerular filtration rate) <30 mL/min/1.73 m^2, dialysis, AKI (acute kidney injury).

ACR (American College of Radiology) group	Agents	NSF risk
I (linear)	Gadodiamide, gadopentetate, gadoversetamide	Highest
II	Gadobenate, gadoteridol, gadoterate, gadobutrol	**Little or none, even in ESRD (end-stage renal disease)**
III	Gadoxetate (liver-specific agent)	Few NSF data; recent ACR editions have moved some agents between groups, so check the current manual

- **Gadolinium deposition:** traces retained in brain (dentate nucleus) and bone, mainly with linear agents; no proven harm. Europe suspended most linear agents (2017).
- **Other:** allergic reactions rarer than with iodinated agents; not nephrotoxic at usual doses; avoid in pregnancy unless essential.
- **Uses:** renal mass, prostate mpMRI (multiparametric MRI), MR urography and angiography.

2. Ultrasound contrast (microbubbles)

- Gas core (perfluorocarbon or sulfur hexafluoride) in a lipid or albumin shell; strong harmonic echoes.
- Purely **intravascular**; gas exhaled by lungs. **No nephrotoxicity or radiation.** Rare cardiopulmonary reactions.
- **CEUS (contrast-enhanced ultrasound):** solid vs cystic renal mass (Bosniak), ablation monitoring, surveillance in CKD (chronic kidney disease). Intravesical use: **voiding urosonography for VUR (vesicoureteral reflux)**; ureteric patency after PCNL (percutaneous nephrolithotomy).

3. Other agents

- **CO_2 angiography:** negative contrast; not allergenic or nephrotoxic. Useful in CKD; **never above the diaphragm** (cerebral embolism).
- **Ferumoxytol** (iron oxide): MRI agent in renal failure. Lymphotropic iron nanoparticles help nodal staging (nodal metastases fail to darken).
- **Barium:** GI (gastrointestinal) studies only; contraindicated if perforation suspected.

- **Also know:** air as negative contrast (historical); radiotracers (^{99m}Tc, PET (positron emission tomography)) are not contrast agents; MRI safety (pacemakers, metal).

Diagram to draw: Chelated gadolinium (macrocyclic vs linear ring).

> **Key points to remember:**
> - GBCAs shorten T1; NSF risk eGFR <30; group II agents almost no risk.
> - Microbubbles: intravascular, no nephrotoxicity; ceVUS for reflux.
> - CO_2 angiography: never above diaphragm.

Health-related quality of life (HRQOL) instruments in urology, including after urinary diversion

Asked as:

- Health related quality of life instruments (HRQOL) in urology. — *Summer 2018 Paper I Q3c (SAQ)*
- Assessment of quality of life after urinary diversion — *Practice Set 3 Paper I Q3c (SAQ)*

Type: SAQ (10 marks)

Definition

- **HRQOL (HRQoL, health-related quality of life):** the parts of life affected by health: physical, emotional and social function and the patient's bother; the patient's view of how disease and treatment affect physical, emotional and social function, plus **bother**. It differs from a **symptom score**, which only counts symptoms.
- Must be **patient-reported**; doctors underestimate dysfunction (urinary, sexual and bowel dysfunction).

Types of instruments

Type	Measures	Examples	Limitation
Generic	Domains common to all diseases; allow comparison across diseases	**SF-36** (Short Form-36), SF-12, **EQ-5D** (EuroQol five-dimension; 5 items + VAS (visual analogue scale); gives utilities for QALYs (quality-adjusted life-years)), Sickness Impact Profile	Insensitive to diversion-specific issues
Cancer-specific	Cancer effects	**FACT-G** (Functional Assessment of Cancer Therapy–General), **EORTC QLQ-C30** (European Organisation for Research and Treatment of Cancer core questionnaire)	General cancer domains
Disease-specific	Domains of one disease; more responsive	See below	
Bladder-cancer-specific		**FACT-BI** (40 items), **EORTC QLQ-BLM30** (muscle-invasive module, 30 items), **BCI (Bladder Cancer Index**, 34 items: urinary, bowel, sexual domains), FACT-VCI (Vanderbilt Cystectomy Index)	Weight symptoms of one diversion, so biased in cross-diversion comparisons
Neobladder-specific		IONB-PRO (23 items)	Not usable across diversions

Urology-specific instruments

- **Prostate cancer: EPIC (Expanded Prostate Cancer Index Composite)**, EPIC-26, UCLA (University of California, Los Angeles) Prostate Cancer Index, FACT-P.
- **Bladder cancer:** FACT-BI, **Bladder Cancer Index**, EORTC QLQ-BLM30 and NMIBC-24, FACT-Vanderbilt Cystectomy Index.
- **Kidney cancer:** FKSI (FACT–Kidney Symptom Index).
- **LUTS (lower urinary tract symptoms): IPSS (International Prostate Symptom Score)** 7 items + 1 quality-of-life item; BPH Impact Index.
- **Incontinence and OAB (overactive bladder): ICIQ** (International Consultation on Incontinence Questionnaire), **OAB-q**, King's Health Questionnaire, I-QOL.

- **Sexual function: IIEF** (International Index of Erectile Function; 15 items), SHIM (5 items; no bother, so not true HRQOL), FSFI (Female Sexual Function Index).
- **Others:** NIH-CPSI (chronic prostatitis), O'Leary–Sant (interstitial cystitis), Wisconsin Stone QOL, USSQ (ureteric stent).
- **PROMIS (Patient-Reported Outcomes Measurement Information System):** free item banks, **computer-adaptive testing**, common T-score.

Psychometric properties (choosing a tool)

- A good tool is **reliable** (Cronbach α >0.7), **valid** in that population and **responsive** to change.

Property	Meaning	Threshold
Internal consistency	Items agree	Cronbach α >0.7 (>0.9 = redundant)
Test–retest reliability	Stable over time	r >0.7
Content (face) validity	Right items present	Expert judgement
Criterion validity	Agrees with gold standard	r ≥0.7
Construct validity	Convergent and divergent	Correlations
Responsiveness	Detects real change	**MCID (minimal clinically important difference)**: AUA (American Urological Association) Symptom Index 3.1 points; otherwise ≥½ SD

- Validity is population-specific; translations need revalidation.
- Development: define domains → generate items → pilot → psychometric testing (Likert scales).

Quality of life after urinary diversion

Domains to assess

- Urinary: continence, catheterisation, stoma problems.
- **Body image**, sexual function, bowel function, social and emotional function, global HRQoL.

Evidence: ileal conduit versus neobladder

- **No diversion proven superior.** Studies are mostly cross-sectional and retrospective, without baseline data or randomisation.
- Verdict depends on instrument: generic tools mostly show **no difference** (largest group of studies); EORTC tools tend to **favour neobladder**; the urinary-focused BCI **penalises neobladder** for leakage.
- **Consistent signal:** body image favours neobladder; urinary function favours conduit.
- Large prospective cohort: global HRQoL exceeded baseline in all diversions by 6 months.
- Neobladder: daytime continence improves over 6–12 months; **night-time incontinence** (up to 50%) is the main bother.
- **Also know:** informed patients have less decision regret; most adapt to their diversion; PROMIS (Patient-Reported Outcomes Measurement Information System).

Practical approach

- Baseline and serial validated questionnaires (3, 6, 12 months), a disease-specific plus a generic tool, and realistic counselling.

Diagram to draw: Pyramid: symptoms → function → bother → overall HRQOL. Bar sketch: body image (neobladder better) versus urinary function (conduit better).

Key points to remember:

- Generic (SF-36, EQ-5D) vs disease-specific (EPIC, IPSS, ICIQ, IIEF) plus bladder-specific (FACT-Bl, BLM30, BCI) tools.
- SHIM lacks bother; I-QOL lacks symptoms.
- Reliable, valid, responsive; Cronbach α >0.7; IPSS MCID ~3 points.
- No diversion proven superior; studies flawed.
- Neobladder better body image; conduit better urinary function.

PET scans in urology.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 1, Q3e

Principle

- **PET (positron emission tomography):** tracer emits a positron → annihilation with an electron → **two 511-keV photons at 180°** → coincidence detection. Fused with CT (computed tomography) or MRI.
- Quantitative (**SUV, standardised uptake value**); costly. Half-lives: ^{18}F 110 min, ^{68}Ga 68 min, ^{11}C 20 min (on-site cyclotron).

Tracers

Tracer	Mechanism	Main use
^{18}F-FDG (fluorodeoxyglucose)	Glucose analogue trapped after phosphorylation (Warburg effect)	Bladder, UTUC (upper tract urothelial carcinoma), penile, seminoma
PSMA (prostate-specific membrane antigen) ligands (^{68}Ga-PSMA-11, ^{18}F-DCFPyL, ^{18}F-PSMA-1007, rhPSMA-7.3)	Bind cell-surface PSMA	Prostate staging and recurrence
^{11}C/^{18}F-choline	Cell membrane synthesis	Prostate recurrence (older)
^{18}F-fluciclovine	Amino acid transport	Prostate recurrence
^{18}F-NaF (sodium fluoride)	Bone turnover	Bone metastases

Uses by organ

Prostate cancer

- FDG is poor (low glucose metabolism). PSMA has specificity >90%.
- **Primary staging (high risk): proPSMA (2020)**: PSMA PET/CT more accurate than CT plus bone scan (92% vs 65%). Nodal sensitivity only 30–40%, so a negative scan does not exclude micrometastases.
- **BCR (biochemical recurrence):** detection rises with PSA (prostate-specific antigen); about half positive below 0.5 ng/mL. PSMA beats fluciclovine (56% vs 26%). CONDOR: management changed in 63.9%.
- **Theranostics:** PSMA-positive mCRPC (metastatic castration-resistant prostate cancer) → **^{177}Lu-PSMA-617 (VISION 2021)**.
- NCCN (National Comprehensive Cancer Network): PSMA PET preferred; outcome benefit still unproven.

Testis

- **Seminoma residual mass >3 cm:** FDG-PET **≥6 weeks after chemotherapy**. Negative scan reliable (observe); positive → consider resection.
- **Not for NSGCT (non-seminomatous germ cell tumour):** teratoma is not FDG-avid.

Bladder and UTUC

- FDG for nodal and distant staging in MIBC (muscle-invasive bladder cancer): modest sensitivity, high specificity; urinary excretion hides the primary. Use when CT is equivocal.

Kidney

- FDG not routine (clear cell RCC (renal cell carcinoma) poorly avid). **^{89}Zr-girentuximab** (anti-CAIX, carbonic anhydrase IX) identifies clear cell RCC (ZIRCON).

Penis

- FDG confirms metastases in **palpable inguinal nodes**; poor for impalpable nodes.
- **Also know:** EMPIRE-1 (fluciclovine-guided salvage radiotherapy); false positives from inflammation; neuroendocrine cancer loses PSMA.

Diagram to draw: Positron annihilation with two 511-keV photons hitting a detector ring.

Key points to remember:

- proPSMA: PSMA PET 92% vs 65% accuracy.
- Seminoma residual >3 cm: FDG-PET at ≥6 weeks; not NSGCT.
- FDG poor in prostate and clear cell RCC.
- VISION: ^{177}Lu-PSMA-617 in mCRPC.

Bacteriuria in pregnancy.

Type: SAQ (10 marks)

Source: 2021-Summer Paper 1, Q3b (also answered in 2019-Summer Paper 3, Q3b)

Definition

- **ASB (asymptomatic bacteriuria):** ≥10^5 CFU (colony-forming units)/mL of one uropathogen in a midstream sample without urinary symptoms (≥10^2 CFU/mL on a catheter sample).

Epidemiology

- Prevalence is **not raised** by pregnancy (about 2–10%); higher with low socioeconomic status, multiparity, sickle cell trait. Organisms as in non-pregnant women (mainly *E. coli*).

Why it matters (pathophysiology)

- **Progesterone** relaxes ureteric smooth muscle; the **uterus compresses the ureters at the pelvic brim** (right more). Dilatation from about 15 weeks; normal by 6 weeks postpartum.
- Result: **stasis and hydroureteronephrosis**; bacteriuria persists and ascends.
- GFR (glomerular filtration rate) rises 30–50%: creatinine and drug levels fall.
- **Untreated ASB → pyelonephritis in 15–45%** (historical), 60–75% in the third trimester. Treatment cuts this to **1–4%**.
- Pyelonephritis causes **preterm birth, low birth weight** and maternal sepsis. **Cochrane 2019:** treating ASB reduces all three.

Screening

- **Urine culture at 12–16 weeks** or the first visit (USPSTF [US Preventive Services Task Force], IDSA [Infectious Diseases Society of America]). Dipstick misses many cases.

Treatment

Safe	Restricted or avoided
β-lactams (amoxicillin, cephalexin): throughout	**Nitrofurantoin:** stop from 35 weeks (neonatal haemolysis, G6PD [glucose-6-phosphate dehydrogenase] deficiency); not for pyelonephritis
Fosfomycin: lower tract only	**Trimethoprim-sulfamethoxazole:** avoid 1st trimester (folate, neural tube); caution near term (kernicterus)
Gentamicin: pyelonephritis	**Fluoroquinolones**, **tetracyclines**, chloramphenicol: avoid

- ASB and cystitis: culture-directed, **4–7 days** (cures more than a single dose); **repeat culture** after treatment.

Pyelonephritis

- **Admit**; IV (intravenous) β-lactam (± gentamicin) until afebrile, then oral to complete ≥14 days. No improvement by **72 h → ultrasound** (stone, obstruction, abscess).

Prophylaxis

- After one pyelonephritis or two cystitis episodes: nightly nitrofurantoin or cephalexin until delivery, or close culture surveillance (Cochrane 2015).
- **Also know:** urinary diversion or augmentation, SCI (spinal cord injury): most treat ASB in pregnancy.

Diagram to draw: right hydroureter compressed at the pelvic brim.

Key points to remember:

- Prevalence unchanged; risk of pyelonephritis raised (15–45% untreated).
- Screen by culture at 12–16 weeks; treat 4–7 days.
- β-lactams safe; no fluoroquinolones or tetracyclines; nitrofurantoin not after 35 weeks.

Recist Criteria.

Type: SAQ (10 marks)

Source: 2024-Winter Paper 1, Q3a

Definition and evolution

- **RECIST (Response Evaluation Criteria In Solid Tumours):** standard imaging rules for tumour response and progression.
- **WHO (World Health Organization) 1981** (bidimensional) → **RECIST 2000** (unidimensional) → **RECIST 1.1 (2009)**, now standard. Used in trials for metastatic RCC (renal cell carcinoma), urothelial and prostate cancer.

Lesions at baseline (RECIST 1.1)

Category	Definition
Measurable lesion	Longest diameter **≥10 mm on CT (computed tomography)**, ≥20 mm on chest X-ray, ≥10 mm with callipers
Measurable node	**≥15 mm short axis**; 10–15 mm pathological but non-measurable; <10 mm normal
Non-measurable	Small lesions, blastic bone metastases, effusions, ascites
Target lesions	Up to **5 (maximum 2 per organ)**; largest, reproducible; **sum of diameters** recorded
Non-target	All other lesions, recorded as present or absent

Response categories

Response	Target lesions
CR (complete response)	All disappear; nodes <10 mm
PR (partial response)	**≥30% fall** in sum versus **baseline**
PD (progressive disease)	**≥20% rise** versus **nadir** and **≥5 mm absolute**, or any **new lesion**
SD (stable disease)	Neither PR nor PD

- **Overall response** combines target, non-target and new lesions; confirm response at ≥4 weeks in single-arm trials.
- **End points:** ORR (objective response rate = CR + PR), PFS (progression-free survival), duration of response.

Limitations

- Size only, ignoring necrosis (TKIs (tyrosine kinase inhibitors)); bone poorly assessed; observer variability.

Modifications

Criteria	Use
iRECIST (2017)	Immunotherapy: first progression is "unconfirmed", confirmed only on the next scan, allowing for **pseudoprogression**
Choi	Density fall counts as response
PCWG3 (Prostate Cancer Working Group 3)	Bone: 12-week scan is the new baseline (flare), then **≥2 new lesions** = progression; PSA (prostate-specific antigen) progression ≥25% and ≥2 ng/mL above nadir; soft tissue by RECIST
PERCIST	PET (positron emission tomography) metabolic response

Diagram to draw: number line of change in sum: −30% (PR), +20% (PD).

Key points to remember:

- RECIST 1.1: ≥10 mm lesion, node ≥15 mm short axis; 5 targets, 2 per organ.
- PR ≥30% fall from baseline; PD ≥20% and ≥5 mm rise from nadir, or new lesion.
- iRECIST for immunotherapy; PCWG3 for prostate.

Outline algorithm for difficult catheter placement.

Type: SAQ (10 marks)

Source: 2024-Winter Paper 1, Q3b

Introduction

- **Normal calibre:** meatus 24 Fr (narrowest), bladder neck 28 Fr, prostatic urethra 32 Fr.
- **Causes:** BPH (benign prostatic hyperplasia, commonest), urethral stricture, false passage, meatal stenosis, bladder neck contracture, urethral trauma, abnormal anatomy.
- **Golden rule:** escalate **blind → wire → vision → suprapubic**; never repeat forceful blind passage.

Before starting

- History (surgery, radiotherapy, AUS (artificial urinary sphincter), reconstruction), DRE (digital rectal examination).
- **Absolute contraindication:** suspected urethral injury (pelvic fracture, blood at meatus, high-riding prostate) → **RUG (retrograde urethrogram)**; pelvic fracture urethral injury → suprapubic tube.
- **Technique:** 10–15 mL lubricant (lidocaine gel), penis stretched, steady pressure; inflate only after urine flows at the hub.

Algorithm (Campbell Fig. 15.6)

```
First attempt fails
 ├─ Prostatic obstruction suspected
 │     → 18-Fr coudé, tip (marker) pointing up
 │     → fails → flexible cystoscopy
 └─ Not suspected (stricture likely)
       → smaller catheter
       → fails → hydrophilic glidewire
             → fails → flexible cystoscopy
At cystoscopy / with wire in bladder:
 ├─ Stricture → wire across → 5-Fr catheter, aspirate urine to confirm
 │      → super-stiff wire → serial dilatation (1 Fr above catheter size)
 │      → Councill (open-tip) catheter over wire
 └─ False passage → pass under vision (scope anterior)
        → wire → Councill over wire
Wire cannot be passed / dense stricture / urethral disruption
 → suprapubic catheter (ultrasound-guided percutaneous, or open)
```

- **Why 18-Fr coudé:** stiffer, larger tip area, curved tip follows the bulbar-prostatic angle; 14–16 Fr usually buckle.
- **Blood on the tip** suggests false passage: stop blind attempts.

Special situations

Situation	Approach
AUS	Deactivate; smallest catheter 12–14 Fr; remove early
Obese or female	Exposure: Trendelenburg, retract pannus, two fingers in vagina to guide
Buried penis	Infrapubic pressure; flexible cystoscopy
Neobladder, continent pouch	Never force; scope + wire; suprapubic only with imaging guidance

Situation	Approach
PUV (posterior urethral valves) neonate	6–8 Fr, or 5-Fr open-ended over glidewire; else suprapubic or vesicostomy

Also know: filiforms, balloon dilators; complications: false passage, haematuria, stricture.

Diagram to draw: coudé catheter following the bulbar-prostatic curve with tip up.

Key points to remember:

- Blind → wire → vision → suprapubic.
- BPH: 18-Fr coudé, tip up.
- Councill catheter over wire after dilatation.
- Suspected urethral injury: RUG first, no blind catheter.

Indications for outpatient cystourethroscopy.

Type: SAQ (10 marks)

Source: 2024-Winter Paper 1, Q3e

Introduction

- **Flexible cystourethroscopy** under local anaesthesia is the **commonest office procedure** in urology; no lithotomy needed.
- **Prerequisites:** consent; treat any UTI (urinary tract infection) first; lidocaine gel with **≥5 minutes dwell** (ideally 10–20); no antibiotic prophylaxis unless risk factors (age, diabetes, immunosuppression, catheters, prior UTI).

Diagnostic indications

Indication	Comment
Haematuria	All **gross haematuria** (with CT (computed tomography) urography, cytology). Microhaematuria by AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) risk: intermediate and high risk; shared decision in low risk
Bladder cancer surveillance	NMIBC (non-muscle-invasive bladder cancer): first check at 3 months; low risk then yearly; high risk 3–4-monthly for 2 years; also upper-tract follow-up of UTUC (upper tract urothelial carcinoma)
Recurrent UTI	Look for stone, diverticulum, stricture, fistula, foreign body
LUTS (lower urinary tract symptoms)	Not routine; before invasive BPH (benign prostatic hyperplasia) treatment (median lobe before urethral lift, bladder-neck contracture, stricture)
Pain syndromes	IC/BPS (interstitial cystitis/bladder pain syndrome): **Hunner lesion**; chronic pelvic pain with bladder pain or low flow
Women	Suspected urethral diverticulum, vesicovaginal fistula, mesh or suture erosion after sling, recurrent UTI after surgery
Others	Urethral stricture assessment and follow-up; abnormal urine cytology; radiation cystitis; foreign body

Therapeutic indications

- **Removal or exchange of ureteric stents** (exchange with fluoroscopy).
- **Biopsy and fulguration** of small (≤1 cm) low-grade recurrences; office laser ablation.
- **Intradetrusor botulinum toxin** injection (OAB (overactive bladder), neurogenic detrusor overactivity).
- Urethral bulking agents; stricture dilatation; small stones or foreign bodies; blue-light cystoscopy where available.

When not indicated or done in theatre

- Not needed for the index stress incontinence patient, initial OAB or uncomplicated LUTS.
- **No surveillance cystoscopy** in chronic indwelling catheters (AUA/SUFU 2021).
- Theatre if TURBT (transurethral resection of bladder tumour) needed, poor tolerance, children, or complex diversions.

Diagram to draw: haematuria risk-group flowchart.

Key points to remember:

- Haematuria and NMIBC surveillance are the two commonest indications.
- Flexible scope, lidocaine dwell ≥5 minutes, no routine prophylaxis.
- Therapeutic: stent removal, botulinum toxin, fulguration of small recurrences.

MR urography

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 1, Q3a

Definition

- MRU (magnetic resonance urography): magnetic resonance imaging of kidneys, ureters and bladder giving anatomy and function **without ionising radiation**.

Techniques

	Static MRU	Excretory MRU
Sequence	Heavily T2-weighted; urine is bright	Gadolinium-enhanced T1
Contrast	None	GBCA (gadolinium-based contrast agent)
Depends on function	No	Yes; low-dose furosemide improves it
Best for	Dilated or obstructed system	Non-dilated system, excretion

- **Functional MRU (children):** hydration, bladder catheter, furosemide 1 mg/kg (maximum 20 mg), T2 images, then dynamic T1 after slow contrast injection. Patlak analysis gives differential function, GFR (glomerular filtration rate) and drainage.

Appearances

- Stones and calcification give **signal voids**; tumour, clot, gas and sloughed papillae are low-signal defects in bright urine.

Indications

- CT (computed tomography) urography contraindicated: contrast allergy, renal insufficiency, **pregnancy** (separates physiological from calculous dilatation; small stones missed).
- **Children:** complex anomalies, ectopic ureter in the incontinent girl, duplex systems, megaureter, crossing vessels in UPJ (ureteropelvic junction) obstruction, ectopic kidneys.
- Hydronephrosis: alternative to MAG3 (mercaptoacetyltriglycine) renography; equivalence not yet proven.
- Upper tract tumours and renal masses.

Advantages and limitations

Advantages	Limitations
No radiation; best anatomical detail	Cost, availability, long study (up to 1 hour)
Anatomy, function and drainage in one test	Sedation often needed under 6 years
As accurate as CTU (CT urography) for obstruction	Poor for stones; complex post-processing
Safe in iodinated contrast allergy	Pacemakers, metal implants, claustrophobia

Gadolinium safety

- **NSF (nephrogenic systemic fibrosis)** was the concern at GFR <30 mL/min/1.73 m². Group II agents (gadobenate, gadoteridol, gadoterate, gadobutrol) carry little or no risk even in end-stage disease.
- **Pregnancy:** gadolinium only if it clearly helps.
- **EAU (European Association of Urology):** CTU is preferred to MRU for upper tract urothelial cancer.

Diagram to draw: coronal static T2 MRU of a hydronephrotic kidney with a dark stone in the ureter.

Key points to remember:

- Static T2 (no contrast, function-independent) versus excretory gadolinium T1.
- Stones are signal voids; CTU preferred for urothelial cancer.
- Functional MRU: furosemide 1 mg/kg, Patlak analysis for split function.
- NSF risk minimal with group II agents.

Urine cytology

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 1, Q3b

Definition and principle

- Microscopy of shed urothelial cells (Papanicolaou, 1945).
- **Principle:** high-grade cells are poorly cohesive, shed readily and look atypical; low-grade cells look almost normal. So cytology detects **high-grade tumours and CIS (carcinoma in situ)** but misses low grade.

Specimens

- Voided urine (not the first morning sample, whose cells have degenerated), at least 30 mL; bladder washing; upper tract washings.

Performance

- Sensitivity 31–62%, specificity 94–100%; strongly grade-dependent.
- **False positives:** instrumentation, stones, infection, radiotherapy, intravesical BCG (bacille Calmette–Guérin), polyomavirus ("decoy" cells), ileal conduit.
- **False negatives:** low grade, low cellularity, degeneration.

The Paris System

- Published 2016; **2nd edition 2022**. Aim: detect **HGUC (high-grade urothelial carcinoma)**. Key criterion: **N:C (nuclear-to-cytoplasmic) ratio**, plus hyperchromasia, coarse chromatin and irregular nuclear membrane.

Category	Criteria
Non-diagnostic	Inadequate
Negative for HGUC	N:C <0.5, or atypia explained by a benign cause
AUC (atypical urothelial cells)	N:C ≥0.5 and one feature
Suspicious for HGUC	N:C ≥0.7, at least two features, few cells (<5–10)
HGUC	N:C ≥0.7, at least two features, many cells
LGUN (low-grade urothelial neoplasm)	Fibrovascular cores; rarely diagnosable
Other malignancy	Squamous, adenocarcinoma, secondary

- It reduces equivocal "atypia" reports; cancer risk rises with each category.

Clinical uses

- **Not** for screening or initial microhaematuria evaluation (AUA/SUFU [American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2020).
- Gross haematuria; NMIBC (non-muscle-invasive bladder cancer) surveillance (mandatory in high risk); suspected CIS; upper tract tumours.
- **Positive cytology, normal cystoscopy:** image upper tracts, selective washings ± ureteroscopy, random bladder and prostatic urethral biopsies.
- **Also know:** FISH (fluorescence in situ hybridisation; chromosomes 3, 7, 17, 9p21), NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), CxBladder: more sensitive, less specific; none replaces cystoscopy.

Diagram to draw: normal urothelial cell beside an HGUC cell with high N:C ratio.

Key points to remember:

- High specificity, low sensitivity; good for high grade and CIS.
- Paris System: N:C ≥0.7 for suspicious and HGUC; AUC N:C ≥0.5.
- Positive cytology, normal cystoscopy: upper tracts, random biopsies.
- Not for screening or initial microhaematuria.

Patient positioning in lap urology

Type: SAQ (10 marks)

Source: Practice Set 2 Paper 1, Q3a (pneumoperitoneum physiology also answered in 2019-Summer Paper 1, Q2)

Principles

- Positioning uses **gravity** to move bowel away and gives port or robot access while keeping ventilation safe.
- Injury comes from **stretch, compression or ischaemia**, so position and duration act together; nerve injury occurs in about 2.8%. Shared team responsibility (ASA [American Society of Anesthesiologists] advisory, 2000).
- Secure IV (intravenous) access, oximetry and BP (blood pressure) cuff before final positioning: tucked arms become inaccessible.

Positions

Position	Used for	Set-up	Hazards
Steep Trendelenburg, low lithotomy or split legs	RARP (robot-assisted radical prostatectomy), cystectomy	Arms tucked and padded; chest strap over egg-crate; **no shoulder braces**; tilt about 22–30°	Brachial plexus; femoral nerve (excess hip flexion); facial and laryngeal oedema; corneal abrasion; tube into right bronchus
Modified flank (30–45°)	Transperitoneal nephrectomy, pyeloplasty	Back on gel rolls; upper arm across chest; wide tape; tilt to test	Pressure points, slippage
Full lateral (90°), table flexed	Retroperitoneoscopy, adrenalectomy	**Axillary roll** 3–4 fingerbreadths below axilla; lower leg flexed 45°, upper straight on pillows	**Rhabdomyolysis**; sciatic stretch
Prone	Posterior adrenalectomy	Pad torso, elbows, hips	Airway, face and eyes

Rhabdomyolysis

- About 0.4% (1% retroperitoneal). Risk: men, BMI (body mass index) ≥25, **kidney rest**, full flexion, long cases. Downside hip pain, brown urine, CK (creatine kinase) >5,000 units. Prevent: limit kidney rest (Campbell gives 20–30 minutes in one place, under 1 hour in another), avoid hypotension, keep cases under 5 hours.

Prevention and physiology

- Abduction ≤90°; pad ulnar groove and fibular head; recheck after every table move.
- **Head-down** raises cardiac output but splints the diaphragm; restrict fluids.
- Nerve injury on day 1: neurology review, conduction studies, physiotherapy.
- **Also know:** forearm compartment syndrome from infiltrated IV lines; Xi side-docking avoids stirrups.

Diagram to draw: full flank position with axillary roll and table break.

Key points to remember:

- Nerve injury about 2.8%; brachial plexus is commonest at risk.
- Never shoulder braces in steep Trendelenburg.
- Axillary roll below, not in, the axilla.
- Rhabdomyolysis: men, BMI ≥25, kidney rest; CK >5,000.

Clinical trial methodology

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 1, Q3a

Definition

- A **clinical trial** is a prospective study that assigns people to an intervention to test its effect on health outcomes. The **RCT (randomised controlled trial)** is the gold standard because randomisation balances known and unknown confounders.

Phases

Phase	Purpose	Subjects
0	Microdose pharmacokinetics	Very few
I	Safety, dose, toxicity	20–80, often volunteers
II	Activity, efficacy signal	Patients, about 100–300
III	Comparative efficacy versus standard	Hundreds to thousands; basis of approval
IV	Post-marketing surveillance	General population

Design

- **Parallel**, **crossover** (own control; stable chronic disease), **factorial** (two questions at once), **cluster** or **adaptive**.
- Hypothesis: **superiority**, **non-inferiority** (pre-set margin) or equivalence.
- Surgical innovation follows the **IDEAL** framework (Idea, Development, Exploration, Assessment, Long-term study).

Randomisation and allocation

- Simple, **block** (equal numbers), **stratified** (by prognostic factors) or **minimisation**.
- **Allocation concealment** (central or web-based) stops selection bias.

Blinding

- Single (patient), double (plus assessor) or triple (plus analyst). Surgery: sham procedures or blinded assessors.

Endpoints

- One **primary endpoint** sets sample size; secondary endpoints are exploratory.
- **Hard** (overall survival) versus **surrogate** endpoints, valid only if they meet the **Prentice criteria**; in prostate cancer only MFS (metastasis-free survival) does.
- **PROs (patient-reported outcomes)** with validated tools; complications graded by **Clavien–Dindo** or CTCAE (Common Terminology Criteria for Adverse Events).

Sample size and analysis

- Set α (type I error, usually 0.05), power (80–90%) and expected effect.
- **ITT (intention-to-treat)** preserves randomisation; per-protocol supports non-inferiority claims.
- Kaplan–Meier curves for time-to-event data.

Ethics and reporting

- Declaration of Helsinki, ICH-GCP (International Council for Harmonisation–Good Clinical Practice), informed consent, ethics committee, DSMB (data safety monitoring board).

- **Registration** before enrolment (CTRI [Clinical Trials Registry–India]).
- **CONSORT** (Consolidated Standards of Reporting Trials): checklist plus **participant flow diagram** (25 items in the 2010 version; updated 2025).
- **Also know:** lead-time bias; Will Rogers phenomenon.

Diagram to draw: CONSORT flow diagram: enrolment → randomisation → allocation → follow-up → analysis.

> **Key points to remember:**
> - Phase III RCT with concealed allocation and blinding is the gold standard.
> - Primary endpoint drives sample size; surrogates need the Prentice criteria.
> - ITT analysis; CONSORT flow diagram for reporting.

Evidence-based medicine and critical appraisal of research papers

Asked as:

- Describe evidence based medicine — *Practice Set 3 Paper I Q3d (SAQ)*
- Criteria for evaluation of quaLliy of medical research papers — *Practice Set 6 Paper I Q3d (SAQ)*

Type: SAQ (10 marks)

(evidence-based medicine also answered in 2020-Summer Paper 4, Q3f)

Assumed meaning: "quaLliy" = quality, i.e. critical appraisal.

Definition

- **EBM (evidence-based medicine)**: "the conscientious, explicit and judicious use of current best evidence in making decisions about the care of individual patients" (**Sackett, 1996**).
- It integrates best research evidence, **clinical expertise** and **patient values**.
- **Critical appraisal:** systematic assessment of a paper's **validity, results and applicability**; the "Appraise" step of EBM.

Steps ("5 As")

1. **Ask** a focused question in **PICO** form (Population, Intervention, Comparison, Outcome).
2. **Acquire**: PubMed, Cochrane Library, guidelines.
3. **Appraise**: validity (randomisation, blinding), effect size (ARR [absolute risk reduction], **NNT [number needed to treat] = 1/ARR**), precision (95% CI [confidence interval]), applicability.
4. **Apply** with expertise and patient preference.
5. **Assess** (audit) the outcome.

Critical appraisal: three core questions

1. Is it valid? (internal validity)

- **PICO question** (Population, Intervention, Comparison, Outcome); suitable design.
- **RCTs (randomised controlled trials):** true randomisation, **allocation concealment**, comparable groups at baseline, **blinding**, complete follow-up (>80%), **ITT (intention-to-treat) analysis**.
- **Bias** checked:

Bias	Meaning
Selection	Groups differ at entry
Confounding	Third factor (comorbidity) linked to exposure and outcome
Performance, detection, attrition	Unequal care, outcome assessment or dropouts
Lead-time, length-time; stage migration (Will Rogers)	Screening or better tests make survival look longer
Publication	Negative studies unpublished

2. What are the results?

- Effect: RR (relative risk), **ARR (absolute risk reduction), NNT (number needed to treat) = 1/ARR**.
- Precision: **95% CI (confidence interval)**; p value; **sample size and power** (usually 80%; α 0.05).

- **Surrogate endpoints** (end points) must meet the **Prentice criteria**; validated outcome tools (validated PROs [patient-reported outcomes]); Clavien–Dindo grading.

3. Does it apply? (external validity)

- Similar patients and setting; benefit outweighs harm.

Levels of evidence (Oxford CEBM [Centre for Evidence-Based Medicine], as adapted/used by EAU [European Association of Urology])

Level	Type of evidence / design
1a	Systematic review or meta-analysis of RCTs (randomised controlled trials)
1b	At least one RCT (single RCT)
2a	Well-designed controlled study without randomisation (controlled non-randomised)
2b	Quasi-experimental (cohort) study
3	Non-experimental: case-control, correlation, case reports, case series
4	Expert committee reports or opinion

- **Grades**: A (consistent level 1), B (level 2–3), C (level 4 or consensus).
- **GRADE** (Grading of Recommendations Assessment, Development and Evaluation): certainty high to very low; RCTs start high, observational studies low; **downgrade** for bias, inconsistency, indirectness, imprecision, publication bias; **upgrade** for large effect or dose–response. Recommendations **strong or weak** (used by EAU). AUA (American Urological Association): evidence strength A/B/C; strong, moderate or conditional recommendations.

Pitfalls in appraisal

- **Bias**: selection, confounding, **lead-time**, **length-time**, stage migration (Will Rogers).
- **Surrogate endpoints** must meet the **Prentice criteria**.

Reporting checklists

Design	Checklist
RCT	**CONSORT** (Consolidated Standards of Reporting Trials)
Systematic review / meta-analysis	**PRISMA** (Preferred Reporting Items for Systematic reviews and Meta-Analyses)
Observational	**STROBE** (Strengthening the Reporting of Observational studies in Epidemiology)
Diagnostic accuracy	**STARD** (Standards for Reporting Diagnostic accuracy studies)
Surgical innovation	**IDEAL** framework (Idea, Development, Exploration, Assessment, Long-term study)

- **Also know:** CONSORT, PRISMA, STROBE reporting checklists; validated PROs (patient-reported outcomes); Cochrane RoB 2 (risk-of-bias) tool; trial registration; conflicts of interest.

Limitations

- Few surgical RCTs; trial patients differ from real ones; publication bias; "cookbook medicine" if patient values are ignored.

Diagram to draw: evidence pyramid and the three overlapping circles of EBM.

Key points to remember:

- Sackett 1996: best evidence + clinical expertise + patient values.
- 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
- Level 1a = systematic review of RCTs; GRADE: strong or weak.
- NNT = 1/ARR; beware lead-time, length-time and confounding bias.
- Three questions: valid, results, applicable.
- Randomisation, concealment, blinding, ITT, follow-up >80%.
- CONSORT (RCT), PRISMA (reviews), STROBE (observational), STARD (diagnostic).

Doppler in urology

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 1, Q3b

Principle

- **Doppler shift:** sound reflected from moving red cells changes frequency in proportion to velocity and the cosine of the insonation angle: $\Delta f = 2f_0 v \cos\theta / c$. Keep the angle **below 60°**.
- **RI (resistive index) = (PSV − EDV) / PSV** (peak systolic and end-diastolic velocity); reflects downstream resistance.

Modes

Mode	Shows	Use
Colour	Velocity and direction (red towards, blue away by convention)	Flow in kidney, testis, ureteric jets
Spectral (duplex)	Waveform at one point; velocities, RI	Penile and renal arteries
Power	Amplitude only; 3–5 times more sensitive, less angle-dependent	Low flow: torsion, prostate lesions

Applications

Organ	Findings
Scrotum	**Torsion:** absent intratesticular flow (compare with the other side); epididymitis: increased flow; **varicocele:** venous reflux on Valsalva; intratesticular RI >0.6 suggests poor spermatogenesis
Penis (after intracavernosal injection)	**PSV <25 cm/s = arterial ED (erectile dysfunction)**, 25–35 equivocal, >35 normal; **EDV >5 cm/s and RI <0.75 = veno-occlusive dysfunction**; Peyronie plaque assessment
Priapism	Ischaemic: absent cavernosal flow; non-ischaemic: high flow (PSV >50 cm/s), locates the fistula
Kidney	Normal RI 0.47–0.70; **raised RI in acute obstruction** (useful in pregnancy, where change in RI is more accurate); renal artery stenosis: high velocity and parvus-tardus waveform; arteriovenous fistula after biopsy or PCNL (percutaneous nephrolithotomy)
Transplant	Arterial thrombosis: absent signal; **venous thrombosis: reversed diastolic flow**; stenosis: high velocity; rejection: loss of diastolic flow, high RI
Bladder	Ureteric jets (absent on the obstructed side)
Tumours	Vascularity of renal and testicular masses; tumour thrombus; hypervascular prostate cancer foci

- **Also know:** intraoperative micro-Doppler in microsurgical varicocelectomy; contrast-enhanced ultrasound and elastography in multiparametric ultrasound.

Limitations

- Operator- and angle-dependent; poor in obesity and deep vessels; early torsion may show collateral flow, so **Doppler cannot exclude torsion**.

Diagram to draw: spectral waveform marking PSV and EDV with the RI formula.

Key points to remember:

- RI = (PSV − EDV)/PSV; normal renal 0.47–0.70.
- Penile PSV <25 cm/s arterial; EDV >5 cm/s veno-occlusive.
- Absent intratesticular flow = torsion, but explore on clinical grounds.

Christian Chaussey

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 1, Q3c

Taken as Christian Chaussy, the Munich urologist who pioneered extracorporeal shock wave lithotripsy; answered as the history and principles of SWL.

Contribution and history

- Engineers at **Dornier** found by chance that shock waves from raindrop and micrometeorite research affected tissue.
- Chaussy, with Brendel and Schmiedt in **Munich**, developed the technique in animals and performed the **first human SWL (shock wave lithotripsy) on 7 February 1980 on the Dornier HM1**. They reported it in the *Lancet* in 1980.
- **HM3** (water bath, electrohydraulic spark, biplanar fluoroscopy): first installations Stuttgart 1983 and Indianapolis 1984. It remains the reference for efficacy.
- SWL made most open stone surgery obsolete.

Principles

- A shock wave is a short acoustic pulse: **peak positive 30–100 MPa, then a negative tail of about −10 MPa**.
- Generated at **F1**, focused onto the stone at **F2** through water or gel coupling (water and tissue have similar acoustic impedance); targeting by fluoroscopy or ultrasound.

Energy source	Feature
Electrohydraulic (HM1, HM3)	Underwater spark, ellipsoid reflector; most effective; electrode wear
Piezoelectric	Ceramic elements; long life, little pain; small focal zone, low energy
Electromagnetic (modern)	Coil and membrane with lens; consistent; dominant today

Mechanisms of fragmentation

- **Spallation:** reflected wave at the back surface creates tension.
- **Shear and tear** inside the stone.
- **Quasistatic squeezing** and **dynamic squeezing** (strongest force in initial breakage).
- **Cavitation:** collapsing bubbles erode the surface; excess bubbles shield the stone and injure tissue.

Practical points

- Better results at **60–90 shocks/min**, with power **ramping** and air-free coupling.
- Poor results: stone >1000 HU (Hounsfield units), skin-to-stone distance >10 cm, lower pole >1 cm.
- **Contraindications:** pregnancy, uncorrected coagulopathy, untreated UTI (urinary tract infection), distal obstruction, aortic aneurysm >4 cm.
- **Also know:** burst wave lithotripsy and ultrasonic propulsion are newer descendants.

Diagram to draw: ellipsoid reflector with spark at F1 and stone at F2.

> **Key points to remember:**
> - Chaussy, Munich: first human SWL 7 February 1980, Dornier HM1.
> - HM3 (1983) is still the efficacy benchmark.
> - Spallation, squeezing, shear, cavitation; electromagnetic sources today.

Flow cytometry

Type: SAQ (10 marks)

Source: Practice Set 5 Paper 1, Q3d

Definition

- **Flow cytometry** measures physical and fluorescent properties of **single cells in suspension** flowing one at a time through a laser beam, thousands per second.

Principle

1. **Sample:** fresh cells, urine, bladder washings, or nuclei from paraffin blocks.
2. **Staining:** fluorochrome-labelled antibodies to CD (cluster of differentiation) antigens, or **DNA-binding dyes** (propidium iodide, DAPI [4′,6-diamidino-2-phenylindole]).
3. **Hydrodynamic focusing** lines the cells up in single file.
4. **Laser interrogation:**
    - **Forward scatter** = cell size.
    - **Side scatter** = internal complexity (granularity).
    - **Fluorescence** = amount of bound antibody or DNA.
5. Photomultiplier detectors → histograms and dot plots; **gating** selects populations.

- **FACS (fluorescence-activated cell sorting)** physically separates selected cells.

DNA content analysis

- **DNA index** = tumour G0/G1 peak ÷ normal diploid peak. DI 1.0 = diploid; about 2.0 = tetraploid; any other = **aneuploid**.
- **S-phase fraction** estimates proliferation.
- Aneuploidy and high S-phase generally mean higher grade and worse prognosis.

Applications in urology

Area	Use
Bladder cancer	Ploidy of bladder washings: aneuploidy with high grade, CIS (carcinoma in situ), recurrence; replaced by **UroVysion FISH (fluorescence in situ hybridisation)**
Upper-tract urothelial cancer	Aneuploidy → poor long-term survival
Prostate cancer	Aneuploid tumours progress more after treatment than diploid
Neuroblastoma	**Hyperdiploid favourable**; diploid or tetraploid adverse; part of COG (Children's Oncology Group) risk
Urinalysis	Automated urine analysers count RBC (red blood cells), WBC (white blood cells) and bacteria
Transplantation	**Flow cytometric crossmatch**: more sensitive than CDC (complement-dependent cytotoxicity) crossmatch for donor-specific antibody; Luminex single-antigen beads; CD3 count to guide antithymocyte globulin
Male infertility	Sperm DNA fragmentation (SCSA [sperm chromatin structure assay]); haploid cells in testicular aspirates
Lymphoma	Immunophenotyping of testicular or renal lymphoma

Advantages and limitations

- **Advantages:** rapid, objective, quantitative, many parameters per cell.

- **Limitations:** architecture lost; near-diploid aneuploidy missed; normal cells dilute tumour peaks; ploidy now adds little independent value.

Diagram to draw: DNA histogram with diploid G0/G1 peak, S-phase and aneuploid peak.

Key points to remember:

- Forward scatter = size; side scatter = granularity; fluorescence = marker or DNA.
- DNA index 1.0 diploid; aneuploidy = worse prognosis.
- Flow crossmatch is more sensitive than CDC crossmatch.

VCUG

Type: SAQ (10 marks)

Source: Practice Set 6 Paper 1, Q3c

Definition

- **VCUG (voiding cystourethrogram; MCU, micturating cystourethrogram):** fluoroscopic study of bladder filling and voiding after catheter instillation of contrast. It shows the bladder, **posterior urethra** and UVJ (ureterovesical junction) competence.
- **Gold standard for VUR (vesicoureteral reflux)** and PUV (posterior urethral valves).

Indications

- **Children:** febrile UTI (urinary tract infection) with abnormal ultrasound or recurrent infection (EAU [European Association of Urology]: also age <12 months); bilateral hydroureteronephrosis (PUV); dilated ureter; ureterocele; neurogenic bladder.
- **Adults:** stricture (with RGU [retrograde urethrogram]), diverticula, fistula, post-operative leak, voiding dysfunction.

Technique

- **Sterile urine** first. After UTI, wait until urine is sterile and the child is improving (about a week); the old 4–6-week delay is unnecessary.
- Scout film → catheter: **5 Fr feeding tube (infants), 6–8 Fr (children), 14–16 Fr (adults)** → confirm tip position.
- **Gravity fill** (≤70 cm H_2O) to expected capacity: <2 years weight (kg) × 7; 2–14 years (age × 30) + 30 mL; >14 years 300–500 mL. (Koff formula: (age + 2) × 30 mL.)
- **Oblique views** (UVJ, prostatic urethra) → remove catheter → **void under fluoroscopy** → **post-void film**.
- **Cyclic VCUG** (2–3 fill–void cycles) detects 12–20% more VUR.

Findings

- **VUR grade (International Reflux Study, 1981):**

Grade	Appearance
I	Ureter only, not dilated
II	Pelvis and calyces, no dilatation
III	Mild–moderate dilatation, minimal blunting
IV	Moderate dilatation and tortuosity
V	Gross dilatation; papillary impressions lost

- Filling-phase reflux has a poorer resolution prognosis than voiding-only reflux.
- **PUV:** dilated, elongated posterior urethra, prominent bladder neck, trabeculation.
- Ureterocele, diverticula, residual urine; post-void pelvic retention suggests PUJ (pelviureteric junction) obstruction.

Limitations and complications

- Invasive, radiation, UTI, catheter trauma; snapshot (occult VUR); interobserver variability; autonomic dysreflexia (injury above T6).

Alternatives

- **Direct radionuclide cystography:** more sensitive, lower dose, poor anatomy.

- **CeVUS (contrast-enhanced voiding urosonography):** no radiation, accuracy close to VCUG.
- **PIC (positional instillation of contrast) cystography:** occult VUR with negative VCUG.

Diagram to draw: VUR grades I–V.

> **Key points to remember:**
> - Gold standard for VUR and PUV; sterile urine first.
> - Capacity (age × 30) + 30 mL (2–14 years); cyclic study detects 12–20% more.
> - Grades I–II non-dilating, III–V dilating.

Pioneers of urology: Hopkins, Huggins, Calne, Gaur and Asopa

Asked as:

- Describe contribution of a. Prof. Herald Hopkins b. Dr Charles Huggins c. Prof. Roy Caine d. Dr D D Gaue — *Practice Set 9 Paper I Q3a (SAQ)*
- Hari Asopa and Charles Huggins — *Practice Set 7 Paper I Q3c (SAQ)*

Type: SAQ (10 marks)

Names read as Prof. Harold Hopkins, Dr Charles Huggins, Prof. Sir Roy Calne and Dr D. D. Gaur. Assumed meaning: contributions of each to urology.

a. Prof. Harold Hopkins (British physicist)

- **Fibreoptics (1954, with Kapany):** coherent bundles of glass fibres transmit an image by total internal reflection (Tyndall's principle). This made flexible endoscopes possible.
- **Rod-lens system (1966):** the old scope used thin glass lenses separated by long air spaces. Hopkins replaced them with **long glass rods separated by short air gaps**. Result: more light transmission, a brighter and wider image, and a **smaller calibre**. Developed commercially with Karl Storz.
- It underpins all modern rigid endoscopes and enabled the first rigid ureteroscope (1979).

b. Dr Charles B. Huggins (1901–1997) (urologist, University of Chicago)

- Canadian-born American urologist, University of Chicago.
- **1941, with (Clarence) Hodges:** showed that prostate cancer is **androgen-dependent** (hormone dependent). Castration (or oestrogen) relieved far-advanced disease and caused **regression of metastatic prostate cancer**, with falling serum acid phosphatase, pain relief and weight gain; injected androgens aggravated it (made it worse). This proved that **prostate cancer is hormone (androgen) dependent**.
- Used serum **acid phosphatase** to monitor response, an early tumour marker (the first tumour marker).
- **1945:** bilateral adrenalectomy for castration-resistant disease (relapse after castration, suspecting adrenal androgens) (modest, brief responses; morbidity high).
- **Nobel Prize in Physiology or Medicine, 1966** (shared with Peyton Rous) for hormonal treatment of prostate cancer: the basis of ADT (androgen deprivation therapy) and of all hormonal cancer therapy.
- **Also know:** canine prostate physiology; adrenalectomy for breast cancer.
- **Legacy:** basis of all ADT (androgen deprivation therapy): orchidectomy, oestrogens, LHRH (luteinising hormone-releasing hormone) agonists and antagonists, and later abiraterone (blocking adrenal androgens).

c. Prof. Sir Roy Calne (Cambridge transplant surgeon)

- Showed in dogs that **6-mercaptopurine and then azathioprine** prolong renal allograft survival; azathioprine with steroids became standard immunosuppression.
- Performed the **first liver transplant in the UK (1968)**.
- **First clinical use of ciclosporin** in kidney transplantation (late 1970s), the calcineurin-inhibitor era; acute rejection fell markedly.

d. Dr D. D. Gaur (Mumbai, India)

- **1992: balloon dissection of the retroperitoneum.** A size 7 surgeon's glove tied on an 8-Fr red rubber catheter, inflated inside the retroperitoneum, creates a working space.
- This made **retroperitoneoscopy** practical and cheap; used for ureterolithotomy, nephrectomy and other renal surgery.

e. Hari S. Asopa (Indian urologist)

1. Asopa dorsal inlay urethroplasty (2001)

- Single-stage graft urethroplasty for anterior urethral stricture by a **ventral sagittal urethrotomy** approach.
- **Steps:** expose the urethra ventrally without circumferential mobilisation; open the urethra ventrally through the stricture; incise the **dorsal urethral plate** in the midline down to the corpora; **inlay** a free graft (buccal mucosa or penile/preputial skin) and quilt it to the tunica albuginea; close the urethra over a catheter.
- **Rationale:** the graft lies on the well-vascularised corporal bed; no mobilisation preserves spongiosal blood supply; suits bulbar and penile strictures.
- Ranks with **Barbagli dorsal onlay** and **Kulkarni one-sided** repair among dorsal graft techniques.

2. Asopa hypospadias repair (1970s)

- One-stage repair using a **vascularised inner preputial island flap** still attached to the outer preputial skin, transferred ventrally to form the neourethra; a forerunner of the Duckett transverse preputial island flap.

Diagram to draw: conventional lens scope versus Hopkins rod-lens scope (glass rods with short air gaps); dorsal inlay: urethra opened ventrally, dorsal plate incised, graft on corpora (cross-section).

Key points to remember:

- Hopkins: rod-lens system 1966; fibreoptics 1954.
- Huggins and Hodges 1941: androgen dependence; castration regresses prostate cancer; Nobel 1966.
- Huggins 1945: adrenalectomy for castration-resistant disease.
- Calne: azathioprine, ciclosporin, first UK liver transplant 1968.
- Gaur 1992: glove-balloon retroperitoneoscopy.
- Asopa: ventral sagittal urethrotomy, dorsal plate incised, free graft inlaid (2001).

Interventional radiology in urology and therapeutic angioembolisation

Asked as:

- Therapeutic angioembolization. — *Practice Set 7 Paper I Q3e (SAQ)*
- Interventional urological treatment in urology — *Practice Set 8 Paper I Q3e (SAQ)*

Type: SAQ (10 marks)

Assumed meaning: interventional radiology (image-guided percutaneous and endovascular) treatment in urology.

Principles

- Guidance by ultrasound, fluoroscopy or CT (computed tomography), usually under local anaesthesia. **Seldinger technique:** needle, guidewire, then devices over the wire.
- **Beforehand:** correct coagulopathy, manage anticoagulants, urine culture, antibiotic prophylaxis.
- **Requirements:** experienced team, resuscitation in the suite; unstable non-responders need surgery.

Non-vascular procedures

Procedure	Indications and points
PCN (percutaneous nephrostomy)	Obstructed infected kidney (as effective as retrograde stent), malignant ureteric obstruction, urinary leak or fistula, access for PCNL (percutaneous nephrolithotomy). Puncture a **posterior calyx** through the avascular Brödel plane. Transfusion 0.5–4%
Antegrade stenting	When retrograde stenting fails: malignant obstruction, ileal conduit, transplant ureter
Antegrade pyelography, Whitaker test	Define strictures; pressure–flow test for obstruction
Percutaneous drainage	Renal or perinephric abscess >3 cm, urinoma, lymphocele
Image-guided biopsy	Renal mass biopsy

Vascular (endovascular) procedures: therapeutic angioembolisation

Definition

- Deliberate **endovascular occlusion** of a target vessel, usually by transfemoral selective catheterisation under fluoroscopy, to stop bleeding, devascularise a tumour or close an abnormal vessel.
- **Superselective** embolisation of a segmental branch preserves most parenchyma.

Embolic agents

Type	Examples	Use
Temporary	Autologous clot; gelatin sponge (2–6 weeks)	Priapism, prostate, bladder bleeding
Permanent particles	PVA (polyvinyl alcohol), microspheres	Tumours, PAE (prostate artery embolisation)
Mechanical	Microcoils, vascular plugs	Pseudoaneurysm, AVF (arteriovenous fistula), varicocele

Type	Examples	Use
Liquid	NBCA (N-butyl cyanoacrylate) glue, absolute ethanol	Tumour ablation, recurrent priapism

Urological indications

Procedure	Indications
Renal angioembolisation	Grade IV–V renal trauma; **pseudoaneurysm or arteriovenous fistula** after PCNL or partial nephrectomy (delayed bleed); bleeding AML (angiomyolipoma, first line); palliative embolisation of RCC (renal cell carcinoma)
Internal iliac (anterior division) embolisation	Intractable bladder haemorrhage; initial control about 90%. Spare the posterior division (gluteal ischaemia)
PAE (prostate artery embolisation)	LUTS (lower urinary tract symptoms) from BPH (benign prostatic hyperplasia); EAU (European Association of Urology) for men accepting lesser outcomes; AUA (American Urological Association) 2023 may offer
Varicocele embolisation	Coils or liquid sclerosants via femoral or jugular vein
High-flow priapism	Superselective cavernosal artery embolisation; resolution ≥75%; recurrence 30–40%
PTRA (percutaneous transluminal renal angioplasty)	Fibromuscular dysplasia; transplant renal artery stenosis

- **Renal trauma:** active contrast extravasation, pseudoaneurysm, AVF; success high in grade IV, lower in grade V. Failure predictors: instability, grade V, urinary extravasation. Repeat embolisation often succeeds.
- **Iatrogenic bleeding** after PCNL (percutaneous nephrolithotomy) or partial nephrectomy: **treatment of choice**; superselective coils nearly 100% technically successful.
- **AML (angiomyolipoma): first line for acute haemorrhage**; prophylactic for >6 cm, aneurysm >5 mm, growth or pregnancy plans. In TSC (tuberous sclerosis complex), embolise then give steroids.
- **RCC:** palliation of haematuria or pain in unfit patients; preoperative use no longer routine.
- **Intractable bladder haemorrhage** (radiation cystitis, cancer): bilateral **anterior division of internal iliac**; avoid the posterior division (gluteal pain).
- **Prostate:** PAE for BPH (benign prostatic hyperplasia) and refractory prostatic haematuria.
- **High-flow priapism:** superselective cavernosal artery embolisation; resolution ≥75%; recurrence → repeat with a non-resorbable agent.
- **Also know:** varicocele (venous coils); lymphatic embolisation for chylous ascites; ADPKD (polycystic kidney) pain.

Ablation

- **Cryoablation or RFA (radiofrequency ablation)**, percutaneous, for cT1a renal masses <3 cm; biopsy first. Higher local recurrence than partial nephrectomy, salvaged by repeat ablation.

Complications

- Contrast nephropathy, access-site haematoma, coil migration.
- **Post-embolisation syndrome:** fever, flank pain, nausea, leucocytosis; up to 72 hours; self-limiting.
- **Non-target embolisation:** bowel, bladder, rectum, gluteal muscle, penile skin.
- Renal infarction or abscess; ED (erectile dysfunction) after priapism embolisation.

Diagram to draw: prone PCN through a posterior calyx along the Brödel line; renal angiogram showing segmental pseudoaneurysm before and after coil occlusion.

Key points to remember:

- PCN equals retrograde stent for decompressing an obstructed infected kidney.
- Superselective coils: treatment of choice for pseudoaneurysm and AVF.
- Embolisation is first line for bleeding AML and post-PCNL pseudoaneurysm or arteriovenous fistula; prophylactic >6 cm or aneurysm >5 mm.
- Bladder bleeding: internal iliac anterior division; spare posterior division.
- Post-embolisation syndrome is common and self-limiting.

Tumour markers in urology

Type: SAQ (10 marks)

Source: Practice Set 18 SAQ Sheet, Q3d

Definition and uses

- A substance made by a tumour, measured in blood, urine or tissue.
- Uses: screening, diagnosis, staging, prognosis, response, and detecting relapse. Ideal: specific, sensitive, proportional to bulk, short half-life.

Prostate

- **PSA (prostate-specific antigen):** a kallikrein serine protease (hK3) from luminal cells; **organ-specific, not cancer-specific** (BPH (benign prostatic hyperplasia), prostatitis, instrumentation raise it). Best marker for post-treatment follow-up.
- **Refinements:** density (PSA ÷ volume; cut-offs 0.08–0.15); %free PSA (lower in cancer; for PSA 4–10 ng/mL); velocity (little value); **PHI (Prostate Health Index) = (p2PSA ÷ free PSA) × √total PSA**; 4Kscore.
- **Urine:** PCA3 (prostate cancer antigen 3) after DRE (digital rectal examination); score = PCA3 mRNA ÷ KLK3 mRNA × 1000.
- **Also know:** tissue genomic classifiers.

Testis (germ cell tumours)

Marker	Half-life	Source	Never raised in
AFP (α-fetoprotein)	5–7 days	Embryonal, yolk sac	Pure seminoma, choriocarcinoma
β-hCG (human chorionic gonadotrophin)	24–36 hours	Choriocarcinoma, embryonal; seminoma in ~15%	Yolk sac tumour
LDH (lactate dehydrogenase)	24 hours	Non-specific; reflects bulk	—

- Measure before and after orchidectomy, on chemotherapy, in follow-up. Post-orchidectomy values give the **S stage** (S1: LDH <1.5 × normal, hCG <5000 mIU/mL, AFP <1000 ng/mL) and the **IGCCCG (International Germ Cell Cancer Collaborative Group)** risk group.
- Raised AFP with "pure seminoma" histology → treat as NSGCT (non-seminomatous germ cell tumour). False positives: liver disease, marijuana, LH (luteinising hormone) cross-reaction.
- **miR-371a-3p:** sensitive new marker; not raised by teratoma.

Bladder and upper tract

- **Urine cytology:** sensitive and specific for high-grade cancer, poor for low grade; for surveillance, not screening.
- NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), ImmunoCyt, UroVysion FISH (fluorescence in situ hybridisation): selected use only.

Others

- **Kidney and penis:** no validated serum marker.
- **Adrenal:** metanephrines, cortisol, aldosterone–renin ratio. **Neuroblastoma:** urinary VMA (vanillylmandelic acid)/HVA (homovanillic acid) raised in 90–95%.

Diagram to draw: marker decline curve after orchidectomy using half-lives.

Key points to remember:

- PSA: organ-specific, not cancer-specific.
- AFP never in pure seminoma; half-lives AFP 5–7 days, hCG 24–36 hours.
- S stage uses post-orchidectomy values.
- Cytology: good for high grade only.

Haemostatic agents in urology

Type: SAQ (10 marks)

Source: Practice Set 18 SAQ Sheet, Q3e

Introduction

- Topical agents are **adjuncts** to surgical haemostasis (suture, clip, energy), used for diffuse ooze. No randomised study shows clear benefit of any single agent in partial nephrectomy.

Classification

Class	Examples	Mechanism / note
Passive (mechanical)	Gelatin sponge (Gelfoam), **oxidised regenerated cellulose (Surgicel)**, microfibrillar collagen, polysaccharide spheres	Matrix for platelets; needs intact clotting. Cellulose is acidic (antibacterial)
Active	Topical thrombin	Converts the patient's fibrinogen to fibrin
Flowable	Gelatin matrix + thrombin (**Floseal**)	Fills irregular beds, swells
Fibrin sealants	Fibrinogen + thrombin (Tisseel, Evicel)	Final clotting step; works despite coagulopathy; also seals leaks
Synthetic sealants and glues	PEG (polyethylene glycol) hydrogel (Coseal), cyanoacrylate, albumin–glutaraldehyde (BioGlue)	Seal; adhesive
Systemic	Tranexamic acid, aminocaproic acid, desmopressin	Antifibrinolytic; desmopressin for uraemic platelets

Urological uses

- **Partial nephrectomy:** renorrhaphy ± sealant or Surgicel bolster; sutureless repair in selected cases.
- **Laparoscopic venous injury:** pressure → sealant → suture.
- PCNL (percutaneous nephrolithotomy) tract sealing in tubeless cases; adrenal and spleen/liver ooze; prostatectomy and lymphadenectomy beds.
- **Fibrin glue:** ureteroscopic fulguration for upper-tract bleeding; fistula plugs.
- **Tranexamic acid:** lowers transfusion at radical cystectomy.

Intravesical agents (haemorrhagic cystitis)

Agent	Note
Alum 1% (50 g in 5 L)	First line; avoid in renal failure (aluminium toxicity)
Silver nitrate 0.5–1%	Chemical coagulation
Aminocaproic acid 200 mg/L irrigation	Thromboembolism and clot retention
Formalin 1–4%	Most effective; under anaesthesia; **cystogram first** to exclude reflux

Complications

- Swelling compresses nerves, ureter or spinal cord (gelatin, cellulose): use the least amount.
- Mimics abscess on imaging; foreign-body reaction; infection.
- Flowables entering open vessels → embolism or thrombosis.
- Bovine thrombin → antibodies and coagulopathy; human products → theoretical viral risk.

- **Also know:** energy adjuncts (bipolar, argon beam, vessel sealers); hyperbaric oxygen in radiation cystitis.

Diagram to draw: clotting cascade showing where passive agents, thrombin and fibrin sealant act.

Key points to remember:

- Agents are adjuncts, not substitutes for sutures.
- Fibrin sealant works despite coagulopathy; passive agents need normal clotting.
- Surgicel: acidic, may mimic abscess.
- Alum first line; formalin last, after cystogram.

Sterilisation of urological instruments

Type: SAQ (10 marks)

Source: Practice Set 19 SAQ Sheet, Q3a

Definitions

- **Cleaning:** removes debris that shields organisms; always first.
- **Disinfection:** kills most microbes but not all spores. HLD (high-level disinfection) kills all except large numbers of spores.
- **Sterilisation:** kills all life, including spores.

Spaulding classification

Category	Contact	Requirement	Urological examples
Critical	Sterile tissue, vessels	Sterilisation	Resectoscope, ureteroscope, nephroscope, laparoscopic and robotic instruments
Semi-critical	Mucosa	HLD at least (sterilise if possible)	Flexible cystoscope, TRUS (transrectal ultrasound) probe
Non-critical	Intact skin	Low-level disinfection	Abdominal ultrasound probe

Reprocessing steps

1. Bedside pre-clean; leak test for flexible scopes.
2. Manual cleaning with enzymatic detergent; brush and flush channels.
3. Rinse → disinfect or sterilise → rinse → dry channels (alcohol, air) → store hanging; record tracking.

Methods

Method	Conditions	Use and drawbacks
Steam autoclave	121 °C for 15 min (15 psi) or 134 °C for 3–4 min	Cheapest, most reliable; metal instruments, autoclavable optics
Dry heat	160 °C for 2 hours	Glass; rarely used
Ethylene oxide	Low temperature	Heat-sensitive scopes, cables; toxic, carcinogenic, long aeration
Hydrogen peroxide gas plasma	Low temperature, short cycle	No toxic residue; not for cellulose; long narrow lumens limited
Peracetic acid (Steris)	Automated liquid, low temperature	Sporicidal; ureteroscopes
Glutaraldehyde 2%	HLD in 20 min; sterilant after about 10 hours	Cheap; irritant, sensitiser; must rinse well
Gamma irradiation	Industrial	Single-use catheters, stents, sutures

Monitoring

- **Physical:** time, temperature, pressure records.
- **Chemical indicators:** autoclave tape; Bowie–Dick test (air removal).
- **Biological indicators:** *Geobacillus stearothermophilus* spores (steam, plasma); *Bacillus atrophaeus* (ethylene oxide, dry heat).

Urology-specific points

- A laser-perforated ureteroscope channel lets fluid in during sterilisation, destroying the scope.
- Poor channel cleaning causes outbreaks (Pseudomonas, mycobacteria).
- **Prions** resist routine methods: single-use or special protocols.
- **Single-use scopes** avoid reprocessing (about $107 per case for reusable ureteroscopes).
- **Also know:** laser fibre re-sterilisation; limited-life robotic instruments.

Diagram to draw: reprocessing cycle flowchart.

> **Key points to remember:**
>
> - Spaulding: critical sterilise, semi-critical HLD, non-critical low-level.
> - Autoclave 121 °C 15 min or 134 °C 3–4 min.
> - Ureteroscopes: ethylene oxide, glutaraldehyde or peracetic acid.
> - Biological indicator: *G. stearothermophilus*.

Ethics and informed consent in urology

Type: SAQ (10 marks)

Source: Practice Set 19 SAQ Sheet, Q3e

Four principles (Beauchamp and Childress, 1979)

Principle	Meaning
Autonomy	Patient's deliberated self-rule; basis of consent and confidentiality
Beneficence	Act for the patient's benefit, judged by the patient's values
Non-maleficence	Avoid harm (*primum non nocere*); requires training and honest risk data
Justice	Fair distribution of resources, benefits and risks

- Hippocratic Oath gave beneficence, non-maleficence, confidentiality; autonomy and justice are modern additions (autonomy is bounded by justice).
- **Jonsen's four-box method:** medical indications; patient preferences; quality of life; contextual features (family, cost, law).

Informed consent: legal evolution

- From battery to negligence. **Schloendorff (1914):** right to bodily integrity. **Salgo (1957):** term "informed consent". **Canterbury v Spence (1972): reasonable-patient** disclosure standard.
- India: **Samira Kohli (2008)**: consent for one procedure does not cover another, except in emergency.

Elements

1. Diagnosis, proposed treatment and alternatives, including no treatment.
2. Risks, benefits and uncertainty compared with alternatives.
3. **Check understanding** (most often missed).
4. Voluntary decision by a patient with capacity; documented.

- Material risks: no agreed cut-off (1% proposed); disclose rarer ones by severity; lay language.
- Urology examples: incontinence and erectile dysfunction after prostatectomy; retrograde ejaculation after TURP (transurethral resection of the prostate); stent symptoms; conversion to open surgery.

Exceptions

- **Emergency** in an incapable patient (presumed consent); a conscious patient may still refuse.
- **No capacity:** surrogate using substituted judgement, then best interests.
- **Minors (<18 years):** parents consent; child gives assent.

Improving consent

- Decision aids, repeat-back, simple written material.
- **Shared decision making:** two-way exchange of options and patient values; vital when several options are reasonable (e.g. localised prostate cancer).

Special areas

- **End of life:** early goals-of-care talks, advance directives, DNR (do-not-resuscitate) orders.

- **Research:** written consent, ethics committee approval, extra protection for vulnerable groups.
- **Also know:** confidentiality; conflicts of interest; futility policies.

Diagram to draw: Jonsen four-box grid.

> **Key points to remember:**
>
> - Four principles: autonomy, beneficence, non-maleficence, justice.
> - Canterbury v Spence: reasonable-patient standard; Samira Kohli in India.
> - Commonest gap: not checking understanding.
> - Exceptions: emergency, incapacity (surrogate), minors.

Operating room set-up

Type: SAQ (10 marks)

Source: Practice Set 19 SAQ Sheet, Q3f (see also 2021-Summer Paper 4, Q3a for operating room safety)

Physical design

- **Zones:** protective (reception, changing) → clean (pre-op, recovery) → sterile (theatre, scrub) → disposal; one-way flow.
- **Ventilation:** positive pressure, about 20 air changes an hour, filtered air; laminar flow for implants.
- **Environment:** temperature about 20–24 °C, relative humidity about 20–60%. Patient warming: even 1–2 °C hypothermia raises SSI (surgical site infection) and blood loss (**Kurz 1996**: 3-fold wound infection).
- Power backup, piped gases, anaesthetic scavenging, radiolucent table.

Equipment layout by procedure

Setting	Arrangement
Endourology (TURP (transurethral resection of the prostate), URS (ureteroscopy), PCNL (percutaneous nephrolithotomy))	Lithotomy or prone; tower and monitor directly ahead; C-arm opposite surgeon; wires, baskets, dilators, spare scopes ready
Laparoscopic, upper abdomen	Surgeon and assistant face the abdomen; monitor opposite; scrub nurse near the foot
Laparoscopic, pelvic	30° Trendelenburg, split legs; surgeon on preferred side, assistant opposite
Robotic	Patient cart, surgeon console, vision cart; Si docks between legs, Xi from the side

- **Checklist before laparoscopic incision:** irrigation–suction, electrosurgical unit, full CO_2 cylinder plus spare, white-balanced camera, insufflator and Veress needle checked, **open tray ready**.
- Cables fixed; camera cable apart from diathermy; insufflator at eye level.

Positioning safety

- Pad bony prominences; axillary roll (lateral); avoid prolonged kidney rest and full flexion (rhabdomyolysis); no shoulder braces in steep Trendelenburg (brachial plexus).

Ergonomics and staff protection

- Monitor straight ahead, top at eye level; elbows 90–120°; table at the surgeon's pubic symphysis (laparoscopy).
- **Radiation:** ALARA (as low as reasonably achievable): time, distance, shielding; lead apron, thyroid shield, glasses; dosimeter.
- Smoke evacuation; sharps safety; vaccination.

Safety systems

- **WHO (World Health Organization) Surgical Safety Checklist** (sign in, time out, sign out); counts; fire-risk assessment (oxidiser, fuel, ignition).

Diagram to draw: theatre plan: table, tower, C-arm, anaesthetist, surgeon, scrub nurse.

Key points to remember:

- Zones with one-way flow; positive pressure, about 20 air changes an hour.
- Monitor directly ahead at eye level; open tray ready for laparoscopy.
- Hypothermia triples wound infection (Kurz).
- ALARA: time, distance, shielding.

Amyloidosis in urology

Type: SAQ (10 marks)

Source: Practice Set 21 SAQ Sheet, Q3c

Definition

- Extracellular deposits of misfolded protein in **β-pleated sheet fibrils**. **Congo red stain gives apple-green birefringence** under polarised light.

Classification

Type	Protein	Cause
AL (amyloid light-chain; primary)	Immunoglobulin light chain	Plasma cell dyscrasia, myeloma
AA (amyloid A; secondary)	Serum amyloid A	Chronic inflammation: TB (tuberculosis), rheumatoid arthritis, osteomyelitis, pressure sores, hidradenitis
ATTR (hereditary transthyretin)	Transthyretin	Autosomal dominant neuropathy
Dialysis-related	β2-microglobulin	Long-term dialysis
Localised	Usually AL	Deposits in one organ only

Urological sites

- **Kidney (systemic):** nephrotic proteinuria, CKD (chronic kidney disease), enlarged kidneys, renal vein thrombosis, proximal tubular defects.
- **Bladder (usually localised): painless gross haematuria**; yellowish plaques or nodules that **mimic carcinoma**. Secondary bladder amyloid can bleed severely; Campbell lists amyloidosis among systemic causes of haemorrhagic cystitis.
- **Ureter and renal pelvis:** stricture or filling defect, often calcified; mimics urothelial cancer; obstruction.
- **Urethra:** rare; mass, obstructive voiding, haematuria.
- **Seminal vesicles:** senile deposits in older men; can mimic cancer invasion on MRI (magnetic resonance imaging).
- **Also know:** ATTR gives neurogenic bladder (underactive detrusor, LUTS [lower urinary tract symptoms] >80%) and erectile dysfunction; adrenal insufficiency; papillary necrosis.

Evaluation

- **Biopsy is diagnostic** (avoids unnecessary nephroureterectomy or cystectomy).
- Exclude systemic disease: serum and urine protein electrophoresis, immunofixation, free light chains, echocardiogram, renal function.

Management

- **Localised bladder:** TUR (transurethral resection) or fulguration; intravesical DMSO (dimethyl sulfoxide) reported; recurrence is common, so follow-up cystoscopy is needed.
- **Ureter:** ureteroscopic resection or segmental excision; stent for obstruction.
- **Urethra:** observe if mild; dilatation or urethroplasty if obstructive.
- **Refractory bleeding:** irrigation, embolisation, cystectomy.
- **Systemic:** AL: chemotherapy (daratumumab with bortezomib regimen, ANDROMEDA trial) or stem cell transplant; AA: treat inflammation (colchicine in familial Mediterranean fever); ATTR: tafamidis, patisiran. AA amyloid can recur in a renal transplant.

Diagram to draw: Congo red stained fibrils with apple-green birefringence.

Key points to remember:

- Congo red, apple-green birefringence.
- Localised bladder amyloid mimics cancer; biopsy then TUR.
- Always exclude systemic AL disease.

Rapid Revision Sheet – Section 1 · Basic Principles

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Iodinated and other contrast agents (use in urology) and their complications (LAQ)

- Ratio 1.5 (ionic), 3 (non-ionic monomer), 6 (iodixanol, 290 mOsm/kg).
- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg, max 1 mg; shellfish allergy is not a risk.
- Premedication: prednisone 50 mg at 13-7-1 h or methylprednisolone 32 mg at 12-2 h, plus diphenhydramine.
- CA-AKI: saline if eGFR <30; bicarbonate and NAC useless; hold metformin only if eGFR <30 or AKI.
- NSF risk at GFR <30; group II agents almost none.

Laparoscopic access (Veress, Hasson cannula, extraperitoneal space), gases, and open vs laparoscopic vs robotic radical prostatectomy (LAQ)

- Veress tests: aspiration, drop, advancement; opening pressure <10 mmHg.
- CO_2 standard (soluble, non-flammable); helium in COPD; N_2O supports combustion.
- Gas embolism: left lateral head-down, oxygen, aspirate.
- Yaxley 2016: equal function and margins; robot less blood loss and stay.
- Hasson 1971: open entry under vision.
- Main indication: previous surgery or adhesions.
- Avoids blind Veress injuries; drawback is gas leak.
- Space is created first: finger, then balloon.
- Gaur balloon: glove on red rubber catheter.
- Balloon trocar seals and prevents emphysema.
- Balloon in muscle causes hernia.

Pneumoperitoneum physiology, complications of retroperitoneoscopy and laparoscopic donor nephrectomy (LAQ)

- Two drivers: pressure and absorbed CO_2; work at 10–12 mmHg.
- Afterload rises at all pressures; output collapses at 40 mmHg.
- Oliguria from renal vein compression; reversible.
- Hypercarbia peaks after extubation; COPD needs arterial gases.
- Gas embolism: left lateral, head down, oxygen, aspirate.
- Retroperitoneoscopy: space is created, not natural: wrong plane gives hernia, peritoneal tear or bleeding.
- Emphysema commonest; balloon-tip port and 12 mmHg prevent it (prevents leak).
- Pneumothorax 1.6–4%, commoner retroperitoneally.
- Rhabdomyolysis about 1%: limit kidney rest to <1 hour.
- Ratner 1995; left kidney preferred; hand-assisted in about two-thirds.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Donor ESRD 90 versus 14 per 10,000 (matched non-donors).

Describe IVU & its applications in Urology. (LAQ)

- Contrast is filtered, not reabsorbed: nephrogram then pyelogram.
- Delayed, increasingly dense nephrogram = acute obstruction; no excretion at 24 h = non-functioning.
- Classic signs: drooping lily, cobra head, fish-hook, paintbrush, ring sign, phantom calyx.
- Severe reaction: IM adrenaline 0.3–0.5 mg; CTU has replaced IVU.

Principles of tissue engineering and its use in urology (LAQ)

- Triad: cells, scaffold, signals; plus bioreactor and blood supply.
- Diffusion limit about 100–200 µm: vascularity is the main barrier.
- Atala 2006 promising; phase II trial showed no gain; conduit half stenosed.
- Raya-Rivera 2011: seeded urethras in 5 boys.

Radionuclide (isotope) scans in urology: principles, renal scans, MAG3 and applications (LAQ)

- ^{99m}Tc: 6 hours, 140 keV; PET: two 511 keV photons.
- DTPA filtered = GFR; MAG3 secreted = obstruction and transplant; DMSA cortical = scars.
- MAG3: tubular secretion, about 60% extraction; best for split function and drainage.
- Split function (DRF) from 1–3 minute counts; normal 45–55%.
- T½ >20 minutes (min) obstructed, <10 min not.
- Hydration, catheter, correct diuretic timing: the well-tempered renogram.
- Captopril renogram: ≥10% improvement off ACE inhibitors.
- Scar persists at 6 months; pyelonephritis resolves.
- PSMA PET preferred; FDG for residual seminoma >3 cm.
- MIBG best in MEN-2; sentinel node with ^{99m}Tc-nanocolloid.
- Sestamibi hot = oncocytoma; ^{177}Lu-PSMA-617 (VISION).

Newer generation CT scan (SAQ)

- Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
- DECT: uric acid stones, virtual unenhanced, iodine maps.
- Enhancement ≥20 HU = solid; deep learning enables ~1 mSv CT.

History of Laparoscopy (SAQ)

- Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
- Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
- da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.

Foley's (urethral) catheters, guidewires and DJ stents with their complications (SAQ)

- Foley balloon catheter: designed for post-prostatectomy haemostasis.
- Fr = 3 × diameter (mm); inflate with sterile water; coudé for prostatic enlargement, Councill over a wire.
- Suspected urethral injury → RUG first; AUS → deactivate, 12–14 Fr.
- CAUTI 5–10% per day; remove early; change the catheter when treating.
- Hydrophilic wire to pass, PTFE wire to stay (safety).
- DJ symptoms in about 80%: trigonal irritation and reflux; α-blockers and antimuscarinics help.
- Long-dwelling stents: monitor for encrustation; encrustation rises with dwell time; FECal I–V.
- Encrusted stent: bladder, then ureter, then kidney; keep a stent register.

Andrew Novick. (SAQ)

- Novick (Cleveland Clinic): nephron-sparing, renovascular and bench surgery.
- PN indications: imperative, relative, elective (cT1a ≤4 cm).
- Warm ischaemia under about 25–30 minutes; ice slush if longer.
- Negative margin is enough; keep ≥20–30% of one kidney.

Proteinuria (SAQ)

- Normal <150 mg/day; nephrotic >3.5 g/day.
- ACR A2 30–300, A3 >300 mg/g.
- Glomerular, tubular, overflow; orthostatic proteinuria is benign.

- Dipstick misses light chains; use sulfosalicylic acid or immunofixation.

Acetylcysteine and Contrast Induced Nephropathy (CIN) (SAQ)

- Thiol; glutathione precursor, antioxidant, mucolytic.
- Antidote for paracetamol overdose; best within 8 h.
- PRESERVE 2018: no benefit in contrast AKI.
- Cystinuria uses tiopronin, not NAC.
- CIN = rise ≥25% or 0.5 mg/dL in 48–72 h; now CA-AKI vs CI-AKI.
- Main risk: eGFR <30 (IV); <45 intra-arterial.
- Isotonic saline only proven; PRESERVE: bicarbonate and NAC useless.
- Metformin: withhold 48 h only if eGFR <30 or AKI.

Non iodinated contrast materials (SAQ)

- GBCAs shorten T1; NSF risk eGFR <30; group II agents almost no risk.
- Microbubbles: intravascular, no nephrotoxicity; ceVUS for reflux.
- CO_2 angiography: never above diaphragm.

Health-related quality of life (HRQOL) instruments in urology, including after urinary diversion (SAQ)

- Generic (SF-36, EQ-5D) vs disease-specific (EPIC, IPSS, ICIQ, IIEF) plus bladder-specific (FACT-Bl, BLM30, BCI) tools.
- SHIM lacks bother; I-QOL lacks symptoms.
- Reliable, valid, responsive; Cronbach α >0.7; IPSS MCID ~3 points.
- No diversion proven superior; studies flawed.
- Neobladder better body image; conduit better urinary function.

PET scans in urology. (SAQ)

- proPSMA: PSMA PET 92% vs 65% accuracy.
- Seminoma residual >3 cm: FDG-PET at ≥6 weeks; not NSGCT.
- FDG poor in prostate and clear cell RCC.
- VISION: ^{177}Lu-PSMA-617 in mCRPC.

Bacteriuria in pregnancy. (SAQ)

- Prevalence unchanged; risk of pyelonephritis raised (15–45% untreated).
- Screen by culture at 12–16 weeks; treat 4–7 days.
- β-lactams safe; no fluoroquinolones or tetracyclines; nitrofurantoin not after 35 weeks.

Recist Criteria. (SAQ)

- RECIST 1.1: ≥10 mm lesion, node ≥15 mm short axis; 5 targets, 2 per organ.
- PR ≥30% fall from baseline; PD ≥20% and ≥5 mm rise from nadir, or new lesion.
- iRECIST for immunotherapy; PCWG3 for prostate.

Outline algorithm for difficult catheter placement. (SAQ)

- Blind → wire → vision → suprapubic.
- BPH: 18-Fr coudé, tip up.
- Councill catheter over wire after dilatation.
- Suspected urethral injury: RUG first, no blind catheter.

Indications for outpatient cystourethroscopy. (SAQ)

- Haematuria and NMIBC surveillance are the two commonest indications.
- Flexible scope, lidocaine dwell ≥5 minutes, no routine prophylaxis.
- Therapeutic: stent removal, botulinum toxin, fulguration of small recurrences.

MR urography (SAQ)

- Static T2 (no contrast, function-independent) versus excretory gadolinium T1.
- Stones are signal voids; CTU preferred for urothelial cancer.
- Functional MRU: furosemide 1 mg/kg, Patlak analysis for split function.
- NSF risk minimal with group II agents.

Urine cytology (SAQ)

- High specificity, low sensitivity; good for high grade and CIS.
- Paris System: N:C ≥0.7 for suspicious and HGUC; AUC N:C ≥0.5.
- Positive cytology, normal cystoscopy: upper tracts, random biopsies.
- Not for screening or initial microhaematuria.

Patient positioning in lap urology (SAQ)

- Nerve injury about 2.8%; brachial plexus is commonest at risk.
- Never shoulder braces in steep Trendelenburg.
- Axillary roll below, not in, the axilla.
- Rhabdomyolysis: men, BMI ≥25, kidney rest; CK >5,000.

Clinical trial methodology (SAQ)

- Phase III RCT with concealed allocation and blinding is the gold standard.
- Primary endpoint drives sample size; surrogates need the Prentice criteria.
- ITT analysis; CONSORT flow diagram for reporting.

Evidence-based medicine and critical appraisal of research papers (SAQ)

- Sackett 1996: best evidence + clinical expertise + patient values.
- 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
- Level 1a = systematic review of RCTs; GRADE: strong or weak.
- NNT = 1/ARR; beware lead-time, length-time and confounding bias.
- Three questions: valid, results, applicable.
- Randomisation, concealment, blinding, ITT, follow-up >80%.
- CONSORT (RCT), PRISMA (reviews), STROBE (observational), STARD (diagnostic).

Doppler in urology (SAQ)

- RI = (PSV − EDV)/PSV; normal renal 0.47–0.70.
- Penile PSV <25 cm/s arterial; EDV >5 cm/s veno-occlusive.
- Absent intratesticular flow = torsion, but explore on clinical grounds.

Christian Chaussey (SAQ)

- Chaussy, Munich: first human SWL 7 February 1980, Dornier HM1.
- HM3 (1983) is still the efficacy benchmark.
- Spallation, squeezing, shear, cavitation; electromagnetic sources today.

Flow cytometry (SAQ)

- Forward scatter = size; side scatter = granularity; fluorescence = marker or DNA.
- DNA index 1.0 diploid; aneuploidy = worse prognosis.
- Flow crossmatch is more sensitive than CDC crossmatch.

VCUG (SAQ)

- Gold standard for VUR and PUV; sterile urine first.
- Capacity (age × 30) + 30 mL (2–14 years); cyclic study detects 12–20% more.
- Grades I–II non-dilating, III–V dilating.

Pioneers of urology: Hopkins, Huggins, Calne, Gaur and Asopa (SAQ)

- Hopkins: rod-lens system 1966; fibreoptics 1954.

- Huggins and Hodges 1941: androgen dependence; castration regresses prostate cancer; Nobel 1966.
- Huggins 1945: adrenalectomy for castration-resistant disease.
- Calne: azathioprine, ciclosporin, first UK liver transplant 1968.
- Gaur 1992: glove-balloon retroperitoneoscopy.
- Asopa: ventral sagittal urethrotomy, dorsal plate incised, free graft inlaid (2001).

Interventional radiology in urology and therapeutic angioembolisation (SAQ)

- PCN equals retrograde stent for decompressing an obstructed infected kidney.
- Superselective coils: treatment of choice for pseudoaneurysm and AVF.
- Embolisation is first line for bleeding AML and post-PCNL pseudoaneurysm or arteriovenous fistula; prophylactic >6 cm or aneurysm >5 mm.
- Bladder bleeding: internal iliac anterior division; spare posterior division.
- Post-embolisation syndrome is common and self-limiting.

Tumour markers in urology (SAQ)

- PSA: organ-specific, not cancer-specific.
- AFP never in pure seminoma; half-lives AFP 5–7 days, hCG 24–36 hours.
- S stage uses post-orchidectomy values.
- Cytology: good for high grade only.

Haemostatic agents in urology (SAQ)

- Agents are adjuncts, not substitutes for sutures.
- Fibrin sealant works despite coagulopathy; passive agents need normal clotting.
- Surgicel: acidic, may mimic abscess.
- Alum first line; formalin last, after cystogram.

Sterilisation of urological instruments (SAQ)

- Spaulding: critical sterilise, semi-critical HLD, non-critical low-level.
- Autoclave 121 °C 15 min or 134 °C 3–4 min.
- Ureteroscopes: ethylene oxide, glutaraldehyde or peracetic acid.
- Biological indicator: *G. stearothermophilus*.

Ethics and informed consent in urology (SAQ)

- Four principles: autonomy, beneficence, non-maleficence, justice.
- Canterbury v Spence: reasonable-patient standard; Samira Kohli in India.
- Commonest gap: not checking understanding.
- Exceptions: emergency, incapacity (surrogate), minors.

Operating room set-up (SAQ)

- Zones with one-way flow; positive pressure, about 20 air changes an hour.
- Monitor directly ahead at eye level; open tray ready for laparoscopy.
- Hypothermia triples wound infection (Kurz).
- ALARA: time, distance, shielding.

Amyloidosis in urology (SAQ)

- Congo red, apple-green birefringence.
- Localised bladder amyloid mimics cancer; biopsy then TUR.
- Always exclude systemic AL disease.

Section 2 · Oncology, DSD, Gender & Ageing — Short answer questions (SAQ)

Oncogenes

Asked as:

- Oncogenes — *Winter 2017 Paper I Q3f (SAQ)*
- Oncogene — *Practice Set 6 Paper I Q3f (SAQ)* (also answered in 2017-Winter Paper 1, Q3f)

Type: SAQ (10 marks)

Definition

- **Proto-oncogene:** a normal gene that promotes cell growth and division (an "accelerator").
- **Oncogene:** its mutated or overactive form that drives (driving) uncontrolled proliferation. The change is **gain of function** and **dominant**: one altered allele is enough.
- The first human oncogene, **HRAS**, was isolated from (found in) the **T24 bladder cancer** cell line.

Mechanisms of activation

Mechanism	Effect	Example
Point (activating) mutation	Constitutively active protein (protein permanently "on")	RAS (bladder); FGFR3 (fibroblast growth factor receptor 3) in low-grade papillary bladder cancer; MET germline (germline MET) in hereditary papillary RCC (renal cell carcinoma)
Gene amplification	Increased copy number (more copies) → overexpression	MYC (prostate, bladder); HER2 (human epidermal growth factor receptor 2); AR (androgen receptor) in castration-resistant prostate cancer
Chromosomal translocation / fusion	Gene placed under a strong promoter, or fusion protein	**TMPRSS2–ERG** (androgen-driven ERG in about half of prostate cancers); BCR–ABL; TFE3 fusions (translocation RCC)

- **Also know:** promoter hypomethylation can switch oncogenes on.

Functional classes

Class	Examples
Growth factors	PDGF (platelet-derived growth factor), sis
Receptor tyrosine kinases	EGFR (epidermal growth factor receptor), HER2, FGFR3, MET, KIT (seminoma)
Signal transducers	**RAS** (binds GTP [guanosine triphosphate]; GTP-binding; mutation locks it "on"), BRAF, PI3K
Transcription factors	**MYC**, ERG, ETS family
Cell cycle / apoptosis regulators	Cyclin D1, BCL-2

Oncogene versus tumour suppressor gene

	Oncogene	Tumour suppressor
Normal role	Accelerator	Brake, DNA repair
Change	Gain of function	Loss of function

	Oncogene	Tumour suppressor
Alleles needed	One	Usually both (**Knudson two-hit**: retinoblastoma)
GU (genitourinary) examples	MYC, MET, FGFR3, ERG	RB1, TP53, VHL, PTEN

Urological significance

- **Bladder:** FGFR3 mutations in up to 80% of low-grade Ta tumours (10–20% of MIBC (muscle-invasive bladder cancer)); FGFR3 and TERT promoter mutations in >60% of low-grade papillary tumours; RAS (HRAS) mutually exclusive with FGFR3; HER2 in 9–12% of MIBC; MYC overexpressed in over half.
- **Kidney:** MET activation in papillary RCC.
- **Prostate:** TMPRSS2–ERG fusion, MYC (8q gain), AR amplification after (under) hormone therapy.
- **Testis:** KIT and KRAS mutations (seminoma).

Targeted therapy ("oncogene addiction")

- **Erdafitinib** (pan-FGFR inhibitor) for FGFR-altered (FGFR3/2-altered) advanced (metastatic) urothelial cancer; MET-active TKIs (tyrosine kinase inhibitors) in papillary RCC; AR-targeted drugs in prostate cancer; HER2-directed drugs (trastuzumab deruxtecan received a tumour-agnostic FDA approval in 2024 for HER2-positive solid tumours; disitamab vedotin is under study in urothelial cancer).

Diagram to draw: growth factor → receptor → RAS → RAF → MEK → ERK → MYC pathway.

Key points to remember:

- Oncogenes: gain of function, one allele; activated by mutation, amplification, translocation.
- HRAS first found in T24 bladder cancer; FGFR3 in up to 80% low-grade Ta.
- TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.

Premalignant conditions in urology (including premalignant lesions of penis)

Asked as:

- Premalignant conditions in urology — *Summer 2021 Paper I Q3a (SAQ)*
- Premalignant lesion of penis — *Practice Set 6 Paper I Q3a (SAQ)*

Type: SAQ (10 marks)

(premalignant conditions also answered in 2021-Summer Paper 1, Q3a)

Definition

- Altered tissue with a raised risk of invasive cancer, often **intraepithelial neoplasia with an intact basement membrane**. Treating it prevents cancer.
- Penis: a lesion with raised risk of invasive SCC (squamous cell carcinoma). The key entity is **PeIN (penile intraepithelial neoplasia)**: dysplastic squamous epithelium with an **intact basement membrane**.

Penis

Classification

1. PeIN (HPV-associated or HPV-independent)

- **HPV (human papillomavirus)-associated** (commonest; basaloid and warty) or differentiated (HPV-independent).
- **Mechanism (HPV):** viral E6 inactivates p53, E7 inactivates Rb (retinoblastoma protein); **p16** is the marker.
- **HPV-independent (differentiated):** arises in lichen sclerosus.
- **WHO (World Health Organization) 2022:** all PeIN is high grade; grades I–III abandoned.

Clinical entity / lesion	Site, appearance / key points	Risk of invasion
Erythroplasia of Queyrat	Glans or prepuce; red velvety (glans) plaque	**Progresses in 10–33%**
Bowen disease	Shaft; scaly red (shaft) plaque	About 5% progress
Bowenoid papulosis	Shaft papules, young men; HPV 16	Does not progress; may regress

2. Other lesions

- **Lichen sclerosus (BXO [balanitis xerotica obliterans]):** SCC (squamous cell carcinoma) in **2.3–9%** (HPV-independent type).
- **Buschke–Löwenstein tumour** (giant condyloma; verrucous carcinoma): HPV 6/11; locally invasive, locally destructive, rarely metastasises; may harbour SCC.
- **Also know:** cutaneous horn; PEKMB (pseudoepitheliomatous keratotic and micaceous balanitis, controversial); condyloma acuminatum; (chronic) phimosis; genital lichen planus; Kaposi sarcoma (HHV-8 [human herpesvirus 8], check HIV [human immunodeficiency virus]).

Evaluation

- Biopsy first: **multiple biopsies of adequate depth** to exclude invasion; p16 staining. Examine inguinal nodes; partner screening for cervical neoplasia.

Treatment (organ-preserving)

Situation	Treatment
Prepuce (preputial lesions)	Circumcision or excision with clear margin
Glans	**Topical 5-FU (5-fluorouracil) or 5% imiquimod**; **laser** (Nd:YAG [neodymium:yttrium-aluminium-garnet] preferred; CO_2 too superficial); **glans resurfacing** with skin graft
Shaft	Excision; Mohs surgery
Lichen sclerosus	Clobetasol 0.05% for 3 months then maintenance; circumcision; lifelong follow-up
Buschke–Löwenstein	Wide excision; avoid radiotherapy (anaplastic change)

- Long-term follow-up (recurrence common after organ-preserving treatment). **Prevention:** neonatal circumcision, HPV vaccination, avoiding smoking.

Bladder

- **CIS** (carcinoma in situ): flat high-grade lesion, umbrella cells lost; untreated, **40–83%** invade muscle. Treat with BCG (bacillus Calmette–Guérin).
- **Urothelial dysplasia:** 14–19% progress; 60% to CIS with prior cancer.
- **Cystitis glandularis** (intestinal type): low risk of adenocarcinoma; resect and survey.
- **Leukoplakia** (keratinising squamous metaplasia): traditionally premalignant; Campbell now says **not** (no TP53 changes). Schistosomiasis and chronic catheters predispose to SCC.

Prostate

- **HGPIN** (high-grade prostatic intraepithelial neoplasia): precursor; isolated HGPIN no longer needs repeat biopsy. IDC-P (intraductal carcinoma of the prostate) usually means invasive cancer nearby.

Testis

- **GCNIS** (germ cell neoplasia in situ): precursor of all postpubertal GCTs (germ cell tumours) except spermatocytic tumour; **50% invasive within 5 years**. Treat by orchidectomy or **radiotherapy ≥20 Gy**.
- **Gonadoblastoma** in dysgenetic gonads with Y material; cryptorchidism.

Kidney and others

- **Also know:** papillary adenoma; acquired cystic kidney disease on dialysis; urethral diverticulum; bowel in augmentation.

Diagram to draw: normal urothelium vs dysplasia vs CIS (umbrella cells lost); penile epithelium with full-thickness atypia above an intact basement membrane.

> **Key points to remember:**
> - PeIN: intact basement membrane; WHO 2022 all high grade.
> - Queyrat 10–33% vs Bowen (about) 5% progression; bowenoid papulosis does not progress.
> - Lichen sclerosus: SCC 2.3–9%.
> - Untreated bladder CIS: 40–83% invade.
> - GCNIS: 50% invasive in 5 years; radiotherapy ≥20 Gy.

46-XX Male

Type: SAQ (10 marks)

Source: 2022-Summer Paper 1, Q3e

Assumed meaning: 46,XX testicular DSD (difference of sex development), "XX male" or de la Chapelle syndrome.

Definition

- Testes and male genitalia in a person with a 46,XX karyotype. Old term "XX sex reversal"; renamed **46,XX testicular DSD** (2006 consensus). First described by **de la Chapelle (1964)**. About **1 in 20,000** males.

Genetics

Type	Mechanism	Phenotype
SRY-positive (about 80–90%)	**SRY** (sex-determining region Y) translocated from Yp to the X chromosome during paternal meiosis	Usually normal male genitalia
SRY-negative	Testis genes switched on without SRY: **SOX9** duplication or upregulation, SOX3, loss of RSPO1 or WNT4	More hypospadias, undescended testes, ovotestis

- SRY activates SOX9, so Sertoli cells form and secrete **AMH (anti-Müllerian hormone)**: no uterus. Leydig testosterone virilises.
- The **AZF (azoospermia factor) region of Yq is absent**, so spermatogenesis fails.

Clinical features

- Normal male identity; often found at **infertility work-up** for azoospermia, or prenatally (cell-free DNA 46,XX but male ultrasound).
- Small firm testes, **azoospermia**, hypergonadotropic hypogonadism, gynaecomastia in some; progressive testicular regression.
- Height normal or **shorter than average** (no Y growth genes), unlike Klinefelter.

Feature	46,XX male	Klinefelter (47,XXY)
Frequency	1 in 20,000	1 in 500–1000
Height	Normal or short	Often tall
Genital anomalies	More common	Rare
Sperm retrieval	Never (no AZF)	Possible by micro-TESE (microdissection testicular sperm extraction)

Evaluation

- Semen analysis (azoospermia); FSH (follicle-stimulating hormone) and LH (luteinising hormone) high, testosterone low or low-normal.
- **Karyotype** 46,XX; **FISH (fluorescence in situ hybridisation) for SRY**; DSD gene panel if SRY-negative; scrotal ultrasound.

Management

- **Testosterone replacement** when hypogonadal (libido, bone density, muscle).
- **Infertility:** testicular sperm extraction is futile; counsel for **donor sperm or adoption**.
- Gynaecomastia: reduction mammoplasty if persistent; hypospadias or undescended testis repaired.
- Genetic counselling and psychological support; bone density monitoring.

Diagram to draw: X and Y chromosomes showing Xp–Yp exchange carrying SRY onto X.

Key points to remember:

- 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.
- Azoospermia always (AZF absent): no sperm retrieval.
- Shorter than Klinefelter; FISH for SRY confirms.

Female circumcision

Type: SAQ (10 marks)

Source: 2023-Winter Paper 1, Q3f (also answered in 2019-Summer Paper 2, Q3f)

Taken as female genital mutilation/cutting (Chapter 58 calls it "female circumcision (infibulation)").

Definition and epidemiology

- **FGM (female genital mutilation):** partial or total removal of, or other injury to, the external female genitalia for non-medical reasons (WHO, World Health Organization).
- About **200 million** females affected; about 3 million African girls a year. Africa, Middle East, Indonesia and Malaysia; some Indian communities.
- Usually at 4–10 years, without anaesthesia, as a rite of passage; predates Islam. Pharaonic type = infibulation.

WHO classification

Type	Description
I (clitoridectomy)	Partial or total removal of the clitoris and/or prepuce
II (excision)	Clitoris and labia minora, ± labia majora
III (infibulation)	Narrowing of the introitus by cutting and joining the labia, ± clitoral excision
IV (other)	Pricking, piercing, incising, scraping, cauterising

- Older Toubia scheme numbers differ: infibulation is Type III in WHO.

Complications

- **Immediate:** pain, bleeding, shock, infection, **urinary retention**, urethral or rectal injury.
- **Urological:** recurrent **UTI** (urinary tract infection), slow stream, meatal obstruction, vaginal stone from urine stasis, **fistula** after obstructed labour.
- **Gynaecological:** retained menses, dysmenorrhoea, cysts, keloid, dyspareunia, apareunia.
- **Obstetric:** obstructed labour, tears, haemorrhage.
- **Psychological:** anxiety, depression, PTSD (post-traumatic stress disorder).

Management

- Sensitive, culturally respectful history; examine and classify; drawings or photographs for consent.
- **Deinfibulation:** simple **midline division of the scar** with subcuticular closure of each edge; exposes meatus and introitus. High success. Best before pregnancy or antenatally.
- **Never reinfibulate.**
- Excise cysts, treat stenosis, repair fistula; counselling.
- Clitoral reconstruction: less predictable, limited follow-up.

Legal aspects

- WHO and UN (United Nations) condemn FGM, including by health workers. Illegal in many countries (UK FGM Act 2003). India has no specific law.

Diagram to draw: vulva with types I–III and the deinfibulation line.

Key points to remember:

- WHO types I–IV; infibulation is type III.
- Retention, UTI, stones, fistula.
- Midline deinfibulation; never reinfibulate.

Familial malignancies in urology

Asked as:

- Familial malignancies in Urology. — *Practice Set 2 Paper I Q3b (SAQ)*
- Familial malignancy in urology — *Practice Set 8 Paper I Q3f (SAQ)*

Type: SAQ (10 marks)

(Practice Set 2: also answered in 2021-Summer Paper 2, Q3f. Practice Set 8: also answered in 2021-Summer Paper 2, Q3f and Practice Set 2 Paper 1, Q3b)

Principles

- **Knudson two-hit hypothesis:** every cell carries one inherited mutant tumour-suppressor allele, so one more hit causes cancer: tumours are **early, bilateral and multifocal**.
- **4–6% of RCC (renal cell carcinoma)** is familial. All familial RCC syndromes are **autosomal dominant**; only HPRC (hereditary papillary renal carcinoma) involves an activated oncogene.

Hereditary kidney cancer syndromes

Syndrome	Gene (chromosome)	Kidney tumour	Other features
VHL (von Hippel–Lindau)	VHL (3p25)	Clear cell, multifocal	Retinal angioma, CNS (central nervous system) haemangioblastoma, phaeochromocytoma
HPRC	MET (7q31)	Multiple bilateral papillary (type 1)	None
HLRCC (hereditary leiomyomatosis and RCC)	FH (fumarate hydratase, 1q42–43)	**Aggressive** papillary, collecting duct	Skin and uterine leiomyomas
BHD (Birt–Hogg–Dubé)	FLCN (17p11.2)	Chromophobe, hybrid oncocytic	Fibrofolliculomas, lung cysts, pneumothorax
TSC (tuberous sclerosis complex)	TSC1 (9q34), TSC2 (16p13.3)	Multiple AMLs (angiomyolipomas); RCC 2–3%	Angiofibromas, cortical tubers

- **Also know:** SDH (succinate dehydrogenase)-deficient RCC; BAP1 (uveal melanoma, mesothelioma); PTEN (Cowden).

Kidney management

- **3 cm rule:** watch tumours <3 cm; nephron-sparing surgery at ≥3 cm (VHL, HPRC, BHD) to save (saving) function for repeat surgery. **Exception: HLRCC**: excise promptly, wide margins.
- TSC AML: everolimus at 3–5 cm (another source: if >3 cm); embolise or excise ≥5 cm. **Belzutifan** (HIF-2α [hypoxia-inducible factor 2α] inhibitor) approved 2021 for VHL (VHL tumours).

Other urological cancers

- **Prostate:** BRCA2, BRCA1, ATM, CHEK2, PALB2, HOXB13. Father and brother affected: 9-fold risk. Earlier screening, cautious active surveillance, **PARP (poly ADP-ribose polymerase) inhibitors**.

- **UTUC (upper tract urothelial carcinoma): Lynch syndrome** (MLH1, MSH2, MSH6, PMS2): about 14-fold risk, 75-fold with MSH2. Screen with MMR (mismatch repair) immunohistochemistry, then germline testing.
- **Testis:** affected brother RR (relative risk) 8–12, father 2–4.
- **Phaeochromocytoma:** about one-third hereditary: MEN-2 (multiple endocrine neoplasia type 2; RET), VHL, NF1 (neurofibromatosis type 1), SDHB (highest malignancy).
- **Also know:** Wilms tumour (WT1: in WAGR [Wilms, aniridia, genitourinary anomalies, retardation] and Denys–Drash; Beckwith–Wiedemann).

Approach

- Young age, multifocality or family history: genetic counselling, germline testing, screen relatives.

Diagram to draw: VHL loss → HIF accumulation → VEGF (vascular endothelial growth factor) → angiogenesis.

> **Key points to remember:**
> - All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
> - VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
> - Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

Secondary malignancies in urology

Asked as:

- Secondary malignancies in Urology. — *Practice Set 3 Paper I Q3f (SAQ)*
- Secondary malignancies in Uro — *Practice Set 9 Paper I Q3c (SAQ)* (also answered in Practice Set 3 Paper 1, Q3f)

Type: SAQ (10 marks)

Interpreted (read) as second cancers caused by urological or oncological treatment (radiotherapy, chemotherapy, urinary diversion, immunosuppression).

Definition

- A **second (treatment-related) malignancy** is a new, histologically distinct cancer caused by treatment of a first disease. It appears after a latency of years, often **in the radiation field**.

Causes and risks

Cause	Second cancer	Key facts
Pelvic radiotherapy	**Bladder cancer**, rectal cancer, sarcoma (in-field)	Bladder RR (relative risk) 2–4 after cervical EBRT (external beam radiotherapy), 1.5–2 after prostate EBRT, smaller after brachytherapy, little with IMRT (intensity-modulated radiotherapy). Latency 15–30 years
Cyclophosphamide	**Bladder cancer**, UTUC (upper tract urothelial carcinoma)	Only chemotherapy proven to cause bladder cancer; metabolites phosphoramide mustard and acrolein. About 4.5-fold, dose-dependent (>20 g cumulative)
Urinary diversion / augmentation	Adenocarcinoma at the uro-intestinal junction	Ureterosigmoidostomy 6–29% after 10–20 years (registry 2.6%); cystoplasty 1.6%; ileal conduit 0.02%. Risk tracks colonic mucosa exposed to urine and faeces
Transplant immunosuppression	Skin cancer, PTLD (post-transplant lymphoproliferative disorder), RCC (renal cell carcinoma), bladder cancer	RCC mostly in native kidneys; bladder cancer about 3-fold and aggressive
Testicular GCT (germ cell tumour) treatment	Solid tumours, leukaemia	60–100% excess after cisplatin or radiotherapy, 200% (highest) after both; etoposide-related leukaemia
Paediatric tumours	In-field solid tumours, leukaemia	Wilms tumour 1% at 10 years

Mechanisms

- Radiation and alkylating (drugs) or topoisomerase-inhibitor chemotherapy: mutagenic DNA damage in normal tissue.
- Immunosuppression: loss of (lost) immune surveillance and oncogenic viruses (EBV [Epstein–Barr virus], HPV [human papillomavirus]).
- Diversion: urine–faeces contact at the anastomosis.

Prevention and surveillance

- IMRT, smaller fields; **mesna** with cyclophosphamide (reduces cystitis).

- **Haematuria after pelvic radiotherapy or cyclophosphamide: always cystoscopy.**
- Endoscopy from 5 years after ureterosigmoidostomy; augmentation surveillance from about 10 years; excise a defunctioned ureterosigmoid anastomosis.
- Testis cancer: fewer treatment modalities; low-dose CT (computed tomography) in surveillance.
- **Also know:** pioglitazone, aristolochic acid and phenacetin as drug-related urothelial carcinogens.

Diagram to draw: ureterosigmoid anastomosis showing (with) adenocarcinoma at the junction.

Key points to remember:

- Pelvic EBRT raises bladder cancer 1.5–4-fold; IMRT little.
- Cyclophosphamide: only chemotherapy proven to cause bladder cancer.
- Ureterosigmoidostomy cancer 6–29% after 10–20 years.
- Transplant recipients: bladder cancer about 3-fold.

Genetic counselling

Type: SAQ (10 marks)

Source: Practice Set 7 Paper 1, Q3d

Assumed meaning: genetic counselling in urological practice.

Definition

- A communication process that helps patients and families understand the **medical, psychological and family implications** of a genetic condition, so they can make informed choices. It is **non-directive, voluntary and confidential**.
- **Germline testing** (blood) finds inherited variants with family implications; **somatic testing** (tumour) guides treatment.

Steps

1. **Three-generation pedigree**.
2. **Risk assessment:** inheritance and recurrence risk (autosomal dominant 50%; autosomal recessive 25%).
3. **Pre-test counselling:** which test; possible results (pathogenic, **VUS [variant of uncertain significance]**, negative); insurance and psychological harms; consent.
4. **Post-test counselling:** interpret (do not act on a VUS); **cascade testing of relatives**; surveillance; targeted therapy.
5. **Reproductive options:** prenatal diagnosis, PGT (preimplantation genetic testing), donor gametes; psychosocial support.

Urological applications

Condition	Genes	Key point
Prostate cancer	BRCA2 (> BRCA1), ATM, CHEK2, PALB2, mismatch repair (Lynch), HOXB13	NCCN (National Comprehensive Cancer Network): test metastatic, high-risk, strong family history; BRCA2: aggressive disease, PARP (poly-ADP-ribose polymerase) inhibitors, earlier screening
Hereditary RCC (renal cell carcinoma)	VHL, MET, FH, FLCN, TSC1/2, SDH, BAP1	Counsel RCC at ≤46 years, bilateral or multifocal, family history
UTUC (upper tract urothelial carcinoma)	Lynch (MSH2 highest)	Screen if <60 years or Lynch-spectrum cancer history
ADPKD (autosomal dominant polycystic kidney disease)	PKD1 (chromosome 16), PKD2 (chromosome 4)	Testing selects living related donors
Cystinuria	SLC3A1, SLC7A9 (recessive)	Screen siblings
Male infertility	Karyotype, AZF (azoospermia factor) deletion, CFTR (cystic fibrosis transmembrane conductance regulator)	AZF deletions pass to sons by ICSI (intracytoplasmic sperm injection); absent vasa: test partner for CFTR
DSD (differences of sex development)	e.g. CAH (congenital adrenal hyperplasia, recessive), AIS (androgen insensitivity, X-linked)	Recurrence and gonadal tumour risk

Ethics and law

- Confidentiality; right not to know.
- **India:** PCPNDT Act (Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994) bans disclosure of fetal sex.

Diagram to draw: three-generation pedigree with standard symbols.

> **Key points to remember:**
>
> - Non-directive; pre-test and post-test counselling; cascade testing.
> - Prostate: BRCA2 most important; test all metastatic men.
> - RCC ≤46 years, ADPKD, cystinuria, AZF/CFTR, DSD are classic urological indications.

Congenital Penile curvature..

Type: SAQ (10 marks)

Source: Practice Set 10 Paper 1, Q3d

Definition and principle

- Curvature of the erect penis present from birth, with an **orthotopic meatus**.
- Rigidity needs **symmetric compliance** of the tunica albuginea; curvature means one aspect is less compliant.

Embryology

- DHT (dihydrotestosterone) drives fusion of the urethral folds and maturation of spongiosum, Buck's and dartos fascia. The urethra may form normally while the mesenchyme stays deficient → dysgenetic, inelastic ventral tissue.
- Normal fetal ventral curvature resolves by 20 weeks.

Devine and Horton classification (1973)

Type	Abnormal tissue
I	Corpus spongiosum deficient (chordee without hypospadias)
II	Dysgenetic Buck's and dartos fascia; normal spongiosum
III	Dartos only
IV	Normal layers; **asymmetric corporal compliance** (true congenital curvature)
V	Congenitally short urethra (very rare; doubted)

Two clinical groups

Feature	Chordee without hypospadias (I–III)	Congenital curvature (IV)
Direction	Ventral ± torsion	Ventral, lateral (often left), rarely dorsal
Erection size	In proportion to flaccid size	**Noticeably large**
Presentation	Childhood	18–30 years, noticed at puberty

- **Versus acquired** (Peyronie disease, fracture): congenital is painless, no plaque, no shortening.

Evaluation

- Home photographs of erection; **artificial erection** (Gittes–McLaughlin saline test or prostaglandin) in theatre.
- Grade: mild <30°, moderate 30–45°, severe ≥45°.

Management

- Surgery if curvature hinders intercourse or distresses.

1. Deglove; excise ventral dysgenetic tissue; mobilise spongiosum from glans to perineum.
2. Residual curve: **midline ventral septotomy** under artificial erection.
3. Still curved: **dorsal plication** contralateral to the curve: Nesbit (excise ellipses, close transversely), Yachia (Heineke–Mikulicz), 16-dot or TAP (tunica albuginea plication). **12 o'clock is nerve-free**, so midline plication is safe.
4. Severe: ventral **corporotomy with graft** (dermis, tunica vaginalis).

- **Rule:** if the meatus reaches the glans tip, do not divide the urethra; the corpora usually limit straightening.

- **Results:** single operation in almost all; recurrence 2–15%, mostly after plication.

Diagram to draw: penile cross-section: nerves at 11 and 1 o'clock, plication at 12.

Key points to remember:

- Devine–Horton I–III = chordee without hypospadias; IV = corporal asymmetry.
- Congenital curvature: young man, large erection, orthotopic meatus.
- Dorsal 12 o'clock plication; never divide a urethra that reaches the glans tip.

Circumcision undone

Type: SAQ (10 marks)

Source: Practice Set 13 Paper 1, Q3d

"Circumcision undone" is read as an incomplete or complicated circumcision needing revision. Foreskin restoration is mentioned briefly at the end.

Introduction

- Circumcision complications: **0.2–5%**, rising **10- to 20-fold after infancy**.
- Most late problems concern **skin quantity** (too little or too much) or the **exposed meatus**.

Inadequate ("undone") circumcision

- **Too little skin removed** or asymmetrical excision: redundant prepuce covering the glans.
- Leads to **secondary (cicatricial) phimosis**, smegma retention, balanitis, adhesions and cosmetic distress.
- A large hydrocele or hernia pushes skin forward and raises this risk.

Other complications

Complication	Management
Bleeding (commonest, about 0.1%), usually frenular	Pressure, cautery, silver nitrate or suture
Glanular adhesions; **skin bridges** (can cause chordee, torsion)	Office lysis or excision; steroids do not help
Excess skin removal, degloving	Heals by secondary intention with ointment; severe: grafts
Trapped penis (cicatrix over glans)	Betamethasone 0.05% and retraction; else relaxing incision, flaps, Z-plasty
Meatal stenosis (almost only after circumcision)	Meatotomy or meatoplasty
Subcoronal fistula; glans amputation; necrosis (cautery on clamp)	Hypospadias-type repair; reattach glans within 8 hours

Evaluation

- Original procedure (age, device); examine skin quantity, meatus, curvature; exclude hypospadias, buried or webbed penis and BXO (balanitis xerotica obliterans).

Circumcision revision

- **Indications:** redundant or asymmetrical skin, phimosis, adhesions, cosmesis.
- **Needs general anaesthesia** in children, unlike neonatal circumcision.
- **Technique (sleeve):** mark inner and outer incisions (inner 5–10 mm from the corona), excise the excess collar, haemostasis, interrupted absorbable sutures.
- Lyse adhesions and skin bridges; correct torsion or chordee; meatotomy if needed; histology if BXO suspected.

Prevention

- Separate prepuce fully and inspect the meatus before using a device (Gomco, Mogen, Plastibell); defer with hypospadias, curvature, buried or webbed penis; trained operator.

Also know: foreskin restoration

- Traction tissue expansion or skin grafting, for men wanting a foreskin; results variable.

Diagram to draw: sleeve revision showing two circumferential incisions.

> **Key points to remember:**
> - Complications 0.2–5%; bleeding commonest.
> - Too little skin: redundancy, phimosis, adhesions; too much: trapped penis, chordee.
> - Meatal stenosis almost only in circumcised boys.
> - Revision needs general anaesthesia.

Noonan syndrome

Type: SAQ (10 marks)

Source: Practice Set 22 SAQ Sheet, Q3a

Definition and genetics

- Noonan syndrome is an **autosomal dominant RASopathy** (disorder of the RAS–MAPK [mitogen-activated protein kinase] signalling pathway) with variable expression; many cases are de novo. Incidence about 1 in 1,000–2,500.
- **Karyotype is normal** (46,XY or 46,XX). The phenotype resembles Turner syndrome, hence the old name "male Turner".
- **Mechanism:** gain-of-function variants increase RAS–MAPK signalling, which disturbs growth and organ development.

Gene	Approximate share	Note
PTPN11 (encodes SHP2 phosphatase)	About 50%	Pulmonary stenosis common
SOS1	About 10%	Milder growth and learning effects
RAF1	About 5–10%	Hypertrophic cardiomyopathy
RIT1, KRAS, others	Smaller	KRAS: more learning difficulty

- Related RASopathies: Costello, cardiofaciocutaneous, NF-1 (neurofibromatosis type 1).

Clinical features

- Short stature; webbed neck; hypertelorism, ptosis, low-set posteriorly rotated ears; pectus deformity.
- **Heart:** pulmonary valve stenosis (commonest), HCM (hypertrophic cardiomyopathy), septal defects.
- **Bleeding tendency:** factor XI deficiency, platelet dysfunction.
- Lymphatic dysplasia: genital (penoscrotal) lymphoedema; mild learning difficulty.

Urological aspects

- **Cryptorchidism:** the main urological feature, in most affected boys (often bilateral).
- **Testicular dysfunction:** primary Sertoli cell dysfunction (low inhibin B, high FSH [follicle-stimulating hormone]) with poor spermatogenesis, even after orchidopexy. Leydig function is usually adequate; puberty is often delayed. Female fertility is usually normal.
- **Renal anomalies (about 10%):** mostly mild renal pelvic dilatation; also duplex system, solitary kidney, hypoplasia.
- Small excess of childhood cancers (JMML [juvenile myelomonocytic leukaemia], embryonal rhabdomyosarcoma).

Evaluation

- Clinical diagnosis (van der Burgt criteria), confirmed by gene panel; karyotype excludes Turner in girls.
- Before surgery: **echocardiography, coagulation screen and factor XI level**, anaesthetic review.
- Renal and bladder ultrasound at diagnosis.

Management

- **Orchidopexy by 18 months** (EAU [European Association of Urology] and AUA [American Urological Association]); laparoscopic for non-palpable testes.

- Follow-up: pubertal hormones, semen analysis, fertility counselling.
- **Also know:** growth hormone for short stature; multidisciplinary cardiac and haematology care.

Diagram to draw: RAS–MAPK cascade with SHP2 (PTPN11) acting upstream of RAS.

Key points to remember:

- Autosomal dominant, normal karyotype; PTPN11 in about half.
- Cryptorchidism plus Sertoli dysfunction: male subfertility.
- Check heart and clotting (factor XI) before orchidopexy.

Rapid Revision Sheet – Section 2 · Oncology, DSD, Gender & Ageing

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Oncogenes (SAQ)

- Oncogenes: gain of function, one allele; activated by mutation, amplification, translocation.
- HRAS first found in T24 bladder cancer; FGFR3 in up to 80% low-grade Ta.
- TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.

Premalignant conditions in urology (including premalignant lesions of penis) (SAQ)

- PeIN: intact basement membrane; WHO 2022 all high grade.
- Queyrat 10–33% vs Bowen (about) 5% progression; bowenoid papulosis does not progress.
- Lichen sclerosus: SCC 2.3–9%.
- Untreated bladder CIS: 40–83% invade.
- GCNIS: 50% invasive in 5 years; radiotherapy ≥20 Gy.

46-XX Male (SAQ)

- 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.
- Azoospermia always (AZF absent): no sperm retrieval.
- Shorter than Klinefelter; FISH for SRY confirms.

Female circumcision (SAQ)

- WHO types I–IV; infibulation is type III.
- Retention, UTI, stones, fistula.
- Midline deinfibulation; never reinfibulate.

Familial malignancies in urology (SAQ)

- All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
- VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
- Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

Secondary malignancies in urology (SAQ)

- Pelvic EBRT raises bladder cancer 1.5–4-fold; IMRT little.
- Cyclophosphamide: only chemotherapy proven to cause bladder cancer.
- Ureterosigmoidostomy cancer 6–29% after 10–20 years.
- Transplant recipients: bladder cancer about 3-fold.

Genetic counselling (SAQ)

- Non-directive; pre-test and post-test counselling; cascade testing.
- Prostate: BRCA2 most important; test all metastatic men.
- RCC ≤46 years, ADPKD, cystinuria, AZF/CFTR, DSD are classic urological indications.

Congenital Penile curvature.. (SAQ)

- Devine–Horton I–III = chordee without hypospadias; IV = corporal asymmetry.
- Congenital curvature: young man, large erection, orthotopic meatus.
- Dorsal 12 o'clock plication; never divide a urethra that reaches the glans tip.

Circumcision undone (SAQ)

- Complications 0.2–5%; bleeding commonest.
- Too little skin: redundancy, phimosis, adhesions; too much: trapped penis, chordee.
- Meatal stenosis almost only in circumcised boys.

- Revision needs general anaesthesia.

Noonan syndrome (SAQ)

- Autosomal dominant, normal karyotype; PTPN11 in about half.
- Cryptorchidism plus Sertoli dysfunction: male subfertility.
- Check heart and clotting (factor XI) before orchidopexy.

Section 3 · Infections & Inflammation — Long answer questions (LAQ)

Genitourinary infections, common organisms, bacteraemia and urosepsis: pathophysiology and management

Asked as:

- Discuss general principles of genitourinary infections, characteristics of common organisms and management of bacteremia and septic shock. — *Summer 2016 Paper I Q2 (LAQ)*
- Describe Pathophysiology of urosepsis and its management. — *Summer 2017 Paper I Q2 (LAQ)*
- Describe pzthophysiology of urosepsis and its management. — *Practice Set 11 Paper I Q1 (LAQ)* (also answered in 2017-Summer Paper 1, Q2) — *"Pzthophysiology" is read as pathophysiology.*

Type: LAQ (25 marks)

Definitions

- **UTI (urinary tract infection):** urothelial inflammation from bacterial invasion, usually with bacteriuria and pyuria. **Sterile pyuria** needs evaluation (tuberculosis, stone, tumour).
- **Uncomplicated:** healthy non-pregnant woman, normal tract. **Complicated:** men, pregnancy, obstruction, stones, catheter, neurogenic bladder, diabetes, immunosuppression.
- **Recurrent:** ≥2 in 6 months or ≥3 in 12 months. Unresolved = never cleared; reinfection = new organism; **bacterial persistence** = same organism from a focus (stone, prostate).
- **Prophylaxis** assumes a sterile tract; **suppression** controls a focus that cannot be eradicated.
- **Urosepsis:** sepsis whose source is the urogenital tract (the urinary tract or male genital organs). It is usually Gram-negative and usually follows obstruction or instrumentation.

Pathogenesis of UTI

- **Routes:** ascending from bowel flora (commonest); haematogenous (*S. aureus*, *Candida*); lymphatic (rare).
- **Virulence (uropathogenic *E. coli*): type 1 pili** (mannose-sensitive) bind bladder uroplakins and cause cystitis. **P pili** (mannose-resistant) bind renal glycolipids and are found in about 90% of pyelonephritis strains. Haemolysin damages tissue. Bacteria invade umbrella cells and form intracellular communities; quiescent reservoirs cause recurrence.
- **Principle:** a strong host is infected only by virulent strains; a compromised host (obstruction, catheter) by almost any organism.
- **Host defences:** voiding washout, lactobacilli, urine osmolality and pH, uromodulin (Tamm–Horsfall protein), urothelial exfoliation.
- **Risk factors: obstruction** (stasis impairs defences and turns UTI life-threatening), VUR (vesicoureteral reflux; with infection it scars the growing kidney), stones, catheters, diabetes, pregnancy, spermicides.
- **Diagnosis:** dipstick (nitrite, leucocyte esterase) rules out better than it rules in. Culture is the gold standard: $\geq 10^5$ CFU (colony-forming units)/mL (Kass); $\geq 10^2$ in a symptomatic woman or a catheter specimen. Image (ultrasound, CT (computed tomography)) in complicated infection or fever beyond 72 hours.

Characteristics of common organisms

Organism	Key features
E. coli	~85% of community and ~50% of hospital UTI; commonest cause of urosepsis; ST131 clone is ESBL (extended-spectrum β-lactamase) and quinolone resistant
Proteus	Urease splits urea into ammonia: alkaline urine (pH >7), **struvite stones**, persistence in stones
Klebsiella	Capsule; urease; ESBL and KPC (*K. pneumoniae* carbapenemase) producers
Pseudomonas	Nosocomial, catheter biofilm; needs ceftazidime, piperacillin, aminoglycoside, fluoroquinolone or meropenem
Enterococcus	**Always resistant to cephalosporins**; amoxicillin, nitrofurantoin; VRE (vancomycin-resistant)
S. saprophyticus	~10% of cystitis in young sexually active women
S. aureus	Haematogenous renal abscess; in urine, look for bacteraemia
Anaerobes, *Candida*	Abscesses, fistulae; *Candida* with catheters and diabetes (*Candida* in catheterised diabetics)

- **Urosepsis organisms:** *E. coli* (commonest), *Proteus*, *Klebsiella*, *Pseudomonas*, *Enterococcus*; ESBL (extended-spectrum β-lactamase) producers rising; *Candida* in catheterised diabetics.
- **Also know:** emerging *Aerococcus urinae*; *Ureaplasma* and *Mycoplasma* with negative routine cultures; *Chlamydia* and tuberculosis do not grow on routine culture (sterile pyuria).
- **Antibiotic principles (see 2015-Summer Paper 1, Q3f):** urine levels matter in cystitis, tissue levels in pyelonephritis and bacteraemia. Use a narrow spectrum and short course; adjust for renal function; **drain obstruction**.

Bacteraemia, sepsis, septic shock and urosepsis

Sepsis-3 (2016)

Term (Sepsis-3, 2016)	Definition
Sepsis	Life-threatening organ dysfunction from a dysregulated host response to infection: acute rise in **SOFA (Sequential Organ Failure Assessment) score ≥2**
qSOFA (quick SOFA)	Respiratory rate ≥22/min, altered mentation, systolic blood pressure ≤100 mmHg; ≥2 = high risk (of death) (prompt only)
Septic shock	Vasopressor needed for MAP (mean arterial pressure) ≥65 mmHg **and** lactate >2 mmol/L despite (adequate) fluids; mortality >40%

- SIRS (systemic inflammatory response syndrome) criteria were dropped from the definition as non-specific.

Causes and risk factors (urological triggers)

- **Urological triggers:** infected obstructed kidney, PCNL (percutaneous nephrolithotomy), high-pressure ureteroscopy, prostate biopsy, TURP (transurethral resection of the prostate), infected stones, emphysematous pyelonephritis, abscess.
- **Obstruction with infection:** stone, PUJ (pelviureteric junction) obstruction, tumour, pregnancy, giving infected hydronephrosis or pyonephrosis. Pyonephrosis, emphysematous pyelonephritis, renal, perinephric and prostatic abscess.
- **Parenchymal infection:** emphysematous pyelonephritis, renal or perinephric abscess, prostatic abscess, Fournier gangrene.
- **Instrumentation:** PCNL (percutaneous nephrolithotomy), ureteroscopy with high irrigation pressure, TURP (transurethral resection of prostate), prostate biopsy (sepsis 0.3–3.1%),

catheters.
- **Host:** diabetes, immunosuppression, old age, neurogenic bladder.

Pathophysiology

- **Pathophysiology (summary):** endotoxin binds TLR4 (Toll-like receptor 4). Cytokines (TNF-α (tumour necrosis factor-α), interleukins IL-1 and IL-6) and nitric oxide cause vasodilatation, capillary leak, myocardial depression and DIC (disseminated intravascular coagulation), then organ failure.

1. Entry of bacteria (and endotoxin) into blood

- Obstruction causes stasis, impairs washout and raises pelvic pressure. Bacteria and endotoxin then enter the blood (circulation) by **pyelovenous and pyelolymphatic backflow**.
- High irrigation pressure at endoscopy does the same, and so does fragmenting infected stones, which release endotoxin (fragmenting infected stones releases endotoxin). Stone culture predicts postoperative sepsis better than bladder urine culture.
- An obstructed infected kidney is (behaves like) an **abscess with a collecting system**. Antibiotics barely reach it, and bladder urine may be sterile.

2. Host response

```
Endotoxin (LPS, lipopolysaccharide) / (or) Gram-positive wall products
  → TLR4 (Toll-like receptor 4) on macrophages → NF-κB (nuclear factor κB)
  → cytokines TNF-α (tumour necrosis factor α), interleukins IL-1, IL-6, IL-8
  → complement + coagulation activation + endothelial injury (activation)
```

- **Vasodilatation:** inducible nitric oxide causes (lowers) low systemic vascular resistance ("warm" shock).
- **Capillary leak:** (endothelial) glycocalyx damage leads to hypovolaemia and tissue oedema.
- **Coagulopathy:** tissue factor release causes microthrombi, platelet consumption and DIC (disseminated intravascular coagulation).
- **Myocardial depression** and **mitochondrial dysfunction (failure):** cells cannot use oxygen, so lactate rises.
- **Immunoparalysis:** a later anti-inflammatory phase predisposes to (invites) secondary infection.

3. Organ dysfunction: AKI (acute kidney injury), ARDS (acute respiratory distress syndrome), encephalopathy, liver dysfunction (failure), DIC, multi-organ failure.

Clinical features

- Fever or hypothermia, rigors, flank pain, tachycardia, tachypnoea, confusion, oliguria, hypotension. Often starts (begins) within hours of an endourological procedure (endourology).

Investigations

- **Lactate**, blood cultures ×2 (two blood cultures), urine culture (and pelvic urine at drainage), blood count, creatinine, bilirubin, platelets, coagulation, arterial (blood) gas, glucose. Procalcitonin helps decide when to stop antibiotics.
- **Imaging:** bedside ultrasound (hydronephrosis, debris); **CT (computed tomography)** for stone, gas or abscess.

Management (Surviving Sepsis Campaign: hour-1 bundle 2018, guidelines 2021)

A. Resuscitation and antibiotics (SSC, Surviving Sepsis Campaign 2021)

```
Recognise (SSC: do not use qSOFA alone for screening)
→ Lactate → blood cultures ×2 + urine culture
→ IV (intravenous) broad-spectrum antibiotics within 1 hour
  (shock or probable sepsis; within 3 h if sepsis only possible)
→ 30 mL/kg (balanced) crystalloid if hypotensive or lactate ≥4 mmol/L
  (within 3 h if hypotensive or hypoperfused)
→ noradrenaline first line for MAP ≥65 (target MAP 65); add vasopressin
→ SOURCE CONTROL: stent or nephrostomy (equally effective),
  drain abscess, change catheter; stone surgery later
→ refractory shock (persistent vasopressor need): hydrocortisone 200 mg/day; ICU (intensive
care unit)
→ re-measure (repeat) lactate, capillary refill
→ de-escalate by culture at 48–72 h
```

- **Empirical antibiotics:** by local resistance: third-generation cephalosporin or piperacillin–tazobactam ± aminoglycoside; carbapenem if ESBL risk. Adjust for renal function.

B. Source control: the key urological step

- **Drain the obstructed infected kidney urgently:** retrograde **double-J stent** or **percutaneous nephrostomy (PCN)**. Randomised trials show equal effect on sepsis (equally effective in a randomised trial (Pearle, 1998)).
  - Stent: suits coagulopathy; avoid high-pressure contrast injection.
  - PCN: better for (suits) large or impacted stones, pyonephrosis; allows (gives) pelvic urine (for) culture.
- **No definitive stone surgery during sepsis.** Treat the stone after (once) the infection clears.
- Other sources: drain renal or perinephric abscess; emphysematous pyelonephritis (percutaneous drainage, nephrectomy if diffuse gas without fluid); prostatic abscess drainage; change or remove (the) catheter; debride Fournier gangrene.

C. Supportive care (Support)

- Oxygen and (or) ventilation, glucose control (insulin if glucose ≥180 mg/dL), transfusion (transfuse) below haemoglobin 7 g/dL, thromboprophylaxis, stress-ulcer prophylaxis, renal replacement if needed. Watch for AKI (acute kidney injury), ARDS (acute respiratory distress syndrome) and DIC.

D. De-escalation and duration

- Narrow antibiotics by (to) culture at 48–72 h. Usual course (usually) 7–10 days (another source: 7–14 days); longer with an undrained focus or *S. aureus* bacteraemia; shorter courses suffice after good source control. Step down (switch) to oral therapy when (once afebrile and) stable.

Special situations

- **Emphysematous pyelonephritis** (diabetics): fluids, glucose control, antibiotics, drain; nephrectomy for diffuse gas without fluid.
- **Pregnancy:** stent or nephrostomy; avoid fluoroquinolones.
- **Fungal urosepsis:** remove catheter, antifungal, drain fungal balls.
- **Prognosis:** urosepsis does better than sepsis from other sites if the source is drained early; delay in drainage (delayed drainage) is the main avoidable cause of death.

Prevention

- Sterile urine before elective endourology; antibiotic prophylaxis; low irrigation pressure and access sheath; short operating time with infected stones; stage the procedure if pus is found.

Diagram to draw: pathway from *E. coli* adherence to bacteraemia and cytokine-driven shock; obstructed kidney leading to backflow, cytokine cascade and shock; obstructed kidney → pyelovenous backflow → cytokine cascade → shock and organ failure.

Key points to remember:

- *E. coli* ~85% community UTI; P pili cause pyelonephritis; *Proteus* causes struvite; *Enterococcus* resists cephalosporins.
- Sepsis = SOFA rise ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2 mmol/L.
- Obstruction plus infection is the emergency; bladder urine may be sterile.
- Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline (SSC 2021).
- Infected obstructed kidney: drain urgently; no definitive stone surgery in sepsis.
- Stent and nephrostomy equally effective (stent = nephrostomy for drainage); delay definitive stone surgery.

Describe clinical implication of AIDS on the Genito-Urinary Tract?

Type: LAQ (25 marks)

Source: 2020-Summer Paper 1, Q1

Introduction

- AIDS (acquired immunodeficiency syndrome) is the late stage of HIV (human immunodeficiency virus) infection. HIV is a retrovirus that destroys CD4 T lymphocytes. It spreads by blood, semen, vaginal fluid and breast milk.

Stage	Features
1. Acute	2–4 weeks after infection; flu-like illness; high viral load; antibody tests may be negative
2. Clinical latency	Years; usually no symptoms
3. AIDS	**CD4 <200 cells/mm³** or an AIDS-defining condition

- **Concept:** genitourinary (GU) disease follows the falling CD4 count. Organisms become opportunistic, infections become chronic and the GU tract becomes a reservoir. With HAART (highly active antiretroviral therapy), the burden shifts to drug toxicity and diseases of ageing.

Diagnosis (urologist's role)

- Test anyone with, or at risk of, an STD (sexually transmitted disease), and every TB (tuberculosis) patient.
- Antigen–antibody immunoassay → antibody differentiation assay; if discordant, HIV RNA (acute infection).

Manifestations by site

Site	Manifestations
Genital skin and STDs	Ulcerative STDs (herpes, syphilis, chancroid) increase HIV acquisition and transmission; HSV-2 (herpes simplex virus 2) in 50–90%. Warts (HPV (human papillomavirus) 6, 11) in about 20%, may be giant; molluscum 5–18%. Perineal cellulitis (cover MRSA (methicillin-resistant *Staphylococcus aureus*)); Fournier gangrene commoner
UTI (urinary tract infection)	About fivefold commoner. Bacteriuria about 30% at CD4 <200, 11% at 200–500, none above 500. *E. coli* and other gram-negatives; *Pseudomonas* up to 30%. *Salmonella* UTI needs lifelong prophylaxis. Culture before treatment
Opportunistic UTI	No response, or CD4 <500: look for fungi (*Candida*, *Cryptococcus*, *Histoplasma*), TB, CMV (cytomegalovirus), adenovirus
Kidney	Opportunistic pyelonephritis (*Pneumocystis*, *Candida*, mycobacteria). AKI (acute kidney injury) from sepsis, dehydration, nephrotoxins (tenofovir, indinavir, amphotericin). Obstruction by stones, lymphoma, fungal balls
GU TB	HIV raises active TB 15–20-fold. Disease is disseminated and bilateral, with **less fibrosis, fewer strictures and fewer contracted bladders**, because fibrosis needs an intact immune response. Mortality is higher. Same 6-month regimen; rifampicin lowers protease inhibitor levels
Prostate	Bacterial, TB and fungal prostatitis; prostatic abscess and urosepsis; relapse up to 70%. Cryptococcal prostatitis is a reservoir for relapsing meningitis, so give long-term antifungals

Site	Manifestations
Testis and epididymis	Recurrent, chronic epididymo-orchitis (including TB). Testis harbours opportunists. Testicular atrophy and hypogonadism
Voiding	LUTS (lower urinary tract symptoms) worsen with disease stage. Retention may herald seroconversion. Neurogenic causes: HIV encephalopathy, toxoplasmosis, **CMV polyradiculopathy** (bladder and bowel dysfunction with sciatic pain: lumbar puncture, treat early), neuropathy
Sexual function	ED (erectile dysfunction) in about half of men under 50; protease inhibitors worsen it. **Protease inhibitors raise sildenafil levels 3–10-fold: start low.** Low libido, ejaculatory disorders

HIVAN (HIV-associated nephropathy)

- Nephrotic-range proteinuria and rising creatinine, often without oedema or hypertension. Typically CD4 <200 and viral load >1000 copies/mL; about 12 times commoner in Black patients.
- Biopsy: **collapsing FSGS (focal segmental glomerulosclerosis)** with tubular microcysts.
- Treatment: HAART plus ACE (angiotensin-converting enzyme) inhibitor. Without it, dialysis follows within about 10 months.

Stones

- **Indinavir** stones in up to 22% of treated patients. Indinavir precipitates in alkaline urine. Stones are radiolucent, and pure stones may be invisible even on CT (computed tomography).
- Stop the drug and hydrate: succeeds in up to 70%. Newer protease inhibitors rarely cause stones.

Malignancy

AIDS-defining	Non-AIDS-defining
Kaposi sarcoma (HHV-8 (human herpesvirus 8)), non-Hodgkin lymphoma, cervical cancer	Penile, anal, testicular, prostate cancers; Hodgkin lymphoma
Falls with HAART	Now about 70% of cancers in treated HIV

- **Kaposi sarcoma:** genital in about 3% of AIDS; glans lesions can block the meatus. Treat with ART (antiretroviral therapy) first. Local excision, laser or intralesional vinblastine; radiotherapy 15–30 Gy; systemic chemotherapy if disseminated.
- **Penile cancer:** HIV raises HPV infection; about fourfold risk of death.

Implications for urological practice

- **Universal precautions** for every patient: double gloves, no hand-to-hand passing of sharps.
- Occupational risk: percutaneous about 0.3%, mucous membrane about 0.09%.
- **PEP (post-exposure prophylaxis)** within 72 hours, for 28 days: tenofovir + emtricitabine plus raltegravir or dolutegravir. HIV test at 6 weeks, 3 and 6 months.
- CD4 <200: TMP-SMX (trimethoprim-sulfamethoxazole) *Pneumocystis* prophylaxis also lowers UTI.
- **Prevention:** male circumcision cuts heterosexual acquisition by 50–60% (three African RCTs (randomised controlled trials)); consistent condom use cuts risk about 80%; PrEP (pre-exposure prophylaxis).

Also know: IRIS (immune reconstitution inflammatory syndrome) on starting ART; serodiscordant couples: suppressed viral load with sperm washing and assisted reproduction;

transplantation is possible with controlled HIV.

Diagram to draw: male GU tract outline labelled with the lesion at each site.

Key points to remember:

- AIDS = CD4 <200; UTI risk rises as CD4 falls.
- HIV-TB: disseminated, less fibrosis, fewer strictures.
- Indinavir stones: radiolucent, may be CT-invisible; stop drug and hydrate.
- PEP within 72 hours for 28 days; circumcision cuts acquisition 50–60%.

Section 3 · Infections & Inflammation — Short answer questions (SAQ)

Principles of antimicrobial therapy in UTI

Type: SAQ (10 marks)

Source: 2015-Summer Paper 1, Q3f

Definitions

- **Uncomplicated** urinary tract infection (UTI): healthy non-pregnant women, normal tract. **Complicated:** men, pregnancy, abnormal tract, catheter, immunosuppression.
- **Recurrent:** ≥2 in 6 months or ≥3 in 12 months.

Principles

- **Site decides the level:** urine in cystitis, tissue elsewhere; nitrofurantoin and fosfomycin lack tissue levels: **cystitis only**.

Killing pattern	Drugs	Dosing goal
Time-dependent (time above MIC [minimum inhibitory concentration])	β-lactams, carbapenems	Frequent doses or extended infusion
Concentration-dependent (peak:MIC)	Aminoglycosides, fluoroquinolones (FQs)	High once-daily dose

- **Resistance thresholds:** trimethoprim–sulfamethoxazole (TMP-SMX) if local resistance ≤20%; oral FQ for pyelonephritis if ≤10%.
- **Collateral damage:** FQs and cephalosporins select resistance and *C. difficile*; avoid them in cystitis.
- Culture all but simple cystitis; **drain** obstructed infected kidneys; shortest course.

Empirical choices

Scenario	Drug	Duration
Cystitis (women)	Nitrofurantoin 100 mg twice daily; fosfomycin 3 g; pivmecillinam 400 mg; TMP-SMX 160/800 mg twice daily	5 d; single dose; 3–5 d; 3 d
Cystitis (men)	TMP-SMX or per culture	7 d (= 14 d, Drekonja 2021)
Pyelonephritis (oral)	Ciprofloxacin 500 mg twice daily; levofloxacin 750 mg daily	7 d; 5 d
Pyelonephritis (intravenous)	Ceftriaxone 1 g daily; gentamicin 5–7 mg/kg once daily	10–14 d; oral when afebrile
Complicated / catheter UTI	Aminoglycoside or 3rd-generation cephalosporin per culture; change catheter	7–14 d
Urosepsis	Broad-spectrum **within 1 h**, source control	–

Asymptomatic bacteriuria

- Treat **only in pregnancy and before mucosa-breaching urological procedures**. Not in elderly, diabetic, catheterised or spinal cord injury patients.
- **Pregnancy:** avoid FQs and TMP-SMX; cephalexin is suitable.

Prophylaxis

- Recurrent: vaginal oestrogen, **methenamine hippurate** (non-inferior to antibiotics, ALTAR 2022), cranberry, post-coital antibiotics.
- Surgical: single dose within 1 hour; ≤24 h for percutaneous nephrolithotomy and prostheses.
- **Also know:** ESBL (extended-spectrum β-lactamase) producers; nitrofurantoin avoided if creatinine clearance <30 mL/min; gepotidacin (US 2025).

Diagram to draw: Concentration–time curve showing peak:MIC and time above MIC.

> **Key points to remember:**
> - Recurrent = ≥2 in 6 months or ≥3 in 12 months.
> - Cystitis: nitrofurantoin 5 d, fosfomycin single, TMP-SMX 3 d (resistance ≤20%).
> - Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d (FQ resistance ≤10%).

Renal and perinephric abscess

Asked as:

- Perinephric abscess — *Summer 2017 Paper I Q3c (SAQ)*
- Treatment of renal abscess — *Winter 2017 Paper I Q3b (SAQ)*

Type: SAQ (10 marks)

Definition and types

- **Renal (cortical/corticomedullary) abscess:** pus confined to the renal parenchyma.
  - **Cortical (renal carbuncle):** haematogenous *Staphylococcus aureus* from a skin source; formerly about 80% of cases.
  - **Corticomedullary (now commonest):** ascending Gram-negative infection (*E. coli*, *Klebsiella*, *Proteus*) in a damaged or obstructed kidney (stones, reflux, diabetes).
- **Perinephric abscess:** pus **outside the renal capsule but within Gerota fascia**. Rupture through Gerota fascia gives a **paranephric** abscess (paranephric beyond it).

Aetiology (perinephric abscess)

- Rupture of a renal cortical abscess or pyonephrosis (commonest); extravasation of infected urine from obstruction (stone).
- **Haematogenous** spread in about a third (skin source, *S. aureus*).
- Infected perirenal haematoma or urinoma (trauma, surgery).
- Spread from bowel (Crohn disease, perforation), pancreas, spine.
- Risk factors: **diabetes (about a third)**, stones, obstruction, polycystic kidneys on dialysis.
- Organisms: *E. coli*, *Proteus*, *S. aureus*; anaerobes with bowel sources.

Clinical features

- Fever, flank pain, leucocytosis.
- Perinephric abscess is **insidious:** symptoms >5 days in most; more than a third afebrile.
- Flank pain, tenderness, flank mass (about half); psoas sign (flexed, externally rotated hip).
- **Suspect it when "pyelonephritis" persists:** pyelonephritis defervesces within 4–5 days; perinephric abscess does not.

	Pyelonephritis	Perinephric abscess
Symptoms before admission	≤5 days	>5 days
Fever on antibiotics	Never >4 days	≥5 days (median 7)

Investigations / diagnosis (needed to plan treatment)

- Leucocytosis, raised creatinine, pyuria; urine culture identifies the organism in only about a third; urine may be sterile (haematogenous or non-communicating abscess). Blood cultures positive in a quarter to a third.
- **CT (computed tomography): investigation of choice.** Renal abscess: round low-attenuation lesion with enhancing "ring sign". It shows the collection, its spread (psoas, flank), the cause (stone, obstruction) and plans drainage.
- Ultrasound: quick; guides drainage. Plain X-ray: loss of psoas shadow, scoliosis.
- AFBN (acute focal bacterial nephritis) is wedge-shaped without a wall and is treated medically.

Treatment

1. Antibiotics immediately (after blood and urine cultures)

Presumed source	Empirical therapy
Haematogenous (Gram-positive)	Penicillinase-resistant penicillin; vancomycin if allergic or MRSA (methicillin-resistant *S. aureus*)
Ascending (Gram-negative)	IV (intravenous) third-generation cephalosporin, antipseudomonal penicillin or aminoglycoside
Perinephric abscess	Aminoglycoside plus antistaphylococcal penicillin; cephalosporin or vancomycin if allergic. Adjust to culture

- Switch to oral once afebrile; continue until clinical and imaging resolution.

2. Size-based drainage (renal abscess)

Size	Management
≤3 cm	Antibiotics alone (stable patient)
3–5 cm	Antibiotics alone or drainage; no consistent pathway
>5 cm	**Percutaneous drainage** (CT/ultrasound-guided) first line; may need several drains
Any size: immunocompromised or not responding	Percutaneous drainage

- **Perinephric abscess (if present) — lower threshold:** small (<3 cm) in an immunocompetent patient may resolve with antibiotics; drain if >3 cm. Larger or non-responding: **CT- or ultrasound-guided percutaneous drainage** (first line).
- Aspiration also distinguishes abscess from a cystic tumour; culture the pus.
- Serial ultrasound or CT until resolution. Worsening suggests perinephric spread, wrong diagnosis or resistant organism.

3. Treat the cause

- Relieve obstruction (stent or nephrostomy); treat stones once infection settles; control diabetes.

4. Surgery

- Open drainage for loculated or failed (percutaneous drainage) cases; **nephrectomy** for a non-functioning, destroyed or pyonephrotic kidney, at the same time or staged.

Outcome

- Perinephric abscess: mortality fell from 40–50% to about 12% with CT; delay in diagnosis causes most deaths.

Diagram to draw: coronal kidney showing renal, perinephric and paranephric spaces / abscess sites.

> **Key points to remember:**
> - Now mostly ascending Gram-negative abscess in a damaged kidney.
> - Perinephric: within Gerota fascia; diabetes in a third.
> - Fever beyond 4–5 days of antibiotics: get a CT.
> - CT is the investigation of choice; follow until resolution.
> - ≤3 cm antibiotics; 3–5 cm either; >5 cm percutaneous drainage.
> - Percutaneous drainage first line; nephrectomy if non-functioning.

Ciprofloxacin

Type: SAQ (10 marks)

Source: 2017-Summer Paper 1, Q3d

Class and mechanism

- Second-generation **fluoroquinolone**, derived from nalidixic acid.
- Inhibits **DNA gyrase** (topoisomerase II, main target in Gram-negatives) and **topoisomerase IV** (Gram-positives). This blocks DNA supercoiling and replication: **bactericidal**.

Spectrum

- Enterobacteriaceae, ***Pseudomonas aeruginosa***, *Haemophilus*, *S. saprophyticus*, *S. aureus* (methicillin-sensitive); *Legionella*, some mycobacteria.
- Weak against streptococci and enterococci; **no anaerobic cover**, so bowel and vaginal flora are partly spared.

Pharmacokinetics

- Oral bioavailability about 70%; IV (intravenous) and oral forms; half-life about 4 h.
- **Concentration-dependent killing:** AUC/MIC (area under curve to minimum inhibitory concentration) predicts effect; post-antibiotic effect.
- High urine and tissue levels; **good prostatic penetration**.
- Renal excretion: reduce dose if creatinine clearance <30 mL/min.

Doses

- Cystitis 250 mg twice daily × 3 days; pyelonephritis 500 mg twice daily × 7 days (or 1 g extended-release daily); IV 400 mg 12-hourly.
- Chronic bacterial prostatitis: 4–6 weeks.

Indications and restrictions

- Pyelonephritis and complicated UTI (urinary tract infection) **only where local resistance ≤10%**; prostatitis; prophylaxis for transrectal biopsy (falling, because of resistance).
- **Not first line for uncomplicated cystitis** (FDA (US Food and Drug Administration) 2016; European restrictions 2019): use only when no alternative.

Resistance

- *gyrA*/*parC* mutations, efflux pumps, reduced porins, plasmid *qnr* genes. Linked to ESBL (extended-spectrum β-lactamase) *E. coli* (ST131). Main cause of post-biopsy sepsis.

Adverse effects

- GI (gastrointestinal) upset (commonest), photosensitivity, *C. difficile*.
- **FDA boxed warnings:** tendinopathy and **Achilles rupture** (2008; risk higher with age, steroids, renal failure); peripheral neuropathy (2013); disabling effects (2016); psychiatric effects and dysglycaemia (2018); aortic aneurysm and dissection (2018).
- CNS (central nervous system): seizures, confusion; QT prolongation.

Interactions

- Antacids, iron, zinc, calcium block absorption (separate by ≥2 h). Raises theophylline and tizanidine levels; potentiates warfarin.

Pregnancy and children

- Avoid: **cartilage damage** in immature animals. Also avoid in breastfeeding.

Key points to remember:

- DNA gyrase and topoisomerase IV; concentration-dependent.
- Use for pyelonephritis only if local resistance ≤10%.
- Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

Catheter Associated Urinary Tract Infections (CAUTI)

Type: SAQ (10 marks)

Source: 2017-Winter Paper 1, Q3c

Definition

- **CAUTI (catheter-associated urinary tract infection):** symptomatic UTI (urinary tract infection) with a catheter in place >2 days (or removed within 48 hours). Distinguish from **asymptomatic bacteriuria**, which needs no treatment.

Epidemiology and risk factors

- Commonest hospital-acquired infection (about 75–80% of hospital UTIs); commonest cause of secondary bacteraemia.
- Bacteriuria develops at about **5–10% per catheter-day**; 100% with long-term catheters. Intermittent catheterisation: 1–3% per catheterisation.
- **Risk factors:** duration (most important), female sex, older age, diabetes, breaks in the closed system, no systemic antibiotics.

Pathogenesis

- Entry at insertion, then **extraluminal** (along the outer surface; commoner in women) or **intraluminal** (from the drainage bag).
- **Biofilm:** bacteria in a matrix on the catheter surface, formed within minutes; it shields them from antibiotics. Urease-producers (*Proteus*) cause encrustation.
- Organisms: *E. coli*, *Pseudomonas*, *Proteus*, *Enterococcus*; long-term catheters polymicrobial.

Diagnosis

- Fever, rigors, flank or suprapubic pain, or delirium with no other source. Pyuria does not distinguish infection.
- Culture from a **freshly replaced catheter**, not the bag. **IDSA (Infectious Diseases Society of America): ≥10^3 CFU/mL** (colony-forming units) with symptoms; surveillance (CDC (Centers for Disease Control and Prevention)) definition uses ≥10^5.

Prevention (bundle)

- **Avoid unnecessary catheters;** review daily, remove early. Aseptic insertion.
- **Closed, dependent drainage**; bag below bladder; no routine irrigation.
- Meatal cleaning with water; no prophylactic antibiotics (delay bacteriuria only 3–4 days, select resistance).
- Alternatives: **clean intermittent catheterisation**, condom or suprapubic catheter.
- Silver-alloy catheters reduce bacteriuria in short-term use; no proven fall in symptomatic CAUTI.
- Do not treat asymptomatic bacteriuria (except pregnancy, before urological surgery).

Treatment

- **Replace or remove the catheter** before taking the culture and starting antibiotics, if in place >2 weeks (biofilm source).
- Empirical, then culture-directed antibiotics: **7 days** if prompt response; **10–14 days** if delayed.

Diagram to draw: catheter in bladder showing extraluminal and intraluminal routes and biofilm.

Key points to remember:

- Bacteriuria 5–10% per catheter-day; duration is the key risk.
- Diagnosis: symptoms + ≥10^3 CFU/mL from a fresh catheter.
- Remove catheter early; closed drainage; treat only symptomatic CAUTI, 7 or 10–14 days.

Complicated urinary tract infections.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 1, Q3f

Definition

- **Complicated UTI (urinary tract infection):** infection in a **structurally or functionally abnormal tract** or a **compromised host**, or with a resistant organism; more likely to fail or recur.
- **FDA (US Food and Drug Administration) 2018:** pyuria + pathogen in urine or blood + fever, chills, flank pain, with an abnormality or catheter.
- **Uncomplicated:** healthy non-pregnant woman, normal tract.

Complicating factors

Category	Examples
Obstruction	BOO (bladder outlet obstruction), PUJ (pelviureteric junction) obstruction, stricture, stone, tumour
Voiding dysfunction	Neurogenic bladder, high PVR (post-void residual), VUR (vesicoureteral reflux)
Foreign body / iatrogenic	Catheter, stent, nephrostomy, recent instrumentation, hospital-acquired
Anatomical	Diverticulum, fistula, cystocele
Host	**Male sex**, pregnancy, diabetes, immunosuppression, renal failure, transplant, elderly
Other	Symptoms >7 days; recent antibiotics

- Each causes stasis, biofilm or weak defences.

Organisms

- ***E. coli*** under half; more *Klebsiella*, *Proteus*, *Pseudomonas*, *Enterococcus*, *Candida*. Often **MDR (multidrug-resistant)** or ESBL (extended-spectrum β-lactamase) producers.

Evaluation

- **Urine culture mandatory before antibiotics**; blood cultures if febrile.
- Renal function, glucose, lactate.
- **Imaging:** ultrasound first; **CT (computed tomography)** if no response in 72 hours, stones, abscess or emphysematous infection suspected.

Treatment

- **Empirical IV (intravenous) therapy** by local resistance (EAU (European Association of Urology)): amoxicillin + aminoglycoside, 2nd-generation cephalosporin + aminoglycoside, or IV 3rd-generation cephalosporin. Avoid fluoroquinolones if used in the last 6 months or resistance >10%. Carbapenem or newer agents for ESBL (ceftolozane–tazobactam, ceftazidime–avibactam).
- **Source control is essential:** drain obstruction (**stent or nephrostomy, equally effective**), change catheter, drain abscess, treat stones later.
- Switch to oral when afebrile and stable; de-escalate by culture.
- **Duration:** usually 7–14 days (14 days in men if prostatitis not excluded); 7 days may suffice once afebrile ≥48 hours.
- **Sepsis:** antibiotics within 1 hour, fluids, vasopressors, ICU (intensive care unit).

- Correct the underlying cause, or infection recurs.

Special forms

- **Also know:** emphysematous pyelonephritis and cystitis (diabetics); pyonephrosis; renal and perinephric abscess; XGP (xanthogranulomatous pyelonephritis); CAUTI (catheter-associated UTI); funguria; unresolved bacteriuria (resistance, staghorn "critical mass").

Diagram to draw: Flowchart: suspect cUTI → culture → imaging → empirical IV → drain obstruction → oral step-down.

Key points to remember:

- Complicated = abnormal tract or compromised host; male sex alone qualifies.
- Culture mandatory; often MDR.
- Drain obstruction: stent = nephrostomy.
- 7–14 days; oral step-down when afebrile.

ESBL producing enterobacterial UTI

Asked as:

- ESBL producing enterobacterial UTI — *Summer 2019 Paper I Q3f (SAQ)*
- ESBL producing enterobacterial UTI. — *Practice Set 5 Paper I Q3c (SAQ)* (also answered in 2019-Summer Paper 1, Q3f)

Type: SAQ (10 marks)

Definition

- **ESBLs (extended-spectrum β-lactamases): plasmid-encoded** enzymes that hydrolyse penicillins, **third-generation cephalosporins** and aztreonam. They are **inhibited by clavulanate** and spare **carbapenems** and cephamycins.
- Mainly *E. coli* and *Klebsiella* (Enterobacterales); first reported 1983.

Mechanism and spread

- Older types **TEM** and **SHV**; now **CTX-M** dominates worldwide. Spread through the community is linked to *E. coli* **ST131**, whose hallmark is fluoroquinolone resistance (fluoroquinolone resistance is its hallmark).
- Plasmids carry co-resistance genes: **fluoroquinolones, aminoglycosides, TMP-SMX (trimethoprim–sulfamethoxazole)**. So oral options are few.
- India has one of the highest prevalence rates.

Risk factors

- Prior antibiotics (especially cephalosporins, quinolones); recent hospital stay; **indwelling catheter**, urological instrumentation; recurrent UTI (urinary tract infection); diabetes; nursing-home residence; travel to high-prevalence areas; transrectal prostate biopsy.

Detection

- Screening: reduced susceptibility to ceftriaxone, cefotaxime or ceftazidime.
- **Phenotypic confirmation:** clavulanate synergy (**combination disc** or double-disc synergy test); automated systems.
- Genotypic PCR (polymerase chain reaction) for CTX-M.

Treatment

Infection	Options
Cystitis	**Nitrofurantoin** 5 days, **fosfomycin** 3 g single dose (E. coli), **pivmecillinam**; TMP-SMX or quinolone only if susceptible; single-dose aminoglycoside
Pyelonephritis, complicated UTI	Quinolone or TMP-SMX if susceptible; else (otherwise) **aminoglycoside**
Urosepsis, bacteraemia	**Carbapenem** (meropenem, ertapenem), then oral step-down by culture

- **MERINO trial (2018):** in ceftriaxone-resistant *E. coli*/*Klebsiella* bacteraemia, **piperacillin–tazobactam** had higher 30-day mortality than meropenem (12.3% vs 3.7%). Avoid pip-taz for serious ESBL infection.
- **Ceftolozane–tazobactam, ceftazidime–avibactam:** active but **reserved** for carbapenem-resistant organisms (stewardship).
- **Source control:** remove or change catheters; drain obstruction.
- Treat **asymptomatic bacteriuria** only before urological procedures or in pregnancy.
- **Febrile prostatitis after transrectal biopsy:** carbapenem is favoured.

Prevention

- **Antibiotic stewardship:** avoid broad-spectrum cephalosporins and quinolones; narrowest spectrum; shortest effective course; culture before treatment.
- Hand hygiene, contact precautions, catheter care.
- Pre-biopsy rectal swab culture for targeted prophylaxis; transperineal biopsy.

Also know: AmpC (not clavulanate-inhibited) and carbapenemases **KPC (*Klebsiella pneumoniae* carbapenemase)**, NDM (New Delhi metallo-β-lactamase) are wider resistance.

Diagram to draw: treatment flowchart from culture result to oral or IV (intravenous) choice.

Key points to remember:

- Plasmid CTX-M; resists 3rd-gen (third-generation) cephalosporins; inhibited by clavulanate (clavulanate inhibits).
- Oral: nitrofurantoin, fosfomycin, pivmecillinam.
- Serious infection: carbapenem; MERINO: pip-taz (piperacillin–tazobactam) inferior.
- Stewardship prevents spread.

Lichen sclerosus (BXO, balanitis xerotica obliterans)

Asked as:

- Lichen sclerosis. — *Summer 2020 Paper I Q3c (SAQ)*
- BXO — *Practice Set 10 Paper I Q3b (SAQ)*

Type: SAQ (10 marks)
(also answered in 2020-Summer Paper 1, Q3c)

Taken as lichen sclerosus (LS), formerly balanitis xerotica obliterans (BXO). BXO (balanitis xerotica obliterans) is taken as genital lichen sclerosus (LS) in the male.

Definition

- Chronic, progressive, **scarring inflammatory disease of genital skin**; possibly premalignant. BXO is its late stage in men.

Epidemiology

- 6–10 times commoner in females (prepubertal and menopausal).
- Men: peak 30–50 years in one chapter, older men in another; also children (boys with phimosis). Commonly found at circumcision for phimosis after infancy. 15–20% have extragenital disease.

Aetiology

- Cause unknown. Autoimmune links (thyroid disease, vitiligo, alopecia); familial clustering.
- **Chronic occlusion by urine** under the foreskin (dribbling, obesity) definitely increases risk.
- Infection and trauma are proposed but unproven.

Clinical features

- White, atrophic, sclerotic plaques with loss of architecture; purpura, fissures; itch, pain, painful erection, poor stream, spraying.
- Sequence: prepuce → **phimosis** → glans → **meatal stenosis** (sentinel site) → fossa navicularis and penile urethral stricture.
- **Spares the proximal (bulbar and posterior) urethra.**
- Buried penis from loss of shaft skin.
- **Mechanism of stricture:** voiding against meatal stenosis forces urine into the glands of Littre → inflammation → spongiofibrosis.

Evaluation

- Clinical diagnosis. **Biopsy** if doubtful or suspicious: hyperkeratosis, epidermal atrophy, basal vacuolation, dermal oedema with homogenised collagen, band-like lymphocytic infiltrate (lymphocytes).
- Uroflowmetry; **RUG (retrograde urethrogram) before meatal surgery**, because a proximal stricture often coexists (may coexist).
- **Differential:** lichen planus, Zoon balanitis, penile intraepithelial neoplasia.

Management

Situation	Treatment
Skin disease	**Clobetasol propionate 0.05%** daily for 1–3 months (another source: for about 3 months), then taper to twice-weekly maintenance; calcineurin inhibitors second line; retinoids for refractory disease

Situation	Treatment
Phimosis	**Circumcision**, usually curative
Meatal stenosis	Dilatation with clobetasol; extended meatotomy; BMG (buccal mucosal graft) meatoplasty
Anterior stricture	**Oral mucosal graft, never genital skin** (LS recurs in skin); staged or single stage
Panurethral or unfit	**Perineal urethrostomy** (easy because the proximal urethra is spared); self-dilatation

- Recurrence 40–50% after medical treatment (recurrence is common); restricture higher than in other aetiologies (restricture rates exceed other aetiologies); treatment follows the patient's goals.

Malignancy

- Penile SCC (squamous cell carcinoma) in **2.3–9%** on long follow-up; HPV (human papillomavirus)-independent type.
- **Lifelong follow-up**; biopsy any non-healing ulcer or nodule.

Diagram to draw: penis with LS at (showing) prepuce, glans, meatus and fossa involved, proximal urethra spared.

> **Key points to remember:**
> - Genital skin disease; meatus is the sentinel site.
> - Spares proximal urethra; perineal urethrostomy easy.
> - Clobetasol first; oral mucosa, never skin, for repair.
> - SCC in 2.3–9%: lifelong follow-up.

Drug Fosfomycin

Type: SAQ (10 marks)

Source: 2022-Summer Paper 1, Q3c

Introduction

- Bactericidal **phosphonic acid derivative** antibiotic, chemically unrelated to other classes. Oral form: **fosfomycin trometamol**; intravenous form: fosfomycin disodium.

Mechanism of action

- Enters bacteria through glycerol-3-phosphate and glucose-6-phosphate transporters.
- Irreversibly inhibits **MurA** (enolpyruvyl transferase), blocking the **first step of peptidoglycan (cell wall) synthesis**. β-lactams and vancomycin act at later steps, so **cross-resistance is limited**.

Pharmacokinetics

- Oral bioavailability is modest, but the drug is excreted **unchanged in urine**. One 3 g dose keeps high urinary concentrations for about 1–2 days, which permits single-dose therapy. Low tissue levels.
- Concentrations fall in renal failure; avoid at very low creatinine clearance.

Spectrum

- **Escherichia coli**, including **ESBL (extended-spectrum β-lactamase)** producers; most Enterobacteriaceae (Klebsiella variable); *Enterococcus* including **VRE (vancomycin-resistant enterococcus)**; staphylococci.
- **Not reliable:** *Pseudomonas aeruginosa*, *Acinetobacter*.

Uses in urology

Indication	Dose
Uncomplicated cystitis in women (first-line, EAU (European Association of Urology) and IDSA (Infectious Diseases Society of America))	3 g oral sachet, single dose
Cystitis in pregnancy, asymptomatic bacteriuria (safe, category B)	3 g single dose
MDR (multidrug-resistant) lower UTI (urinary tract infection)	Oral option avoiding IV (intravenous) therapy; repeated doses used off-label
Prophylaxis for transrectal prostate biopsy	As effective as fluoroquinolone
Recurrent UTI prophylaxis	3 g every 10 days (EAU)

- **Not for pyelonephritis** or urosepsis orally (poor tissue levels). Emerging data suggest IV fosfomycin may help in bacteraemic MDR E. coli UTI.
- Not needed as prophylaxis for flexible cystoscopy.

Adverse effects

- Diarrhoea, GI (gastrointestinal) upset, headache, vaginitis; rare hypersensitivity. IV form: **sodium load**, hypokalaemia.

Resistance

- Transporter mutations, MurA changes, plasmid **fosA** enzymes. Resistance develops quickly in vitro, so systemic infections use combinations.

Evidence

- Huttner (2018) RCT (randomised controlled trial): nitrofurantoin 5 days gave higher clinical resolution than single-dose fosfomycin (70% vs 58%).

Diagram to draw: peptidoglycan synthesis showing fosfomycin (MurA), β-lactams and vancomycin sites.

> **Key points to remember:**
>
> - Inhibits MurA: first step of cell wall synthesis; no cross-resistance.
> - 3 g single dose for cystitis; active against ESBL E. coli and VRE.
> - Not for pyelonephritis; poor Pseudomonas cover.

Chronic Prostatitis : diagnosis, classification and management.

Type: SAQ (10 marks)

Source: 2024-Winter Paper 1, Q3c (also answered in 2018-Summer Paper 4, Q3d)

Classification: NIH (National Institutes of Health), 1999

Category	Description
I	Acute bacterial prostatitis
II	**Chronic bacterial prostatitis (CBP)**: recurrent UTI (urinary tract infection), **same organism**; well between episodes
IIIA	**CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome)**, inflammatory: leucocytes in EPS (expressed prostatic secretion), VB3 (post-massage urine) or semen
IIIB	CP/CPPS, non-inflammatory (split has no treatment value)
IV	Asymptomatic inflammatory (incidental)

- **CP/CPPS:** pelvic pain **≥3 of the last 6 months**, no uropathogens; 90–95% of cases. Infection, neurogenic inflammation, **pelvic floor dysfunction**, stress and **central sensitisation** interact.

Diagnosis

- **Symptoms:** perineal (commonest), testicular, suprapubic, penile or ejaculatory pain; LUTS (lower urinary tract symptoms); sexual dysfunction.
- **NIH-CPSI (Chronic Prostatitis Symptom Index):** pain, urinary and quality-of-life domains.
- **DRE (digital rectal examination):** prostate tender in under half; palpate pelvic floor for **trigger points**.
- **Localisation tests:**
  - **Meares–Stamey 4-glass:** VB1 (urethra), VB2 (bladder), EPS, VB3 (prostate).
  - **2-glass** (pre- and post-massage urine): matches 4-glass in >95%.
- Urinalysis, culture, PVR (post-void residual), flow; cytology if irritative symptoms; cystoscopy if low flow or bladder pain.
- **UPOINT phenotyping:** Urinary, Psychosocial, Organ-specific, Infection, Neurological/systemic, Tenderness of muscles.

Management

Category II (CBP): fluoroquinolone (lipid-soluble zwitterion, enters prostate) for **4–6 weeks** (EAU (European Association of Urology)); Chlamydia: azithromycin; resistant: TMP-SMX (trimethoprim-sulfamethoxazole) 3 months; refractory: suppression or TURP (transurethral resection of prostate).

Category III: multimodal, UPOINT-directed (monotherapy fails)

Domain	Therapy
Urinary	α-blocker if recent onset and α-blocker naive
Psychosocial	CBT (cognitive behavioural therapy), stress reduction
Organ	5-ARI (5α-reductase inhibitor) if BPH (benign prostatic hyperplasia); quercetin
Infection	Quinolone ≥6 weeks only if treatment-naive, <1 year; no repeat courses
Neurological	Pregabalin, amitriptyline, duloxetine
Tenderness	**Pelvic floor physiotherapy, myofascial release**, biofeedback

- **Others:** acupuncture (61.5% vs 38.3% response with sham), aerobic exercise, PTNS (percutaneous tibial nerve stimulation). **Avoid** TURP or prostatectomy for pain.

Diagram to draw: UPOINT snowflake.

> **Key points to remember:**
> - NIH I–IV; III = pain ≥3 of 6 months, no infection.
> - CBP: same organism, fluoroquinolone 4–6 weeks.
> - CP/CPPS: NIH-CPSI to score, UPOINT to treat.

Lymphogranuloma venereum

Type: SAQ (10 marks)

Source: Practice Set 2 Paper 1, Q3c

Definition

- **LGV (lymphogranuloma venereum)** is an invasive, lymphotropic STD (sexually transmitted disease) caused by ***Chlamydia trachomatis*** **serovars L1, L2 and L3**.
- Of 15 serovars, A–C cause trachoma and D–K anogenital infection. The L serovars invade lymphatics, causing lymphadenitis and scarring.

Epidemiology

- Endemic in the tropics, including India; increasing as **proctitis in MSM (men who have sex with men)**. Not more frequent or virulent in HIV (human immunodeficiency virus).

Clinical stages

Stage	Timing	Features
Primary	3–30 days	Small painless papule or ulcer, often unnoticed. Rectal inoculation → **proctitis** (bleeding, pain, tenesmus, discharge), now the commonest presentation
Secondary (inguinal syndrome)	2–6 weeks later	Fever, malaise; **painful, matted, unilateral inguinal or femoral buboes** that may form fistulae. **Groove sign**: nodes above and below the inguinal ligament
Tertiary	Untreated	Fibrosis and lymphatic block: **genital elephantiasis** (esthiomene in women), perirectal abscess, fistulae, **rectal stricture**

- Inguinal disease is commoner in men: the cervix and vagina drain to pelvic nodes.

Differential diagnosis of genital ulcer

- Syphilis (painless ulcer and nodes); herpes (painful vesicles); chancroid (painful ulcer, suppurative nodes); donovanosis (beefy ulcer, no nodes); **LGV: transient painless lesion, then painful matted nodes**.

Investigations

- **NAAT (nucleic acid amplification test)** of ulcer, rectal or node aspirate; LGV-specific genotyping confirms where available. Rectal chlamydia NAAT in all proctocolitis.
- Serology not routine. Screen for HIV, syphilis, gonorrhoea; offer PrEP (pre-exposure prophylaxis).

Treatment

- **Doxycycline 100 mg twice daily for 21 days** (first line).
- Alternatives: erythromycin 500 mg four times daily for 21 days, or azithromycin 1 g weekly for 3 weeks (pregnancy; consider test of cure by NAAT).
- Fluctuant buboes: aspirate through healthy skin to prevent sinus formation.
- **Late disease:** stricture dilatation or resection, fistula repair, reduction of elephantiasis.

Diagram to draw: groin with buboes above and below the inguinal ligament (groove sign).

Key points to remember:

- Serovars L1–L3; invasive and lymphotropic.
- Proctitis is now the commonest presentation.
- Groove sign; esthiomene; rectal stricture.
- Doxycycline 100 mg twice daily for 21 days.

Xanthogranulomatous pyelonephritis.

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 1, Q3a

Definition

- XGP (xanthogranulomatous pyelonephritis) is a rare, severe, **chronic** infection causing diffuse destruction of an enlarged, non-functioning kidney, with **lipid-laden foamy macrophages** (xanthoma cells).
- Up to a fifth of nephrectomies for pyelonephritis. Peak age 50–70 years; women; diabetes is a risk factor.

Pathogenesis

- **Triad: stone, obstruction, infection.** Stones in up to 80% (half staghorn).
- Low-virulence organisms (**Proteus** commonest, *E. coli*) in an obstructed, ischaemic kidney with an inadequate acute inflammatory response.

Pathology

- Diffuse (>80%) or segmental (focal).
- Gross: yellow nodules, pus-filled dilated calyces, a **contracted pelvis** encased by fibrosis, frequent **perinephric extension**.
- Microscopy: sheets of foamy histiocytes with lymphocytes, plasma cells and giant cells.
- **Malek staging:** I confined to kidney; II into perinephric fat; III into paranephric tissue or retroperitoneum.

Clinical features

- Flank pain, fever (about two-thirds), flank mass (about 60%), weight loss; persistent bacteriuria.
- Labs: anaemia, raised inflammatory markers, pyuria; liver dysfunction in up to half. One-third have **sterile urine** (prior antibiotics).

Imaging

- **CT (computed tomography), most useful:** enlarged kidney, little or no function, central calculus in a contracted pelvis; low-density cavities with rim enhancement (**"bear paw" sign**); extrarenal spread.
- Ultrasound: enlarged kidney with hypoechoic areas. DMSA (dimercaptosuccinic acid) scan quantifies lost function.
- **Differentials:** RCC (renal cell carcinoma), pyonephrosis, malakoplakia, lymphoma, tuberculosis.

Management

- Antibiotics to stabilise; drain if septic.
- **Nephrectomy** (with perinephric fat) for diffuse disease; partial nephrectomy for focal disease.
- Surgery is difficult: dense adhesions to psoas, great vessels, bowel and diaphragm. Remove the whole inflammatory mass, because simple drainage leads to **nephrocutaneous fistula**.
- Laparoscopic nephrectomy is feasible; conversion rates are high.
- Frozen section can mimic clear cell carcinoma, and XGP can **coexist with RCC** or urothelial carcinoma. If cancer is not excluded, do nephrectomy.

Diagram to draw: CT sketch of an enlarged kidney with central staghorn and "bear paw" cavities.

Key points to remember:

- Stone, obstruction, Proteus infection → foamy macrophages.
- CT: non-functioning enlarged kidney, contracted pelvis, "bear paw" sign.
- Mimics RCC; treat by complete nephrectomy, not drainage.

EPN. Prognostic factors & conserv. mgt.

Type: SAQ (10 marks)

Source: Practice Set 6 Paper 1, Q3b (prognostic indicators also answered in 2024-Winter Paper 3, Q3c)

Assumed meaning: EPN = emphysematous pyelonephritis.

Definition

- Acute **necrotising** infection of renal parenchyma and perirenal tissue by **gas-forming organisms** (*E. coli*, *Klebsiella*, *Proteus*). Mostly **diabetics**; obstruction in about a quarter. Mortality 19–43%.
- **Mechanism:** high tissue glucose fermented to CO_2 in poorly perfused tissue.
- **CT (computed tomography)** is the imaging of choice.

Prognostic factors

1. Huang–Tseng CT classification (2000)

Class	Gas location
1	Collecting system only
2	Parenchyma only
3A	Perinephric space
3B	Pararenal space
4	Bilateral or solitary kidney

2. Wan CT pattern

- **Type I:** streaky gas, **no fluid**: mortality 50–60%.
- **Type II:** fluid with bubbly or loculated gas: <20%. Fluid means a contained collection that can be drained.

3. Clinical risk factors (Huang–Tseng): thrombocytopenia, AKI (acute kidney injury), altered sensorium, shock.

- **Also adverse:** hypoalbuminaemia, bacteraemia, need for haemodialysis, polymicrobial infection.

Conservative management (now standard)

1. Resuscitation: fluids, inotropes, ICU (intensive care unit) if shocked; **tight glucose control**; correct electrolytes.

2. Antibiotics: broad-spectrum IV (intravenous), adjusted to culture; usually several weeks in total.

3. Drainage

- **Relieve obstruction:** DJ (double-J) stent or PCN (percutaneous nephrostomy).
- **PCD (percutaneous catheter drainage)** of gas and fluid collections under CT or ultrasound; multiple drains if needed; repeat CT to check.

Algorithm

```
All: resuscitation + glucose control + IV antibiotics + relieve obstruction
Class 1–2                      -> antibiotics ± PCD
Class 3–4, <2 risk factors     -> antibiotics + PCD
Class 3–4, >=2 risk factors,
or PCD failure                 -> nephrectomy (emergency or delayed)
```

- Emergency nephrectomy carried 40–50% mortality, so drainage-first care replaced it. Delayed nephrectomy for a non-functioning kidney on renal scan.
- **Emphysematous pyelitis** (class 1) often responds to antibiotics alone.

Diagram to draw: CT sketches of Huang–Tseng classes.

> **Key points to remember:**
>
> - Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral/solitary.
> - Risk factors: thrombocytopenia, AKI, altered sensorium, shock.
> - Wan type I (no fluid) 50–60% mortality; drainage first, nephrectomy for failure.

Papillary necrosis.

Type: SAQ (10 marks)

Source: Practice Set 7 Paper 1, Q3a

Definition

- **RPN (renal papillary necrosis):** ischaemic necrosis of the renal papillae (inner medulla), which may slough into the collecting system.

Pathophysiology

- The papilla is vulnerable: it depends on the **vasa recta**, has **low oxygen tension** and a **hypertonic** environment. Almost every cause reduces medullary blood flow or oxygenation.
- **Sickle cell:** hypoxia and hypertonicity make cells sickle and block the vasa recta, causing microinfarcts.
- **Analgesics:** NSAIDs (non-steroidal anti-inflammatory drugs) block vasodilator prostaglandins; metabolites concentrate in the medulla.
- **Diabetes:** microangiopathy plus infection.

Causes ("POSTCARDS")

Letter	Cause
P	Pyelonephritis
O	Obstruction
S	Sickle cell disease or trait
T	Tuberculosis
C	Cirrhosis
A	Analgesic abuse
R	Renal transplant rejection; renal vein thrombosis
D	**Diabetes mellitus**
S	Systemic vasculitis

- **Also know:** infant dehydration and hypoxia; shock; renal candidiasis.

Clinical features

- **Acute:** fever, flank pain, haematuria, **ureteric colic from a sloughed papilla**, sepsis; bilateral cases may cause AKI (acute kidney injury).
- **Chronic or incidental:** polyuria (poor concentration), sterile pyuria, recurrent UTI (urinary tract infection), CKD (chronic kidney disease). Retained papillae may **calcify** and harbour infection, including fungi.

Investigations

- Urine microscopy and culture; **strain urine** and send tissue for histology; creatinine, glucose, haemoglobin electrophoresis.
- **CT urography or IVU (intravenous urography):** early contrast tracking into the papilla ("**ball-on-tee**", "**lobster claw**"); "**ring sign**" (contrast around a detached papilla); clubbed calyces; filling defects; triangular ring-shaped calcification.
- Retrograde pyelography when renal function is poor.

- **Differential diagnosis:** tuberculosis, medullary sponge kidney, calyceal diverticulum, upper tract urothelial carcinoma.

Management

- **Treat the cause:** stop analgesics, control diabetes, hydrate, manage sickle crisis.
- **Culture-guided antibiotics.**
- **Obstruction by a sloughed papilla:** urgent drainage (stent or nephrostomy), then ureteroscopic extraction.
- Remove calcified retained papillae that cause persistent infection.
- **Follow-up:** renal function; analgesic abusers risk urothelial cancer.

Diagram to draw: calyx showing ring sign and lobster-claw appearance.

> **Key points to remember:**
> - Vasa recta, low oxygen and hypertonicity make the papilla vulnerable.
> - POSTCARDS; diabetes, analgesics, sickle cell, obstruction with infection.
> - Ring sign; sloughed papilla causes colic and obstruction: drain urgently.

Cephalsporins

Type: SAQ (10 marks)

Source: Practice Set 8 Paper 1, Q3c

Assumed meaning: cephalosporins (generations, spectrum, urological use).

Introduction and mechanism

- **β-lactam** antibiotics. They bind **PBPs (penicillin-binding proteins)** and block peptidoglycan cross-linking, so the cell wall fails: **bactericidal**, time-dependent killing.
- Mostly renal excretion (adjust dose in renal failure); **ceftriaxone** also has biliary excretion (no adjustment).
- Resistance: β-lactamases (**ESBL [extended-spectrum β-lactamase]**, AmpC), altered PBPs, reduced porin entry.

Generations

Generation	Examples	Spectrum	Urological use
First	Cefazolin (intravenous), cephalexin, cefadroxil (oral)	Gram-positive (not MRSA [methicillin-resistant *Staphylococcus aureus*]); *E. coli*, *Klebsiella*, *P. mirabilis*	**Surgical prophylaxis (cefazolin)**; cystitis in pregnancy; paediatric prophylaxis
Second	Cefuroxime, cefaclor; cefoxitin, cefotetan	Wider Gram-negative; cefoxitin and cefotetan add **anaerobes**	Prophylaxis for vaginal and bowel surgery; SWL (shock-wave lithotripsy)
Third	Ceftriaxone, cefotaxime, cefixime, cefpodoxime; **ceftazidime**	Most Enterobacterales; **only ceftazidime covers *Pseudomonas***	Pyelonephritis, urosepsis; **gonorrhoea (ceftriaxone 500 mg IM [intramuscular])**; prostate biopsy prophylaxis
Fourth	Cefepime	Gram-negatives incl. *Pseudomonas*, stable to AmpC; Gram-positive	Complicated, nosocomial UTI (urinary tract infection)
Fifth	Ceftaroline	MRSA + Gram-negatives	Limited urological role

- **Newer combinations** for complicated UTI with multidrug resistance: ceftazidime–avibactam, ceftolozane–tazobactam, cefiderocol.

Gaps in spectrum

- **Enterococci** (intrinsically resistant; never treat with cephalosporins), *Listeria*, MRSA (except ceftaroline), atypical organisms.
- ESBL and AmpC producers hydrolyse third-generation agents.

Adverse effects

- Hypersensitivity, less than penicillins. Cross-reactivity with third-generation agents is <1%; avoid after immediate (IgE [immunoglobulin E]-mediated) penicillin reaction.
- ***C. difficile* colitis**; "collateral damage" (selection of ESBL organisms).
- Positive Coombs test; bleeding with cefotetan and cefamandole.
- Ceftriaxone: biliary sludge; avoid in jaundiced neonates and with intravenous calcium.

Paediatric use

- Febrile UTI in neonates: **ampicillin + third-generation cephalosporin** (covers enterococcus). Oral cefixime 8 mg/kg/day. A first-generation agent is a good narrow-

spectrum choice for many outpatient UTIs.

Diagram to draw: β-lactam ring binding a PBP and blocking peptidoglycan cross-links.

Key points to remember:

- Cefazolin = standard surgical prophylaxis.
- Ceftriaxone misses *Pseudomonas*; ceftazidime and cefepime cover it.
- No cephalosporin covers enterococci; ESBL hydrolyses third-generation agents.

Fungal UTI

Type: SAQ (10 marks)

Source: Practice Set 18 SAQ Sheet, Q3a

Introduction and organisms

- Fungal UTI (urinary tract infection) is rising with antibiotics, catheters and immunosuppression. **Candida** is commonest: *C. albicans*, then *C. glabrata* (often fluconazole-resistant); *C. krusei* is intrinsically resistant.
- Others in AIDS (acquired immunodeficiency syndrome) or transplant patients: Aspergillus, Cryptococcus, Mucorales, Histoplasma.

Risk factors and routes

- Antibiotics, indwelling catheters and stents, diabetes, steroids, neutropenia, prematurity, parenteral nutrition, obstruction.
- **Ascending** (catheter) causes most candiduria. **Haematogenous** spread from candidaemia: in disseminated candidiasis the kidney is the organ most often involved.

Clinical spectrum

- Asymptomatic candiduria (most) → cystitis → pyelonephritis → **fungus ball (bezoar)** causing obstruction → renal abscess or candidaemia.

Diagnosis

- Microscopy: budding yeasts and pseudohyphae (may mimic red cells). Repeat culture: contamination is common.
- Children: treat when repeated cultures exceed **10,000–15,000 CFU (colony-forming units)/mL**.
- Persistent funguria: ultrasound or CT (computed tomography) for bezoar; blood cultures if high risk.

Management

Situation	Treatment
Asymptomatic	Remove or change catheter, stop antibiotics; no antifungal
Treat asymptomatic only if	Neutropenia, very-low-birth-weight infant, before a urological procedure
Cystitis	Fluconazole 200 mg (3 mg/kg) daily for 2 weeks
Pyelonephritis	Fluconazole 200–400 mg (3–6 mg/kg) daily for 2 weeks
Fluconazole-resistant (*C. glabrata*, *C. krusei*)	Amphotericin B deoxycholate 0.3–0.6 mg/kg/day for 1–7 days ± flucytosine
Resistant cystitis	Amphotericin B bladder irrigation 50 mg/L
Fungus ball	PCN (percutaneous nephrostomy) drainage + systemic antifungal + local amphotericin irrigation; endoscopic or open removal if persistent

- Fluconazole concentrates in urine; **echinocandins and lipid amphotericin do not.**
- **Before procedures:** IDSA (Infectious Diseases Society of America) 2016 treats all candiduria (fluconazole 400 mg daily for several days before and after). AUA (American Urological Association) Best Practice Statement 2019 gives single-dose antifungal prophylaxis only for urothelium-breaching surgery. The two disagree.
- **Also know:** alkalinisation for bezoar; fluconazole caution under 6 months.

Diagram to draw: budding yeast with pseudohyphae; fungus ball in renal pelvis with nephrostomy.

Key points to remember:

- Candida commonest; *C. glabrata* often fluconazole-resistant.
- Asymptomatic candiduria: remove catheter, no antifungal (except neutropenia, neonate, procedure).
- Fluconazole 2 weeks; echinocandins fail in urine.
- Fungus ball: drain, antifungal, remove.

Antimicrobial resistance

Type: SAQ (10 marks)

Source: Practice Set 18 SAQ Sheet, Q3b (also answered in part in 2015-Summer Paper 1, Q3f)

Definition

- AMR (antimicrobial resistance): a microbe grows despite drug levels achievable in the patient. Reported as S (sensitive), I (intermediate), R (resistant) with the MIC (minimum inhibitory concentration).
- One of the most serious global health threats.

Types

Type	Mechanism	Relevance
Intrinsic	Species lacks the target	Predictable (enterococci and cephalosporins)
Acquired chromosomal	Mutants selected when urine level is close to MIC	Underdosing, poor compliance
Plasmid (extrachromosomal)	Horizontal transfer in bowel flora	**Commonest**; multidrug resistance after antibiotics

Mechanisms

- **Enzymes:** β-lactamases; aminoglycoside-modifying enzymes.
- **Altered target:** penicillin-binding protein; DNA gyrase mutation (quinolones).
- **Reduced entry** (porin loss) and **efflux pumps**.
- **Bypass:** enterococci scavenge folate (escape TMP-SMX (trimethoprim–sulfamethoxazole)).

Key resistant organisms

- **ESBL (extended-spectrum β-lactamase)** (plasmid, 1983): resists 3rd-generation cephalosporins; often co-resistant to fluoroquinolones and aminoglycosides. *E. coli* ST131 carries it.
- **AmpC; KPC (*Klebsiella pneumoniae* carbapenemase, 2001); NDM (New Delhi metallo-β-lactamase)**: carbapenem resistance.
- MRSA (methicillin-resistant *S. aureus*), VRE (vancomycin-resistant enterococci).

Drivers in urology

- Overuse of fluoroquinolones and cephalosporins; treating ASB (asymptomatic bacteriuria); long prophylaxis; catheters, stents, stones (biofilm).
- Transrectal biopsy: fluoroquinolone-resistant rectal *E. coli* causes sepsis.

Prevention: stewardship

1. Culture before treatment; follow the **local antibiogram**.
2. TMP-SMX only if local resistance ≤20%; oral fluoroquinolone for pyelonephritis only if ≤10%.
3. Prefer narrow, urine-confined drugs for cystitis (nitrofurantoin, fosfomycin, pivmecillinam: low resistance).
4. Shortest course; de-escalate to culture.
5. Do not treat ASB, except in pregnancy and before mucosa-breaching procedures.
6. Single-dose surgical prophylaxis; transperineal biopsy or rectal-swab-targeted prophylaxis.
7. Infection control and hand hygiene.

- **Also know:** FDA (Food and Drug Administration) 2016 fluoroquinolone warning; *C. difficile*; ESBL: carbapenem, or fosfomycin/nitrofurantoin for cystitis.

Recent advances

- New β-lactam/β-lactamase inhibitor combinations (ceftazidime–avibactam, meropenem–vaborbactam); cefiderocol; gepotidacin.

Diagram to draw: Gram-negative cell showing β-lactamase, porin loss, efflux pump, altered target.

Key points to remember:

- Plasmid-mediated resistance is commonest.
- ESBL: plasmid, co-resistant; KPC/NDM: carbapenem resistance.
- TMP-SMX if resistance ≤20%; fluoroquinolone for pyelonephritis if ≤10%.
- Stewardship: culture, narrow, short, no ASB treatment.

Biofilm

Type: SAQ (10 marks)

Source: Practice Set 18 SAQ Sheet, Q3c (also answered in part in 2015-Summer Paper 3, Q3d)

Definition

- A **structured microbial community in a self-made polymeric matrix**, adherent to a living or inert surface. The matrix contains exopolysaccharide, extracellular DNA and proteins.

Stages of formation

1. **Conditioning film:** host proteins coat the device within minutes.
2. **Attachment:** reversible, then irreversible (pili, adhesins, flagella).
3. **Matrix production and microcolonies.**
4. **Maturation:** 3-D (three-dimensional) structure with water channels; cell-to-cell signalling (quorum sensing).
5. **Dispersion:** free bacteria seed new infection.

Why antibiotics and immunity fail

- Poor drug penetration through the matrix.
- Slow-growing, dormant cells resist drugs that need active growth.
- Altered genes, envelope and targets.
- Survive **1000–1500 times** the usual killing concentration. Neutrophils are excluded.

Urological examples

Site	Point
Urethral catheter	Forms within minutes; **the catheter, not the bladder, is infected**. Proteus urease makes crystalline biofilm → encrustation and blockage
Ureteric stent, nephrostomy	Stent colonised more often than urine shows; nephrostomy changed every 3 months
Penile prosthesis, AUS (artificial urinary sphincter)	Revision of an "uninfected" device: about 10% infection
Infection stones	Bacteria inside struvite; antibiotics cannot sterilise
Chronic bacterial prostatitis	Biofilm commoner in prostatitis *E. coli*; needs longer, penetrating therapy
Recurrent cystitis	**Intracellular bacterial communities**: biofilm-like pods inside umbrella cells

- Common organisms: Pseudomonas, Proteus, *E. coli*, Enterococcus, coagulase-negative staphylococci, Candida.

Prevention and management

- Avoid devices; shortest dwell; aseptic insertion; closed dependent drainage.
- **Change the catheter** when treating symptomatic CAUTI (catheter-associated UTI (urinary tract infection)); remove infected stent or implant.
- Silver-alloy catheters reduce bacteriuria, not clinical UTI.
- Implants: antibiotic-coated devices, no-touch technique, **antiseptic washout** at revision (infection about 10% → 3%).
- Stones: complete clearance. Chemodissolution for catheter encrustation. Macrolides penetrate biofilm (prostatitis).

- **Also know (research):** quorum-sensing inhibitors, bacteriophages, new coatings.

Diagram to draw: five-stage biofilm cycle on a catheter surface.

> **Key points to remember:**
> - Biofilm: microbes in self-made matrix on a surface.
> - Survives 1000–1500 times normal antibiotic levels.
> - Treat CAUTI by changing the catheter.
> - Revision washout cuts prosthesis infection from about 10% to 3%.

Renal mucormycosis

Type: SAQ (10 marks)

Source: Practice Set 19 SAQ Sheet, Q3c

Introduction

- Mucormycosis: an **angioinvasive** fungal infection by Mucorales (*Rhizopus* commonest; *Mucor*, *Lichtheimia*).
- Kidney involvement occurs in **disseminated** disease, or as **isolated renal mucormycosis**, reported particularly from India, sometimes in apparently immunocompetent young adults. Cases rose during the 2021 COVID-19 (coronavirus disease 2019) wave.

Risk factors

- Uncontrolled diabetes, especially **ketoacidosis**; corticosteroids; neutropenia and haematological cancer; transplantation; iron overload or deferoxamine; intravenous drug use; AIDS (acquired immunodeficiency syndrome).

Pathogenesis

- Spores inhaled or ingested → **haematogenous** seeding of the kidney.
- Hyphae invade arteries and veins → **thrombosis → infarction and haemorrhagic necrosis**.
- Acidosis and free iron favour growth (hence the ketoacidosis risk).

Clinical features

- Fever, flank pain, tender flank mass, haematuria, pyuria with sterile bacterial cultures.
- **Bilateral** disease → AKI (acute kidney injury) or anuria.
- Suspect when "pyelonephritis" fails antibiotics in a high-risk patient.

Investigations

- **CECT (contrast-enhanced computed tomography):** enlarged kidney(s), wedge-shaped or diffuse **non-enhancing** areas (infarcts), perinephric stranding or collection, renal vessel thrombosis.
- **Tissue diagnosis** (biopsy or nephrectomy specimen): **broad, ribbon-like, aseptate or pauciseptate hyphae branching at right angles**, invading vessels.
- KOH (potassium hydroxide) mount, calcofluor stain, fungal culture (often negative), PCR (polymerase chain reaction).
- Differential: *Aspergillus* has narrow septate hyphae with acute-angle branching.
- Image sinuses, lungs, brain.

Management (three pillars)

1. **Reverse predisposition:** control glucose, correct ketoacidosis, taper steroids and immunosuppression, stop deferoxamine.
2. **Early surgical debridement:** nephrectomy for a unilateral necrotic kidney; drain collections.
3. **Antifungal: liposomal amphotericin B 5 mg/kg/day** first line (ECMM/MSG ERC (European Confederation of Medical Mycology/Mycoses Study Group Education and Research Consortium) guideline, 2019). Step-down or salvage: **posaconazole or isavuconazole**.

- **Fluconazole, voriconazole and echinocandins are inactive.**

- Bilateral: antifungal plus dialysis.

Prognosis

- Mortality is high, especially with bilateral or disseminated disease and delayed diagnosis.

Diagram to draw: right-angle branching aseptate hypha invading a renal vessel.

> **Key points to remember:**
> - Angioinvasion → infarction; diabetic ketoacidosis is the key risk.
> - CECT: non-enhancing enlarged kidney; histology: broad aseptate right-angle hyphae.
> - Liposomal amphotericin B 5 mg/kg/day plus nephrectomy and glucose control.
> - Voriconazole and echinocandins do not work.

Hydatid disease of kidney

Type: SAQ (10 marks)

Source: Practice Set 19 SAQ Sheet, Q3d

Introduction

- Cystic echinococcosis (CE) by *Echinococcus granulosus*, a cestode. Dog: definitive host; sheep and humans: intermediate.
- Ingested eggs → liver (commonest), lung, then **kidney (third site, about 2–3%)** via the systemic circulation. Usually single, unilateral, cortical.

Pathology

- Wall: **pericyst** (host fibrous layer), **laminated membrane** (ectocyst), **germinal layer** (endocyst) producing brood capsules, scolices and daughter cysts ("hydatid sand").
- **Closed** versus **open** cyst (ruptured into the collecting system → **hydatiduria**).

Clinical features

- Grows 1–2 cm a year; long asymptomatic; flank mass, dull pain.
- **Hydatiduria** ("grape-skin" membranes in urine; pathognomonic) with colic, in a minority.
- Infection; rupture (anaphylaxis). Renal function usually preserved.

Investigations

- **Serology** (ELISA (enzyme-linked immunosorbent assay)) is adjunctive only (sensitivity 60–90%); eosinophilia; Casoni test obsolete.
- **Ultrasound and CT (computed tomography):** thick-walled cyst, wall calcification, daughter cysts, detached membranes ("water-lily sign"). Stage guides treatment.

WHO (World Health Organization) stage	Appearance	Activity
CE1	Unilocular, double-line wall	Active
CE2	Multivesicular, daughter cysts	Active
CE3a / CE3b	Detached membranes / daughter cysts in solid matrix	Transitional
CE4 / CE5	Solid "ball of wool" / calcified wall	Inactive

- Retrograde pyelogram shows communication; image liver and lungs. **Never aspirate blindly** (anaphylaxis, seeding).

Management

- **Albendazole 400 mg twice daily**, 1–6 months: alone for small or inoperable cysts; around surgery to reduce seeding (± praziquantel).
- **Surgery is the mainstay** (open, laparoscopic or robotic), nephron-sparing when possible:
  1. Pack the field with scolicide-soaked swabs (hypertonic saline, povidone-iodine).
  2. Inject scolicide only if **not** communicating (chemical injury to the collecting system).
  3. **Pericystectomy, partial nephrectomy or deroofing**; close communications; omentoplasty or drain.
  4. **Nephrectomy** for a non-functioning kidney.
- **PAIR (puncture, aspiration, injection, re-aspiration):** chapter says liver only, not kidney; some series report percutaneous renal treatment.
- Calcified inactive cysts (CE4, CE5) may be observed.

Diagram to draw: layers of a hydatid cyst with daughter cysts and a communication into a calyx.

Key points to remember:

- Kidney: third site (about 2–3%).
- Hydatiduria (grape-skin membranes) is pathognomonic of a communicating cyst.
- Albendazole 400 mg twice daily around surgery; no scolicide if communicating.

Rapid Revision Sheet – Section 3 · Infections & Inflammation

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Genitourinary infections, common organisms, bacteraemia and urosepsis: pathophysiology and management (LAQ)

- *E. coli* ~85% community UTI; P pili cause pyelonephritis; *Proteus* causes struvite; *Enterococcus* resists cephalosporins.
- Sepsis = SOFA rise ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2 mmol/L.
- Obstruction plus infection is the emergency; bladder urine may be sterile.
- Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline (SSC 2021).
- Infected obstructed kidney: drain urgently; no definitive stone surgery in sepsis.
- Stent and nephrostomy equally effective (stent = nephrostomy for drainage); delay definitive stone surgery.

Describe clinical implication of AIDS on the Genito-Urinary Tract? (LAQ)

- AIDS = CD4 <200; UTI risk rises as CD4 falls.
- HIV-TB: disseminated, less fibrosis, fewer strictures.
- Indinavir stones: radiolucent, may be CT-invisible; stop drug and hydrate.
- PEP within 72 hours for 28 days; circumcision cuts acquisition 50–60%.

Principles of antimicrobial therapy in UTI (SAQ)

- Recurrent = ≥2 in 6 months or ≥3 in 12 months.
- Cystitis: nitrofurantoin 5 d, fosfomycin single, TMP-SMX 3 d (resistance ≤20%).
- Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d (FQ resistance ≤10%).

Renal and perinephric abscess (SAQ)

- Now mostly ascending Gram-negative abscess in a damaged kidney.
- Perinephric: within Gerota fascia; diabetes in a third.
- Fever beyond 4–5 days of antibiotics: get a CT.
- CT is the investigation of choice; follow until resolution.
- ≤3 cm antibiotics; 3–5 cm either; >5 cm percutaneous drainage.
- Percutaneous drainage first line; nephrectomy if non-functioning.

Ciprofloxacin (SAQ)

- DNA gyrase and topoisomerase IV; concentration-dependent.
- Use for pyelonephritis only if local resistance ≤10%.
- Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

Catheter Associated Urinary Tract Infections (CAUTI) (SAQ)

- Bacteriuria 5–10% per catheter-day; duration is the key risk.
- Diagnosis: symptoms + ≥10^3 CFU/mL from a fresh catheter.
- Remove catheter early; closed drainage; treat only symptomatic CAUTI, 7 or 10–14 days.

Complicated urinary tract infections. (SAQ)

- Complicated = abnormal tract or compromised host; male sex alone qualifies.
- Culture mandatory; often MDR.
- Drain obstruction: stent = nephrostomy.
- 7–14 days; oral step-down when afebrile.

ESBL producing enterobacterial UTI (SAQ)

- Plasmid CTX-M; resists 3rd-gen (third-generation) cephalosporins; inhibited by clavulanate (clavulanate inhibits).
- Oral: nitrofurantoin, fosfomycin, pivmecillinam.
- Serious infection: carbapenem; MERINO: pip-taz (piperacillin–tazobactam) inferior.
- Stewardship prevents spread.

Lichen sclerosus (BXO, balanitis xerotica obliterans) (SAQ)

- Genital skin disease; meatus is the sentinel site.
- Spares proximal urethra; perineal urethrostomy easy.
- Clobetasol first; oral mucosa, never skin, for repair.
- SCC in 2.3–9%: lifelong follow-up.

Drug Fosfomycin (SAQ)

- Inhibits MurA: first step of cell wall synthesis; no cross-resistance.
- 3 g single dose for cystitis; active against ESBL E. coli and VRE.
- Not for pyelonephritis; poor Pseudomonas cover.

Chronic Prostatitis : diagnosis, classification and management. (SAQ)

- NIH I–IV; III = pain ≥3 of 6 months, no infection.
- CBP: same organism, fluoroquinolone 4–6 weeks.
- CP/CPPS: NIH-CPSI to score, UPOINT to treat.

Lymphogranuloma venereum (SAQ)

- Serovars L1–L3; invasive and lymphotropic.
- Proctitis is now the commonest presentation.
- Groove sign; esthiomene; rectal stricture.
- Doxycycline 100 mg twice daily for 21 days.

Xanthogranulomatous pyelonephritis. (SAQ)

- Stone, obstruction, Proteus infection → foamy macrophages.
- CT: non-functioning enlarged kidney, contracted pelvis, "bear paw" sign.
- Mimics RCC; treat by complete nephrectomy, not drainage.

EPN. Prognostic factors & conserv. mgt. (SAQ)

- Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral/solitary.
- Risk factors: thrombocytopenia, AKI, altered sensorium, shock.
- Wan type I (no fluid) 50–60% mortality; drainage first, nephrectomy for failure.

Papillary necrosis. (SAQ)

- Vasa recta, low oxygen and hypertonicity make the papilla vulnerable.
- POSTCARDS; diabetes, analgesics, sickle cell, obstruction with infection.
- Ring sign; sloughed papilla causes colic and obstruction: drain urgently.

Cephalsporins (SAQ)

- Cefazolin = standard surgical prophylaxis.
- Ceftriaxone misses *Pseudomonas*; ceftazidime and cefepime cover it.
- No cephalosporin covers enterococci; ESBL hydrolyses third-generation agents.

Fungal UTI (SAQ)

- Candida commonest; *C. glabrata* often fluconazole-resistant.
- Asymptomatic candiduria: remove catheter, no antifungal (except neutropenia, neonate, procedure).
- Fluconazole 2 weeks; echinocandins fail in urine.

- Fungus ball: drain, antifungal, remove.

Antimicrobial resistance (SAQ)

- Plasmid-mediated resistance is commonest.
- ESBL: plasmid, co-resistant; KPC/NDM: carbapenem resistance.
- TMP-SMX if resistance ≤20%; fluoroquinolone for pyelonephritis if ≤10%.
- Stewardship: culture, narrow, short, no ASB treatment.

Biofilm (SAQ)

- Biofilm: microbes in self-made matrix on a surface.
- Survives 1000–1500 times normal antibiotic levels.
- Treat CAUTI by changing the catheter.
- Revision washout cuts prosthesis infection from about 10% to 3%.

Renal mucormycosis (SAQ)

- Angioinvasion → infarction; diabetic ketoacidosis is the key risk.
- CECT: non-enhancing enlarged kidney; histology: broad aseptate right-angle hyphae.
- Liposomal amphotericin B 5 mg/kg/day plus nephrectomy and glucose control.
- Voriconazole and echinocandins do not work.

Hydatid disease of kidney (SAQ)

- Kidney: third site (about 2–3%).
- Hydatiduria (grape-skin membranes) is pathognomonic of a communicating cyst.
- Albendazole 400 mg twice daily around surgery; no scolicide if communicating.

Section 4 · Paediatric Urology 1 — Long answer questions (LAQ)

Primary (congenital) VUR: VUJ embryology, pathophysiology, natural history and management

Asked as:

- Describe embryological development of VU Junction and outline management of primary VUR? — *Summer 2020 Paper I Q2 (LAQ)*
- B/l congenital VUR - Pathophysiology and natural history — *Practice Set 1 Paper I Q3f (SAQ)* (VUR management also answered in 2015-Summer Paper 3, Q2 and 2020-Summer Paper 1, Q2)

Type: LAQ (25 marks); also asked as SAQ (10 marks)

"B/l congenital VUR" is read as bilateral primary (congenital) VUR (vesicoureteral reflux), often detected antenatally.

Part A: Embryology of the VUJ (vesicoureteric junction)

1. **Ureteric bud (weeks 4–6)** arises from the caudal mesonephric (Wolffian) duct: ureter and duct form a "Y", the common nephric duct is the stem.
2. **Reciprocal induction:** the bud enters the metanephric blastema, branches into the collecting system and induces nephrons.
3. **Weeks 5–6:** the urorectal septum divides the cloaca into urogenital sinus (UGS) and anorectal canal.
4. **Absorption:** the common nephric duct is taken into the UGS (current view: removed by **apoptosis**), so ureter and duct open separately.
5. **Rotation (Mackie and Stephens):** the ureteric orifice moves **cranially and laterally** to the bladder base; the duct stays caudal as the ejaculatory duct (Gartner duct in females).
6. **Trigone:** classically ureteral; lineage tracing shows mostly bladder muscle. **Chwalla membrane** over the orifice ruptures; persistence explains ureterocele.
7. **Maturation:** ureteric muscle joins the detrusor as **Waldeyer's sheath**; the tunnel lengthens with growth, so VUR (vesicoureteral reflux) resolves with age.

Bud origin	Orifice	Result
Normal	Trigone corner	Tunnel length to diameter about **5:1** (Paquin); refluxing junctions about 1.4:1
Too low (early contact)	High, **lateral**	Short tunnel → **primary VUR**; abnormal bud–blastema contact → dysplasia
Too high (late contact)	Caudal, **medial**	**Ectopic ureter** (bladder neck, urethra, Wolffian structures), often obstructed
Two buds	Duplex	**Weigert–Meyer rule:** upper pole caudal-medial (ectopia, ureterocele); lower pole lateral (reflux)

- **Antireflux mechanism:** passive flap valve, ureteric peristalsis, low bladder pressure.
- **Concept:** bud position fixes both orifice and kidney, so high-grade VUR and dysplasia coexist.

Part B: Pathophysiology of (bilateral) primary VUR

- **Embryology:** a ureteric bud arising too early over-rotates, so the orifice lies **high and lateral** with a short tunnel. Its abnormal contact with the metanephros causes **renal**

dysplasia, so high-grade VUR and dysplasia coexist.

- **Failed UVJ (ureterovesical junction) flap valve:** filling normally compresses the intramural ureter against detrusor backing. Tunnel length to diameter is **5:1** normally versus 1.4:1 in refluxing units (Paquin). High infant male voiding pressure and BBD (bladder and bowel dysfunction) add a secondary component.

Renal damage	Reflux-associated dysplasia	Acquired scar
Cause	Abnormal bud	Infected urine plus VUR
Typical patient	Boy, grade IV–V, antenatal hydronephrosis	Girl, recurrent febrile UTI (urinary tract infection), BBD
Preventable	No	Yes

- **Intrarenal reflux** through compound polar papillae needs only 2 mmHg in neonates, so scars are polar. **Sterile reflux rarely scars** unless pressure is high.
- **Also know:** massive bilateral VUR gives the megacystis–megaureter association.

Diagram to draw: long oblique tunnel versus short lateral one.

Part C: Management of primary VUR

- **Primary:** congenitally short tunnel with a normal bladder. **Secondary:** high bladder pressure (PUV (posterior urethral valves), neurogenic bladder, BBD (bladder and bowel dysfunction)).
- In about 30% of children with UTI (urinary tract infection) and about 30% of siblings.
- **Infection, not reflux, injures the kidney.** The aim is to prevent febrile UTI and scarring while VUR resolves.

International grading (VCUG, voiding cystourethrogram)

Grade	Findings
I	Ureter only, not dilated
II	Up to pelvis and calyces; no dilatation (undilated)
III	Mild–moderate dilatation; minimal forniceal blunting
IV	Moderate dilatation and tortuosity
V	Gross dilatation and tortuosity; **papillary impressions lost**

Evaluation

- Urine culture; BP (blood pressure), creatinine, urine protein; BBD history.
- **RBUS (renal and bladder ultrasound)**; VCUG (gold standard); nuclear cystogram for follow-up; **DMSA (dimercaptosuccinic acid) scan** for scars.

Natural history

- Resolution comes from **elongation of the intravesical ureter** with growth. Grade II resolves in 63–85%; grade III in about 50%; **grades IV–V and bilateral grade III rarely** (International Reflux Study). Younger age resolves better.
- **Bilaterality and older age delay resolution:** persistence is highest in bilateral grade IV, and in bilateral grade III at 6–10 years. Infants with grade IV–V may still resolve.
- **Consequences of bilateral damage:** adult hypertension in 34–38% of scarred children; CKD (chronic kidney disease), with reflux nephropathy a leading paediatric transplant diagnosis; poor growth; pre-eclampsia (worse with bilateral scars). Correction cannot reverse dysplasia.

Risk stratification: high risk = female, under 1 year, bilateral, grade III–V, BBD, scars, multiple febrile UTIs.

Medical management

- **Treat BBD first:** timed voiding, constipation treatment, anticholinergics.
- **CAP (continuous antibiotic prophylaxis):** one low dose at night. Trimethoprim or amoxicillin under 2 months; TMP-SMX (trimethoprim-sulfamethoxazole) or nitrofurantoin over 2 months.
- Boys with phimosis: circumcision or topical steroid (about 4 circumcisions prevent one UTI).
- **RIVUR (Randomized Intervention for Children with Vesicoureteral Reflux, 2014):** recurrent UTI fell from 23.6% to 12.9% on TMP-SMX; **scarring unchanged**.
- **Swedish Reflux Trial** (grade III–IV): febrile UTI in girls 19% on CAP, 23% after endoscopy, 57% on surveillance; no difference in boys.
- IRSC (International Reflux Study in Children), Birmingham: surgery equalled medical care for scarring.
- Low-risk children: observation without CAP.

Indications for correction: breakthrough febrile UTI on CAP; new scars or falling function; persistent high grade; non-adherence; parental choice.

Endoscopic injection: Dx/HA (dextranomer/hyaluronic acid) by STING (subureteric Teflon injection) or HIT (hydrodistension implantation technique); about 90% for low grade after one injection (see 2020-Summer Paper 4, Q3c).

Ureteric reimplantation: open success **95.9%** (Elder 1997).

Technique	Tunnel	Note
Cohen (intravesical)	Cross-trigonal	Commonest; over 98%; later ureteroscopy difficult
Politano–Leadbetter	New hiatus above old	Long tunnel; blind hiatus risks bowel
Glenn–Anderson	Advanced towards bladder neck	Least kinking; short tunnel
Lich–Gregoir (extravesical)	Detrusorotomy	No haematuria or spasm; **retention up to 10% after bilateral**
RALUR-EV (robotic extravesical)	Y-shaped detrusorotomy	Less pain; approaching open results

- Ultrasound at 6–12 weeks. Contralateral VUR in about 1 in 10, usually resolves.
- **Guidelines:** AUA (American Urological Association) 2010/2017 and EAU/ESPU (European Association of Urology/European Society for Paediatric Urology): treat BBD, CAP for high risk, correct after breakthrough infection.
- **Follow-up:** BP, growth; scarred women risk pre-eclampsia.

```
Treat BBD -> low risk: observe | high risk: CAP
Breakthrough UTI or new scar -> injection or reimplant
```

Diagram to draw: "Y" of ureteric bud and mesonephric duct with orifice migration; grades I–V.

Key points to remember:

- Low (early) bud → lateral orifice → VUR plus dysplasia; high (late) bud → ectopia.
- One embryological fault: lateral orifice, short tunnel, dysplasia.
- Tunnel length to diameter 5:1 (Paquin).
- Dysplasia (boys, congenital) versus scar (girls, infection).
- Grade IV–V and bilateral grade III rarely resolve.
- Scarred children: 34–38% adult hypertension.
- RIVUR: CAP halves recurrent UTI, not scarring.
- Open reimplantation 95.9%; Cohen over 98%.

Urological evaluation and management of pediatric kidney transplant patient.

Type: LAQ (25 marks)

Source: 2022-Summer Paper 1, Q2

Introduction

- Children needing KRT (kidney replacement therapy) often have CAKUT (congenital anomalies of the kidney and urinary tract). The urologist must ensure that the bladder that destroyed the native kidneys does not destroy the graft.
- **Five goals:** unobstructed drainage; a capacious, low-pressure reservoir with social continence; volitional emptying; no infection; the fewest operations.

Causes of ESKD (end-stage kidney disease) in children

- Commonest age at transplant 13–17 years.
- **Younger:** dysplasia, hypoplasia, aplasia and obstructive uropathy, mainly PUV (posterior urethral valves). **Older:** FSGS (focal segmental glomerulosclerosis) and neuropathic bladder.

Pretransplant evaluation

- **Every child:** urological history, renal ultrasound, uroflow with PVR (post-void residual) if toilet-trained.
- **VCUG (voiding cystourethrogram)** only with neuropathic bladder, PUV, VUR (vesicoureteral reflux), previous lower-tract surgery, recurrent febrile UTI (urinary tract infection) or unexplained hydronephrosis.
- **Urodynamics** for neuropathic bladder, PUV or large ureterocele, refractory voiding dysfunction, high-grade hydronephrosis or recurrent UTI. They measure capacity, compliance, detrusor activity, emptying and sphincter function, and are repeated.

```
History positive → ultrasound + VCUG → urodynamics
  Poor capacity/compliance → anticholinergic, CIC, cycling → still poor → augment
  Poor emptying → CIC ± catheterisable stoma
  Adequate → clear for transplant
History negative → ultrasound + PVR → normal → clear
```

- An anuric child with no urological history needs no further tests.

Pretransplant management

1. Bladder optimisation (drugs → CIC → cycling → augmentation)

- Commonest problem: low capacity, detrusor overactivity, poor compliance. High pressure harms the kidney even without reflux.
- **Target:** age-expected capacity at detrusor pressure **<40 cm H_2O**.
- Antimuscarinics (oxybutynin titrated; solifenacin), mirabegron, intradetrusor botulinum toxin; α-blockers aid emptying. Better storage may worsen emptying, so add **CIC (clean intermittent catheterisation)**. Prove volitional emptying by repeated low PVRs.
- A family's adherence to CIC predicts adherence to immunosuppression.

2. Valve bladder versus neuropathic bladder

- PUV bladder moves from overactivity to **myogenic failure**: postpone augmentation. Neuropathic bladder may deteriorate under high post-transplant urine output: reconstruct earlier.

3. Defunctionalised bladder (anuria or diversion)

- **Bladder cycling:** catheter instillation of increasing saline volumes, with drugs, until age-expected capacity; confirm by urodynamics. **Never use the graft to test the bladder.**

4. Augmentation

- Indication: cannot reach **>75% expected capacity at <40 cm H_2O** with CIC every 3–4 hours despite maximal therapy. It does not clearly raise transplant complications.

Segment	Point
Dilated ureter (ureterocystoplasty)	Explore first: urothelium, no mucus or acidosis
Ileum, sigmoid	Most used; mucus, stones, infection, malignancy; lifelong follow-up
Stomach	Haematuria–dysuria syndrome in anuric patients; rarely used

- Continent stoma (Mitrofanoff) if the urethra is sensate or difficult; site it medial to the planned incision. Allow **6–12 weeks** healing before transplant.

5. Infection, nephrectomy and dialysis

- Treat stasis (double voiding, CIC); distinguish colonisation from UTI; consider circumcision in boys with valves.
- **Native nephrectomy** (default is to keep): uncontrolled hypertension, nephrotic syndrome, massive polyuria, recurrent pyelonephritis, large stones, malignancy risk, very large kidneys, massive VUR. Retroperitoneal route keeps PD (peritoneal dialysis).
- Keep a normal native ureter for later ureteroureterostomy. Intraperitoneal surgery on PD needs temporary haemodialysis.

At transplantation

- Retroperitoneal Gibson incision; children **<15 kg** may need aorta–caval anastomosis, sometimes transperitoneal.
- **Ureter:** extravesical non-refluxing ureteroneocystostomy (3–4 cm detrusorotomy) preferred; Barry tunnel; ureteroureterostomy to a normal native ureter.
- **Stent:** selective; double-J removed at 4–6 weeks, or early removal at 5–7 days.

Post-transplant complications

Complication	Rate	Management
Ureteric obstruction	6%	Usually distal ischaemic stenosis; >half within 100 days. Stent or nephrostomy; balloon dilatation; open repair (native ureter, bladder flap)
Urine leak	3%	Early; drain creatinine; catheter, nephrostomy, stent, then reimplantation
Graft VUR	2% (significant)	Febrile UTI → reimplant; asymptomatic → observe; injection works in one-third
Stones	2%	Treat if >1 cm and growing, obstructing or infected

- **Rising creatinine with hydronephrosis:** catheter drainage 1–2 weeks (bladder cause?) → biopsy if rejection is suspected → stent or nephrostomy. Obstruction and rejection can coexist.
- Also: lymphocele, **PTLD (post-transplant lymphoproliferative disorder)**, graft thrombosis (a leading cause of early graft loss), late infection reflecting bladder dysfunction.

Follow-up

- Lifelong ultrasound, PVR and bladder surveillance; recurrent UTIs lower graft GFR (glomerular filtration rate).

Diagram to draw: extravesical ureteroneocystostomy with detrusor flaps closed over the ureter.

Key points to remember:

- Bladder target: age-expected capacity at <40 cm H_2O; augment if <75% despite CIC.
- PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.
- Never test a bladder with a graft; cycle the defunctionalised bladder.
- Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis

Asked as:

- Write a note on the embryological descent of the gonads, explaining the various phases of descent. Outline the evaluation and management of bilateral undescended testis in a 3 year old child. — *Winter 2024 Paper I Q2 (LAQ)* (also answered in part in 2018-Summer Paper 3, Q3a, the treatment algorithm)
- Describe developmental embryology of testes and its physiological descent. Mention various abnormalities associated with testicular descent. — *Practice Set 14 Paper I Q1 (LAQ)* (descent also answered in 2024-Winter Paper 1, Q2; maldescent in Practice Set 5 Paper 3, Q3b)

Type: LAQ (25 marks)

Development of the testis

- **Indifferent gonad:** the gonadal (germinal) ridge forms at **4 weeks** on the medial urogenital ridge. It has four components: coelomic epithelium, underlying mesenchyme, adjacent mesonephros and **PGCs (primordial germ cells)**.
- PGCs arise in the **yolk sac wall in week 3** and migrate along the dorsal mesentery of the hindgut to the ridge (about 6 weeks).
- **SRY (sex-determining region Y)** on the short arm of Y, partly activated by SF1 (steroidogenic factor 1), switches on **SOX9** in pre-Sertoli cells. Sertoli cells then appear (about 40 days). Without SRY no Sertoli cells form and the gonad becomes an ovary.
- Sertoli cells enclose germ cells as **testicular cords (about 6 weeks)**, which canalise into seminiferous tubules.
- **Rete testis** forms with the cords and joins **5–12 mesonephric tubules**, which become efferent ductules. Full continuity (tubules, rete, efferent ductules, epididymis) is established by the **third month**.
- The testis turns round and a thick **tunica albuginea** forms. Testes become spherical after 7–8 weeks; ovaries stay oblong.

Hormone	Cell	Action	Male remnants
AMH (anti-Müllerian hormone)	Sertoli	Müllerian regression, weeks 8–10	Appendix testis, prostatic utricle
Testosterone	Leydig (from 8 weeks; driven by placental hCG (human chorionic gonadotrophin), later fetal LH (luteinising hormone))	Wolffian duct to epididymis, vas, seminal vesicle (weeks 8–12)	Appendix epididymis, paradidymis
DHT (dihydrotestosterone)	Formed locally by 5α-reductase type 2	External genitalia, prostate, scrotum (fused by 10–12 weeks)	–
INSL3 (insulin-like peptide 3)	Leydig	Gubernacular swelling (via RXFP2 receptor)	–

Gonadal (physiological) descent: starting point

- At **7–8 weeks** the gonad (testis) lies near the kidney (upper lumbar), held by the **cranial suspensory ligament (CSL)** above and the **gubernaculum** below. The gubernaculum appears at 7 weeks, attaches to the **epididymis** and runs to the future inguinal canal.

- **Female:** CSL persists; the gubernaculum becomes the ovarian and round ligaments; the ovary descends only into the pelvis.

Phases of testicular descent (classic two-phase model)

	Transabdominal phase	Inguinoscrotal phase
Timing	Weeks 10–15	From the 7th month; inguinal passage rarely before 22 weeks, most testes scrotal after 27 weeks
Events	**CSL regresses**; **gubernaculum swells** (outgrowth); its cord shortens, holding the testis at the internal ring while the kidney and abdomen grow away (the kidney moves cranially)	Gubernaculum migrates **3–5 cm** to the scrotum; **processus vaginalis** forms within it; bulb then regresses; processus closes
Control	CSL regression by **androgen**; swelling by **INSL3 (insulin-like peptide 3)** from Leydig cells, via its receptor RXFP2 (occurs even in androgen resistance)	**Androgen**, mainly (mostly) indirectly via the **genitofemoral nerve (GFN)** releasing **CGRP (calcitonin gene-related peptide)**; intra-abdominal pressure
Failure gives	Intra-abdominal testis	Canalicular or ectopic testis

- **Modern view:** the "transabdominal descent" is largely relative, not true migration. Barteczko and Jacob describe **five phases** (5 weeks to 28 weeks). The gubernaculum **swells to make room** (outgrowth), then **shrinks to let the testis pass (follow)** (regression). Testosterone (peak 14–16 weeks) and INSL3 (15–17 weeks) act together on gubernacular and cremaster growth (myogenesis).
- Spontaneous postnatal descent occurs mainly in the first 3–6 months.
- **Also know:** AMH (anti-Müllerian hormone) defects (persistent Müllerian duct syndrome) cause cryptorchidism; Insl3-knockout mice are cryptorchid; scrotum forms by 10–12 weeks under DHT (dihydrotestosterone).

Failed step	Result
CSL retained	Intra-abdominal testis
Abnormal or ectopic gubernacular attachment	Canalicular testis; ectopic (superficial inguinal pouch, perineal, femoral, prepenile)
Processus not closed	Hernia, hydrocele (patent in 87% of unilateral cases)
Fibrous remnant of processus / processus not fully involuted	Acquired (ascending) testis; acquired cryptorchidism

Abnormalities of descent

Type	Description
Undescended (cryptorchid)	Arrested on the normal path: abdominal, "peeping" at the ring, canalicular, prescrotal. 1–4% of term boys, up to 45% preterm; 75–80% palpable; 60–70% unilateral
Ectopic	Off the path: **superficial inguinal pouch** (commonest), perineal, femoral, prepubic, penile, contralateral scrotum
Retractile	Scrotal testis pulled up by the cremaster; stays down after sustained traction; some become acquired
Acquired (ascending)	Documented scrotal earlier; diagnosed at 8–11 years; fibrous processus remnant
Recurrent, secondary, iatrogenic	Re-ascent after spontaneous descent; after hernia repair or orchidopexy

Type	Description
Vanishing vs agenesis	Lost after formation (perinatal torsion; monorchia, anorchia) vs never formed

Associated anomalies and sequelae

- **Testicular: TTE (transverse testicular ectopia)**, polyorchidism, splenogonadal fusion; epididymal non-fusion.
- **PMDS (persistent Müllerian duct syndrome):** mutation of AMH or its receptor (AMHR2); bilateral cryptorchidism or TTE.
- **Genital and DSD (differences of sex development):** hypospadias (12–24%); bilateral non-palpable testes need karyotype and 17-OHP (17-hydroxyprogesterone) to exclude **CAH (congenital adrenal hyperplasia)**.
- **Syndromes:** prune belly (all), Klinefelter, androgen insensitivity, hypogonadotrophic hypogonadism, cerebral palsy, gastroschisis, omphalocele, myelomeningocele, PUV (posterior urethral valves).
- **Sequelae:** subfertility (worse if bilateral), **TGCT (testicular germ cell tumour) 2–5-fold**, torsion, hernia.
- **Treatment in brief:** orchidopexy from 6 months, by 18 months; laparoscopy for non-palpable; no imaging, no hormones.

Bilateral undescended testis at 3 years: evaluation

- This is **late presentation**; surgery is advised from 6 months, preferably by 12 months and by 18 months at the latest.
- **History:** prematurity, testes once scrotal (acquired), hernia, family history, salt-wasting illness.
- **Examination:** warm room; supine, then cross-legged or squatting. Sweep from the anterior superior iliac spine down the canal to find the lowest position; check the superficial inguinal pouch. **Retractile** testis stays in the scrotum after gentle traction. Note penis size, meatus (hypospadias), scrotal development, syndromic features.
- **Bilateral palpable:** no investigations needed; plan orchidopexy.
- **Bilateral non-palpable (or with hypospadias or micropenis): treat as DSD (difference of sex development) until proven otherwise.**
  - **Karyotype**; electrolytes and **17-OHP (17-hydroxyprogesterone)** to exclude **CAH (congenital adrenal hyperplasia)** in a virilised 46,XX child.
  - **Hormones:** LH (luteinising hormone), FSH (follicle-stimulating hormone), testosterone, **AMH and inhibin B**. Measurable AMH or inhibin B confirms testicular tissue; high gonadotrophins with absent AMH suggests **anorchia**. The **hCG (human chorionic gonadotrophin) stimulation test** (testosterone rise) is now less used.
  - Pelvic ultrasound for Müllerian structures.
- **Imaging to localise the testis is not recommended:** ultrasound sensitivity about 45%; it never removes the need for surgery.

Management

- **No hormonal therapy to induce descent:** LHRH (luteinising hormone-releasing hormone) works in about 20%; hCG may increase germ-cell apoptosis. Adjuvant LHRH (buserelin) in bilateral cases is experimental.
- **Palpable:** inguinal orchidopexy (free the cord, ligate the patent processus, retroperitoneal mobilisation, Prentiss manoeuvre if needed, **subdartos pouch**) or scrotal (Bianchi) orchidopexy for low testes. Both sides usually in one sitting.
- **Non-palpable:** examination under anaesthesia, then **diagnostic laparoscopy**:

```
Laparoscopy
 ├─ Testis seen → laparoscopic orchidopexy
 │     vessels short (testis not within 2–4 cm of ring) → Fowler–Stephens
 │     (clip testicular vessels; testis lives on vasal collaterals;
 │      one-stage, or two-stage with second stage at 6 months) or Shehata traction
 ├─ Vessels and vas enter ring → inguinal exploration, excise nubbin
 └─ Blind-ending vessels → stop (vanished testis)
```

- **Both testes very high or of doubtful viability:** stage the sides, the second at 6 months, to avoid bilateral atrophy. Primary orchidopexy (>90% success) beats Fowler–Stephens (60–97%).
- **Biopsy** only if DSD or dysgenesis is suspected. **Bilateral anorchia:** testosterone at puberty, prostheses later.

Follow-up and prognosis

- Check position and size at 6 months.
- **Fertility:** worse when bilateral (oligospermia 42%, azoospermia 31% in one large review).
- **Cancer:** about **3-fold** risk (another source: TGCT 2–5-fold); teach self-examination after puberty.

Diagram to draw: sagittal fetus showing CSL, gubernaculum, processus vaginalis and both phases.

Key points to remember:

- SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
- Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
- Inguinoscrotal: androgen via genitofemoral nerve (GFN) and CGRP.
- Most testes scrotal after 27 weeks.
- Ectopic = off the path; commonest site superficial inguinal pouch.
- Bilateral non-palpable = DSD work-up first (karyotype, 17-OHP, AMH).
- Laparoscopy decides; no imaging, no hormones.

IVC anomalous development and retrocaval ureter

Asked as:

- Describe embryology & clinical implications of IVC anomalous development. — *Practice Set 2 Paper I Q2 (LAQ)*
- Retrocaval ureter — *Practice Set 11 Paper I Q3a (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(retrocaval ureter also answered in 2017-Winter Paper 3, Q3d)

Introduction

- The **IVC (inferior vena cava)** forms between the 6th and 8th weeks from three pairs of fetal veins by anastomosis, regression and persistence. The final IVC is **right-sided**, so long left veins must cross the aorta to reach it.
- Most anomalies are silent CT (computed tomography) findings. They **bleed at surgery, mimic lymph nodes, trap the ureter and alter IVC filter placement**.

Embryology

Three paired venous systems

- Fetal veins form a collar around each ureter: posterior cardinal and supracardinal (dorsal), subcardinal (ventral).

Vein	Position	Adult derivatives
Posterior cardinal (first)	Dorsolateral	Iliac confluence; mostly regresses
Subcardinal	Ventromedial	Right: **suprarenal IVC**; intersubcardinal anastomosis: **left renal vein**; left: adrenal and gonadal veins
Supracardinal (last)	Dorsomedial	Right: **infrarenal IVC**; thoracic parts: **azygos and hemiazygos**

Segments of the definitive IVC (top to bottom)

1. **Hepatic**: right vitelline vein.
2. **Suprarenal**: right subcardinal vein (via the subcardinal–hepatic anastomosis).
3. **Renal**: subcardinal–supracardinal anastomosis.
4. **Infrarenal**: right supracardinal vein.
5. **Iliac confluence**: posterior cardinal veins.

Renal collar

- A venous ring encircles the aorta at renal level. The **ventral (pre-aortic) limb** persists as the left renal vein; the **dorsal (retro-aortic) limb** regresses.
- The right supracardinal vein lies **dorsal** to the ureter, so the normal ureter lies in front of the IVC. Normally the right **supracardinal** vein, **dorsal** to the ureter, forms the infrarenal IVC.

```
Left-sided channels regress -> right-sided persist
  -> single right IVC + pre-aortic left renal vein
```

Diagram to draw: three stages: paired cardinal veins, anastomoses with renal collar, definitive IVC.

Anomalies

Anomaly	Embryological error	Key features
Retrocaval ureter ("preureteral vena cava")	Infrarenal IVC forms from a vein **ventral** to the ureter	Right only; 1 in 1500 autopsies; male 3:1; presents 3rd–4th decade. **Type I** low loop, fishhook, obstructs; **type II** high loop, sickle, mild
Circumaortic left renal vein	Dorsal limb persists with ventral	Commonest (2–16%); posterior limb runs lower into lumbar IVC
Retroaortic left renal vein	Ventral limb regresses, dorsal persists	Less common; may be compressed ("posterior nutcracker")
Left IVC	Left supracardinal persists, right regresses	Crosses to the right at renal level, in front of the aorta
Duplicated IVC	Both supracardinals persist	Left IVC joins left renal vein; double right cava can trap the ureter
Azygos continuation	Subcardinal–hepatic anastomosis fails	Large azygos; hepatic veins drain directly to right atrium; polysplenia, heterotaxy
Absent infrarenal IVC	Supracardinal and posterior cardinal failure	Deep vein thrombosis in the young

- For retrocaval ureter Campbell names a persistent **lumbar subcardinal** (Chapter 52) or **posterior cardinal** vein (Chapter 93, most texts); both mean a ventral vein forms the IVC.

Retrocaval ureter

Definition

- The **right ureter** passes **behind the IVC (inferior vena cava)**, then crosses in front of it from medial to lateral before resuming a normal course.
- Better named **"preureteral vena cava"**: the anomaly is venous; the ureter is trapped.

Embryology

- If the **lumbar subcardinal vein** (ventral) persists as the IVC, the ureter is trapped behind it.
- Variant: double right IVC, with the ureter between the limbs.

Incidence and presentation

- About **1 in 1500** autopsies; **male 3:1**; presents in the **third–fourth decade** with right flank pain, UTI (urinary tract infection), stones or incidental hydronephrosis.

Types (Bateson and Atkinson)

	Type I (low loop), commoner	Type II (high loop)
Crossing	Lower, L3–L4	Higher, at renal pelvis level
Shape	**J or "fishhook"**	Smooth horizontal curve
Obstruction	Yes, at iliopsoas edge	Mild or none

Investigations

- **CT urography (computed tomography urography): investigation of choice**; MRI (magnetic resonance imaging) alternative.
- IVU (intravenous urography): ureter not seen beyond the J hook. Retrograde: **S-curve** at L3–L4. **Diuretic renogram**: degree of obstruction, split function.
- Pitfall: mistaken for **PUJ (pelviureteric junction) obstruction**, so pyeloplasty fails.

Management

- **Non-obstructed (usually type II):** observe.
- **Obstructed:** divide the ureter, transpose it **anterior to the IVC**, and do a spatulated **ureteroureterostomy** (or ureteropyelostomy) over a **double-J stent**. Excise or bypass the **aperistaltic retrocaval segment**; respect the ureteric blood supply.
- **Laparoscopic or robotic** (transperitoneal or retroperitoneal) now preferred. Nephrectomy if non-functioning.
- **Also know:** retroiliac ureter (L5–S1); bilateral cases in situs inversus.

Diagram to draw: right ureter hooking behind the IVC with a fishhook pelvis.

Clinical implications

1. Diagnosis

- Venous-phase CT or MRI (magnetic resonance imaging). A left or duplicated IVC mimics **para-aortic lymphadenopathy**; a large azygos mimics a mediastinal mass, risking needless biopsy.
- Retrocaval ureter: S-shaped or fishhook ureter at L3–L4 on retrograde pyelography; CT urography confirms; diuretic renography grades obstruction.

2. RPLND (retroperitoneal lymph node dissection)

- Preoperative CT must look for circumaortic or retroaortic veins and retrocaval ureter. A left or duplicated IVC lies in the left template; a retroaortic limb tears when lumbar vessels are divided.

3. Radical nephrectomy

- The hidden posterior limb of a circumaortic vein bleeds when the artery is approached from behind: ligate both limbs. Caval thrombus in a duplicated IVC needs rerouting of the left IVC.

4. Living donor nephrectomy

- The left kidney is preferred for its long vein. A circumaortic or retroaortic vein shortens it and adds bleeding risk; CT angiography is mandatory, and the right kidney may be chosen.

5. Retrocaval ureter repair

- Observe non-obstructed type II. For obstruction, **divide the ureter, transpose it in front of the IVC and re-anastomose** over a stent, excising the aperistaltic segment; laparoscopic or robotic. A pyeloplasty that misses it fails.

6. Interventional and vascular

- **IVC filter:** with duplication, emboli bypass a single filter, so filter both cavae or place it suprarenally.
- Aortic surgery risks retroaortic vein injury. A retroaortic vein can cause haematuria or left varicocele.
- **Also know:** variable left adrenal vein entry (adrenal vein sampling); right gonadal vein into right renal vein in up to 10%.

Key points to remember:

- Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
- Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
- Retrocaval ureter: venous anomaly (persistent subcardinal vein); right side only; 1 in 1500; male 3:1; transpose anterior to IVC.
- Type I fishhook obstructs; type II does not.
- CT urography; anterior transposition and ureteroureterostomy.
- Duplicated IVC: filter both cavae; do not mistake it for nodes.

Section 4 · Paediatric Urology 1 — Short answer questions (SAQ)

Glandular hypospadias

Type: SAQ (10 marks)

Source: Practice Set 11 Paper 1, Q3b

"Glandular" is read as glanular hypospadias: meatus on the glans.

Definition

- The mildest **distal (anterior)** hypospadias: the meatus opens on the ventral glans, proximal to the normal tip.
- Hypospadias occurs in 1 in 150–300 male births; distal forms are 70–85%.
- Usual features: **dorsal hooded foreskin**, ventral glans tilt, little or no curvature. **MIP (megameatus intact prepuce)** variant (up to 5%): wide meatus with normal foreskin.

Embryology

- Urethral folds close from proximal to distal during weeks 8–16 under **DHT (dihydrotestosterone)**.
- Glanular urethra: ectodermal ingrowth (classic) or endodermal plate ("two-zipper", current view). Arrest at the last step gives a glanular meatus.

Evaluation

- Examine meatus, plate, glans size, curvature, torsion; retract the foreskin before any **newborn circumcision**.
- **GMS (glans–meatus–shaft) score** (each 1–4, total 3–12) predicts complications.
- No workup needed for isolated distal hypospadias; DSD (difference of sexual development) workup if a testis is undescended.

Indications and timing

- Deflected or spraying stream, meatal stenosis, curvature, cosmesis. A straight penis that voids standing may need no surgery; counsel parents.
- **Age 6–18 months** (AAP, American Academy of Pediatrics, 6–12 months).

Operations

Procedure	Suitable for	Key point
Urethromeatoplasty	Stenotic meatus, blind pit	Heineke–Mikulicz: vertical incision, horizontal closure
MAGPI (meatal advancement and glanuloplasty, Duckett 1981)	Glanular meatus, mobile skin	Advance meatus, two-layer glanuloplasty; retracts if used more proximally
MIV (M inverted V) glansplasty	Glanular or coronal	Avoids MAGPI retraction
GAP (glans approximation)	Wide deep groove, fish-mouth meatus (MIP)	Tubularise and close glans wings
TIP (tubularised incised plate, Snodgrass 1994)	Narrow plate	Most used; slit-like meatus
Mathieu flip-flap; Thiersch–Duplay	Selected	Stenosis, fistula

- Preputioplasty or circumcision; dartos barrier layer.

Complications

- Meatal stenosis or retraction, fistula, **glans dehiscence**, residual curvature, poor cosmesis. Distal repairs succeed in about 80%.

Diagram to draw: MAGPI steps (meatal advancement, glans wings closed beneath).

> **Key points to remember:**
> - Glanular = most distal hypospadias; dorsal hood, minimal curvature.
> - MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
> - Repair at 6–18 months; may be left alone if function is normal.

TIP repair in hypospadias

Type: SAQ (10 marks)

Source: Practice Set 12 Paper 1, Q3b (also answered in 2023-Winter Paper 3, Q3c; extended indications in Practice Set 5 Paper 3, Q3c)

Introduction

- **TIP (tubularised incised plate) urethroplasty**, Snodgrass (1994), built on Rich's hinging incision (1989).
- Used for **>90% of distal** and **about 80% of midshaft** repairs.

Principle

- A **deep midline relaxing incision** widens a narrow urethral plate so it can be tubularised **without tension**.
- The raw incised area **heals by re-epithelialisation** without a graft, giving a **vertical slit-like meatus**.

Indications and prerequisites

- Distal and midshaft hypospadias; redo cases with a healthy plate; curvature **<30°** corrected by degloving or dorsal plication.
- **Glans width <14 mm** raises complications: consider preoperative testosterone.
- **POST (plate objective scoring tool):** a poor plate favours a dorsal inlay graft.
- Not for curvature needing plate division; proximal TIP largely abandoned.

Operative steps

1. Glans traction suture; tourniquet.
2. U-incision around plate and meatus; deglove; artificial erection.
3. Mobilise glans wings deeply off the plate.
4. **Midline plate incision** from meatus to plate end, down to corpora.
5. Tubularise over a **6–8 Fr stent** with subepithelial 6-0/7-0 absorbable sutures, distal to proximal, leaving an oval meatus.
6. **Dartos flap** (dorsal preputial) over the neourethra as a barrier layer.
7. Two-layer glansplasty; mucosal collar; Byars flaps or preputioplasty.
8. Stent 5–7 days; dressing; oxybutynin for bladder spasm.

Complications and outcomes

- Pfistermuller meta-analysis (2015, 4,675 boys): complications 10.6%, reoperation 4.5%; **fistula 5.7%**, **meatal stenosis 3.6%**, stricture 1.3%.
- Glans dehiscence; meatal stenosis commoner than after Mathieu.
- Proximal and redo cases do worse; a dartos layer reduces fistula.

Modifications

- **Dorsal inlay graft (Snodgraft)**: inner preputial graft in the incised bed for a shallow plate.

Diagram to draw: cross-sections of plate incision, tubularisation and dartos cover.

Key points to remember:

- Snodgrass 1994: incise, tubularise, heals by epithelialisation.
- Glans <14 mm and poor plate raise complications.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).

Vanishing testes

Type: SAQ (10 marks)

Source: Practice Set 13 Paper 1, Q3b

Definition

- A testis that **developed normally and was then lost**, usually from **torsion or a vascular accident in fetal life**, after genital masculinisation but before scrotal fixation.
- Unilateral loss gives **monorchia**; bilateral loss gives **anorchia** (testicular regression syndrome), which is rare.
- Excised "nubbins" contain haemosiderin.

Incidence

- **15–40% of nonpalpable testes** at surgery.

Clinical features

- **Contralateral compensatory hypertrophy:** volume >2 mL (sensitivity 72%, specificity 100%) or length about 19–20 mm suggests a solitary testis.
- No palpable intrascrotal "appendage" tissue; sometimes a scrotal nubbin.
- These aid counselling but **do not replace exploration**.
- **Anorchia:** micropenis in nearly half.

Evaluation

- **Imaging** (ultrasound, MRI [magnetic resonance imaging]) is not recommended: it does not change the operation.
- **Bilateral nonpalpable:** exclude DSD (difference of sex development) and CAH (congenital adrenal hyperplasia) first.
  - Anorchia: high FSH (follicle-stimulating hormone) and LH (luteinising hormone); low AMH (anti-Müllerian hormone) and inhibin B; no rise after hCG (human chorionic gonadotrophin). Most reliable during mini-puberty.

Management: diagnostic laparoscopy (after examination under anaesthesia)

Laparoscopic finding	Action
Blind-ending vessels and vas above a closed ring	Diagnosis confirmed; stop
Vessels and vas entering the internal ring	Inguinal or scrotal exploration; excise the nubbin
Testis seen	Orchidopexy (single stage or Fowler–Stephens)
Neither vas nor vessels	Search up to the kidney (agenesis very rare)

- **Excise the remnant:** viable germ cells or tubules in 5–15%.
- **Scrotal-first exploration** is reasonable with a palpable nubbin and contralateral hypertrophy (length ≥1.8 cm); laparoscopy is then needed in about a quarter.
- **Contralateral fixation:** debated; not required by the AUA (American Urological Association) algorithm.
- **Prosthesis:** offer at least 6 months after scrotal surgery or after puberty, inguinal approach.
- **Anorchia:** testosterone from puberty. **Also know:** agenesis shows ipsilateral Müllerian persistence.

Diagram to draw: laparoscopic view of blind-ending vessels at a closed ring.

Key points to remember:

- Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
- Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
- Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
- Anorchia: high FSH/LH, low AMH and inhibin B.

Pelvicalyceal anomalies other than PUJ obstruction

Type: SAQ (10 marks)

Source: Practice Set 14 Paper 1, Q3d

Introduction

- Congenital pelvicalyceal variants occur in about 4% of people.
- **Key question:** is there obstruction? Dilatation without obstruction needs no treatment; stasis causes stones and infection.

Anomalies

Anomaly	Features	Management
Calyceal diverticulum	Transitional-lined intrarenal cavity joined to a calyx (Type I, commoner, small) or pelvis (Type II, larger, symptomatic) by a narrow neck; 4.5 per 1000; congenital (persistent 3rd–4th generation ureteric branches) or acquired. **Stones in almost 40%**; milk of calcium	Asymptomatic: observe. Pain, infection, stones: ureteroscopic neck incision, **percutaneous ablation with fulguration**, laparoscopic marsupialisation; SWL (shock wave lithotripsy) ineffective
Hydrocalycosis	Dilated major calyx connected to the pelvis; infundibular obstruction by vessel, stenosis, scarring or achalasia	Symptomatic only: infundibulotomy or dilatation
Megacalycosis	**Non-obstructive**; malformed papillae; **12–20 calyces**; pelvis normal; male 6:1; normal diuretic renogram	**None**; may coexist with segmental megaureter
Infundibular / infundibulopelvic stenosis	Narrow infundibula ± small pelvis; usually bilateral, with VUR (vesicoureteric reflux), dysplasia; over a third reach renal failure (hyperfiltration)	Monitor creatinine and eGFR (estimated glomerular filtration rate); ACE (angiotensin-converting enzyme) inhibitors; surgery only if obstruction harms function: endoscopic, calicocalicostomy, ureterocalicostomy
Extrarenal pelvis	Large pelvis outside the sinus; mimics dilatation	None
Bifid pelvis	About 10%; normal variant	None
Unipapillary kidney	Single papilla and calyx; rare; usually hypoplastic kidney	None

- **Also know:** acquired infundibular stenosis (tuberculosis, post-PCNL (percutaneous nephrolithotomy)); Fraley syndrome (crossing artery on upper infundibulum).

Evaluation

- **Ultrasound:** cyst-like space; milk of calcium layers and shifts with position.
- **CT (computed tomography) urography or MRU (magnetic resonance urography): delayed contrast pooling** in a diverticulum.
- **Diuretic renogram (MAG3 (mercaptoacetyltriglycine)):** separates obstruction (hydrocalycosis) from non-obstruction (megacalycosis, extrarenal pelvis).
- **Retrograde pyelography:** maps the neck.

Differential diagnosis

- Obstructive hydronephrosis, papillary necrosis, pyelonephritic clubbing, tuberculosis, renal cyst.

- **Stones:** retrograde first (find, dilate or laser-incise the ostium); percutaneous for lower-pole diverticula.

Diagram to draw: kidney showing diverticulum with narrow neck, hydrocalyx and megacalyces.

> **Key points to remember:**
> - Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
> - Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
> - Infundibulopelvic stenosis: bilateral, renal failure risk.

Methods of orchidopexy

Type: SAQ (10 marks)

Source: Practice Set 21 SAQ Sheet, Q3f (treatment algorithm of undescended testis answered in 2018-Summer Paper 3, Q3a)

Principles

- Operate **after 6 months, ideally before 18 months** (Kollin RCT [randomised controlled trial]: surgery at 9 months gave better testicular growth than at 3 years). No hormones.
- Goals: tension-free **dependent scrotal** testis with intact blood supply; ligate the patent processus.

Palpable testis

- **Inguinal orchidopexy:** groin incision → open external oblique (spare ilioinguinal nerve) → free cord, divide gubernaculum → separate and **ligate sac at internal ring** → retroperitoneal mobilisation, divide lateral bands → **Prentiss manoeuvre** (medial transposition under inferior epigastric vessels) → **subdartos pouch**. Transalbugineal sutures unnecessary.
- **Trans-scrotal (Bianchi) orchidopexy:** high scrotal incision; sac ligated through it; inguinal conversion 3.5–4.4%; atrophy 0.3–0.6%; less pain.

Non-palpable testis: examination under anaesthesia, then laparoscopy

Method	Principle	Success
Primary laparoscopic	Mobilise vessels, preserve vasal collaterals, new hiatus medial to epigastric vessels	>90% (97%)
Open transabdominal (Jones and Bagley)	Preperitoneal high mobilisation	95%
Fowler–Stephens (one- or two-stage)	Divide testicular vessels high; testis survives on **vasal and gubernacular collaterals**; second stage after 6 months	60–97% (one-stage 63%, two-stage 68%)
Shehata traction	Fix testis near opposite anterior superior iliac spine; vessels lengthen; orchidopexy at 12 weeks	84–85%; atrophy 10%
Microvascular autotransplantation	Testicular vessels to inferior epigastric vessels	96%
Orchidectomy	Postpubertal, dysgenetic or atrophic testis	—

- Fowler–Stephens for testes **not within 2–4 cm of the internal ring**.
- **Also know:** staged (two-stage) inguinal orchidopexy; Dessanti PTFE (polytetrafluoroethylene) sleeve traction; gubernaculum-sparing technique; redo approaches (Redman lateral, Cartwright fascial patch).

Complications

- **Atrophy** (highest after Fowler–Stephens), retraction or reascent, vas or vessel injury, haematoma, infection; torsion very rare after subdartos fixation.

Follow-up

- Size and position at 6 months; counsel on fertility and cancer risk; self-examination after puberty. Offer prosthesis for an atrophic testis.

Diagram to draw: Fowler–Stephens collateral supply: testicular artery clipped, deferential and cremasteric arteries intact.

Key points to remember:

- Orchidopexy at 6–18 months; subdartos pouch.
- Prentiss manoeuvre gives extra length.
- Fowler–Stephens relies on vasal collaterals.

Hypospadias repairs – single-stage versus staged techniques

Type: SAQ (10 marks)

Source: Practice Set 22 SAQ Sheet, Q3e (present concepts also in Practice Set 6 Paper 4, Q1)

Introduction

- Choice depends on **curvature after degloving**, plate quality and glans size, not on meatal position alone.
- Algorithm: deglove, artificial erection. Straight or <30°: dorsal plication, single-stage. >30°: divide plate, ventral corporotomy ± graft, then staged.

Single-stage techniques

Group	Examples	Indication
Advancement	**MAGPI** (meatal advancement and glanuloplasty, Duckett 1981); MIV (M inverted V)	Glanular meatus, no curvature
Tubularisation	**TIP** (tubularised incised plate, Snodgrass 1994); Thiersch–Duplay; GAP (glans approximation procedure)	Distal–midshaft; TIP in >90% of distal
Flap	Mathieu (flip-flap); **OIF** (onlay island flap)	Thin plate
Proximal one-stage	TPIF (transverse preputial island flap, Duckett tube); proximal OIF	Curvature corrected; good tissue
Graft	Dorsal buccal inlay	Narrow plate

- Pros: one anaesthetic. Proximal TIP is abandoned: devascularised plate, stricture, recurrent curvature.

Staged techniques

- Now the approach of choice for most proximal hypospadias. **Rationale:** a urethroplasty over a fresh corporal graft fails; stage 1 builds a supple, vascular bed.

Type	Stage 1	Stage 2 (≥6 months)
Bracka / STAG (staged tubularised autograft)	Correct curvature, divide plate, quilt inner preputial or buccal graft	Tubularise, glansplasty, tunica vaginalis cover
Byars flaps	Split dorsal prepuce, transpose ventrally	Tubularise
Distal-first (Ulaanbaatar)	Curvature and distal urethra	Join proximal urethra

- Repeat artificial erection before stage 2.

Comparison of outcomes

Measure	Single-stage	Staged
Distal success	83–95%	–
Proximal complications	About 60% (long TIP, mixed)	STAG about 23%; Byars 43–81%
Recurrent curvature	Higher after plication	0–10%
Penile length, perception	Lower	Better
Extra procedures	Fewer planned	25–75% need more

- Fistula is the commonest complication. Long, rigid neourethra without spongiosum raises resistance (Poiseuille's law), causing fistula and diverticulum.

Diagram to draw: STAG stage 1 graft and stage 2 tubularisation.

Key points to remember:

- Curvature >30° after degloving: stage.
- TIP for distal; STAG has lowest proximal complications.
- Stage 2 after ≥6 months.

Rapid Revision Sheet – Section 4 · Paediatric Urology 1

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Primary (congenital) VUR: VUJ embryology, pathophysiology, natural history and management (LAQ)

- Low (early) bud → lateral orifice → VUR plus dysplasia; high (late) bud → ectopia.
- One embryological fault: lateral orifice, short tunnel, dysplasia.
- Tunnel length to diameter 5:1 (Paquin).
- Dysplasia (boys, congenital) versus scar (girls, infection).
- Grade IV–V and bilateral grade III rarely resolve.
- Scarred children: 34–38% adult hypertension.
- RIVUR: CAP halves recurrent UTI, not scarring.
- Open reimplantation 95.9%; Cohen over 98%.

Urological evaluation and management of pediatric kidney transplant patient. (LAQ)

- Bladder target: age-expected capacity at <40 cm H_2O; augment if <75% despite CIC.
- PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.
- Never test a bladder with a graft; cycle the defunctionalised bladder.
- Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis (LAQ)

- SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
- Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
- Inguinoscrotal: androgen via genitofemoral nerve (GFN) and CGRP.
- Most testes scrotal after 27 weeks.
- Ectopic = off the path; commonest site superficial inguinal pouch.
- Bilateral non-palpable = DSD work-up first (karyotype, 17-OHP, AMH).
- Laparoscopy decides; no imaging, no hormones.

IVC anomalous development and retrocaval ureter (LAQ)

- Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
- Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
- Retrocaval ureter: venous anomaly (persistent subcardinal vein); right side only; 1 in 1500; male 3:1; transpose anterior to IVC.
- Type I fishhook obstructs; type II does not.
- CT urography; anterior transposition and ureteroureterostomy.
- Duplicated IVC: filter both cavae; do not mistake it for nodes.

Glandular hypospadias (SAQ)

- Glanular = most distal hypospadias; dorsal hood, minimal curvature.
- MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
- Repair at 6–18 months; may be left alone if function is normal.

TIP repair in hypospadias (SAQ)

- Snodgrass 1994: incise, tubularise, heals by epithelialisation.
- Glans <14 mm and poor plate raise complications.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).

Vanishing testes (SAQ)

- Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
- Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
- Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
- Anorchia: high FSH/LH, low AMH and inhibin B.

Pelvicalyceal anomalies other than PUJ obstruction (SAQ)

- Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
- Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
- Infundibulopelvic stenosis: bilateral, renal failure risk.

Methods of orchidopexy (SAQ)

- Orchidopexy at 6–18 months; subdartos pouch.
- Prentiss manoeuvre gives extra length.
- Fowler–Stephens relies on vasal collaterals.

Hypospadias repairs – single-stage versus staged techniques (SAQ)

- Curvature >30° after degloving: stage.
- TIP for distal; STAG has lowest proximal complications.
- Stage 2 after ≥6 months.

Section 5 · Paediatric Urology 2 — Long answer questions (LAQ)

Describe embryology of bladder. What are the proposed theories for extrophy epispadias complex? Describe skeletal abnormalities.

Type: LAQ (25 marks)

Source: Practice Set 6 Paper 1, Q2 (bladder embryology also answered in 2021-Summer Paper 3, Q1)

A. Embryology of the bladder

1. Division of the cloaca (weeks 5–6)

- The **urorectal septum** (Tourneux and Rathke folds) grows caudally and divides the endodermal cloaca into a dorsal **anorectal canal** and a ventral **UGS (urogenital sinus)**. It meets the cloacal membrane, which perforates to form the urogenital and anal openings.
- The **entry of the Wolffian (mesonephric) ducts** divides the UGS:
  - **Cranial vesicourethral canal** (continuous with the allantois) → bladder and pelvic urethra.
  - **Caudal (definitive) UGS** → phallic urethra in males; vestibule and distal vagina in females.

2. Wall layers

- Urothelium from UGS endoderm. Lamina propria, detrusor and adventitia from surrounding splanchnic mesenchyme, by epithelial–mesenchymal signalling.

3. Trigone (weeks 5–6)

- The common excretory ducts dilate and are absorbed into the UGS. **Apoptosis** separates the ureters from the Wolffian ducts.
- Ureteric orifices migrate **cranially and laterally**; Wolffian ducts end as ejaculatory ducts at the verumontanum.
- Classic view: trigone muscle is ureteric. Current view (mouse lineage tracing): **mostly bladder smooth muscle**.
- **Weigert–Meyer:** bud too low → lateral orifice → VUR (vesicoureteral reflux); too high → caudal, medial, ectopic orifice.

4. Urachus: the bladder descends into the pelvis (months 4–5); the allantois obliterates into the **urachus** (median umbilical ligament).

5. Function: urine from about week 10; after week 16 amniotic fluid is mainly fetal urine.

B. Exstrophy–epispadias complex (EEC): theories

- **Spectrum:** glanular epispadias → classic bladder exstrophy (CBE, >50%; about 1 in 46,000; male:female 2.3:1) → cloacal exstrophy.
- **Core defect:** the **cloacal membrane is not reinforced by mesodermal ingrowth**. This mesoderm normally forms the lower abdominal muscles and pelvic bones, so the thin membrane ruptures. **Timing and extent of rupture** decide the variant.

Theory	Proposed defect	Result
Cloacal membrane overdevelopment (Marshall–Muecke)	Abnormally large membrane in week 4 blocks medial mesenchymal migration	Rupture before septum descent → cloacal exstrophy; after → bladder exstrophy; limited → epispadias

Theory	Proposed defect	Result
Lateral body-wall fold failure	Folds fail to meet in midline	Abdomen and pelvis → cloacal; pelvis alone → classic
Caudal genital hillocks	Genital tubercles fuse **below** rather than above the membrane	Dorsal urethral defect (epispadias)
Abnormal body-stalk insertion	Caudal insertion prevents mesenchyme interposition	Superficial unstable membrane disintegrates
Bony pelvis first	Pelvic ring primordium fails to rotate	Pelvic structures cannot meet → bladder herniates

- All theories converge on **absent infraumbilical midline mesoderm**.
- **Also know (aetiology):** multifactorial; recurrence in offspring of an affected parent 1 in 70; 22q11.2 duplication (commonest variant), ISL1 susceptibility gene, p63 under-expression; IVF (in vitro fertilisation) and high-dose early progesterone raise risk; maternal smoking linked to cloacal exstrophy.

C. Skeletal abnormalities

1. Classic bladder exstrophy (3D [three-dimensional] CT [computed tomography] studies)

- **Rotational:** posterior pelvis externally rotated (mean 12° each side); anterior segment externally rotated (mean 18° each side); sacroiliac joint rotated; acetabular and femoral retroversion.
- **Dimensional: pubic diastasis** (mean 4.8 cm); **anterior pubic segment shortened by 30%**; intertriradiate distance increased by about 31%; posterior segment length normal; larger sacrum.
- Bone histology is normal, so growth potential is normal.
- **Clinical effects:** short pendular penis (anterior segment short and splayed); wide hips, waddling gait and out-toeing in early childhood. These cause little disability and improve with growth.
- **Pelvic floor:** levator ani displaced posteriorly, flattened; anterior anus; later pelvic organ prolapse.
- **Spine:** minor variants (spina bifida occulta, lumbarisation, sacralisation) fairly common; scoliosis and dysraphism uncommon.

2. Cloacal exstrophy

- Widest, **asymmetric** diastasis (about twice CBE), sacroiliac malformation, hip dislocation.
- **Spinal anomalies in 85–100%** (lumbar myelodysplasia commonest; lipomeningocele); MRI (magnetic resonance imaging) for all.
- Limb defects: **clubfoot** (commonest), limb shortening, tibial and fibular deformity.

3. Isolated epispadias: male symphysis widened (less than CBE); female symphysis usually closed or narrow.

Surgical significance: pelvic osteotomy

- Closing the pelvic ring places the bladder neck deep in the pelvis, reduces tension and helps continence.
- **Indications:** diastasis >4 cm or poorly malleable pelvis; closure after 72 hours; failed closure; all cloacal exstrophy. Under 72 hours with malleable pubes: closure without osteotomy and traction.
- **Combined bilateral anterior innominate + vertical iliac osteotomy** (most used) with external fixator.
- Epispadias repair and continence surgery: see Practice Set 6 Paper 3, Q2.

Diagram to draw: cloacal membrane with urorectal septum and failed mesodermal ingrowth; exstrophy pelvis with diastasis.

Key points to remember:

- Vesicourethral canal → bladder; urachus from allantois.
- Marshall–Muecke: overdeveloped cloacal membrane; rupture timing decides variant.
- CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis ~4.8 cm.
- Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.

Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management.

Type: LAQ (25 marks)

Source: Practice Set 7 Paper 1, Q1 (also answered as an SAQ in 2015-Summer Paper 3, Q3f)

Definition

- **PBS (prune belly syndrome)**, also called Eagle–Barrett or triad syndrome: **deficient abdominal wall muscles**, **bilateral intra-abdominal testes** and a **dilated, poorly contracting urinary tract**.
- 1 in 29,000–44,000 live births; **95% male**.
- **Survival depends on renal dysplasia and lung hypoplasia, not on the abdominal wall.**

Antenatal diagnosis

- **Ultrasound** may suggest PBS from 11–14 weeks, but the classic signs are not consistent even at 30 weeks.
- **Findings:** huge bladder filling the abdomen; bilateral hydroureteronephrosis; urachal diverticulum; dilated posterior urethra; echogenic kidneys with cortical cysts (dysplasia); irregular abdominal circumference; fetal ascites; oligohydramnios when severe.
- **Differential diagnosis:** PUV (posterior urethral valves), urethral atresia, megacystis–megaureter.
- Fetal dilatation does **not** predict postnatal renal function.
- **Prenatal management:** intervention has no proven renal benefit. Vesicoamniotic shunting is justified only for **urethral atresia with progressive oligohydramnios**, or to decompress the abdomen and avoid dystocia. Termination is hard to justify except with early, severe oligohydramnios. Plan delivery in a centre with neonatal intensive care.

Pathophysiology

Theory (none proven)	Mechanism
Early urethral obstruction	Back-pressure dilates the tract and stretches the wall; but most have no obstruction after birth
Lateral plate mesoderm defect	This mesoderm forms ureters, bladder, prostate, urethra and gubernaculum, so one defect explains the triad
Intrinsic urinary tract defect	Primary ureteric dilatation and fetal ascites
Yolk sac defect	Primary defect of the yolk sac

- **Genetics:** mostly sporadic, normal karyotype; smooth-muscle genes **MYOCD (myocardin)** and **FLNA (filamin A)** in multiplex families.
- **Kidneys:** dysplasia in about 50% (Potter type II from mesenchymal defect; type IV cystic from obstruction).
- **Ureters:** long and tortuous, worse distally. Collagen replaces smooth muscle, so walls do not coapt and peristalsis fails: **a failed pump, not a blocked pipe**. The result is stasis and infection. **VUR (vesicoureteric reflux) in about 75%.** The proximal ureter has better muscle.
- **Bladder:** large, smooth, collagen-rich; urachal pseudodiverticulum; splayed trigone. Normal compliance but poor contractility, so voiding is "unbalanced", with residual urine.
- **Prostate and urethra:** prostatic hypoplasia dilates the posterior urethra; distal obstruction in about 20%; **megalourethra** (scaphoid: spongiosum deficient; fusiform: all

erectile bodies, more severe).
- **Testes:** intra-abdominal; azoospermia; paternity only by sperm retrieval and ICSI (intracytoplasmic sperm injection).
- **Extra-genitourinary (about 75%):** pulmonary hypoplasia; cardiac about 10%; gastrointestinal at least 30% (malrotation); orthopaedic up to 65% (talipes, hip dysplasia).

Classification (Woodard)

Category	Features	Management
I	Oligohydramnios, pulmonary hypoplasia, dysplasia, urethral atresia	Supportive; bladder drainage only
II	Full triad, moderate or unilateral dysplasia	Individualised (controversial group)
III	Mild or incomplete; stable function (majority)	Surveillance

- **Also know:** RUBACE score (Renal, Ureter, Bladder, Abdominal wall, Cryptorchidism, Extra-genitourinary) grades morbidity in survivors; pseudo-prune presents late with renal failure.

Postnatal management

Neonatal (neonatology, nephrology, urology, cardiology):

1. Heart and lungs first: **chest radiograph** for pneumothorax and hypoplasia.
2. Creatinine, BUN (blood urea nitrogen), electrolytes; follow the **trend**, because the first value reflects the mother.
3. Renal and bladder ultrasound once stable.
4. Antibiotic prophylaxis; **circumcision**; strict asepsis.
5. **VCUG (voiding cystourethrogram)** only with renal insufficiency or suspected outlet obstruction.
6. **DMSA (dimercaptosuccinic acid) at 4–6 weeks** for split function; MAG3 (mercaptoacetyltriglycine) for drainage.
7. Outlet obstruction: catheter or suprapubic tube.

Urinary tract:

- **Conservative** for category III: surveillance, treat infection.
- **Surgery for:** recurrent upper tract infection, progressive dilatation or renal decline, high-grade VUR, true obstruction.
- **Temporary diversion:** wide **cutaneous vesicostomy (Blocksom)**. Avoid high ureterostomy, which wastes the good proximal ureter.
- **Reconstruction after 3–6 months** (lung maturity): excise distal ureter, taper and reimplant the healthier proximal ureter; reduction cystoplasty (capacity recurs). **CIC (clean intermittent catheterisation)** or Mitrofanoff for poor emptying.

Orchidopexy: bilateral transabdominal at about 6 months, often with other surgery; later, **staged Fowler–Stephens** (better than single-stage) or laparoscopic.

Abdominoplasty (cosmesis; better cough, voiding, defecation):

Technique	Key feature
Randolph	Transverse; excises worst lower wall
Ehrlich	Vertical; umbilicus kept; vest-over-pants overlap
Monfort	Vertical; umbilicus kept; central fascial plate with lateral flaps overlapped; transperitoneal access for combined surgery

- **Also know:** Furness (extraperitoneal plication); Dénes; corset.

Prognosis and follow-up

- About 20% die perinatally; **up to 30% develop chronic renal failure**. **Nadir creatinine <0.7 mg/dL at 1 year** predicts stable function. Lifelong timed voiding, cultures and imaging; transplant when needed.

Diagram to draw: fetal ultrasound sketch: huge bladder, dilated posterior urethra, bilateral hydroureteronephrosis.

Key points to remember:

- Triad; 95% male; dysplasia and lungs decide survival.
- Ureter is a failed pump (collagen); VUR about 75%; keep the proximal ureter.
- Woodard I–III; vesicostomy for drainage; orchidopexy at about 6 months.
- Nadir creatinine <0.7 mg/dL is favourable; up to 30% reach renal failure.

Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children

Type: LAQ (25 marks)

Source: Practice Set 12 Paper 1, Q1 (frequency and urgency in a child also in 2017-Summer Paper 3, Q3c)

Introduction

- The LUT (lower urinary tract) has two opposite functions: **storage** at low pressure with a closed outlet, and **emptying** by a coordinated detrusor contraction with a relaxed outlet. Bladder and sphincter act reciprocally.

Peripheral innervation

	Parasympathetic	Sympathetic	Somatic
Origin	**S2–S4** (sacral parasympathetic nucleus)	Thoracolumbar (T10–L2)	**Onuf nucleus** (S2–S4 ventral horn)
Nerve	Pelvic	Hypogastric	Pudendal
Bladder	Contracts (ACh [acetylcholine] → **M3**; ATP)	Relaxes (noradrenaline → **β3**)	–
Outlet	Relaxes urethra (**NO**, nitric oxide)	Contracts bladder neck (**α1**)	Contracts EUS (external urethral sphincter; nicotinic)
Role	Voiding	Storage	Voluntary closure

- All three nerves carry afferents. **Aδ fibres** (tension receptors in muscle) signal normal fullness. **Unmyelinated C fibres** are "silent" normally; after inflammation, SCI (spinal cord injury) or obstruction they are recruited and drive overactivity (basis of capsaicin and resiniferatoxin).
- Detrusor has more M2 receptors, but **M3 mediates contraction**.

Reflex and central control

- **Storage (spinal reflexes):** low afferent activity → sympathetic outflow (detrusor inhibition, bladder-neck closure) and the **guarding reflex** (pudendal → rising sphincter activity with filling).
- **Voiding (spinobulbospinal reflex):** high afferent activity → **PAG (periaqueductal grey)** → **PMC (pontine micturition centre, Barrington nucleus)** → sphincter relaxes first, then detrusor contracts. PMC is an on–off switch.
- **Forebrain decides:** insula senses fullness; **medial prefrontal cortex** tonically holds the switch "off" until voiding is socially appropriate.
- Lesion patterns: above pons → overactivity, coordinated sphincter; between pons and sacral cord → overactivity with **DSD (detrusor–sphincter dyssynergia)**; sacral/peripheral → underactive bladder.
- **Central transmitters:** glutamate excitatory; GABA (γ-aminobutyric acid) and glycine inhibitory; serotonin inhibits voiding; dopamine tonically inhibits the PMC (lost in Parkinson disease).

```
Filling → Aδ afferents → sacral cord → PAG
  low traffic: spinal storage reflexes (sympathetic + guarding)
  high traffic + cortex "permits": PMC → EUS relaxes → detrusor contracts
```

Maturation in children

- Bladder visible on fetal ultrasound by 14 weeks. Infants void reflexly but **wake before voiding** (cortical arousal gates voiding).
- Infant voiding is often interrupted (transient dyscoordination) with detrusor pressure often **>100 cmH2O**; frequent small voids largely disappear by **2 years**.
- Capacity rises: **EBC (expected bladder capacity) = (age + 2) × 30 mL** (infants: weight in kg × 7 mL).
- Toilet training: bowel control → daytime bladder → night-time bladder. Persisting dyscoordination after toddlerhood is dysfunction, not immaturity.

Types of voiding dysfunction in children

A. Non-neurogenic (functional) LUT dysfunction (ICCS [International Children's Continence Society])

Type	Key feature
OAB (overactive bladder)	Urgency ± frequency (≥8 voids a day), urgency incontinence; tower flow
Voiding postponement	Habitual holding; holding manoeuvres; infrequent voids (≤3 a day)
Dysfunctional voiding	Habitual sphincter/pelvic-floor contraction during voiding in a neurologically normal child; **staccato** flow
Underactive bladder	Needs abdominal straining to void; interrupted flow
Giggle incontinence	Complete emptying with laughter
Vaginal reflux	Small leak just after voiding in girls
Enuresis	Wetting in sleep, ≥5 years; monosymptomatic or non-monosymptomatic

- **BBD (bladder and bowel dysfunction):** LUT dysfunction plus constipation; rectal loading worsens symptoms and UTI (urinary tract infection).
- **Hinman (non-neurogenic neurogenic bladder):** severe end of dysfunctional voiding → trabeculated bladder, VUR (vesicoureteric reflux), renal damage.

B. Neurogenic LUT dysfunction

- Causes: **myelomeningocele** (commonest), occult dysraphism/**tethered cord**, sacral agenesis, anorectal malformation, cerebral palsy, SCI, transverse myelitis, pelvic tumour surgery (sacrococcygeal teratoma).
- Pattern = detrusor (overactive or underactive) × outlet (competent, incompetent or dyssynergic). Bony level does not predict bladder function.
- **Hostile bladder: DLPP (detrusor leak point pressure) ≥40 cmH2O** (McGuire 1981), poor compliance or NDO (neurogenic detrusor overactivity) with DSD.

C. Anatomical: PUV (posterior urethral valves) "valve bladder": overactivity → poor compliance → myogenic failure; ectopic ureter.

Evaluation

- Bladder and bowel diary, Bristol stool chart, spine and neurological examination, urinalysis.
- **Uroflowmetry with PVR (post-void residual) and patch EMG (electromyography)**: bell (normal), tower (OAB), staccato (dysfunctional voiding), interrupted (underactive/DSD), plateau (obstruction).
- Invasive (video)urodynamics only for neurogenic, anatomical or refractory cases.

Management principles

- **Treat constipation first**; standard urotherapy (timed voiding, posture, fluids) for 4 weeks.

- Dysfunctional voiding: **biofeedback**; α-blocker; sphincter botulinum toxin if refractory.
  OAB: **oxybutynin 0.3–0.6 mg/kg/day** (maximum 15 mg); mirabegron; botulinum toxin; parasacral TENS (transcutaneous electrical nerve stimulation).
- Enuresis: alarm or desmopressin.
- Neurogenic: **CIC (clean intermittent catheterisation; Lapides 1971) + antimuscarinic** → intradetrusor botulinum toxin → augmentation.

Diagram to draw: bladder with pelvic, hypogastric and pudendal nerves, plus brainstem loop (PAG–PMC).

Key points to remember:

- S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
- PMC = switch; medial prefrontal cortex holds it off.
- EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
- DLPP ≥40 cmH2O = hostile; treat bowel first.

Section 5 · Paediatric Urology 2 — Short answer questions (SAQ)

Embryology of Posterior Urethral Valves.

Type: SAQ (10 marks)

Source: 2016-Summer Paper 1, Q3b

Introduction

- PUV (posterior urethral valves): commonest cause of bladder outlet obstruction in boys (1.4–2.2 per 10,000 births). Morgagni described them (1769); Hugh Hampton Young named and classified them (1919).

Normal development

- Weeks 5–6: the **urorectal septum** divides the cloaca into the anorectal canal and the **urogenital sinus**.
- The **Wolffian (mesonephric) ducts** join the cloaca just before division. Their orifices normally migrate to the **verumontanum**, leaving small folds (plicae colliculi).
- The **urogenital membrane** normally disappears completely.
- Fetal urine starts at 10–14 weeks; after 16 weeks amniotic fluid is mainly fetal urine. So obstruction in this period damages bladder, kidneys and lungs.

Young types and their origin

Type	Anatomy	Embryology
I (~95%)	Leaflets from the verumontanum fuse anteriorly just proximal to the external sphincter	Abnormal insertion of the Wolffian duct ends into the anterolateral cloacal wall: hypertrophied inferior urethral crest
II	Folds from verumontanum to bladder neck	Not real (Stephens 1955): secondary bladder-neck hypertrophy
III (5–10%)	Annular diaphragm with a central hole	**Persistent urogenital membrane**

Other theories

- Hypertrophied mucosal folds (oldest); cloacal remnants; Wolffian–Müllerian maldevelopment.
- **COPUM (congenital obstructing posterior urethral membrane, Dewan):** one oblique membrane, torn by the first catheter into apparent type I leaflets. Plausible but unproven.
- **Cobb's collar (Moormann's ring):** a congenital ring-like narrowing of the proximal bulbar urethra, just distal to the external sphincter; some authors group it with type III valves.
- **Genetics:** mostly sporadic; chromosome 11 duplications; copy number variants (worse renal outcome); TBX5, PTK7.

Consequences

- Thick bladder, dilated posterior urethra (**keyhole sign**), hydroureteronephrosis, **renal dysplasia**, oligohydramnios, pulmonary hypoplasia. Pop-offs (VUR (vesicoureteral reflux), urinoma, ascites) relieve pressure.

Diagram to draw: posterior urethra with type I leaflets versus a type III membrane.

Key points to remember:

- Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
- Urorectal septum divides the cloaca at weeks 5–6.
- COPUM: one membrane torn into leaflets by catheterisation.
- Dysplasia is fixed in utero; ablation cannot reverse it.

Anomalies detected on Prenatal scan between 16th and 20th weeks of gestation

Type: SAQ (10 marks)

Source: 2016-Summer Paper 1, Q3d

Introduction

- The **anomaly scan is done at 18–20 weeks** (ACOG (American College of Obstetricians and Gynecologists)); by 20 weeks most urological anomalies are visible.
- **Why this window:** after about 16 weeks amniotic fluid is mainly fetal urine, so **oligohydramnios reflects renal failure or outlet obstruction**. The bladder should be seen filling and emptying.
- Survey kidneys, collecting system, ureters, bladder, genitalia, amniotic fluid, abdominal wall and spine. AFI (amniotic fluid index): normal 8–18 cm; oligohydramnios <5–6 cm.

UTD (urinary tract dilation) risk, 16–27 weeks

Parameter	A1 (low)	A2–3 (any one)
APRPD (anteroposterior renal pelvic diameter)	4 to <7 mm	≥7 mm
Calyces	Central or none	Peripheral
Parenchyma, ureter, bladder	Normal	Abnormal
Oligohydramnios	No	Unexplained

- A1: rescan after 32 weeks. A2–3: scan every 4 weeks. Both: postnatal scan after 48 hours.
- Causes: transient (50–70%), PUJ (pelviureteric junction) obstruction, VUR (vesicoureteral reflux), megaureter, duplex, LUTO (lower urinary tract obstruction).

Anomalies and signs

Anomaly	Signs
PUV (posterior urethral valves)	Male; thick bladder; **keyhole sign**; bilateral hydroureteronephrosis; oligohydramnios
Bilateral renal agenesis	**Anhydramnios after 16 weeks**, no bladder, no renal arteries: lethal Potter sequence
MCDK (multicystic dysplastic kidney)	Non-communicating cysts, no central pelvis
ARPKD (autosomal recessive polycystic kidney disease)	Large, homogeneously **bright** kidneys; oligohydramnios
Bladder exstrophy	**Bladder never seen**, low cord insertion, lower abdominal mass, wide pubis
Prune belly	Thin-walled large bladder, empty scrotum
Myelomeningocele	Spinal defect, lemon and banana signs

- **Also know:** duplex kidney with ureterocele (intravesical cyst); cloacal exstrophy (adds omphalocele, spinal defects); hydrocolpos; ambiguous genitalia; renal ectopia.

Implications

- Severe LUTO: fetal MRI (magnetic resonance imaging), urine biochemistry, vesicoamniotic shunt or fetal cystoscopy at 18–24 weeks. **PLUTO** (shunt trial): better survival, not renal function. Plan delivery with a multidisciplinary team.

Diagram to draw: fetal keyhole sign.

Key points to remember:

- Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
- UTD A2–3 = APRPD ≥7 mm or any abnormal feature.
- Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.
- ARPKD large bright kidneys; MCDK non-communicating cysts.

USG in neonates

Type: SAQ (10 marks)

Source: Practice Set 8 Paper 1, Q3d

Assumed meaning: ultrasonography (USG) of the neonatal urinary tract, including antenatal hydronephrosis follow-up.

Why ultrasound first

- No radiation or sedation; repeatable. Shows **anatomy, not function**.
- **Technique:** higher-frequency probes than in adults; kidneys supine and prone; length against age nomograms; bladder full and after voiding; Doppler.

Normal neonatal kidney (pitfalls)

- Cortex **iso- or hyperechoic** to liver; **prominent hypoechoic pyramids mimic dilated calyces or cysts**.
- **Dilation is under-read in the first 48 hours** (physiological dehydration).

Antenatal hydronephrosis follow-up

- **First scan after 48 hours**; earlier if solitary kidney or suspected PUV (posterior urethral valves).
- Normal postnatal **APD (anteroposterior pelvic diameter) <10 mm**, no calyceal or ureteric dilation.

UTD (urinary tract dilation) risk	Findings	Action
P1 (SFU [Society for Fetal Urology] 1–2)	APD 10–15 mm, central calyces	Repeat US at 3–6 months; no VCUG or prophylaxis
P2 (SFU 3)	APD ≥15 mm, peripheral calyces or dilated ureter	Repeat US; VCUG (voiding cystourethrogram) and prophylaxis at discretion
P3 (SFU 4)	Abnormal parenchyma or bladder	Repeat US at 1 month; VCUG and prophylaxis

- **SFU grades:** 0 none; 1 pelvis only; 2 major calyces; 3 major and minor calyces; 4 grade 3 plus parenchymal thinning.

Other neonatal indications

Condition	Ultrasound finding
PUV	Thick bladder, **keyhole sign**, bilateral hydroureteronephrosis
MCDK (multicystic dysplastic kidney)	Non-communicating cysts of varying size, scant parenchyma
ARPKD (autosomal recessive polycystic kidney disease)	Large, bilaterally echogenic kidneys
Masses	Mesoblastic nephroma; adrenal haemorrhage; renal vein thrombosis (large kidney, hypertension)
Duplex anomalies	Ureterocele, ectopic ureter, megaureter
Spine	Occult dysraphism before arch ossification (up to 6 months)
DSD (differences of sex development)	Müllerian structures, gonads

Also know: hydrometrocolpos, scrotal masses, contrast-enhanced voiding urosonography.

Limitations

- Normal US (ultrasound) does not exclude VUR (vesicoureteral reflux). Function needs MAG3 (mercaptoacetyltriglycine) renography (after renal maturation) or DMSA (dimercaptosuccinic acid).

Diagram to draw: SFU grades 1–4 as kidney outlines.

> **Key points to remember:**
> - First postnatal scan after 48 h; normal APD <10 mm.
> - Echogenic cortex and prominent pyramids are normal in neonates.
> - Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

Nocturnal enuresis

Type: SAQ (10 marks)

Source: Practice Set 10 Paper 1, Q3a (also answered in Practice Set 1 Paper 3, Q3e)

Definition (ICCS [International Children's Continence Society])

- Intermittent incontinence **during sleep** in a child **aged 5 years or more**.
- **Primary:** never dry. **Secondary:** wetting after at least **6 months** dry (look for stressors, diabetes, UTI [urinary tract infection], neurological disease).
- **MNE (monosymptomatic):** no daytime LUTS (lower urinary tract symptoms). **NMNE (non-monosymptomatic):** with daytime LUTS.

Epidemiology

- About **5–10% at 7 years**; falls with age; about 2% of adults; familial.

Pathophysiology: three-way mismatch

1. **Nocturnal polyuria** (poor night-time ADH [antidiuretic hormone] rise).
2. **Small nocturnal bladder capacity** or DO (detrusor overactivity).
3. **Failure to arouse** from sleep.

- **Why sleep apnoea matters:** obstructed breathing makes arousal difficult and releases atrial natriuretic peptide, which raises urine output.
- Comorbidities: constipation, sleep-disordered breathing, ADHD (attention deficit hyperactivity disorder).

Evaluation

- History, examination (spine, genitalia, abdomen), urinalysis.
- **Bladder diary:** voided volumes against EBC (expected bladder capacity) = (age + 2) × 30 mL; night output by weighing nappies.
- Uroflowmetry, PVR (post-void residual), ultrasound only for NMNE or before anticholinergics.

Management

- **Treat daytime LUTS and constipation first.** Education; waking, fluid restriction and punishment do not work.

	Alarm	Desmopressin
Target	Arousal (conditioning)	Night urine output
Use	Nightly; until 14 dry nights	0.2–0.4 mg tablet or 120–240 µg melt, 1 hour before bed
Response	50–70%; durable	Rule of thirds; relapses off drug
Caution	Motivated family	Limit evening fluid (hyponatraemia)

- **Failure:** switch or combine. **Second line:** add oxybutynin after checking PVR and bowels. **Resistant:** imipramine (cardiotoxic; ECG [electrocardiogram]), off-label β3-agonists, adenotonsillectomy for sleep apnoea.
- **Adults:** new enuresis may signal high-pressure chronic retention: catheterise and treat obstruction.

Diagram to draw: triad: nocturnal polyuria, small capacity, arousal failure.

Key points to remember:

- ≥5 years; secondary after 6 months dry.
- Alarm (durable) and desmopressin (rule of thirds) are first line.
- Treat daytime symptoms and constipation first.

Neurogenic bladder in myelodysplasia

Type: SAQ (10 marks)

Source: Practice Set 22 SAQ Sheet, Q3b (prenatal and neonatal assessment also in 2016-Summer Paper 3, Q3e)

Introduction

- Myelodysplasia (NTD [neural tube defect], mainly **myelomeningocele**) is the commonest cause of neurogenic LUT (lower urinary tract) dysfunction in children. Lesions: lumbosacral 47%, lumbar 26%, sacral 20%.
- **Bony or motor level does not predict bladder function**; the lesion can progress (re-tethering), so urodynamic testing is essential.
- About 10% of deaths in SB (spina bifida) are urological. Folic acid 400 µg daily before conception prevents 50–70% of NTDs.

Patterns

- Detrusor: overactive (NDO, neurogenic detrusor overactivity) or underactive. Outlet: competent, incompetent or dyssynergic.
- Newborns: most have detrusor contractions; over a third are acontractile. Needle EMG (electromyography): intact sacral arc about 40%, partial denervation 25%, complete 35%.

Evaluation (EAU [European Association of Urology] 2022)

- Renal and bladder ultrasound after birth, then serially; **UDS (urodynamics) within 3 months**; baseline DMSA (dimercaptosuccinic acid); creatinine (cystatin C better with low muscle mass); VCUG (voiding cystourethrogram) or video-UDS for VUR (vesicoureteric reflux), which is dilating in 7%.

Risk (McGuire 1981)	Finding
Hostile	DLPP (detrusor leak point pressure) or end-filling pressure ≥40 cm H_2O
Intermediate	NDO, poor compliance, 25–39 cm H_2O
Safe	Below 25 cm H_2O

Management (kidneys first, then continence)

- **Early CIC (clean intermittent catheterisation; Lapides 1971) plus oxybutynin** for hostile bladders. Hydronephrosis 0–15% versus 18–80% expectant; ESRD (end-stage renal disease) 0–1.6% versus about 30%.
- Others: solifenacin, mirabegron; **botulinum toxin A** about 10 U/kg (maximum 360 U), continence 65–87%.
- Surgery: vesicostomy (bridge); **augmentation cystoplasty**; Mitrofanoff channel; bladder-neck sling, reconstruction or artificial sphincter.
- VUR: CIC and antimuscarinic first; avoid Credé; reimplant during augmentation if grade ≥III.
- **Also know:** latex allergy; bowel programme; sexual dysfunction (about 75% of men have ED [erectile dysfunction]; sildenafil helps); precocious puberty.

Diagram to draw: UDS trace with DLPP above 40 cm H_2O.

> **Key points to remember:**
>
> - Level does not predict bladder: UDS within 3 months.
> - DLPP ≥40 cm H_2O = hostile (McGuire 1981).
> - Early CIC plus antimuscarinic protects kidneys.

Rapid Revision Sheet – Section 5 · Paediatric Urology 2

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Describe embryology of bladder. What are the proposed theories for extrophy epispadias complex? Describe skeletal abnormalities. (LAQ)

- Vesicourethral canal → bladder; urachus from allantois.
- Marshall–Muecke: overdeveloped cloacal membrane; rupture timing decides variant.
- CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis ~4.8 cm.
- Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.

Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management. (LAQ)

- Triad; 95% male; dysplasia and lungs decide survival.
- Ureter is a failed pump (collagen); VUR about 75%; keep the proximal ureter.
- Woodard I–III; vesicostomy for drainage; orchidopexy at about 6 months.
- Nadir creatinine <0.7 mg/dL is favourable; up to 30% reach renal failure.

Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children (LAQ)

- S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
- PMC = switch; medial prefrontal cortex holds it off.
- EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
- DLPP ≥40 cmH2O = hostile; treat bowel first.

Embryology of Posterior Urethral Valves. (SAQ)

- Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
- Urorectal septum divides the cloaca at weeks 5–6.
- COPUM: one membrane torn into leaflets by catheterisation.
- Dysplasia is fixed in utero; ablation cannot reverse it.

Anomalies detected on Prenatal scan between 16th and 20th weeks of gestation (SAQ)

- Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
- UTD A2–3 = APRPD ≥7 mm or any abnormal feature.
- Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.
- ARPKD large bright kidneys; MCDK non-communicating cysts.

USG in neonates (SAQ)

- First postnatal scan after 48 h; normal APD <10 mm.
- Echogenic cortex and prominent pyramids are normal in neonates.
- Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

Nocturnal enuresis (SAQ)

- ≥5 years; secondary after 6 months dry.
- Alarm (durable) and desmopressin (rule of thirds) are first line.
- Treat daytime symptoms and constipation first.

Neurogenic bladder in myelodysplasia (SAQ)

- Level does not predict bladder: UDS within 3 months.
- DLPP ≥40 cm H_2O = hostile (McGuire 1981).

- Early CIC plus antimuscarinic protects kidneys.

Section 6 · Female Urology — Long answer questions (LAQ)

Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia

Asked as:

- Describe female pelvic anatomy in relation to prolapse and SUI — *Summer 2015 Paper I Q2 (LAQ)*
- Describe anatomy of pelvic floor in females. Discuss fertility sparing treatment options for procedentia. — *Practice Set 2 Paper I Q1 (LAQ)*
- Describe anatomy of pelvic floor in females. Discuss fertility preserving treatment options for procidentia — *Practice Set 9 Paper I Q2 (LAQ)*
- Describe anatomy of pelvic floor in females and its importance in surgeries described for SUI — *Practice Set 11 Paper I Q2 (LAQ)*

Type: LAQ (25 marks)

(pelvic floor anatomy also answered in 2017-Winter Paper 3, Q2 and 2015-Summer Paper 1, Q2; also answered in Practice Set 2 Paper 1, Q1)

"Procedentia" is read as procidentia (complete uterine prolapse); "fertility sparing" / "fertility preserving" means uterus-preserving treatment for a woman who wants future pregnancy.

Introduction

- **SUI (stress urinary incontinence):** involuntary leakage on effort, coughing or sneezing.
- Support comes from the **levator ani, endopelvic fascia and ligaments, perineal membrane and perineal body**, anchored to the **bony pelvis** (pelvic organs are supported by the **levator ani**, the **endopelvic fascia and its ligaments** and the **perineal membrane and perineal body**). Failure causes **pelvic organ prolapse (POP)** and **stress urinary incontinence (SUI)**. Every SUI operation uses one of these anchors.

A. Bony pelvis and surgical anchors

- **Bony landmarks** and dangers: **sacrospinous ligament** (pudendal bundle), **Cooper's ligament** (bleeding), **sacral promontory** (left common iliac vein, right ureter).
- **Sacrospinous ligament (SSL)**, 7–8 cm, ischial spine to sacrum: the pudendal nerve and vessels pass behind the spine, the sacral plexus just posterior.
- **Sacral promontory**: left common iliac vein and right ureter each about 2.7 cm away.

B. Pelvic floor muscles: pelvic diaphragm (levator ani + coccygeus)

Muscle	Attachment	Function
Puborectalis	Back of pubis, loops behind anorectal junction (sling behind anorectal junction)	Anorectal angle; anorectal sling
Pubococcygeus (pubovaginal part)	Posterior pubis and arcus tendineus → viscera (vagina, perineal body, anus), anococcygeal raphe (coccygeus is separate)	Slings urethra and vagina; closes hiatus, lifts urethra
Iliococcygeus	ATLA (arcus tendineus levator ani), ischial spine → raphe, coccyx	Horizontal **levator plate**
Coccygeus (not levator)	Ischial spine → sacrum, coccyx	Lies on sacrospinous ligament

- Levator tone (tonic contraction) closes the **urogenital hiatus**, through which all POP passes; the uterus and upper vagina rest on the levator plate.
- Nerve to levator ani **(S3–S5)** runs on the pelvic surface; **pudendal nerve (S2–S4)** supplies the sphincters and perineal muscles.
- **Muscle fails first, fascia second:** once injury (birth injury, avulsion) opens the hiatus, ligaments carry the load and fail; the fascia then carries the whole load and stretches.
- Levator defects in up to **20% of primiparous women**; avulsion predicts advanced prolapse.

C. Endopelvic fascia and ligaments

- **Cardinal ligament** (side wall to cervix; carries the uterine artery) and **uterosacral ligament** (sacrum to cervix): the **cardinal–uterosacral complex** suspends the apex; **rectovaginal fascia** posteriorly. The ureter is 4 cm from the uterosacral ligament near the sacrum but 0.9 cm near the cervix (ureter close to uterosacral at the cervix).
- **Pubocervical (pubovesical) fascia** attaches laterally to the **ATFP (arcus tendineus fasciae pelvis)**, the "white line" from the lower pubis to just above the ischial spine.
- **Rectovaginal fascia** supports the posterior wall; **pubourethral ligaments** suspend the urethra from the pubis / anchor the mid-urethra; **urethropelvic ligaments** anchor it to the ATFP.

D. Perineal membrane, perineal body and urethra

- The perineal membrane spans the pubic arch, pierced by urethra and vagina (urethra and vagina pierce it); above it lies the striated urogenital sphincter. The perineal body anchors the distal vagina.
- Continence zone: **proximal and mid-urethra**; striated sphincter (compressor urethrae, urethrovaginal sphincter) plus oestrogen-sensitive submucosa (~30% of closure).
- **Striated urogenital sphincter**: sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**.
- Urethra 4–5 cm; continence zone is the **proximal and mid-urethra**. The **oestrogen-dependent (oestrogen-sensitive) vascular submucosa** gives up to 30% (~30%) of closure force.

E. DeLancey's levels of support (1992)

Level	Region	Structures	Defect causes / loss produces
I – suspension	Cervix/upper third	Cardinal–uterosacral complex	**Uterine or vault prolapse**, enterocele
II – attachment	Middle third	Paravaginal attachment to ATFP, levator fascia	Cystocele (central or paravaginal), rectocele; anterior or posterior wall prolapse
III – fusion	Distal third	Urethra, perineal membrane, perineal body	**Urethral hypermobility**, perineal descent, distal rectocele

- **Central cystocele:** midline weakness, rugae lost. **Paravaginal cystocele:** ATFP detachment (Richardson 1981), rugae preserved.
- Level I is the keystone: always correct the apex.

Diagram to draw: sagittal female pelvis with levator plate and DeLancey levels I–III; hammock cross-section.

F. Theories of continence and the operations they produced

Theory	Mechanism	Implication / surgical answer
Pressure transmission (Enhorning 1961)	Hypermobile urethra leaves (drops out of) the abdominal pressure zone	Restore position: Burch, MMK (Marshall–Marchetti–Krantz)
Hammock (DeLancey 1994)	Urethra lies on anterior vaginal wall and endopelvic fascia attached to ATFP and levator; abdominal pressure compresses it against this layer (a firm backing); repositioning not essential	Restore backing
Integral theory (Petros & Ulmsten 1990)	Mid-urethral pubourethral ligament laxity (at the **mid-urethra**)	Tension-free mid-urethral sling
Intrinsic sphincter deficiency (ISD) (McGuire 1978)	Urethral function fails (urethra fails to close); low VLPP (Valsalva leak point pressure) <60 cmH_2O	Sling, bulking, AUS (artificial urinary sphincter)

- ISD: Valsalva leak point pressure <60 or maximal urethral closure pressure <20 cmH_2O. Most SUI has (includes) some ISD. So slings work in hypermobility, but Burch does poorly in severe ISD / fails in ISD with a fixed urethra.

G. Innervation

- **Pudendal nerve (S2–4):** sphincters and perineum (perineal muscles); through Alcock's canal near the ischial spine; stretched in delivery.
- **Nerve to levator ani (S3–5):** runs on the muscle's pelvic surface, separate from pudendal; neuropathy in about a quarter of primiparous women.
- Autonomic pelvic splanchnic and hypogastric nerves: injured in radical pelvic surgery.

H. Why prolapse occurs: risk factors (Bump & Norton)

- Muscle fails first (birth trauma, levator avulsion, denervation). The hiatus widens, the ligaments carry the whole load and stretch.
- Collagen disorders (Ehlers–Danlos, Marfan) add risk; other risks: vaginal birth, forceps, age, obesity, constipation.
- Predisposing: genetics, collagen disorders (Ehlers–Danlos, Marfan). Inciting: **vaginal delivery** (strongest; forceps worse), hysterectomy. Promoting: obesity, constipation, cough. Decompensating: ageing (age).

I. POP-Q (Pelvic Organ Prolapse Quantification, Bump 1996)

- Nine points in cm relative to the **hymen (0)**: inside negative, outside positive; measured on maximal strain, recorded on a 3 × 3 grid.

Point	Definition
Aa / Ap	Anterior / posterior wall 3 cm inside hymen (−3 to +3)
Ba / Bp	Most distal point of anterior / posterior wall (−3 if no prolapse)
C	Cervix or vault cuff
D	Posterior fornix (omitted after hysterectomy)
gh, pb, tvl	Genital hiatus, perineal body, total vaginal length (at rest)

Stage	Most distal point
0	No prolapse
I	>1 cm above hymen
II	Within ±1 cm of hymen
III	>1 cm below hymen, < (tvl – 2) cm
IV	Complete eversion (≥ tvl – 2)

- **Also know:** Baden–Walker grades 0–4; stage I–II often normal (POSST study).

J. Examination

- Lithotomy and **standing** with Valsalva; Sims speculum for each compartment; lateral sulci (paravaginal defect); rectovaginal examination.
- **Cough stress test**, repeated with prolapse reduced (**occult SUI**); **Q-tip** angle change >30° = hypermobility; levator strength and avulsion; urodynamics if ISD or mixed symptoms.

K. How anatomy guides surgery

Defect	Operation	Principle
SUI	**Mid-urethral sling**: retropubic (Ulmsten 1995, TVT [tension-free vaginal tape]) or transobturator (Delorme 2001)	Mid-urethral support
SUI	**Burch colposuspension**	Fascia to Cooper's ligament
SUI – ISD	Autologous sling at bladder neck, bulking	Compression
Paravaginal / central cystocele	Paravaginal repair / anterior colporrhaphy	Reattach to ATFP / midline plication
Apical	**Sacrocolpopexy** (most durable)	Mesh to ligament over S1
Apical (vaginal)	**Sacrospinous fixation** 1.5–3 cm medial to spine; uterosacral suspension (cystoscopy for ureters)	Avoids pudendal bundle
Rectocele	Posterior or site-specific repair, perineorrhaphy	Levels II–III
Elderly, not sexually active	Colpocleisis	Obliteration

- **April 2019:** the US FDA (Food and Drug Administration) stopped transvaginal mesh for anterior POP; slings unaffected.

Importance in SUI surgery

Operation	Anatomy used	Anatomical hazards
Burch colposuspension	Space of Retzius (avascular: symphysis in front, bladder behind, endopelvic fascia below). Paravaginal fascia at mid-urethra and bladder neck sutured to **Cooper's (iliopectineal) ligament**	Bleeding from Cooper's ligament and perivesical veins; ureteric kinking (cystoscopy); needs a mobile vagina
Paravaginal repair / VOS (vagino-obturator shelf)	Lateral vaginal sulcus to **ATFP** or obturator fascia (level II)	Obturator vessels and nerve
Retropubic MUS (mid-urethral sling), TVT	Tape under mid-urethra (integral theory), through **endopelvic fascia** and Retzius; suprapubic exits about 2 cm from midline	Bladder perforation (4.5%), vessels, bowel

Operation	Anatomy used	Anatomical hazards
TOT (transobturator tape)	Through obturator membrane, **obturator internus and externus**; exit at adductor longus tendon notch	Groin pain (6.4%), obturator neurovascular bundle; bladder injury rare (0.6%)
SIMS (single-incision mini-sling)	Anchors in obturator internus or endopelvic fascia	Works by restricting mobility
Autologous PVS (pubovaginal sling)	Rectus fascia at the **bladder neck** (proximal urethra), arms to rectus	Voiding dysfunction; best for ISD
Bulking agents	**Submucosa** coaptation	Repeated injections

- **Why mid-urethra:** it is the zone of maximal closure pressure and pubourethral support. A loose tape lets the posterior urethral wall be compressed (and kinked) during stress.
- **TOMUS (2010):** retropubic 80.8% vs transobturator 77.7% objective success, not different.
- **SISTEr:** autologous sling more successful than Burch at 2 years, with more voiding dysfunction.
- **Also know:** needle suspensions abandoned (sutures pulled through); anterior colporrhaphy is not an SUI operation; levator palpable through the lateral vaginal walls for PFMT (pelvic floor muscle training) assessment; uncorrected level I loss causes recurrent prolapse after SUI repair.

Diagram to draw: cross-section of urethra on the vaginal hammock (DeLancey) with Burch, bladder-neck sling and mid-urethral sling sites.

L. Procidentia: definition and assessment

- **Procidentia**: the uterus lies wholly outside the hymen — complete uterine (or vault) prolapse outside the hymen, **POP-Q (pelvic organ prolapse quantification) stage 4**.
- Assess all compartments, occult SUI (stress urinary incontinence) with prolapse reduced, PVR (post-void residual), hydronephrosis (ureteric kinking at the hiatus), cervical length and ulceration.
- Abnormal bleeding or cervical pathology needs gynaecological evaluation first, because it may argue against keeping the uterus.

M. Fertility-sparing (fertility-preserving) options

Principle: restore **level I** support while keeping uterus and cervix. Defer definitive surgery until the family is complete where possible, because pregnancy and delivery cause recurrence.

1. Conservative (first line)

- PFMT (pelvic floor muscle training), weight loss, treat constipation.
- **Pessary**: space-filling type (Gellhorn, cube) for procidentia; removed for conception; regular review about 6-monthly for discharge and erosion.

2. Uterus-preserving surgery

Operation	Technique	Fertility points
Sacrospinous hysteropexy (vaginal)	Cervix sutured to (into) the SSL **1.5–3 cm medial to the ischial spine**, into the ligament	Non-inferior to vaginal hysterectomy with uterosacral suspension (SAVE U trial); later data suggest superiority. Buttock pain, pudendal injury
Uterosacral hysteropexy (vaginal or laparoscopic)	Shorten and reattach uterosacral ligaments to cervix, sutures lateral to medial	No mesh; cystoscopy for ureteric kinking

Operation	Technique	Fertility points
Sacrohysteropexy (open, laparoscopic, robotic)	Mesh from posterior cervix to anterior longitudinal ligament over the promontory	Most durable apical support; later pregnancy (delivery) usually by caesarean
Manchester (Fothergill)	Cervical amputation, cardinal ligaments sutured in front of the stump, anterior colporrhaphy	Suits elongated cervix, but risks **cervical stenosis, incompetence, miscarriage, preterm labour**: avoid if pregnancy wanted

- **Also know:** Indian abdominal (cervicopexy) slings for young nulliparous prolapse (Shirodkar, Purandare, Khanna, Virkud). **Le Fort colpocleisis** keeps the uterus but not fertility (elderly only).

3. Choosing

```
Procidentia, wants pregnancy -> PFMT + pessary
  -> fails -> sacrospinous hysteropexy or
     laparoscopic sacrohysteropexy (avoid Manchester)
  -> recurrence after childbearing -> hysterectomy
     + vault suspension or sacrocolpopexy
```

N. Complications and follow-up

- Recurrence (mostly posterior), de novo SUI, mesh exposure (3.4–5.4% after sacrocolpopexy; another source: 3.4–5.4% vaginal mesh exposure in meta-analyses), ureteric kinking. The retained uterus needs cervical screening.

Key points to remember:

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey: I suspension (uterosacral–cardinal; failure causes uterine prolapse), II attachment (ATFP), III fusion (urethral hypermobility); level I is the keystone.
- Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
- Integral theory → mid-urethral sling; hammock → backing, not repositioning.
- Burch → Cooper's ligament; TOT → obturator foramen; PVS → bladder neck.
- Central cystocele rugae lost, paravaginal preserved; POP-Q II = ±1 cm of hymen.
- Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
- Manchester amputates the cervix: poor choice for future pregnancy.

Section 6 · Female Urology — Short answer questions (SAQ)

Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications

Asked as:

- VLPP in SUI — *Summer 2019 Paper I Q3b (SAQ)*
- Leak point pressures-clinical implication. — *Summer 2020 Paper I Q3e (SAQ)*

Type: SAQ (10 marks)

Definition

- **LPP (leak point pressure):** pressure at which leakage is first seen, spontaneous or provoked. Two independent types:

Feature	ALPP (abdominal LPP), incl. VLPP (Valsalva LPP) and CLPP (cough LPP)	DLPP (detrusor LPP)
Definition	Total Pves (vesical pressure) at leak during raised abdominal pressure, **no detrusor contraction**	Lowest Pdet (detrusor pressure) at leak, **without contraction or straining**
Measures	Sphincter strength (sphincter vs abdominal pressure)	Bladder pressure against (vs) outlet resistance
Bad result	**Low** (weak sphincter) (**low = bad**)	**High** (danger to upper tracts) (**high (>40 cmH_2O) = bad**: upper-tract risk (McGuire 1981))
Used in	SUI (stress urinary incontinence)	Neurogenic or low-compliance bladder

- **VLPP (Valsalva leak point pressure):** the **intravesical pressure** at which urine leaks during Valsalva, **without a detrusor contraction**.
- It is a type of **ALPP (abdominal leak point pressure)**, which measures the urethra's ability to resist abdominal pressure. Lower value = weaker sphincter.
- Seen only in SUI (stress urinary incontinence); **no normal value** (continent women never leak).

VLPP: technique

1. Small dual-lumen urethral catheter plus rectal catheter; zero to atmosphere.
2. Patient **standing or sitting**; fill to **150 mL** (original method; some use 200 mL).
3. Ask for a **graded Valsalva**; record the **lowest total Pves (vesical pressure)** at visible leak (include baseline, not the change).
4. No leak: repeat every 50 mL to capacity; then (at capacity) **remove the urethral catheter** and read pressure from the rectal line.
5. Report volume, catheter size and provocation (Valsalva or cough).

VLPP: interpretation and clinical implications (McGuire 1993)

VLPP (cmH_2O)	Inference (McGuire 1993)
<60	**ISD (intrinsic sphincter deficiency)**
60–90	Equivocal; mixed
>90 (≥90)	**Urethral hypermobility**, little ISD

- Low VLPP (ISD) traditionally favours a sling or bulking agent over colposuspension.

Limitations

- Not standardised: falls with **larger catheters** and **higher volumes**.
- **VLPP is lower than CLPP (cough leak point pressure)** in the same woman (VLPP < CLPP); state the method.
- Invalid during DO (detrusor overactivity) or poor compliance; prolapse raises it (reduce it).
- ISD and hypermobility coexist; VLPP **does not predict surgical outcome** (poor predictor of surgical outcome).
- **VALUE trial (Nager 2012):** for uncomplicated SUI, office evaluation was non-inferior to urodynamics before surgery. AUA/SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2023 / AUA (American Urological Association) 2023 SUI guideline: urodynamics optional in the index patient.
- **Guidelines:** EAU (European Association of Urology): do not use LPP to grade severity.

DLPP: clinical implications

- **McGuire (1981):** DLPP **>40 cmH_2O** in myelodysplastic children → hydronephrosis or VUR (vesicoureteral reflux) in **85%**.
- A strong outlet keeps the patient dry but transmits pressure to the kidneys; a weak outlet "pops off" safely.
- **Hostile bladder** (≥40 cmH_2O) → CIC (clean intermittent catheterisation) plus antimuscarinic; botulinum toxin, augmentation or vesicostomy if needed.
- **Caveat:** 40 is not a validated "safe" value; 20 cmH_2O is more sensitive in children.

Also know: DO leak point pressure (ICS (International Continence Society) 2017); MUCP (maximum urethral closure pressure) ≤20 cmH_2O suggests ISD.

Diagram to draw: Pves, Pabd and Pdet traces (tracing) marking ALPP and DLPP, with leak marked on Valsalva.

> **Key points to remember:**
>
> - VLPP: Pves at leak with Valsalva, no detrusor contraction; 150 mL.
> - ALPP low = weak sphincter; DLPP high = upper-tract danger.
> - VLPP <60 ISD; 60–90 equivocal; >90 (≥90) hypermobility.
> - VLPP < CLPP; catheter size and volume alter it.
> - DLPP >40: upper-tract danger; 85% upper-tract change (McGuire 1981).
> - Lower DLPP is safer; no validated safe cut-off.

Burch colposuspension

Type: SAQ (10 marks)

Source: 2022-Summer Paper 1, Q3b (related: 2017-Winter Paper 3, Q2)

Definition

- Retropubic suspension (**Burch, 1961**) of the paraurethral and para-bladder-neck vaginal wall to **Cooper's (iliopectineal) ligament**; Tanagho (1976) modified it.

Principle

- SUI (stress urinary incontinence) from **urethral hypermobility**: the urethra drops out of the retropubic position. Elevation restores **pressure transmission** and a backstop; scarring holds it.
- It treats hypermobility, not ISD (intrinsic sphincter deficiency).

Indications

- SUI or stress-predominant MUI (mixed urinary incontinence) with hypermobility, especially with **concomitant abdominal surgery** (sacrocolpopexy), **mesh refusal**, or poor vaginal access.
- **Contraindications:** immobile urethra or vagina, ISD alone, pure urgency incontinence.

Work-up

- Cough stress test, PVR (post-void residual), urinalysis; urodynamics only if SUI is not demonstrated.

Steps (open)

1. Low lithotomy, Foley; Pfannenstiel incision; enter the **space of Retzius**.
2. Vaginal finger elevates the vagina; sweep bladder medially. **Avoid midline dissection** over urethra.
3. Two to four 0 or 1-0 delayed-absorbable or non-absorbable sutures each side through vaginal wall (not lumen): distal at mid-urethra, proximal at bladder neck, placed laterally.
4. Pass through Cooper's ligament; tie with vaginal elevation, without tension; leave about **two fingerbreadths** between bladder neck and pubis.
5. **Cystoscopy:** ureteric efflux, no suture in bladder.

- Laparoscopic and robotic versions match open short-term results.

Outcomes

- Cure 85–90% at 1 year, about 70% at 5 years (Cochrane).
- **SISTEr (2007):** autologous fascial sling more effective but more voiding dysfunction and UTI.
- **Ward and Hilton:** TVT (tension-free vaginal tape) equivalent to colposuspension; meta-analyses favour midurethral slings.
- **CARE (2006):** prophylactic Burch at sacrocolpopexy reduced postoperative SUI.

Complications

- Voiding dysfunction 2–27%, de novo storage symptoms 8–27%, **de novo prolapse** 2.5–27%; bladder or ureteric injury; bleeding.
- **Also know:** MMK (Marshall–Marchetti–Krantz; osteitis pubis 0.7–2.5%), paravaginal repair, VOS (vagino-obturator shelf); failure predictors: BMI (body mass index) >25, age >60, prior surgery.

Diagram to draw: sagittal pelvis with sutures from paravaginal fascia to Cooper's ligament.

Key points to remember:

- Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
- Cure 85–90% at 1 year, ~70% at 5 years.
- Niche: concomitant abdominal surgery, mesh avoidance.

Management of perforated mid urethral sling

Type: SAQ (10 marks)

Source: 2023-Winter Paper 1, Q3a

Terminology (IUGA/ICS, International Urogynecological Association / International Continence Society)

- **Perforation:** mesh in a hollow organ (bladder, urethra). **Exposure:** mesh seen through unhealed vaginal epithelium. **Extrusion:** passage out of a tissue. "Erosion" is no longer used.
- Two settings: **intraoperative trocar injury** and **late mesh perforation**.

A. Intraoperative trocar injury

- Bladder injury: 4.5% retropubic versus **0.6% transobturator**; registries 2.7–3.8%.
- **Detection:** cystoscopy with a **70° lens and full bladder distension** after every trocar pass.

Injury	Management
Bladder	Withdraw trocar, **repass more laterally**, confirm on cystoscopy, complete the sling; catheter overnight to 72 h. Benign; no later mesh perforation.
Urethra	**Do not place mesh** (AUA/SUFU, American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction). Repair, catheter drainage; **staged** anti-incontinence surgery after healing.

B. Late perforation of bladder or urethra

- **Presentation:** recurrent UTI (urinary tract infection), haematuria, urgency, pain, stones on mesh, voiding dysfunction, fistula. Often months later.
- **Evaluation:** examination, **cystourethroscopy**, translabial ultrasound (best imaging for mesh); exclude fistula.

Management

- **Bladder:** endoscopic excision (scissors, resection or holmium laser) flush with the wall; or transvesical laparoscopic or robotic excision; open cystotomy if large.
- **Urethra:** transvaginal excision of mesh in the field and layered urethral repair; **Martius flap** if complex or fistula.
- **Continence after excision:** SUI recurs in many (44–100% after synthetic sling revision). Options: concomitant **autologous pubovaginal sling** (Shah 2013: good results) or staged. A damaged urethra predicts poor continence.
- Avoid urethral dilation after mesh slings; it can cause perforation.

Prevention

- Empty bladder; trocar hugging the pubis; bottom-up retropubic route; transobturator route when suitable.

Also know: Amid type I macroporous polypropylene integrates best; chronic pain or infection may need excision from both sides of the bone.

Diagram to draw: retropubic and transobturator trocar paths with bladder at risk.

Key points to remember:

- Bladder trocar injury: repass laterally, continue.
- Urethral injury: abandon mesh; stage the repair.
- Late perforation: excise mesh, repair, consider autologous sling.

TOT in SUI

Type: SAQ (10 marks)

Source: Practice Set 11 Paper 1, Q3d (also answered in 2020-Summer Paper 2, Q3f)

Introduction

- **TOT (transobturator tape):** a **MUS (mid-urethral sling)** of macroporous monofilament polypropylene passed through the obturator foramina instead of the retropubic space. Delorme (2001) designed it to avoid bladder, bowel and vessel injury.

Mechanism

- **Integral theory:** pubourethral ligaments and the suburethral hammock maintain continence at the mid-urethra.
- A loose tape supports the posterior urethral wall, which is compressed against it on straining. It anchors in **obturator internus and externus**.

Indications and evaluation

- **SUI (stress urinary incontinence)** or stress-predominant mixed incontinence after failed PFMT (pelvic floor muscle training).
- History, examination, **objective stress test**, urinalysis, PVR (post-void residual); urodynamics only if complicated. Counsel about permanent mesh and alternatives.

Technique (outside-in)

1. Lithotomy, hips hyperflexed; 1.5 cm midline vaginal incision 1.5 cm from the meatus; dissect to the ischiopubic ramus.
2. Groin stab incision at the notch between inferior pubic ramus and **adductor longus tendon**.
3. Trocar perforates the obturator membrane (felt as a give), turns medially onto the finger and exits the vaginal incision.
4. Cystourethroscopy; check the vaginal sulci.
5. Tension-free: scissors pass easily between tape and urethra.

- **Inside-out** (de Leval): fewer vaginal perforations, more voiding dysfunction.

Outcomes

- **TOMUS (2010):** objective success 77.7% vs 80.8% retropubic, not significantly different.
- Poorer with **ISD (intrinsic sphincter deficiency)** or a **fixed urethra**: retropubic sling preferred.

	TOT	Retropubic
Bladder perforation	0.6%	4.5%
Groin pain	6.4%	1.3%
Voiding dysfunction	Less	More
Long-term repeat surgery	Possibly more	Less

Complications

- Groin or thigh pain, vaginal mesh exposure, voiding dysfunction (sling incision), de novo urgency, UTI (urinary tract infection), rarely thigh abscess.
- **Also know:** SIMS (single-incision mini-sling); alternatives: Burch colposuspension, autologous fascial sling, bulking agents.

Diagram to draw: pelvis with tape under the mid-urethra through both obturator foramina.

Key points to remember:

- Delorme 2001; anchors in obturator muscles.
- TOMUS: equivalent to retropubic (77.7% vs 80.8%).
- Fewer bladder injuries, more groin pain; weaker in ISD.

Describe clinical presentation of pelic floor prolapse. diagnostic evaluation and brief discussion of various surgical options.

Type: SAQ (10 marks)

Source: Practice Set 12 Paper 1, Q3d *Assumed meaning: pelvic organ prolapse (POP) in women.* (pelvic floor anatomy also in Practice Set 2 Paper 1, Q1)

Definition

- POP (pelvic organ prolapse): descent of the anterior wall, posterior wall, uterus/cervix or vault. Anterior > posterior > apical. It is a disease only if symptomatic.
- Cause: loss of **DeLancey support** (level I uterosacral–cardinal apex; II paravaginal to the ATFP [arcus tendineus fasciae pelvis]; III perineal body) and levator ani injury. Risks: vaginal delivery, parity, age, obesity.

Clinical presentation

- **Vaginal bulge** (most specific), pressure, heaviness, worse by evening.
- **Splinting/digitation** to void or defecate (high-grade POP).
- Urinary: voiding difficulty (urethral kinking), OAB (overactive bladder); SUI (stress urinary incontinence) in ~40%, may be **occult**.
- Bowel (rectocele): obstructed defecation, stool trapping. Sexual dysfunction, discharge, ulceration; procidentia may cause hydronephrosis.

Diagnostic evaluation

- Examine supine and standing on strain; split speculum per compartment.
- **POP-Q (Pelvic Organ Prolapse Quantification, 1996):** points Aa, Ba, C, D, Ap, Bp measured from the hymen (above negative), plus gh, pb, tvl.

Stage	Leading edge
0	No prolapse
I	< −1 cm
II	−1 to +1 cm
III	> +1 cm but < (tvl − 2) cm
IV	Complete eversion

- Urinalysis, PVR (post-void residual), diary; **cough test or urodynamics with prolapse reduced** for occult SUI; renal ultrasound (procidentia); dynamic MRI or translabial ultrasound if complex or recurrent; pessary trial confirms POP causes the symptoms.

Surgical options

Type	Operations
Restorative (native tissue)	Anterior colporrhaphy; paravaginal repair; posterior midline plication; perineorrhaphy
Apical (vaginal)	**Uterosacral suspension**, **sacrospinous fixation** (OPTIMAL: similar), vaginal hysterectomy; uterus-sparing hysteropexy, Manchester
Compensatory (abdominal)	**Sacrocolpopexy** (open, laparoscopic, robotic): most durable; mesh exposure ~3–10%
Obliterative	**Le Fort / total colpocleisis**: elderly, not sexually active

- Transvaginal synthetic mesh for POP is withdrawn (FDA order, 2019). Add a sling for occult SUI (OPUS, CARE trials).

Diagram to draw: DeLancey three levels of support.

Key points to remember:

- Bulge is most specific; splinting means high grade.
- POP-Q stage II = −1 to +1 cm.
- Sacrocolpopexy most durable; colpocleisis for elderly.

Female urethral diverticula

Type: SAQ (10 marks)

Source: Practice Set 13 Paper 1, Q3e

Definition

- **UD (urethral diverticulum):** a urine-filled periurethral sac within the periurethral fascia, connected to the urethra by an **ostium**.
- Third to seventh decades; prevalence up to 1–6%.

Aetiology

- Mostly **acquired**. **Routh theory** (most accepted): infection of **periurethral glands** leads to duct obstruction, suburethral abscess, rupture into the lumen and an epithelialised sac.
- Hence the ostium is **posterolateral at mid or distal urethra** in ≥90%.
- **Also know:** childbirth trauma theory (but 20–30% are nulliparous).

Clinical features

- "Three Ds" (dysuria, dyspareunia, post-void dribbling) in only about 5%.
- Commonest: frequency, urgency, pain, **recurrent UTI (urinary tract infection)**.
- Tender anterior vaginal wall mass; about one-third incontinent.
- **Complications:** stones (4–10%); 10% show metaplasia, dysplasia or carcinoma (**adenocarcinoma** commonest).

Evaluation

- Urine culture; **cystourethroscopy** (ostium seen in under half).
- **Imaging sensitivity: MRI (magnetic resonance imaging) 97%**, double-balloon urethrography 95%, ultrasound 82%, VCUG (voiding cystourethrogram) 74%.
- MRI shows saddlebag or circumferential extension.
- **Videourodynamics:** true SUI (stress urinary incontinence) versus dribbling from the sac.
- **Differential diagnosis:** Skene cyst (distal, distorts meatus, no communication), Gartner cyst, leiomyoma, caruncle, bulking agent.

Classification

- **L/N/S/C3 (Leach):** location, number, size, configuration, communication, continence.
- **Also know:** Leng–McGuire (absent fascia = pseudodiverticulum).

Management

- **Asymptomatic or unfit:** suppressive antibiotics, post-void stripping; counsel about cancer.
- **Symptomatic: transvaginal diverticulectomy.**
- **Principles:**
  1. Sterile urine; topical oestrogen if atrophic.
  2. **Inverted-U** anterior vaginal wall flap.
  3. **Preserve periurethral fascia** as a separate layer.
  4. Excise the **entire sac** and close the ostium.
  5. **Watertight** urethral closure over a catheter (as small as 12 Fr).
  6. **Multilayered, non-overlapping** closure; obliterate dead space.
  7. **Martius flap** if poor tissue.
- **Bothersome SUI:** concomitant **autologous pubovaginal sling**; **never synthetic mesh**.

- **Also know:** Spence–Duckett marsupialisation for distal lesions only.
- Pericatheter VCUG at 14–21 days.

Complications

- Recurrence 10–22%; de novo SUI 7–16%; urethrovaginal fistula 1–8%; stricture; recurrent UTI.

Diagram to draw: sagittal section of a mid-urethral sac and its ostium.

> **Key points to remember:**
> - Periurethral gland infection (Routh); posterolateral ostium.
> - MRI most sensitive (97%).
> - Preserve fascia; non-overlapping layers; Martius flap.
> - No synthetic mesh at diverticulectomy.

Endoscopic procedure of SUI

Type: SAQ (10 marks)

Source: Practice Set 13 Paper 1, Q3f

Read as endoscopic injection of urethral bulking agents.

Introduction

- **SUI (stress urinary incontinence)** results from urethral hypermobility and **ISD (intrinsic sphincter deficiency)**.
- **Bulking agents**, injected submucosally under cystoscopic vision, are the least invasive surgical option.

Mechanism

- Bulk at the proximal urethra improves **mucosal coaptation** and resistance.
- It does not correct hypermobility, but most SUI includes some ISD.

Indications

- ISD with a **fixed, immobile urethra** (low VLPP [Valsalva leak point pressure] <60 cmH_2O, historically).
- Women wanting a **low-risk** procedure; elderly or unfit; future pregnancy.
- Persistent leakage after a sling; neurogenic SUI (safer for bladder compliance than a sling).
- **Avoid in:** active UTI (urinary tract infection), untreated detrusor overactivity.

Agents

Agent	Note
Polyacrylamide hydrogel (Bulkamid)	Non-particulate
Polydimethylsiloxane (Macroplastique)	Silicone particles
Calcium hydroxylapatite; carbon-coated zirconium beads	Particulate
Bovine collagen (skin test); PTFE (polytetrafluoroethylene); autologous fat	Withdrawn or abandoned: allergy, migration, granuloma, embolism

Technique

1. Sterile urine; local anaesthesia; cystoscope (0–30° lens) in the urethra.
2. **Transurethral:** needle into the submucosa about 1–1.5 cm distal to the bladder neck; or **periurethral:** needle alongside the urethra, watched cystoscopically.
3. Inject at 2–4 sites until the mucosal cushions **coapt in the midline**.
4. Avoid an indwelling catheter (it moulds the bulk); use CISC (clean intermittent self-catheterisation) for retention.

Outcomes

- Early improvement is common, but cure is lower than with slings and **declines with time**; repeat injections are often needed.
- **Cochrane review (Kirchin 2017):** limited evidence of benefit.
- **AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction):** an option, with counselling on durability.
- **Men:** EAU (European Association of Urology): **do not offer** for post-prostatectomy incontinence.

Complications

- Transient retention, UTI, dysuria, haematuria, urgency.
- Sterile abscess, migration, erosion; may mimic a urethral diverticulum on imaging.

Diagram to draw: cystoscopic view of coapted mucosal cushions after injection.

Key points to remember:

- Bulking improves coaptation; does not correct hypermobility.
- Best for fixed urethra with ISD, or unfit patients.
- Less durable than slings; repeat injections common.
- Not recommended for post-prostatectomy incontinence.

Eponymous techniques of VVF repair

Type: SAQ (10 marks)

Source: Practice Set 22 SAQ Sheet, Q3c (management of VVF also in Practice Set 3 Paper 3, Q2)

Introduction

- VVF (vesicovaginal fistula): an epithelialised bladder–vagina tract. **Principles:** exclude ureteric involvement and cancer; tension-free, multilayer, non-overlapping, watertight closure; interposition when needed; drainage ≥10–14 days. **The first repair has the best chance.**

Transvaginal techniques

- **Sims:** first successful systematic vaginal repair (silver wire).
- **Flap-splitting (Raz)**:
  - Stent ureters if close; Foley through the tract for traction.
  - J or U incision with a 1–3 mm cuff; raise anterior and posterior vaginal flaps.
  - Layer 1: bladder (3-0 or 4-0), with the tract edge. Layer 2: perpendicular perivesical layer. Fill test 200–300 mL.
  - Optional Martius or peritoneal flap; advance the opposite vaginal flap ≥2 cm beyond. Closure ≥90%.
- **Latzko (1942), high partial colpocleisis**:
  - Denude vaginal epithelium 1–2 cm around the tract; the bladder is **not opened or dissected**.
  - Close vaginal walls over the tract in layers.
  - ≥90% success in non-irradiated post-hysterectomy apical VVF; favoured near ureters or after radiation. Drawback: **vaginal shortening and dyspareunia**.
- **Also know:** vaginal cuff excision (all apical scar excised as an inverted cone).

Abdominal technique

- **O'Conor and Sokol (1951)**, suprapubic extraperitoneal:
  - Midline cystotomy at the dome, **bivalved down** to the fistula; ureteric catheters.
  - Excise the tract; separate bladder from vagina 2–3 cm beyond.
  - Close vagina; omentum or peritoneum between; close bladder in layers; suprapubic catheter.
- Modifications: limited-cystotomy transvesical; extravesical. Laparoscopic (1994) and robotic (2005) versions give equal closure with less morbidity.

Interposition flaps

Flap	Blood supply	Use
Martius (1928), labial fibrofatty pad	Internal pudendal (inferior), external pudendal (superior)	Mid–distal VVF; does not reach the apex
Peritoneal	Local	High apical VVF
Omental J-flap	Gastroepiploic	Abdominal, radiation
Gracilis, Singapore	Myocutaneous	Refractory cases

- RCT (randomised controlled trial): Martius adds nothing in simple VVF.

Diagram to draw: Latzko denuded area versus flap-splitting layers.

Key points to remember:

- Latzko: colpocleisis, bladder not opened, vagina shortened.
- O'Conor: bivalve bladder to the fistula, interpose omentum.
- Martius for distal, omentum or peritoneum for apical VVF.

Endometriosis in urology

Type: SAQ (10 marks)

Source: Practice Set 22 SAQ Sheet, Q3d

Definition and epidemiology

- Endometriosis is endometrial glands and stroma outside the uterus. Urinary tract involvement occurs in about 1–5% of women with endometriosis, mostly with deep disease.
- Sites: **bladder** (commonest, about 80%) > ureter (about 15%) > kidney, urethra (rare).

Pathogenesis

- Retrograde menstruation (Sampson), coelomic metaplasia, lymphovascular spread, and **iatrogenic implantation** at pelvic surgery (e.g. caesarean section).
- Bladder lesions start on the **serosa** (vesicouterine pouch) and invade the detrusor.
- Ureteric disease: **extrinsic** (periureteric fibrosis, commoner) or intrinsic (muscularis or mucosa); usually distal, left more often.

Clinical features

- **Bladder:** cyclical haematuria or frequency, urgency and dysuria with menses; suprapubic pain. Differential: BPS (bladder pain syndrome), vesicouterine fistula (menouria).
- **Ureter:** often **silent**; hydronephrosis and loss of renal function may go unnoticed.

Evaluation

- Transvaginal and transabdominal ultrasound gauge size and depth; **renal ultrasound in every woman with deep endometriosis**.
- **MRI (magnetic resonance imaging):** extent; high T1 signal from blood products.
- **Cystoscopy:** limited value (disease is extravesical); may show a bluish submucosal lesion. Biopsy excludes carcinoma.
- CTU (computed tomography urography) and MAG3 (mercaptoacetyltriglycine) renogram for obstruction and split function.
- Staging: rASRM (revised American Society for Reproductive Medicine) I–IV; #Enzian records bladder and ureter.

Management (with gynaecologist)

- **Medical:** progestins (dienogest), combined pill, GnRH (gonadotrophin-releasing hormone) analogues. Symptomatic relief; not for obstructive ureteric disease.
- **Bladder: partial cystectomy** (full-thickness excision with margin), laparoscopic or robotic; stent ureters if near. TUR (transurethral resection) is incomplete, with recurrence and perforation risk.
- **Ureter:**
  - Extrinsic, mild: ureterolysis.
  - Intrinsic or stricture: segmental resection with ureteroureterostomy, or **reimplantation with psoas hitch or Boari flap**.
  - Non-functioning kidney: nephrectomy.
- Treat all pelvic disease at the same operation to lower recurrence.
- **Also know:** risk factor for VVF (vesicovaginal fistula) and ureteric injury at hysterectomy; rare malignant change (clear-cell, endometrioid).

Diagram to draw: endometriotic nodule in the vesicouterine pouch invading the detrusor.

Key points to remember:

- Bladder commonest; cyclical haematuria.
- Ureteric disease silent: screen kidneys.
- Partial cystectomy, not TUR.

Rapid Revision Sheet – Section 6 · Female Urology

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia (LAQ)

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey: I suspension (uterosacral–cardinal; failure causes uterine prolapse), II attachment (ATFP), III fusion (urethral hypermobility); level I is the keystone.
- Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
- Integral theory → mid-urethral sling; hammock → backing, not repositioning.
- Burch → Cooper's ligament; TOT → obturator foramen; PVS → bladder neck.
- Central cystocele rugae lost, paravaginal preserved; POP-Q II = ±1 cm of hymen.
- Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
- Manchester amputates the cervix: poor choice for future pregnancy.

Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications (SAQ)

- VLPP: Pves at leak with Valsalva, no detrusor contraction; 150 mL.
- ALPP low = weak sphincter; DLPP high = upper-tract danger.
- VLPP <60 ISD; 60–90 equivocal; >90 (≥90) hypermobility.
- VLPP < CLPP; catheter size and volume alter it.
- DLPP >40: upper-tract danger; 85% upper-tract change (McGuire 1981).
- Lower DLPP is safer; no validated safe cut-off.

Burch colposuspension (SAQ)

- Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
- Cure 85–90% at 1 year, ~70% at 5 years.
- Niche: concomitant abdominal surgery, mesh avoidance.

Management of perforated mid urethral sling (SAQ)

- Bladder trocar injury: repass laterally, continue.
- Urethral injury: abandon mesh; stage the repair.
- Late perforation: excise mesh, repair, consider autologous sling.

TOT in SUI (SAQ)

- Delorme 2001; anchors in obturator muscles.
- TOMUS: equivalent to retropubic (77.7% vs 80.8%).
- Fewer bladder injuries, more groin pain; weaker in ISD.

Describe clinical presentation of pelic floor prolapse. diagnostic evaluation and brief discussion of various surgical options. (SAQ)

- Bulge is most specific; splinting means high grade.
- POP-Q stage II = −1 to +1 cm.
- Sacrocolpopexy most durable; colpocleisis for elderly.

Female urethral diverticula (SAQ)

- Periurethral gland infection (Routh); posterolateral ostium.
- MRI most sensitive (97%).
- Preserve fascia; non-overlapping layers; Martius flap.
- No synthetic mesh at diverticulectomy.

Endoscopic procedure of SUI (SAQ)

- Bulking improves coaptation; does not correct hypermobility.
- Best for fixed urethra with ISD, or unfit patients.
- Less durable than slings; repeat injections common.
- Not recommended for post-prostatectomy incontinence.

Eponymous techniques of VVF repair (SAQ)

- Latzko: colpocleisis, bladder not opened, vagina shortened.
- O'Conor: bivalve bladder to the fistula, interpose omentum.
- Martius for distal, omentum or peritoneum for apical VVF.

Endometriosis in urology (SAQ)

- Bladder commonest; cyclical haematuria.
- Ureteric disease silent: screen kidneys.
- Partial cystectomy, not TUR.

Section 7 · Voiding Dysfunction — Long answer questions (LAQ)

Describe pathophysiology of OAB. Briefly outline management &describe nonmedical non-pharmacological management

Type: LAQ (25 marks)

Source: Practice Set 4 Paper 1, Q2 (neuromodulation also answered in 2025-Winter Paper 3, Q3e and 2023-Winter Paper 2, Q3b)

Definition

- **OAB (overactive bladder), ICS (International Continence Society):** urgency, usually with frequency and nocturia, with or without UUI (urgency urinary incontinence), without UTI (urinary tract infection) or other obvious pathology.
- OAB is a **symptom** diagnosis. DO (detrusor overactivity) is a **urodynamic** finding; only 44–69% of patients with urgency show DO.

Pathophysiology

All mechanisms converge on **more afferent signal reaching awareness**: the bladder turns the signal up, or the CNS (central nervous system) fails to turn it down.

Mechanism	Explanation
Afferent (sensory)	Aδ fibres normally sense filling. Unmyelinated **C fibres** are "silent", but become spontaneously active and hypersensitive in pathological states
Urothelial	Urothelium acts as a transducer: mechanical, inflammatory or chemical stimuli release ATP (adenosine triphosphate), acetylcholine and other mediators onto suburothelial afferents
Myogenic	Patchy denervation causes receptor upregulation and spontaneous contraction; exaggerated **micromotions** cause urgency, and coordinated spread across the wall gives DO
Neurogenic / central	Forebrain and PAG–PMC (periaqueductal grey–pontine micturition centre) normally inhibit voiding; loss of this inhibition lets "afferent noise" be felt as urgency
Other	Bladder–bowel crosstalk (constipation), ischaemia–reperfusion, chronic inflammation, altered urinary microbiome

- **Risk factors:** ageing, female sex, BOO (bladder outlet obstruction: an animal model, but only an indirect cause in humans; DO persists in up to 60% after TURP [transurethral resection of the prostate]), anxiety and depression, metabolic syndrome, poor toileting habits.

Brief outline of management

- **Assessment:** history, examination, **urinalysis** (minimum); bladder diary (3–7 days); questionnaires; PVR (post-void residual) when indicated. Urodynamics only if it will change an invasive decision.
- **Ladder (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction]):** not a compulsory sequence.

```
Education + behavioural therapy (first line)
        ↓  (4–12 weeks)
Antimuscarinic or β3-agonist (mirabegron, vibegron)
        ↓
PTNS / onabotulinumtoxinA / SNM (third line)
        ↓
Augmentation cystoplasty or urinary diversion
```

- Antimuscarinics: dry mouth, constipation; **dementia risk with use >3 months**, so β3-agonists are preferred in the elderly.

Non-pharmacological management

1. Education and lifestyle

- Explain normal bladder function; set realistic goals.
- Reduce fluid intake by about 25% if excessive; cut caffeine; treat constipation.
- **Weight loss:** about 8 kg loss in obese women reduced UUI episodes.

2. Bladder training (BT) (breaks the cycle of urgency, frequent voiding, smaller capacity)

1. Keep a bladder diary.
2. Start a fixed interval (often 1 hour while awake).
3. Void by the clock, not by urge.
4. Increase by 15–30 minutes a week as control improves.
5. Target interval 3–4 hours.

- Works by **uncoupling the void from the urge**.

3. PFMT (pelvic floor muscle training) and urge suppression

- Kegel exercises: quick 1–2 s flicks and 5–10 s holds, twice daily.
- A pelvic floor contraction **reflexly inhibits the detrusor**. Urge suppression: stop, stay still, squeeze quickly several times, relax, let the urge pass, then walk calmly to the toilet.
- PFMT and BT reduce incontinence episodes by **50–80%**.
- **Also know:** biofeedback helps patients who cannot find the muscle; electrical stimulation at 5–20 Hz inhibits the detrusor; timed and prompted voiding for cognitively impaired patients.

4. PTNS (percutaneous tibial nerve stimulation)

- 34-gauge needle 3–4 cm above the medial malleolus; 20 Hz, 200 µs; confirm great-toe flexion or sole tingling. **30 min weekly for 12 weeks**, then maintenance.
- Posterior tibial nerve (L4–S3) shares sacral roots with the bladder. **SUmiT trial:** 55% vs 21% (sham) responders.

5. SNM (sacral neuromodulation)

- Mechanism: stimulates **afferents** that modulate brainstem and forebrain control.
- **Test phase:** PNE (percutaneous nerve evaluation, 5–7 days) or **staged tined lead** at **S3** (anal "bellows" and great-toe flexion). ≥50% improvement on diary → implant pulse generator in the buttock.
- **InSite trial:** 61% vs 42% (standard medical therapy). Revision >30% at 5 years; MRI (magnetic resonance imaging)-conditional rechargeable devices now available.

6. Intradetrusor onabotulinumtoxinA

- Cleaves SNAP-25 (synaptosomal-associated protein 25), blocking acetylcholine release; also reduces afferent signalling.
- **100 U in 10 mL at 20 sites** cystoscopically; effect lasts about 6–9 months.
- About 25% become completely dry. Risks: UTI about 20%, retention needing CISC (clean intermittent self-catheterisation). **ROSETTA:** greater UUI reduction than SNM at 6 months,

but more UTI.

7. Last resort: augmentation cystoplasty or urinary diversion.

Diagram to draw: bladder showing urothelial afferents, C fibres and micromotions, with the brainstem and forebrain inhibitory pathways.

Key points to remember:

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- BT and PFMT with urge suppression first: 50–80% fewer leaks.
- PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

Section 7 · Voiding Dysfunction — Short answer questions (SAQ)

Detrusor–external sphincter dyssynergia (DESD / DSD)

Asked as:

- External Sphincter Dyssynergia — *Summer 2016 Paper I Q3a (SAQ)*
- Detrusor sphincter dyssynergia — *Practice Set 10 Paper I Q3f (SAQ)* (also answered in 2016-Summer Paper 1, Q3a)

Type: SAQ (10 marks)

Definition

- **DESD / DSD (detrusor–external sphincter dyssynergia):** the striated sphincter contracts involuntarily (involuntary contraction) during detrusor contraction, causing functional obstruction, high pressure and poor emptying.
- **It requires a lesion between the pons and the sacral cord:** suprasacral spinal cord injury (SCI), MS (multiple sclerosis), transverse myelitis, myelomeningocele. Brain lesions (stroke, Parkinson's) do not cause it (**never with brain lesions**).
- Lesions (at or) above T6 may add bladder-neck (**smooth sphincter**) dyssynergia and autonomic dysreflexia.

Blaivas classification

Type	EMG (electromyography) pattern	Lesion
1	Activity rises with detrusor pressure, then relaxes suddenly at peak	Incomplete
2	Sporadic contractions throughout voiding	Complete
3	Crescendo–decrescendo contraction; obstruction throughout	Complete

- Alternative: intermittent versus continuous (continuous means a complete lesion). Type has no relation to lesion level (type is unrelated to level).

Diagnosis

- **Pressure–flow study with EMG:** EMG rises during the detrusor contraction, with high pressure and interrupted flow. Needle EMG is more reliable than surface.
- **VUDS (videourodynamics):** narrowed (narrow) membranous urethra with dilated posterior urethra; shows VUR (vesicoureteral reflux).
- **Exclude:** guarding reflex (normal); **pseudodyssynergia** (straining or suppressing a contraction); dysfunctional voiding (learned, no neurological lesion; Hinman syndrome in children).

Significance

- Worse when complete, continuous and **in men**: about half of untreated men get VUR, hydronephrosis, stones or urosepsis. Other risk factors: high filling pressure, indwelling catheter.
- **Management logic:** DSD is both a storage and an emptying failure. Lower bladder pressure, then empty it.
- DLPP (detrusor leak point pressure) ≥40 cmH_2O endangers the kidneys.

Management

- **First line:** CIC (clean intermittent catheterisation) plus antimuscarinic or β3-agonist; then intradetrusor botulinum toxin.
- **CIC impossible (men):** sphincterotomy or sphincteric botulinum toxin, with condom drainage. The goal is DLPP <40; DLPP ≥40 afterwards means failure; repeat procedures are common.
- Others: suprapubic catheter, Brindley stimulator with deafferentation, diversion. Urethral stents are abandoned; oral drugs disappoint. Credé and Valsalva voiding contraindicated.
- **Also know:** α-blockers for bladder-neck dyssynergia; lifelong upper-tract surveillance.

Diagram to draw: urodynamic trace with rising detrusor pressure (Pdet), EMG bursts and interrupted flow.

> **Key points to remember:**
>
> - Lesion between pons and sacral cord; never brain lesions.
> - Blaivas 1 incomplete; 2 and 3 complete.
> - Worst in men; DLPP ≥40 cmH_2O hostile.
> - CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

Compare and contrast various detrusor relaxants in management of OAB

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 1, Q3e (mirabegron also answered in 2024-Winter Paper 3, Q3d)

Introduction and mechanisms

- In OAB (overactive bladder), drugs follow behavioural therapy. Two classes: **antimuscarinics** and **β3-adrenoceptor agonists**.
- **Antimuscarinics:** acetylcholine acts on **M3** receptors (raising intracellular calcium) to contract the detrusor; the commoner M2 enhances it. Blockade reduces involuntary contractions and afferent signalling.
- **β3 agonists:** β3 is the commonest detrusor adrenoceptor. Activation raises **cAMP (cyclic adenosine monophosphate)** and relaxes the detrusor during storage, sparing the voiding contraction.

Agents

Drug	Distinguishing feature	Usual dose
Oxybutynin	Non-selective; direct relaxant; metabolite causes dry mouth	5 mg two to three times daily; extended-release (ER), patch
Tolterodine	Bladder over salivary selective	ER 4 mg daily
Fesoterodine	Prodrug of tolterodine's active metabolite	4–8 mg daily
Solifenacin	10-fold M3 > M2; long half-life	5–10 mg daily
Darifenacin	About 30-fold M3-selective	7.5–15 mg daily
Trospium	Quaternary; does not cross the blood–brain barrier	20 mg twice daily
β3: mirabegron	Inhibits CYP2D6 (a liver enzyme): metoprolol interaction	25–50 mg daily
β3: vibegron	No CYP2D6 interaction	75 mg daily

Comparison

Feature	Antimuscarinics	β3 agonists
Efficacy	Similar	Similar (TAURUS: mirabegron ≈ tolterodine)
Dry mouth, constipation	Common; main reason for stopping	Near placebo
Cognition	Central M1 blockade; **dementia risk 10–65% higher with use >3 months**	No anticholinergic burden
Retention	Up to 3% in men	Low
Contraindications	Urinary or gastric retention, uncontrolled narrow-angle glaucoma	Uncontrolled hypertension (≥180/110 mmHg)

- **Choice:** in the elderly, prefer a β3 agonist or ER trospium, darifenacin or fesoterodine; ER forms cause less dry mouth.

Combination

- Solifenacin plus mirabegron beats each alone (**SYNERGY**); in non-responders it equals solifenacin 10 mg with less dry mouth (**BESIDE**). In men, add to an α-blocker; watch PVR

(post-void residual).

- **Also know:** intravesical oxybutynin; imipramine; refractory OAB: onabotulinumtoxinA 100 U, neuromodulation.

Diagram to draw: detrusor cell: acetylcholine → M3 → contraction; noradrenaline → β3 → cAMP → relaxation.

Key points to remember:

- M3 mediates contraction; β3 raises cAMP to relax.
- Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
- Solifenacin + mirabegron (SYNERGY) works.

EMG

Type: SAQ (10 marks)

Source: Practice Set 9 Paper 1, Q3b

EMG is taken as urodynamic electromyography of the striated urethral/anal sphincter and pelvic floor.

Definition and principle

- EMG (electromyography) records the electrical potentials of depolarising striated muscle membranes, here the **external urethral sphincter and pelvic floor**, during filling and voiding.
- It shows **coordination** between detrusor and sphincter, not sphincter strength. The bladder neck (smooth muscle) is invisible to EMG.
- External anal sphincter activity mirrors the urethral sphincter, except in some neurological diseases.

Electrodes

Surface (patch)	Concentric needle
Perianal patches at 3 and 9 o'clock (ICS [International Continence Society] standard: two skin electrodes plus reference)	Needle into the sphincter (girls beside the meatus; boys transperineal)
Comfortable; poorer signal; movement artefact	Best signal; painful; dislodges
Coordination only	Also motor unit analysis: acute versus chronic denervation

Normal pattern

- Filling: activity **rises gradually** (recruitment, guarding reflex).
- Voiding: EMG **falls silent before** detrusor pressure rises and stays silent until flow ends.
- **Lag time** (uroflow with patch EMG in children): relaxation to flow 2–6 s is normal; <2 s suggests DO (detrusor overactivity); >6 s suggests primary bladder neck dysfunction.

Abnormal patterns

Finding	Meaning
DSD (detrusor–sphincter dyssynergia)	Sphincter contracts during detrusor contraction; needs a **suprasacral** cord lesion (SCI [spinal cord injury], MS [multiple sclerosis]). Blaivas types 1–3
Dysfunctional voiding	Same tracing without neurological lesion (learned)
Pseudodyssynergia	EMG rise from straining or trying to suppress a contraction
Denervation potentials	Sacral cord or cauda equina lesion; MSA (multiple system atrophy)
Complex repetitive discharges, decelerating bursts	**Fowler syndrome** (retention in young women); predicts SNM (sacral neuromodulation) response

Clinical uses

- Neurogenic bladder (spina bifida, SCI); voiding dysfunction in children; female retention; before sphincter surgery.
- **Limitation:** surface EMG misdiagnoses some dysfunctional voiders; **VUDS (videourodynamics)** is more accurate.

Diagram to draw: tracing of Pdet, flow and EMG: EMG silent during normal voiding versus active in DSD.

Key points to remember:

- EMG judges coordination, not strength.
- True DSD needs a suprasacral lesion.
- Lag time 2–6 s normal.
- Fowler syndrome: complex repetitive discharges on needle EMG.

Botulinum toxin in urology: mechanism, indications and doses

Asked as:

- Botulimum toxin — *Practice Set 9 Paper I Q3e (SAQ)*
- MOA of botulinum toxin. Indications and dosages in UROLOGY — *Practice Set 13 Paper I Q3a (SAQ)*

Type: SAQ (10 marks)

(Practice Set 9: also answered in 2025-Winter Paper 2, Q3d. Practice Set 13: also answered in 2025-Winter Paper 2, Q3d and 2017-Winter Paper 3, Q3b)

"Botulimum" is read as botulinum toxin.

Introduction

- Neurotoxin of *Clostridium botulinum*, serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**, the only licensed formulation: a heavy chain (100 kDa) and a zinc-dependent light chain (50 kDa).

Mechanism of action

1. The heavy chain binds the cholinergic nerve terminal and the toxin is internalised.
2. The light chain (a zinc protease) **cleaves SNAP-25** (synaptosomal-associated protein 25), a SNARE docking protein. Vesicles cannot fuse, so **acetylcholine release stops**: chemical denervation.

- **Afferent action:** less ATP (adenosine triphosphate), substance P and CGRP (calcitonin gene-related peptide) release; fewer TRPV1 (capsaicin) and P2X3 (purinergic) sensory receptors, so urgency falls (urgency and pain fall).
- Effect wanes (lasts 6–9 months) as nerve terminals sprout (6–9 months; until terminals sprout), so injections are repeated; repeat injections stay effective.

Technique

- Cystoscopic (intra)detrusor injection, **20 sites (100 U, in 10 mL) or 30 sites (200 U, in 30 mL)**, usually sparing trigone (trigone usually spared); check PVR (post-void residual) at 2 weeks.

Urological uses: indications and doses

Indication	Dose and evidence
Idiopathic OAB (overactive bladder), third line	**100 U** (EMBARK). ABC trial: equal to antimuscarinic, more complete dryness. ROSETTA: comparable to SNM (sacral neuromodulation), more UTI (urinary tract infection)
NDO (neurogenic detrusor overactivity): SCI (spinal cord injury), MS (multiple sclerosis)	**200 U** (DIGNITY): less incontinence, larger capacity, lower pressure; works with or without antimuscarinics
Paediatric NDO (spina bifida)	About **10 U/kg (maximum 360 U)**, about 30 sites, in published series (Campbell series); FDA (Food and Drug Administration) label 6 U/kg, maximum 200 U
DSD (detrusor–sphincter dyssynergia)	Intrasphincteric injection, mainly men
IC/BPS (interstitial cystitis/bladder pain syndrome)	Intradetrusor, often trigonal; AUA option, off-label (option)
BPH (benign prostatic hyperplasia)	100–300 U intraprostatic: **no benefit** over placebo; large RCTs (randomised controlled trials) equal to saline; not used

- **Also know:** pelvic floor spasm; chronic orchialgia (RCT negative).

Complications and contraindications

- **Retention needing CISC (clean intermittent self-catheterisation)**, commoner (more) in NDO; each 10 mL of baseline PVR raises the risk about 27%. The patient must accept CISC beforehand.
- UTI (urinary tract infection), haematuria; rarely generalised weakness. Contraindications (avoid in): active UTI, myasthenia gravis, pregnancy.

Diagram to draw: cholinergic synapse showing (with) SNAP-25 cleavage by the light chain.

> **Key points to remember:**
> - Cleaves SNAP-25; efferent and afferent action.
> - 100 U idiopathic OAB; 200 U NDO.
> - Children 10 U/kg (maximum 360 U); FDA label 6 U/kg (maximum 200 U).
> - Main risks: retention needing CISC (the main risk), UTI.

Rapid Revision Sheet – Section 7 · Voiding Dysfunction

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Describe pathophysiology of OAB. Briefly outline management &describe nonmedical non-pharmacological management (LAQ)

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- BT and PFMT with urge suppression first: 50–80% fewer leaks.
- PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

Detrusor–external sphincter dyssynergia (DESD / DSD) (SAQ)

- Lesion between pons and sacral cord; never brain lesions.
- Blaivas 1 incomplete; 2 and 3 complete.
- Worst in men; DLPP ≥40 cmH_2O hostile.
- CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

Compare and contrast various detrusor relaxants in management of OAB (SAQ)

- M3 mediates contraction; β3 raises cAMP to relax.
- Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
- Solifenacin + mirabegron (SYNERGY) works.

EMG (SAQ)

- EMG judges coordination, not strength.
- True DSD needs a suprasacral lesion.
- Lag time 2–6 s normal.
- Fowler syndrome: complex repetitive discharges on needle EMG.

Botulinum toxin in urology: mechanism, indications and doses (SAQ)

- Cleaves SNAP-25; efferent and afferent action.
- 100 U idiopathic OAB; 200 U NDO.
- Children 10 U/kg (maximum 360 U); FDA label 6 U/kg (maximum 200 U).
- Main risks: retention needing CISC (the main risk), UTI.

Section 9 · Andrology & Sexual Medicine — Long answer questions (LAQ)

Spermatogenesis, azoospermia, non-obstructive azoospermia and maturation arrest

Asked as:

- Describe physiology of spermatogenesis and discuss management of azoospermia. — *Winter 2017 Paper I Q2 (LAQ)*
- Non obstructive azoospermia — *Summer 2019 Paper I Q3d (SAQ)*
- Maturation Arrest. — *Practice Set 10 Paper I Q3e (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Maturation Arrest taken as spermatogenic maturation arrest on testicular histology.

Hormonal control: HPG (hypothalamic–pituitary–gonadal) axis

- Hypothalamus releases **GnRH (gonadotropin-releasing hormone)** in pulses; pulsatility is essential (constant agonist exposure shuts the axis down). Kisspeptin triggers puberty.
- **LH (luteinising hormone)** → Leydig cells → testosterone (about 5 mg/day). **FSH (follicle-stimulating hormone)** → Sertoli cells.
- Feedback: testosterone (and oestradiol) on LH; **inhibin B** from Sertoli cells on FSH. So FSH and inhibin B report the tubules, LH reports Leydig cells.
- **Intratesticular testosterone** is about 50-fold higher than serum (held by **ABP (androgen-binding protein)**). This is why exogenous testosterone suppresses spermatogenesis. Intratesticular testosterone is essential for postmeiotic maturation.
- Testis is kept 2–4 °C below core temperature by the pampiniform countercurrent exchange.

Sertoli cells and blood–testis barrier

- Nurse cells: nourish germ cells, secrete ABP, inhibin, fluid; phagocytose residual bodies.
- **Sertoli–Sertoli tight junctions** divide the tubule into a **basal** compartment (spermatogonia) and an **adluminal** compartment (spermatocytes, spermatids). Post-meiotic cells are "foreign" to the immune system; the barrier and local tolerance protect them.

Phases of spermatogenesis (about 64 days; 42–76 days)

Phase	Events
1. Proliferation (spermatocytogenesis)	Dark type A (Ad, reserve stem cells) self-renew; pale type A (Ap) divide every 16 days into type B spermatogonia → primary spermatocytes. Daughter cells stay joined by cytoplasmic bridges
2. Meiosis	Primary spermatocytes cross into the adluminal compartment. Meiosis I (reduction division; crossing-over at synapsis) → secondary spermatocytes → meiosis II → 4 haploid round spermatids
3. Spermiogenesis	No division: acrosome from Golgi, flagellum from centriole, nucleus condenses to about 10%, mitochondria form the midpiece, cytoplasm shed
4. Spermiation	Sertoli cell releases sperm into the lumen

- **Cycle, stage, wave:** six stages in humans; cycles arranged as a helical wave, so production is continuous (about 123 million sperm a day).

- Defective meiotic recombination causes maturation arrest and sperm aneuploidy.

Epididymal maturation and transport

- Testicular sperm are immotile. During 2–12 days of epididymal transit they gain **progressive motility** and fertilising ability; stored in the cauda.
- **Capacitation** and the acrosome reaction occur in the female tract.
- **Clinical relevance:** a new semen sample reflects events about 3 months earlier.

Azoospermia: definition and classification

- No sperm in the **centrifuged** pellet (ejaculate) on **two** semen analyses (samples); 10–15% of infertile men.
- **NOA (non-obstructive azoospermia):** due to **failure of spermatogenesis** (testicular failure). About 60% of azoospermic men.

Type	Causes	Clues
Pre-testicular	Kallmann syndrome, pituitary tumour, hyperprolactinaemia, **exogenous testosterone/anabolic steroids**	Low FSH, LH, testosterone
Testicular (NOA (non-obstructive azoospermia))	Klinefelter, Y microdeletion, cryptorchidism, mumps, chemotherapy/radiation	Small testes, high FSH
Post-testicular (OA (obstructive azoospermia))	CBAVD (congenital bilateral absence of the vas), epididymal obstruction, vasectomy, hernia repair, EDO (ejaculatory duct obstruction)	Normal testes and FSH
Ejaculatory	Retrograde ejaculation, anejaculation	Low volume

Causes of NOA

- **Genetic: Klinefelter (47,XXY)**, the commonest; **Y microdeletion** (AZF (azoospermia factor) a, b, c); other karyotype anomalies.
- **Acquired:** cryptorchidism, mumps orchitis, torsion, **chemotherapy or radiation**, trauma, varicocele.
- **Idiopathic**: commonest overall.
- *Note:* hypogonadotropic hypogonadism is pre-testicular and treatable medically (hCG (human chorionic gonadotropin) ± FSH (follicle-stimulating hormone)).

Evaluation

```
Azoospermia (2 centrifuged samples)
 ├─ Vasa absent → CBAVD → CFTR (cystic fibrosis gene) testing, both partners; renal scan
 ├─ Volume <1 mL → post-ejaculate urine: sperm = retrograde; none → TRUS (transrectal
ultrasound) for EDO
 └─ Normal volume: FSH ≤7.6 IU/L + testis long axis >4.6 cm → obstructive
                   FSH >7.6 + testis ≤4.6 cm → spermatogenic failure
```

- History (testosterone use, surgery, chemotherapy), examination (testes, vasa, epididymis, varicocele), hormones (FSH, LH, testosterone, oestradiol, prolactin).
- NOA: small soft testes, palpable vasa, normal ejaculate volume. **FSH >7.6 IU/L and testis long axis ≤4.6 cm** point to NOA (FSH ≤7.6 with normal testes suggests obstruction).
- **Genetics (AUA/ASRM (American Urological Association/American Society for Reproductive Medicine)):** karyotype and **Y microdeletion** in all NOA (in all azoospermic men); counselling. AZF (azoospermia factor) **a or b complete deletion: no sperm, do not operate**; AZFc: sperm often found.
- TRUS: seminal vesicle >1.5 cm wide or midline cyst suggests EDO.
- **Histology patterns (three patterns in NOA):**

Pattern	Histology / meaning	Micro-TESE retrieval
Hypospermatogenesis	All stages (present), fewer cells	Best
Maturation arrest	Halts at one (a germ-cell) stage	About 40%
SCO (Sertoli cell–only)	No germ cells	About 30%

Maturation arrest

Definition

- A histological pattern of **spermatogenic failure** in which germ cells develop normally up to one stage and **the sequence halts** at a premeiotic, meiotic or postmeiotic step. No mature sperm, or very few, are made.

Classification

- **Early:** arrest at spermatogonia or **primary spermatocyte** (meiotic failure).
- **Late:** arrest at the **spermatid** stage.
- **Complete** (all tubules) versus **incomplete** (foci progress to sperm): incomplete gives better retrieval.

Causes

- **Genetic:** complete **AZFb (azoospermia factor b) deletion** → arrest; meiotic recombination defects in about half of men with maturation arrest; karyotype anomalies.
- **Acquired:** heat (varicocele, cryptorchidism), chemotherapy, radiation, drugs, orchitis; idiopathic often.

Clinical features and diagnosis

- Azoospermia or severe oligospermia, normal semen volume, palpable vasa.
- **Testis size and FSH (follicle-stimulating hormone) are often normal**, so it can mimic obstruction (FSH ≤7.6 IU/L, long axis >4.6 cm).
- **Diagnosis needs testicular biopsy** (Bouin's fixative, never formalin), usually done as sperm extraction.
- **Karyotype and Y microdeletion** testing in all azoospermic men (AUA/ASRM).

Management of maturation arrest

- Optimise: treat varicocele, stop exogenous testosterone; off-label hormonal stimulation (clomiphene, hCG [human chorionic gonadotropin]) unproven.
- **Micro-TESE (microdissection testicular sperm extraction)** with ICSI (intracytoplasmic sperm injection): retrieval **about 40%** (versus 30% in SCO).
- **Complete AZFb deletion: retrieval highly unlikely; do not operate.**
- No sperm: donor sperm or adoption. Round spermatid injection is experimental.
- Genetic counselling: sons inherit Y deletions.

Management of azoospermia

1. Pre-testicular (hypogonadotropic)

- Stop exogenous testosterone. **hCG (human chorionic gonadotropin) 1500–5000 IU 2–3 times weekly ± FSH**; pulsatile GnRH in hypothalamic failure; dopamine agonist for prolactinoma.

2. Obstructive

- Microsurgical **vasovasostomy** or **vasoepididymostomy** (clear watery fluid or sperm parts in vasal fluid → vasovasostomy; thick toothpaste-like fluid without sperm → vasoepididymostomy).
- **TURED (transurethral resection of the ejaculatory ducts)** for EDO.

- CBAVD or unreconstructable: **MESA (microsurgical epididymal sperm aspiration)** or PESA (percutaneous epididymal sperm aspiration) + **ICSI (intracytoplasmic sperm injection)**; freeze sperm at reconstruction.

3. Non-obstructive

- **Optimisation (before retrieval):** stop exogenous testosterone; repair a clinical varicocele (no definitive evidence in NOA); optimise hormones: hormonal therapy off-label (clomiphene, anastrozole, hCG for low testosterone).
- **Micro-TESE (microdissection testicular sperm extraction)** + ICSI.
  - **Principle:** spermatogenesis in NOA is **focal**. Under ×20–25 magnification, **larger, opaque** (dilated opaque) **tubules** are chosen under the microscope; least tissue removed, blood supply preserved.
  - Sperm found in **about 50%** (about half); better than conventional TESE (Klinefelter 68%, Sertoli-cell-only 30%).

Predictor	Retrieval (%)
Post-orchidopexy	74
AZFc deletion	70
Klinefelter	68
Post-chemotherapy	53
AZFa or AZFb complete deletion	**0: do not operate**

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- FSH and testis size **do not** reliably predict retrieval.
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- **ICSI (intracytoplasmic sperm injection):** fresh or frozen testicular sperm. Live birth per cycle modest; miscarriage higher (sperm DNA damage). Male offspring of AZF deletion carriers inherit it.

4. Ejaculatory

- Retrograde: sympathomimetics or sperm from alkalinised urine. Anejaculation: vibratory or electroejaculation.

5. Failure (no sperm): donor sperm or adoption; genetic counselling.

- **Also know:** spermatogonial stem cell renewal by GDNF (glial cell line–derived neurotrophic factor); in-vitro spermatogenesis and stem-cell transplantation are experimental.

Diagram to draw: cross-section of a seminiferous tubule showing basal and adluminal compartments with germ cells maturing towards the lumen. For NOA: opened testis under microscope: plump tubules versus thin Sertoli cell–only tubules. For maturation arrest: seminiferous tubule showing germ cells up to spermatocytes with no spermatids or sperm.

Key points to remember:

- Spermatogenesis about 64 days; Ap divide every 16 days; 6 stages.
- Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
- FSH ≤7.6 IU/L and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in all.
- AZFa/b deletion: do not operate (no retrieval); micro-TESE finds sperm in about half of NOA; SCO 30%.
- Maturation arrest: arrest at one stage; early (spermatocyte) or late (spermatid).
- FSH and testis size may be normal: biopsy diagnoses.
- Micro-TESE retrieval in maturation arrest about 40%; none with complete AZFb deletion.

Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED

Asked as:

- Write in detail about physiology of erection. How will you evaluate a diabetic with ED? — *Practice Set 5 Paper I Q2 (LAQ)*
- Neurotransmitters in erectile function — *Practice Set 12 Paper I Q3e (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(erection physiology also in Practice Set 5 Paper 1, Q2)

Introduction

- Erection is **cavernous smooth-muscle relaxation**: low cytosolic Ca^{2+} dilates helicine arteries and sinusoids; expanding sinusoids compress emissary veins (veno-occlusion).
- Pathways: parasympathetic **S2–S4** cavernous nerves (erection), sympathetic **T11–L2** (detumescence), pudendal (sensation, ischiocavernosus rigidity).

Functional anatomy

- **Corpora cavernosa:** sinusoids lined by endothelium, separated by **trabecular smooth muscle**. Flaccid tone depends on smooth-muscle contraction.
- **Tunica albuginea:** inner circular and outer longitudinal layers. **Emissary veins** pass obliquely between them and are compressed when the corpora expand: the **veno-occlusive mechanism**.
- **Arteries:** internal pudendal → common penile → **cavernous (helicine branches)**, dorsal and bulbourethral arteries. Accessory pudendal arteries in about one-third.
- Spongiosum and glans have a thin or absent tunica, so no veno-occlusion; they stay low-pressure.

Haemodynamic phases

Phase	Events	Intracavernous pressure
Flaccid	Smooth muscle contracted; low inflow	Low
Latent (filling)	Arterial dilation; **inflow peaks**	Barely rises
Tumescence	Sinusoids fill; venous outflow compressed	Rising
Full erection	Maximal veno-occlusion	About 100 mmHg
Rigid	**Ischiocavernosus** contraction (pudendal nerve)	Several hundred mmHg
Detumescence	Initial, slow, fast phases	Falls to baseline

Molecular mechanism and peripheral neurotransmitters

- **Relaxation (erection):** NO (nitric oxide) from **nNOS (neuronal nitric oxide synthase)** in cavernous nerves initiates erection; endothelial NOS (eNOS) sustains it.
  - NO → soluble guanylyl cyclase → **cGMP (cyclic guanosine monophosphate)** → protein kinase G (PKG) → K^+ channels open, Ca^{2+} channels close, Ca^{2+} sequestered → **low cytosolic Ca^{2+}** → relaxation.
  - Second route: PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide) → **cAMP (cyclic adenosine monophosphate)** → protein kinase A.
  - **PDE5 (phosphodiesterase type 5)** degrades (breaks down) cGMP and ends relaxation (ends / terminates erection); PDE5 inhibitors block it.

Peripheral pro-erectile transmitters

Transmitter	Source / mechanism
NO (nitric oxide)	Principal. **nNOS (neuronal NOS)** initiates; **eNOS (endothelial)** sustains. NO → soluble guanylyl cyclase → **cGMP** → PKG → K^+ channels open, Ca^{2+} channels close → relaxation
ACh (acetylcholine)	Indirect: endothelial NO release, inhibits adrenergic nerves; atropine does not block erection
PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide), CGRP (calcitonin gene-related peptide)	**cAMP** → PKA (protein kinase A); VIP role minimal in man
CO, H2S, EDHF (endothelium-derived hyperpolarising factor)	Gaseous/endothelial relaxants via cGMP or K^+ channels

- O2 is a NOS substrate: low flaccid Po2 (~35 mmHg) limits NO; erect Po2 ~90 mmHg.
- **Also know:** carbon monoxide and hydrogen sulfide also raise cGMP; acetylcholine acts indirectly through endothelial NO (atropine does not block erection).

Anti-erectile transmitters (flaccidity and detumescence)

- **Contraction (flaccidity): noradrenaline (α1)** (principal), endothelin-1, angiotensin II → raised Ca^{2+} → calmodulin → myosin light-chain kinase → contraction.
  - **Noradrenaline (α1)**: principal; α2 prejunctional reduces noradrenaline and NO release.
  - **Endothelin-1** (ET-A), angiotensin II (AT1), PGF2α, thromboxane A2: Gq → IP3 → Ca^{2+} rise → Ca^{2+}–calmodulin → MLC (myosin light-chain) kinase → contraction.
- **Rho-kinase** inhibits myosin (MLC) phosphatase (calcium sensitisation) and maintains (sustaining) tone.
- **Detumescence:** NO release stops, PDE5 degrades cGMP and sympathetic discharge contracts the smooth muscle, so emissary veins reopen.
- Gap junctions (connexin-43) synchronise the smooth muscle.

Neural control

Pathway	Origin	Role
Parasympathetic	**S2–S4** → pelvic plexus → cavernous nerves	Erection
Sympathetic	**T11–L2** → hypogastric nerves	Detumescence, emission
Somatic	Pudendal nerve (Onuf nucleus)	Penile sensation; ischiocavernosus rigidity

- **Three types: psychogenic** (brain, chiefly by reducing sympathetic tone), **reflexogenic** (genital touch, sacral arc; preserved in upper cord injury), **nocturnal** (REM [rapid eye movement] sleep).
- **Central:** MPOA (medial preoptic area) and **PVN (paraventricular nucleus)**. Dopamine and oxytocin promote; serotonin, prolactin, GABA (γ-aminobutyric acid) and opioids inhibit.

Central neurotransmitters (PVN [paraventricular nucleus], MPOA [medial preoptic area])

Promote	Inhibit
Dopamine (D2 in PVN; apomorphine)	**Serotonin** (5-HT1A, 5-HT2; but 5-HT2C pro-erectile)
Oxytocin	GABA (γ-aminobutyric acid)
Central NO	Opioids (high dose), cannabinoids
Central noradrenaline; melanocortins (MC4R)	**Prolactin**

- Dopamine and oxytocin converge on NO in the PVN. Noradrenaline is pro-sexual centrally but anti-erectile peripherally.

Clinical relevance

- PDE5 inhibitors, intracavernosal alprostadil (cAMP), papaverine, phenylephrine for priapism (α1), SSRIs (selective serotonin reuptake inhibitors) cause dysfunction, hyperprolactinaemia lowers libido.

Diagram to draw: NO–cGMP pathway in the cavernous smooth-muscle cell.

Evaluation of a diabetic with ED (erectile dysfunction)

Why diabetics: ED is commoner, earlier and more severe in DM (diabetes mellitus). Mechanisms: endothelial NO loss, autonomic neuropathy, arteriopathy, smooth-muscle loss and fibrosis (veno-occlusive failure), AGEs (advanced glycation end products) and low testosterone. ED is a **sentinel of cardiovascular disease**.

1. History

- Onset, duration (≥3 months), severity, morning erections, libido, ejaculation; partner.
- **IIEF-5 (International Index of Erectile Function, SHIM):** 5–7 severe, 8–11 moderate, 12–16 mild-to-moderate, 17–21 mild, 22–25 no ED.
- Diabetes: type, duration, **glycaemic control**, complications (retinopathy, nephropathy, neuropathy, foot disease), autonomic symptoms (postural giddiness, gastroparesis, bladder dysfunction).
- Cardiovascular: angina, exercise tolerance, smoking, hypertension, dyslipidaemia, nitrates.
- Drugs: antihypertensives, antidepressants. Depression screen.

2. Examination

- BMI, waist, blood pressure, peripheral pulses.
- Genitalia: Peyronie plaques, testicular size, gynaecomastia.
- Neurological: genital sensation, **bulbocavernosus reflex**, foot sensation. Penile neuropathy can exist without foot neuropathy.

3. Laboratory

- **HbA1c (glycated haemoglobin)** and fasting glucose, **lipid profile**, creatinine/eGFR (estimated glomerular filtration rate), urine albumin.
- **Morning total testosterone**; if low, repeat with free testosterone, LH (luteinising hormone), prolactin. TSH (thyroid-stimulating hormone) if indicated.

4. Cardiovascular risk stratification (Princeton Consensus)

- **High risk** (unstable angina, recent infarction, uncontrolled hypertension, some arrhythmias): defer sexual activity and treatment; cardiology referral. **Indeterminate:** cardiac assessment (e.g. stress test), then restratify. **Low:** start treatment.

5. Specialised tests (selected cases)

- **CIS (combined intracavernous injection and stimulation):** alprostadil 10–20 µg; a rigid erection excludes major vascular disease.
- **Penile duplex Doppler:** PSV (peak systolic velocity) **<25 cm/s = arterial insufficiency** (>35 normal); EDV (end-diastolic velocity) **>5 cm/s** and RI (resistive index) **<0.75 = veno-occlusive dysfunction**.
- **NPTR (nocturnal penile tumescence and rigidity), RigiScan:** separates psychogenic from organic ED.
- **Neurophysiological:** bulbocavernosus reflex latency, dorsal nerve conduction velocity, biothesiometry.
- DICC (dynamic infusion cavernosometry and cavernosography) and angiography: rarely, as diffuse diabetic vascular disease is not suited to revascularisation.

```
Diabetic with ED → history + IIEF-5 + examination
   → HbA1c, lipids, testosterone, renal function
   → Princeton cardiac risk → treat risk factors
   → PDE5 inhibitor trial → failure → duplex/CIS → ICI or prosthesis
```

Key points to remember:

- nNOS initiates, eNOS maintains; PDE5 ends (terminates) erection.
- Noradrenaline α1 is the main anti-erectile transmitter.
- Dopamine/oxytocin promote; serotonin/prolactin inhibit.
- Rigid phase: ischiocavernosus, suprasystolic pressure.
- Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
- Diabetic ED: check HbA1c, testosterone and cardiac risk.

Section 9 · Andrology & Sexual Medicine — Short answer questions (SAQ)

Male reproductive immunology

Type: SAQ (10 marks)

Source: 2015-Summer Paper 1, Q3e

Introduction

- Haploid germ cells appear at puberty, after self-tolerance is established, so their antigens are "foreign". The testis is therefore **immune-privileged**.

Immune privilege

Anatomical barrier: blood–testis barrier (BTB)

- **Tight junctions between Sertoli cells**, formed at puberty, divide a **basal compartment** (spermatogonia, plasma-like fluid) from an **adluminal compartment** (later spermatocytes, spermatids).

Functional barrier

- A small antigen leak at the **rete testis, efferent ductules and epididymis** induces suppressor T cells and tolerance; a large load provokes immunity.
- Sertoli and local mediators: galectin-1, TGF-β (transforming growth factor-β), interferon-γ, soluble Fc receptor, **androgens**.
- Insult before puberty does not induce antibodies; the barrier also limits chemotherapy (isolated testicular relapse).

Antisperm antibodies (ASA)

- **Causes:** obstruction (**vasectomy** commonest), torsion, trauma, biopsy, infection, heat (varicocele, cryptorchidism), genetic (human leukocyte antigen HLA-B28).
- **Effects:** agglutination, poor motility, failure to cross cervical mucus and bind the zona; head-bound antibody matters most.
- **Indications to test:** isolated asthenozoospermia, agglutination, unexplained infertility.

Test	Method
Mixed antiglobulin reaction (MAR)	Fresh semen + immunoglobulin (Ig) G/A-coated beads + antiglobulin; beads stick to antibody-coated motile sperm
Immunobead test	Washed sperm + beads coated with anti-human IgG/IgA; gives % and site (head/tail) of binding

- **≥50% of motile sperm bound** is significant (WHO [World Health Organization] 2010).

Treatment	Comment
Corticosteroids	Trials conflict; risk of avascular necrosis; not recommended
Sperm washing + intrauterine insemination	Bypasses cervical mucus; modest benefit (bound antibody remains)
IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)	Preferred; fertilisation largely unaffected

Infection and inflammation

- **Leukocytospermia:** >1 × 10^6 peroxidase-positive leukocytes/mL. Reactive oxygen species and cytokines damage sperm.
- **Orchitis:** mumps (post-pubertal, atrophy); bacterial (*Chlamydia*/gonococcus if young, *E. coli* if older); tuberculous; **autoimmune**, primary (antibodies with azoospermia) or secondary to vasculitis.
- **Also know:** sperm DNA fragmentation testing; prostatitis harms sperm via oxidative stress and ASA; seminal plasma suppresses female tract immunity.

Diagram to draw: Seminiferous tubule with Sertoli tight junctions and compartments.

Key points to remember:

- BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
- ASA: vasectomy commonest; ≥50% bound significant; IVF/ICSI.
- Leukocytospermia >1 million/mL.

Management of Peyronie's disease

Type: SAQ (10 marks)

Source: 2017-Winter Paper 1, Q3a

Definition

- **PD (Peyronie's disease)** (AUA (American Urological Association) 2015): acquired fibrosis of the tunica albuginea, with pain, deformity, ED (erectile dysfunction) and/or distress. Prevalence 3–20%; up to half worsen untreated.

Phases

- **Acute (active, first 12–18 months):** pain, worsening deformity; aim to stop progression. **Stable:** pain gone, deformity unchanged for ≥3–6 months; aim to straighten.

Evaluation

- History: pain, deformity, Dupuytren disease, ability to penetrate. **Key question:** "If your penis were straight, would your current rigidity allow intercourse?" (predicts postoperative function).
- Plaque, SPL (stretched penile length); **office intracavernosal injection ± duplex Doppler ultrasound** before any invasive treatment (curvature by goniometer, calcification, vascular status).
- **Kelami grading:** ≤30° mild, 31–60° moderate, >60° severe; note hourglass or hinge. PDQ (Peyronie's Disease Questionnaire).

Non-surgical treatment

Option	Evidence/position
NSAIDs (non-steroidal anti-inflammatory drugs)	For pain (only oral agent AUA supports)
Oral vitamin E, tamoxifen, potaba, pentoxifylline	No reliable curvature benefit; not recommended
CCH (collagenase *Clostridium histolyticum*)	Stable disease, curve 30–90°, intact erections. **IMPRESS I/II:** 34% improvement vs 18% placebo. Up to 4 cycles (2 injections of 0.58 mg) with modelling. Risk: corporal fracture
Verapamil, interferon α-2b (intralesional)	May offer; stabilise and modestly improve
Traction therapy	Gains length, reduces curvature; useful in acute phase
ESWT (extracorporeal shock wave therapy)	Pain only; no effect on curvature

Surgery

- **Indications:** stable disease (≥12 months, stable ≥6 months), deformity preventing intercourse, failed conservative therapy.

```
Erection adequate (± PDE5I (phosphodiesterase-5 inhibitor))?
 ├─ No  → penile prosthesis (inflatable) + manual modelling;
 │        residual >30° → tunical incision; graft if defect >2 cm
 └─ Yes → curve <60–70°, no hourglass/hinge → plication (Nesbit, 16-dot, tunica albuginea
plication)
          curve >60–70°, hourglass, hinge, short penis → plaque incision/partial
          excision + graft (pericardium, small intestinal submucosa)
```

- **Plication** shortens the convex side: simple, low ED risk, but loses length.
- **Grafting** restores length and girth but carries higher ED risk (5–54%).

Diagram to draw: Nesbit wedge excised opposite the plaque.

> **Key points to remember:**
> - Ask the "straight penis" question; duplex before surgery.
> - CCH: IMPRESS, stable 30–90° curve, intact erections.
> - Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

Haematospermia

Asked as:

- Hematospermia — *Winter 2025 Paper I Q3c (SAQ)*
- Hematospermia — *Practice Set 9 Paper I Q3d (SAQ)* (also answered in 2025-Winter Paper 1, Q3c)

Type: SAQ (10 marks)

Definition

- Blood in the ejaculate. It alarms patients but is usually **benign and self-limiting**, most often from non-specific prostatic or seminal vesicle inflammation.
- **Primary (idiopathic):** no cause found, often young men. **Secondary:** a cause is identified.
- **Pseudo-haematospermia:** blood from the partner (partner's blood; condom test) or haematuria.

Causes

Group	Examples
Iatrogenic (commonest identifiable)	Prostate biopsy (about 10–50%, lasts about a month), TURP (transurethral resection of the prostate), radiotherapy, BPH (benign prostatic hyperplasia) procedures
Inflammation and infection	Prostatitis, vesiculitis, urethritis, STI (sexually transmitted infection), **genitourinary tuberculosis**, schistosomiasis
Obstruction and cysts	Ejaculatory duct obstruction or calculi; seminal vesicle cyst (Zinner syndrome) or utricle cyst
Tumours	**Prostate cancer**; rarely urethral, bladder or seminal vesicle
Vascular	Prostatic varices, BPH vessels, AV (arteriovenous) malformation
Systemic	Hypertension, bleeding disorders, anticoagulants, liver disease; also abstinence, trauma

Evaluation

- **History:** episodes, recent biopsy, LUTS (lower urinary tract symptoms), haematuria, TB (tuberculosis) contact, anticoagulants.
- **Examination:** blood pressure, genitalia, **DRE (digital rectal examination)**.
- **Basic tests:** urinalysis and culture, semen culture, STI screen, **PSA (prostate-specific antigen)** in men over 40, coagulation if indicated.

Who needs full work-up (full work-up if): age **≥40**, persistent beyond several weeks or recurrent, haematuria, abnormal DRE or PSA, risk factors for TB or cancer, or a distressed patient.

- **TRUS (transrectal ultrasound):** cysts, calculi, dilated seminal vesicles (>1.5 cm) or ejaculatory ducts (>2.3 mm) suggest obstruction.
- **MRI (magnetic resonance imaging) pelvis:** best for seminal vesicle haemorrhage, cysts and prostate cancer.
- **Cystoscopy and urine cytology**; early-morning urine for acid-fast bacilli.

Management

- **Young, single episode, normal tests: reassure**.

- **Treat the cause:** antibiotics or antitubercular drugs; control BP (blood pressure); correct coagulopathy.
- **Persistent prostatic bleeding:** 5-ARI (5-alpha-reductase inhibitor), e.g. finasteride.
- **Refractory or obstructive:** TURED (transurethral resection of ejaculatory ducts), cyst unroofing or aspiration, **seminal vesiculoscopy** (4–6 Fr small scope through the ejaculatory duct or utricle, irrigation, laser for stones).

Diagram to draw: algorithm: <40 and single episode → reassure; ≥40 or persistent → PSA, TRUS/MRI, cystoscopy.

Key points to remember:

- Usually benign; prostate biopsy is the commonest identifiable cause.
- Full work-up if ≥40, persistent or recurrent, or associated haematuria (or with haematuria).
- Exclude prostate cancer and genitourinary tuberculosis.
- TRUS: seminal vesicle >1.5 cm or ejaculatory duct >2.3 mm suggests obstruction.

Testicular biopsy and TESA (testicular sperm aspiration)

Asked as:

- TESA — *Practice Set 5 Paper I Q3b (SAQ)*
- Testicular biopsy — *Practice Set 11 Paper I Q3e (SAQ)*

Type: SAQ (10 marks)
(also answered in 2019-Summer Paper 2, Q3c) (also answered in 2021-Summer Paper 4, Q2)

Definition

- **TESA (testicular sperm aspiration):** percutaneous needle aspiration of seminiferous tubules under local anaesthesia to retrieve sperm for **ICSI (intracytoplasmic sperm injection)**, which needs one sperm per oocyte.

Indications

Testicular biopsy

- **Azoospermia** with **normal testicular size, palpable vasa and normal FSH (follicle-stimulating hormone)**: separates **OA (obstructive azoospermia)** from **NOA (non-obstructive azoospermia)** (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine).
- Clues to obstruction: FSH ≤7.6 IU/L and testis long axis >4.6 cm.
- Because ICSI (intracytoplasmic sperm injection) needs one sperm, biopsy is **diagnostic and therapeutic**: plan it as **TESE (testicular sperm extraction)** and **cryopreserve** sperm.
- **Not needed:** CBAVD (congenital bilateral absence of the vas deferens) with normal FSH; high antisperm antibodies after iatrogenic vasal injury. Do not operate with complete AZFa or AZFb (azoospermia factor) deletion.

TESA

- **OA (obstructive azoospermia) with normal spermatogenesis:** CBAVD (congenital bilateral absence of the vas deferens), unreconstructable obstruction, failed vasectomy reversal, or failed epididymal aspiration.
- Anejaculation (spinal cord injury) after failed vibratory stimulation or electroejaculation.
- **NOA (non-obstructive azoospermia):** poor yield; **micro-TESE (microdissection testicular sperm extraction)** preferred.

Types / comparison of retrieval methods

Type	Method	Use	Limits
Open biopsy	Small albugineal incision, pea-sized sample	**Gold standard** for histology and retrieval	Open procedure
Percutaneous core	14 G biopsy gun, cord block	Fresh retrieval in OA	Blind; epididymis or testicular artery injury
Aspiration (TESA, testicular sperm aspiration)	23 G needle, suction, 3–4 passes	OA; backup to epididymal retrieval	Poor architecture; low yield in NOA
Micro-TESE (microdissection)	Operating microscope, plump tubules	NOA	Long operation

Technique	Site	Best for	Notes
PESA (percutaneous epididymal sperm aspiration)	Caput epididymis	OA	Motile but few sperm; may obstruct epididymis
MESA (microsurgical epididymal sperm aspiration)	Single tubule at 16–25×	OA	Most motile sperm; many frozen vials
TESA	Testis, needle	OA	Least invasive; few, immotile sperm
Percutaneous biopsy	14-G biopsy gun	OA	Core of tissue; blind
TESE / micro-TESE	Open, larger tubules under microscope	**NOA**	Most sperm, least tissue

TESA technique

1. Spermatic cord block with local anaesthetic.
2. Testis fixed; **23-G needle** (or angiocatheter) on a high-suction syringe.
3. Multiple passes in different directions; tubules drawn into the needle (often 10–20 passes needed; another source: 3–4 passes).
4. Immediate wet examination for sperm; repeat until sperm are found. **Cryopreserve** spare sperm.

Open biopsy technique

1. Local anaesthesia with **spermatic cord block** (or general/spinal); choose the larger testis.
2. 2 cm transverse scrotal incision; open tunica vaginalis.
3. Stay sutures; 3–4 mm albugineal incision avoiding surface vessels (loupes).
4. Excise extruding tubules with iris scissors, **no forceps**.
5. Fix in **Bouin's, Zenker's or glutaraldehyde, never formalin** (distorts histology).
6. **Touch prep**, then wet squash prep for sperm; negative → contralateral biopsy.
7. Close tunica vaginalis separately; no antibiotics.

Histology

Pattern	Meaning
Normal spermatogenesis	Obstruction
Hypospermatogenesis	All stages, fewer cells
Maturation arrest	Stops at a germ-cell stage
SCO (Sertoli cell–only)	No germ cells
Tubular sclerosis	End-stage (e.g. Klinefelter)

TESA: advantages and limitations

- **Advantages:** office procedure, no microsurgery, cheap, repeatable.
- **Limitations:** few tubules with poor architecture (not diagnostic histology); small numbers of immature, immotile sperm; often too few to freeze; blind.

Complications

- **Haematoma** (commonest), pain, testicular vascular injury, infection (rare), epididymal injury; **inadvertent epididymal biopsy** (most serious, causes obstruction).

Outcome

- OA: retrieval almost always succeeds; ICSI is essential for testicular sperm.

- NOA: fine-needle aspiration yields far fewer sperm than micro-TESE.
- Before ICSI: karyotype, Y-microdeletion and CFTR (cystic fibrosis transmembrane conductance regulator) testing as indicated.

Diagram to draw: testis with 23-G needle on a suction syringe and multiple passes; open biopsy: small albugineal window with extruding tubules snipped off.

Key points to remember:

- TESA: 23-G needle, local anaesthesia, OA with normal spermatogenesis.
- NOA: micro-TESE best; TESA yield poor.
- Testicular sperm always need ICSI; cryopreserve extras.
- Biopsy indication: azoospermia, normal testes, vasa and FSH.
- Bouin's, never formalin; no forceps; touch prep.
- Haematoma commonest; epididymal biopsy most serious.

Premature Ejaculation

Type: SAQ (10 marks)

Source: Practice Set 11 Paper 1, Q3f (also answered in 2023-Winter Paper 3, Q3a)

Definition

- **ISSM (International Society for Sexual Medicine) unified definition (2014):** all three needed:
  1. Ejaculation always or nearly always within **about 1 minute** of vaginal penetration (lifelong), or a bothersome fall in latency, often to **≤3 minutes** (acquired).
  2. Inability to delay ejaculation on all or nearly all penetrations.
  3. Negative personal consequences: distress, frustration, avoidance.
- **Types:** lifelong, acquired, variable and subjective.

Physiology and aetiology

- Ejaculation = **emission** (sympathetic T10–L2) then **expulsion** (somatic S2–S4, pudendal). **Serotonin brakes and dopamine accelerates**.
- **Lifelong PE (premature ejaculation):** neurobiological and genetic; 5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity.
- **Acquired:** performance anxiety, relationship problems, **ED (erectile dysfunction)**, chronic prostatitis, hyperthyroidism, drug withdrawal.

Evaluation

- Diagnosis is by **history**: latency, control, bother.
- **IELT (intravaginal ejaculatory latency time):** self-estimate in clinic; stopwatch in trials (population median 5.4 minutes).
- **PEDT (Premature Ejaculation Diagnostic Tool)**; IIEF-5 (International Index of Erectile Function) for ED; examination; TSH (thyroid-stimulating hormone) only if suggested.

Management

- **Psychosexual education** for all; behavioural therapy (stop–start, squeeze).
- Acquired PE: treat the cause first (ED, prostatitis, thyroid).

Drug	Dose and timing
Dapoxetine (short-acting SSRI [selective serotonin reuptake inhibitor])	30–60 mg, 1–3 h before
Paroxetine (strongest delay) 10–40 mg, sertraline, fluoxetine, citalopram	Daily, off-label; full effect 2–3 weeks; do not stop abruptly
Clomipramine	12.5–50 mg daily or 3–4 h before
Topical lidocaine–prilocaine spray	At least 5 min before; condom avoids partner numbness
PDE5Is (phosphodiesterase type 5 inhibitors)	With ED, alone or with SSRI

- SSRIs raise synaptic serotonin and delay ejaculation. Avoid in bipolar disorder.
- **Also know:** tramadol (addiction risk); high dropout from drug therapy.

Diagram to draw: office algorithm: confirm PE → secondary to ED? → treat cause; else classify lifelong or acquired.

Key points to remember:

- ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.
- Lifelong is neurobiological; acquired is usually secondary.
- Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

Avanafil

Type: SAQ (10 marks)

Source: Practice Set 21 SAQ Sheet, Q3e

Introduction

- Second-generation oral **PDE5I (phosphodiesterase type 5 inhibitor)**; **FDA (US Food and Drug Administration) approved 2012**; Europe 2013. Structure differs most from the other PDE5Is.

Mechanism

- Sexual stimulation releases NO (nitric oxide), which raises cGMP (cyclic guanosine monophosphate) and relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; avanafil blocks this.
- **Enhances, does not initiate, erection**: stimulation is needed.
- **More selective for PDE5** over PDE6 (retina) and PDE1 than sildenafil, so **no visual disturbance** in the comparison table.

Pharmacokinetics (Campbell Table 72.7)

Feature	Avanafil	Sildenafil	Tadalafil
Time to peak	**0.3–0.5 h (fastest)**	0.8 h	2 h
Half-life	3–5 h	3–5 h	17.5 h
Fatty food	Reduces absorption	Reduces	No effect
Doses	**50, 100, 200 mg**	25–100 mg	5–20 mg
Blue vision	No	Yes	Rare

- Taken about 15–30 minutes before sex; start 100 mg; maximum once daily. Metabolised by CYP3A4 (cytochrome P450 3A4).

Efficacy

- Effective in general ED (erectile dysfunction), **diabetes** and **after nerve-sparing radical prostatectomy** (REVIVE trial programme): improves IIEF (International Index of Erectile Function) erectile scores and successful-intercourse rates, some within 15 minutes. Efficacy equivalent across PDE5Is; no head-to-head superiority.

Adverse effects

- Headache, flushing, nasal congestion, dyspepsia; back pain rare.

Contraindications and interactions

- **Nitrates in any form**: absolute; no nitrate within 12 hours of a dose.
- Avoid strong CYP3A4 inhibitors (ketoconazole, ritonavir); maximum 50 mg with moderate inhibitors.
- **α-blockers:** start 50 mg once stable (additive hypotension).
- Caution: aortic stenosis, recent myocardial infarction or stroke, hypotension, retinitis pigmentosa. Not recommended in severe renal impairment, dialysis or severe hepatic impairment.
- **Also know:** proposed prophylaxis for recurrent ischaemic priapism; can cause priapism rarely.

Advantages

- **Fastest onset**, fewer visual and muscle effects; suits on-demand spontaneous use.

Diagram to draw: NO–cGMP pathway in cavernosal smooth muscle with the PDE5 block.

> **Key points to remember:**
> - Peak 0.3–0.5 h; doses 50, 100, 200 mg.
> - Highly PDE5-selective: no blue vision.
> - Nitrates absolutely contraindicated.

Rapid Revision Sheet – Section 9 · Andrology & Sexual Medicine

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Spermatogenesis, azoospermia, non-obstructive azoospermia and maturation arrest (LAQ)

- Spermatogenesis about 64 days; Ap divide every 16 days; 6 stages.
- Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
- FSH ≤7.6 IU/L and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in all.
- AZFa/b deletion: do not operate (no retrieval); micro-TESE finds sperm in about half of NOA; SCO 30%.
- Maturation arrest: arrest at one stage; early (spermatocyte) or late (spermatid).
- FSH and testis size may be normal: biopsy diagnoses.
- Micro-TESE retrieval in maturation arrest about 40%; none with complete AZFb deletion.

Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED (LAQ)

- nNOS initiates, eNOS maintains; PDE5 ends (terminates) erection.
- Noradrenaline α1 is the main anti-erectile transmitter.
- Dopamine/oxytocin promote; serotonin/prolactin inhibit.
- Rigid phase: ischiocavernosus, suprasystolic pressure.
- Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
- Diabetic ED: check HbA1c, testosterone and cardiac risk.

Male reproductive immunology (SAQ)

- BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
- ASA: vasectomy commonest; ≥50% bound significant; IVF/ICSI.
- Leukocytospermia >1 million/mL.

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- Ask the "straight penis" question; duplex before surgery.
- CCH: IMPRESS, stable 30–90° curve, intact erections.
- Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

Haematospermia (SAQ)

- Usually benign; prostate biopsy is the commonest identifiable cause.
- Full work-up if ≥40, persistent or recurrent, or associated haematuria (or with haematuria).
- Exclude prostate cancer and genitourinary tuberculosis.
- TRUS: seminal vesicle >1.5 cm or ejaculatory duct >2.3 mm suggests obstruction.

Testicular biopsy and TESA (testicular sperm aspiration) (SAQ)

- TESA: 23-G needle, local anaesthesia, OA with normal spermatogenesis.
- NOA: micro-TESE best; TESA yield poor.
- Testicular sperm always need ICSI; cryopreserve extras.
- Biopsy indication: azoospermia, normal testes, vasa and FSH.
- Bouin's, never formalin; no forceps; touch prep.
- Haematoma commonest; epididymal biopsy most serious.

Premature Ejaculation (SAQ)

- ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.

- Lifelong is neurobiological; acquired is usually secondary.
- Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

Avanafil (SAQ)

- Peak 0.3–0.5 h; doses 50, 100, 200 mg.
- Highly PDE5-selective: no blue vision.
- Nitrates absolutely contraindicated.

Section 10 · Testis, Penis & Urethra — Long answer questions (LAQ)

Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias

Asked as:

- Describe the embryology and anatomy of male Urethra. Describe different endoscopic management techniques of treatment of urethral strictures. — *Summer 2019 Paper I Q1 (LAQ)*
- Discuss anatomy of urethra. Classify its congenital anomalies and discuss management of epispadias. — *Practice Set 5 Paper I Q1 (LAQ)* (male urethral anatomy also answered in 2019-Summer Paper 1, Q1; isolated epispadias in 2022-Summer Paper 3, Q3f)

Type: LAQ (25 marks)

Embryology

- **Cloaca** is divided by the **urorectal septum** (weeks 4–7) into the anterior **UGS (urogenital sinus)** and the posterior anorectal canal.
- The UGS has three parts: **vesical** (bladder), **pelvic** (prostatic and membranous urethra) and **phallic** (urethral plate of the penis).
- **Posterior urethra:** from the pelvic UGS. Prostatic buds appear at 10 weeks; the **utricle** (Müllerian remnant) and ejaculatory ducts (Wolffian) open on the verumontanum.
- **Anterior urethra:** Leydig cells make testosterone from about 8 weeks; type 2 5α-reductase converts it to DHT (dihydrotestosterone), which masculinises the genital tubercle. Germ layers: ectoderm gives skin and prepuce, mesoderm the corpora, endoderm the **urethral plate**.
- **Penile urethra: "double zipper"**
  1. **Opening zipper:** the urethral plate canalises proximal to distal into a urethral groove between the urethral folds.
  2. **Closing zipper:** the folds fuse in the ventral midline, forming the tube and the median raphe. Folds fused by 16 weeks; urethra complete by about 18 weeks.

Glanular urethra	Ectodermal ingrowth theory (Glenister)	Endodermal theory (current)
Formed by	Surface ectoderm grows into the glans to meet the endodermal urethra	Direct canalisation of the urethral plate inside the glans
Epithelium	Mixed ectoderm and endoderm	Endoderm (UGS) along the whole urethra

- **Clinical link:** the shaft–glans junction (change of mechanism) is the commonest hypospadias site. **Also know:** urethral duplication; PUV (posterior urethral valves); utricular cyst.

Anatomy of the male urethra

- About 18–20 cm long in adults; normal diameter 8–9 mm (wide). **Posterior urethra** runs from the bladder neck to the **perineal membrane**; **anterior urethra** from there (perineal membrane) to the meatus, inside the corpus spongiosum.

Segment	Length / features	Lining
Prostatic	Length of prostate; urethral crest, **verumontanum** with utricle (Müllerian remnant) and ejaculatory ducts; angle divides preprostatic and prostatic parts	Urothelium
Membranous	2–2.5 cm; shortest and narrowest; striated sphincter	Pseudostratified columnar
Bulbar	Widens as the bulb; **Cowper ducts at 3 and 9 o'clock**; lies **eccentric, close to (nearer) the dorsum** of spongiosum	Pseudostratified columnar
Penile	About 15 cm; central, uniform calibre; glands of Littré	Pseudostratified columnar
Fossa navicularis	Glanular; lacuna magna on the roof	Stratified squamous

- **Sphincters:** (1) **internal (preprostatic) smooth muscle sphincter** at the bladder neck, autonomic, gives continence at rest; (2) **external striated rhabdosphincter**, horseshoe-shaped around the membranous urethra, supplied by the pudendal nerve (mainly S3, some S2; Onuf nucleus), plus its intrinsic smooth muscle.
- **Blood supply (arteries):** internal pudendal → common penile artery → artery to the bulb (proximal; bulbourethral branches) and **retrograde flow via the dorsal arteries through the glans**. This **dual supply** allows bulbar transection in anastomotic urethroplasty.
- **Venous** (veins) to pudendal and prostatic plexuses; **lymph** to internal and external iliac nodes.
- **Female urethra:** about 4 cm; smooth muscle plus striated sphincter over the distal two-thirds; **Skene glands** (prostate homologue) near the meatus; oestrogen-dependent mucosal seal.

Endoscopic management of stricture

Principle: healing after incision or dilation is by **secondary intention**, a race between epithelialisation and wound contraction. No endoscopic method removes **spongiofibrosis**, so cure is limited to short, superficial bulbar strictures.

Favourable	Unfavourable
Bulbar, **<1.5–2 cm**	Penile, fossa, longer
No dense spongiofibrosis	Straddle injury, LS (lichen sclerosus)
First procedure	Repeat; luminal narrowing >74%

1. Dilation

- Stretches the scar without tearing it (bleeding means tearing).
- **Safest: balloon dilation over a guidewire** under vision; blind filiforms condemned. Sounds or self-dilation for palliation.

2. DVIU (direct vision internal urethrotomy)

1. RUG (retrograde urethrogram) and VCUG (voiding cystourethrogram) define site and length.
2. Guidewire across; **normal saline** irrigation.
3. **Sachse cold knife** at 12 o'clock, through scar to healthy tissue; avoid deep cuts (corpora: ED (erectile dysfunction), fistula).
4. Catheter **3–7 days**; longer adds nothing.

- **Laser urethrotomy** (holmium, others): **no advantage over cold knife**.
- Results: **Steenkamp and Heyns** (randomised): dilation equals urethrotomy; a second attempt rarely and a third never succeeds. **Santucci (2010):** about 8% stricture-free long

term.

3. Adjuncts against recurrence

- **CISC (clean intermittent self-catheterisation)** tapering over 3–6 months: delays recurrence, which returns on stopping.
- **Mitomycin C** injection reduces recurrence; steroid injection, colchicine: weak data.

4. Urethral stents

- **UroLume** (permanent): **withdrawn** (ingrowth). **Memokath**, removable stents: small series only.

5. Optilume (paclitaxel-coated balloon)

- FDA (Food and Drug Administration) 2023. **ROBUST III:** 74.6% vs 26.8% stricture-free at 1 year after failed endoscopy.

Complications: recurrence (commonest), bleeding with erections, irrigant extravasation, UTI (urinary tract infection) and sepsis, ED, spongio-cavernous fistula, incontinence.

Guideline (AUA (American Urological Association) 2016): urethrotomy or dilation for short (<2 cm) bulbar strictures; **urethroplasty, not repeat endoscopy**, after failure. The reconstructive ladder is abandoned.

Classification of congenital urethral anomalies

Group	Anomalies
Abnormal meatal position	**Hypospadias** (ventral); **epispadias** (dorsal)
Obstructive	**PUV (posterior urethral valves)**, Young types I–III; **anterior urethral valves** (with diverticulum); urethral atresia or hypoplasia (congenital stricture); congenital meatal stenosis
Dilatation	**Megalourethra**: scaphoid (spongiosum deficient) or fusiform (all erectile tissue deficient, more severe); anterior urethral diverticulum; Cowper duct cyst (syringocele); enlarged prostatic utricle
Duplication	**Effmann classification**: type I blind incomplete; type II complete (IIA2 "Y-type"); type III with caudal duplication. The ventral channel is the functional one
Accessory openings	Congenital urethrocutaneous fistula; parameatal urethral cyst
Female	Epispadias; urethral prolapse; persistent urogenital sinus

Epispadias: definition and classification

- A **dorsal urethral wall defect**, replaced by a broad dorsal mucosal strip (urethral plate); part of the **exstrophy–epispadias complex**, mostly seen with exstrophy.
- Embryology: an overdeveloped cloacal membrane blocks mesenchymal ingrowth; genital tubercles fuse caudal to it.

	Male	Female
Incidence	About 1 in 117,000	About 1 in 484,000
Types	Glanular, penile, **penopubic** (about 70%, incontinent)	**Davis:** vesicular (patulous meatus), subsymphyseal, retrosymphyseal (incontinent)
Genitalia	Short, upturned penis with **dorsal chordee**	**Bifid clitoris**, flat glabrous mons
VUR (vesicoureteric reflux)	30–40%	30–75%

- The more proximal the meatus, the worse the continence (glanular never incontinent). Pubic diastasis is less than in exstrophy.

Evaluation

- Renal ultrasound; VCUG (voiding cystourethrogram) for reflux; cystoscopy and **bladder capacity** under anaesthesia.
- **Key decision:** use the exstrophy pathway (osteotomy and closure) if pubic diastasis >3 cm, poor intrasymphyseal bar, absent sphincter tissue or tiny bladder.

Management

Aims: continence with safe upper tracts, and a straight penis of adequate length that works for intercourse.

- **Preoperative testosterone** (2 mg/kg intramuscular, 5 and 2 weeks before).

Stage 1: urethroplasty and penile reconstruction (6–12 months)

1. Release dorsal chordee: divide suspensory ligaments; free corpora from the inferior pubic rami.
2. **Modified Cantwell–Ransley:** an 18-mm plate strip is outlined; reverse MAGPI (meatal advancement and glanuloplasty) brings the meatus to the tip; the plate is tubularised over an 8-Fr stent and moved **ventral to the corpora**, which are rotated medially over it (diamond corporotomies if curvature persists).
3. Glans wings closed; skin flaps resurface the dorsum; base Z-plasty; stent 10–12 days.

- **Mitchell–Bägli complete penile disassembly:** reliable chordee correction, but often leaves hypospadias.
- Urethroplasty adds outlet resistance, so the bladder grows before stage 2.

Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)

- **Young–Dees–Leadbetter** repair with Marshall–Marchetti–Krantz suspension and **ureteric reimplantation**. Osteotomy is rarely needed.
- **Good capacity at presentation:** single-stage repair is possible.
- **Female:** urethroplasty and genitoplasty (clitoral halves and labia joined, mons reconstructed) at 6–12 months; bladder neck repair at 4–5 years.

Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%, females 70–87%**; capacity is the main predictor.
- Early fistula about 1 in 6 (penile base); stricture rare; neourethra catheterisable; erections and paternity preserved.
- Better than exstrophy: the bladder was never exposed or scarred. Failures: injectables, augmentation.

Diagram to draw: sagittal male urethra with segments, sphincters and a 12 o'clock urethrotomy; Cantwell–Ransley with plate transposed under the corpora.

Key points to remember:

- Double zipper: canalisation, then fold fusion; glanular urethra endodermal.
- Bulbar urethra eccentric (dorsal); Cowper ducts 3 and 9 o'clock; dual retrograde glanular blood supply.
- DVIU: bulbar, <1.5–2 cm, first attempt; Santucci about 8% long-term.
- Optilume ROBUST III 74.6% vs 26.8%.
- Megalourethra: fusiform worse than scaphoid; duplication: ventral channel functional.
- About 70% of male epispadias are penopubic; diastasis >3 cm → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL.

Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications

Asked as:

- Lymphatic drainage of testis and principles for designing templates for RPLND. — *Practice Set 7 Paper I Q2 (LAQ)*
- Lymphatic drainage of testes and principles of designing templates of RPLND — *Practice Set 12 Paper I Q2 (LAQ)*
- Anatomy of lymphatic drainage of testis and scrotum. Describe management implications of penile malignancies. — *Summer 2021 Paper I Q1 (LAQ)*

Type: LAQ (25 marks)

(lymphatic drainage also answered in 2018-Summer Paper 2, Q2 [testis and scrotum]; minimally invasive RPLND (MI-RPLND) in 2025-Winter Paper 3, Q3f; Practice Set 12 Paper 1, Q2 is the same question as Practice Set 7 Paper 1, Q2; 2017-Summer Paper 2, Q1 [inguinal nodes in penile cancer])

Second part (Summer 2021) taken as the management implications of lymphatic drainage in penile cancer, i.e. management of the inguinal and pelvic nodes.

Embryological basis

- **Why retroperitoneal:** the testis develops at L1–L2 beside the mesonephros (near the kidney) and descends, carrying its vessels and lymphatics (its artery, vein and lymphatics stay retroperitoneal). So its primary drainage (its lymph returns) is to the **retroperitoneal (lumbar) nodes near the renal hilum**, not the groin. The scrotum is skin and drains to the **groin**.

Lymphatic drainage of the testis (lymphatic pathway)

- **Route:** lymphatics of the testis and the caput and corpus of the epididymis run (up the **spermatic cord**) in the spermatic cord with the testicular (gonadal) vessels, through the internal ring and along the gonadal vessels to the lumbar nodes near the renal hilum (**para-aortic and interaortocaval nodes**). Lymph capillaries do not enter seminiferous tubules.
- Onward: lumbar trunks join at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta, behind the right crus; hence retrocrural nodes), then the **thoracic duct** to the left innominate vein. This explains **left supraclavicular** metastases.
- Lymph flows (drifts) **cephalad and right to left**.
- **Epididymis:** caput and corpus with the testis; **cauda epididymis** drains with the vas to the **external iliac** nodes.

Retroperitoneal nodal regions

Region	Extent
Right paracaval (with precaval, retrocaval)	IVC (inferior vena cava) midline to right ureter
Interaortocaval	IVC midline to aortic midline
Left para-aortic (with preaortic/pre-aortic, retroaortic)	Aortic midline to left ureter

- **Also know:** ten mapped regions include right and left common iliac, interiliac and right and left suprahilar.

Primary landing zones

	Right testis	Left testis
Primary (landing zone)	**Interaortocaval** (below renal vessels)	**Left para-aortic** (left hilar)
Secondary / next echelon	Right paracaval, precaval, then para-aortic	Interaortocaval; pre-aortic
Crossover / contralateral	Small but real to left para-aortic (another source: common; right to left occurs)	None significant (another source: rare; implies bulk; rare to right, bulky disease only)

- **Crossover is only right to left**, matching the overall lymph flow (because lymph drifts right to left; lymph flows that way).
- **Donohue mapping (1982):** contralateral and suprahilar spread is rare but rises with bulk; interiliac spread is very rare.
- **Pelvic, iliac and inguinal nodes** are involved only when drainage is altered (or with retrograde spread from bulky disease): previous scrotal or inguinal surgery, scrotal violation, or tumour invading the tunica vaginalis (reaching the tunica/scrotum). Scrotal skin drains to ipsilateral superficial inguinal nodes. **Hence trans-scrotal orchidectomy is contraindicated** (never biopsy or remove a testis trans-scrotally).
- The retroperitoneum is the first metastatic site in 70–80% (choriocarcinoma spreads by blood). CT (computed tomography) understages 25–35% of stage I NSGCT (non-seminomatous germ cell tumour); 5–9 mm landing-zone nodes are suspicious.

Lymphatic drainage of the scrotum

- Scrotal lymphatics **do not cross the median raphe** (nothing crosses the raphe); they drain to the **ipsilateral superficial inguinal nodes**.

Principles of designing RPLND (retroperitoneal lymph node dissection) templates

1. Map-based design (mapping studies): follow the side-specific landing zones to remove likely disease and spare the rest.

- Historical dissection: bilateral infrahilar + suprahilar + interiliac: most men anejaculatory.
- **Ray (1974)** first templated landing zones; **Donohue (1982)** mapping confirmed side-specific spread; **Weissbach and Boedefeld (1987)** reduced left template with frozen section.
- So a template must include the primary landing zone, the next echelon and the known crossover route.

2. Fixed (standard) boundaries/limits

- **Superior:** renal hila (renal vessels above). **Lateral:** ureters laterally. **Inferior:** ureter crossing the common iliac artery below. Plus always remove the **gonadal vein and cord stump** to the internal ring.

3. Omit suprahilar and interiliac regions without bulky disease (in non-bulky disease, as spread there is rare). This reduces (less) chylous ascites, renal, pancreatic and duodenal injury, and protects the hypogastric plexus.

4. Templates: unilateral modified templates leave contralateral sympathetic fibres untouched. Because of right-to-left crossover, both templates take the para-aortic nodes **above the IMA (inferior mesenteric artery)**.

Template	Includes / content
Right modified	(Right) paracaval, precaval, retrocaval, interaortocaval, right common iliac; para-aortic **above IMA**; omit para-aortic below IMA
Left modified	Left para-aortic, preaortic, retroaortic, left common iliac; interaortocaval above IMA; omit paracaval/precaval/retrocaval
(Full) Bilateral infrahilar	All three regions both sides (paracaval, precaval, retrocaval, interaortocaval, pre-aortic, para-aortic, retroaortic, both common iliacs), between renal vessels, ureters and ureteric crossing of iliacs
Historical extended	Plus suprahilar and interiliac (thoracoabdominal); most anejaculatory

- Omitting para-aortic above the IMA (right) or interaortocaval (left) is controversial.

5. Nerve sparing / preserve ejaculation: antegrade ejaculation (emission and bladder-neck closure) depends (depend) on **L1–L4 postganglionic sympathetic fibres** (about three each side) joining the **superior hypogastric plexus at or below the IMA** and running over the sacral promontory. Identify them prospectively before dividing tissue.

- **Unilateral templates** leave contralateral fibres untouched (antegrade ejaculation 75–97%). **Prospective nerve sparing** (Jewett 1988) identifies fibres: 90–100%.

6. Completeness ("duplicate open RPLND"): split-and-roll over the great vessels, **division of (divide) lumbar vessels** to clear retrocaval and retroaortic tissue, clip lymphatics at the cephalad limit (left renal vein) to prevent chylous ascites (chylous leak). Incomplete RPLND or mass excision gives worse outcomes (GETUG T93BP; 33% local relapse after post-chemotherapy mass-only resection).

7. Extend if needed: enlarged nodes at operation → full bilateral template (frozen section, Weissbach 1987). Convert to bilateral if nodes are suspicious at surgery or the primary has somatic-type malignancy.

Template by clinical setting (setting decides extent)

Setting	Template
Primary RPLND (high-risk stage I or low-volume stage II NSGCT [non-seminomatous germ cell tumour], normal markers)	Default: modified unilateral nerve-sparing **or** full bilateral infrahilar nerve-sparing; no randomised trial. Modified allowed: low-volume, clinically negative nodes. Main risk: out-of-template nodes 3–23% (right > left)
PC-RPLND (post-chemotherapy), residual mass >1 cm, normal markers	**Full bilateral infrahilar + (residual) mass** (standard). Main risk: crossover with bulk; viable tissue unrecognisable
Modified PC template allowed only if	Mass ≤5 cm in landing zone, normal markers, IGCCCG (International Germ Cell Cancer Collaborative Group) good or intermediate risk

- Against unilateral templates: 3–23% of node-positive men have nodes outside the modified template, more with right tumours.
- *Sources differ:* Campbell testis-neoplasm chapter calls full bilateral nerve-sparing the standard; the surgery chapter accepts either.

Outcomes and complications

- Primary RPLND: pN+ about 20–30%; cancer-specific survival (CSS) about 98.8–100%; retroperitoneal recurrence <2% after full template.
- **Antegrade ejaculation >90%** with templates plus nerve sparing (modified template alone: early 75–87%, refined 97%).
- Complications: ileus (commonest), wound infection, chylous ascites (0.2–2.1% primary; 2–12% post-chemotherapy), pulmonary (bleomycin lung toxicity; low FiO_2), ejaculatory loss.

- **Also know:** nephrectomy needed at PC-RPLND (left-sided mass >10 cm); GETUG T93BP showed incomplete RPLND predicts relapse.

Recent advances

- **Robotic RPLND** for low-volume disease (uses the same templates, same rules).
- RPLND for low-volume (small-volume) stage II seminoma (**SEMS, PRIMETEST**); serum **miR-371a-3p** for selection.

Clinical implications: testis and scrotum

- **Never operate trans-scrotally** on a testicular tumour. Scrotal violation opens skin lymphatics, so tumour can reach inguinal and pelvic nodes; use **radical inguinal orchidectomy** with high cord clamping.
- Inguinal or pelvic spread from the testis means altered drainage: scrotal violation, previous orchidopexy or hernia repair, scrotal invasion.
- RPLND (retroperitoneal lymph node dissection) templates and para-aortic or "dog-leg" radiotherapy fields follow the landing zones.
- Scrotal skin cancer is staged at the ipsilateral groin.

Lymphatic drainage of the penis

Superficial system	Deep system
Prepuce and shaft skin	Glans, frenulum, corpora
Dorsal network divides at the base → both superficial inguinal groups	Trunks circle the corona, run under Buck's fascia → presymphyseal plexus → femoral triangle nodes

- **Two rules:** drainage is **bilateral and unpredictable in side** (presymphyseal crossover), but **orderly in sequence**: superficial inguinal → deep inguinal → external iliac and obturator. **No skip metastases.**
- **Superficial nodes** (4–25, above fascia lata) lie in **Daseler's five zones**; sentinel nodes are mostly **superomedial**, then central. **Deep nodes** (1–3) lie medial to the femoral vein; **Cloquet's node** is the highest.

Management implications in penile cancer

1. Prognosis and timing

- Nodal status is the **strongest prognostic factor**. Penile SCC (squamous cell carcinoma) has a long **locoregional phase**, so the groin is a **curable station**.
- **Kroon (2005):** early ILND (inguinal lymph node dissection) for impalpable nodes gave 3-year disease-specific survival about **84% vs 35%** when delayed until palpable.
- Manage both groins **within 3 months**; remove **at least 8 nodes per side**.

2. Bilateral management

- Crossover means **both groins** are staged or treated; with one palpable groin, the other holds micrometastasis in about 30%.

3. Impalpable groin (cN0): risk-based

Risk	EAU-ASCO (European Association of Urology–American Society of Clinical Oncology) 2023	Groin
Low	pTis, pTa, pT1a G1	Surveillance
Intermediate	pT1a G2	Surveillance if compliant, or surgical staging
High	**pT1b or higher**, G3	**DSNB (dynamic sentinel node biopsy)**; ILND if DSNB unavailable

- NCCN (National Comprehensive Cancer Network): pT1b G1–2 is intermediate risk and is staged surgically (ILND first).
- **DSNB:** ultrasound ± FNAC (fine-needle aspiration cytology) first, because a tumour-replaced node diverts flow (false negative). Then technetium-99m nanocolloid, blue dye and gamma probe.
- **No-skip spread and superomedial sentinels** justify **modified ILND (Catalona)**: smaller field, saphenous vein preserved; positive frozen section → complete radical ILND.

4. Palpable nodes (cN+)

- About half are reactive, but do not delay with antibiotics. Ultrasound-guided **FNAC**; positive → **radical ILND** on that side, stage the other.
- **Bulky (≥4 cm), fixed or bilateral nodes:** neoadjuvant **TIP** (paclitaxel, ifosfamide, cisplatin) × 4, then surgery in responders.

5. Pelvic nodes

- Stepwise spread means a negative groin needs no pelvic dissection.
- **PLND (pelvic lymph node dissection)**, ipsilateral, if **more than 2 inguinal nodes or ENE (extranodal extension)**. ENE in any node is pN3.

6. Complications of ILND

- Wound infection, **flap necrosis**, **lymphocele**, **lymphoedema**, DVT (deep vein thrombosis). Reduced by modified templates, saphenous sparing and **VEIL (video-endoscopic inguinal lymphadenectomy)**.

Also know: adjuvant therapy for pN2–3; Cabanas sentinel biopsy abandoned.

Diagram to draw: great vessels (retroperitoneum showing) with the three nodal regions, right (interaortocaval) and left (para-aortic) landing zones, and the right-to-left crossover arrow; outline the two modified templates (right and left), IMA level and hypogastric plexus. Beside it, the femoral triangle with Daseler's zones and Cloquet's node.

Key points to remember:

- Right testis to (→) interaortocaval; left to left para-aortic; crossover only right to left.
- Limits/boundaries: renal vessels (hila), ureters, ureteric crossing of iliacs (ureter–iliac crossing); omit suprahilar and interiliac.
- Nerve-sparing of L1–L4 fibres gives antegrade ejaculation >90%; the fibres meet at the IMA: unilateral templates spare ejaculation.
- PC-RPLND (post-chemotherapy): full bilateral template remains standard unless mass ≤5 cm in landing zone.
- Scrotum and penis → inguinal nodes; never operate trans-scrotally.
- Penile drainage is bilateral and stepwise without skips: stage both groins, PLND only if inguinal nodes positive.
- pT1b or higher: DSNB or ILND within 3 months; Kroon 84% vs 35%.

Enlist various grafts used in urology and role of BMG in urology

Type: LAQ (25 marks)

Source: 2023-Winter Paper 1, Q2 (BMG uses also in 2015-Summer Paper 3, Q3b; vascular grafts in 2021-Summer Paper 3, Q2)

Graft versus flap

	Graft	Flap
Blood supply	Separated; gets a new supply from the bed	Moved with its own pedicle
Depends on	A well-vascularised host bed	Its pedicle
Example	BMG (buccal mucosal graft)	Penile fasciocutaneous island flap

Graft take (about 96 hours)

Phase	Duration	Mechanism
Imbibition	about 48 h	Graft "drinks" nutrients from the bed; graft is cooler than core
Inosculation	about 48 h	Bed and graft vessels connect; true microcirculation returns

- Take fails with poor beds (scar, radiation), haematoma, seroma, infection or shear. Hence **quilting or pie-crusting** and **immobilisation for 5–10 days** (bolster or vacuum dressing).

Grafts used in urology

Group	Graft	Main uses
Skin	**STSG** (split-thickness skin graft), meshed	Staged urethroplasty only; penile and scrotal cover (Fournier gangrene, buried penis)
	FTSG (full-thickness skin graft)	Fossa navicularis, penile urethra (avoid in lichen sclerosus)
	Posterior auricular (Wolfe) FTSG	Staged hypospadias repair
Mucosa	**Buccal, lingual, labial (oral mucosa)**	Urethra, hypospadias, ureter
	Bladder epithelium	Urethra, hypospadias
	Rectal (colonic) mucosa	Panurethral stricture when oral mucosa is unavailable
Other autologous	Dermis	Tunica albuginea in Peyronie disease
	Tunica vaginalis	Peyronie disease, urethral cover; easy harvest
	Saphenous vein	Peyronie defects; vascular repair
	Rectus fascia, fascia lata	Pubovaginal sling
Allograft / xenograft	Pericardium, porcine SIS, porcine dermis, cadaveric fascia	Peyronie grafting ("off the shelf"); slings
Synthetic	Polypropylene (macroporous type I mesh)	Mid-urethral sling, prolapse
	Dacron, PTFE (polytetrafluoroethylene)	Vascular grafts; not advised for Peyronie (infection, fibrosis)
Newer	Collagen fleece (TachoSil); acellular matrix, tissue-engineered oral mucosa	Peyronie; experimental urethra

- Vein is not a true graft: it survives by direct endothelial perfusion and its vasa vasorum.
- **Peyronie grafts:** pericardium (thin, strong, does not contract) and SIS (cut about 25% oversize; contracts) are commonest; AUA gives no preference.
- **STSG versus FTSG:** STSG carries little dermis, takes easily, can be meshed (up to sixfold) but contracts. FTSG carries full dermis, contracts less, looks better but is more "fastidious" in take.

Role of BMG

Why it works

- Thick, non-keratinised, elastin-rich epithelium used to a **wet environment**; resists infection.
- Thin, **highly vascular lamina propria** with a panlaminar plexus, so imbibition and inosculation are rapid.
- Hairless; little contraction; handles and holds sutures well; low donor morbidity.

Harvest

1. Nasotracheal intubation; mouth gag; mark the graft (often about 4–5 × 2 cm per cheek).
2. Stay 1 cm from the commissure and avoid **Stensen's (parotid) duct** opposite the upper second molar.
3. Infiltrate, incise, raise the mucosa off buccinator; haemostasis.
4. Donor site: leave open or close; both acceptable.
5. Back table: **defat** the graft, cut about 10% oversize, fenestrate with a needle.

- **Lingual** (under the tongue) and **labial** grafts are alternatives. Buccal and lingual give equal urethroplasty success; lip mucosa is thinner, suits the glans.

Uses in urethroplasty

Technique	Principle	Best for
Dorsal onlay (Barbagli, 1995)	Dorsal urethrotomy; graft spread-fixed on corpora	Bulbar; reliable bed
Ventral onlay	Ventral urethrotomy; graft covered by spongioplasty	Bulbar with healthy spongiosum
Dorsal inlay (Asopa)	Ventral approach; dorsal plate incised; graft inlaid	Penile and bulbar
Augmented anastomosis	Excise narrowest part, floor anastomosis, graft roof	Long, dense bulbar stricture
Kulkarni (2012)	One-sided dissection; penile invagination; dorsal graft	Panurethral, one stage
Staged (Johanson, Bracka)	Stage 1 graft; tubularise after 6 months or more	Lichen sclerosus, failed hypospadias
BMG perineal urethrostomy	Graft completes dorsal wall to skin	Panurethral or failed repairs

- Dorsal and ventral results are similar; onlay beats tubes. **Skin is avoided in lichen sclerosus** because the disease recurs in skin.

Other uses

- **Ureter:** BMG onlay ureteroplasty (first human report 1999), now robotic with an **omental wrap**; for long proximal or mid strictures in dense scar. CORRUS (Collaborative of Reconstructive Robotic Ureteral Surgery) failure about 15–25% by site.
- Redo and staged hypospadias; urethral closure in rectourethral fistula; bladder neck contracture; female urethral stricture; Peyronie grafting (small defects only).

Complications

- **Donor:** pain, swelling (first month), numbness (up to 6 months), mouth tightness, salivary duct injury.
- **Recipient:** graft failure, restricture, sacculation, fistula, ED.

Outcomes

- Bulbar BMG urethroplasty about 85–90% success; most failures within 12 months.

Diagram to draw: bulbar cross-section showing dorsal onlay, ventral onlay and dorsal inlay positions.

> **Key points to remember:**
>
> - Take: imbibition 48 h, then inosculation 48 h; graft lives on its bed.
> - BMG: non-keratinised, vascular lamina propria, wet-adapted.
> - Avoid Stensen's duct; buccal equals lingual in success.
> - Staged BMG, not skin, in lichen sclerosus.

Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy.

Type: LAQ (25 marks)

Source: 2024-Winter Paper 1, Q1 (related: 2019-Summer Paper 1, Q2; 2020-Summer Paper 1, Q3b)

Anatomy: definition and boundaries

- The space behind the parietal peritoneum. **Anterior:** posterior peritoneal reflection; **posterior:** posterior abdominal wall; **cranial:** diaphragm; **caudal:** extraperitoneal pelvis.

Group	Contents
Primarily retroperitoneal (formed there)	Kidneys, ureters, adrenals, aorta, IVC (inferior vena cava)
Secondarily retroperitoneal (mesentery fused to the wall)	Ascending and descending colon, duodenum D2–D3, pancreas (not the tail)
Vessels, lymphatics, nerves	Lumbar nodes, cisterna chyli, sympathetic trunks, autonomic plexuses, lumbosacral plexus

Posterior abdominal wall

- **Flank layers:** external oblique, internal oblique, transversus abdominis, transversalis fascia. **Posterior:** psoas, quadratus lumborum, erector spinae.
- **Lumbodorsal (thoracolumbar) fascia:** three layers that fuse into one at the **tip of the 12th rib**, the standard entry point for retroperitoneoscopy.
- **Ribs:** access above the 12th rib risks the pleura; the intercostal bundle runs under the rib above.

Fasciae and spaces (three embryological strata)

Space	Boundaries	Contents
Anterior pararenal	Posterior peritoneum and anterior renal fascia	Colon, duodenum, pancreas
Perirenal	Anterior (Toldt) and posterior (Zuckerkandl) laminae of Gerota's fascia	Kidney, adrenal, perirenal fat, hilum, upper ureter, gonadal vessels
Posterior pararenal	Posterior lamina and transversalis fascia	Fat only

- The laminae fuse laterally as the **lateroconal fascia**. The perirenal cone is **open inferiorly**, so urine, blood or pus track into the pelvis.
- Perirenal fat is fine and pale; pararenal fat is coarse and yellow-orange.

Vessels, lymphatics and nerves

- **Aorta** enters at T12. Unpaired anterior branches supply the gut (coeliac, SMA (superior mesenteric artery), IMA (inferior mesenteric artery)); paired lateral branches supply adrenals, kidneys (L1–L2) and gonads; lumbar arteries go to the wall.
- At the hilum the **renal vein lies anterior to the artery**. The **left renal vein is long** (gonadal, adrenal and lumbar tributaries); the right is short.
- Lymph flows right to left; **cisterna chyli** at L1–L2.

- **Nerves at risk:** sympathetic trunk (medial to psoas), genitofemoral (on psoas), subcostal, iliohypogastric, ilioinguinal (on quadratus); hypogastric plexus (ejaculation).

Principles of laparoscopic retroperitoneal surgery

1. **No natural cavity:** the working space must be created (finger and balloon dissection) before insufflation.
2. **Psoas is the key landmark:** it forms the floor. Keep it horizontal on screen; the kidney in Gerota's fascia lies anterior, and the peritoneum is pushed medially.
3. **Direct posterior route to the hilum:** the renal artery, posterior to the vein, is met first, allowing early arterial control.
4. **Peritoneum stays intact:** no bowel mobilisation or adhesiolysis; no spill of tumour, urine or pus into the peritoneum.
5. **Small space:** closely spaced ports, limited triangulation, lens smudging.
6. **Gas behaviour:** CO_2 (carbon dioxide) absorbs through fat, so hypercarbia and subcutaneous emphysema are commoner; haemodynamic swings are smaller than transperitoneal.

Preferred for	Limitations
Adhesions, peritoneal dialysis, morbid obesity, pregnancy; posterior tumours, pyeloplasty, cyst surgery, renal biopsy	Small space, unfamiliar landmarks, harder suturing, learning curve

- **Evidence:** randomised trials and meta-analysis: blood loss, complications and oncology similar to transperitoneal; the consistent advantage is **shorter operative time**.

Salient steps: retroperitoneal laparoscopic nephrectomy

1. **Position:** full flank (90°), table flexed, axillary roll; surgeon behind the patient.
2. **Access (open Hasson):** 1.5–2.5-cm incision just below the 12th rib tip; split muscle, open the lumbodorsal fascia; finger confirms psoas behind and Gerota-covered lower pole in front.
3. **Space creation:** balloon dilator (commercial, or Gaur glove balloon with 600–800 mL saline) anterior to psoas, posterior to Gerota's fascia; blunt-tip balloon-cuff trocar seals the port.
4. **Ports:** two more trocars under vision, one off the 12th rib tip, one posterior and superior to the camera.
5. **Dissection:** along psoas, find the ureter and gonadal vein medially; lift ureter and lower pole to put the hilum on stretch; see arterial pulsation.
6. **Hilum:** clip or staple the **artery first**, then the vein; on the left a **lumbar vein** usually needs division.
7. **Mobilise:** upper pole (adrenal preserved or taken), then lateral and anterior attachments; radical = outside Gerota's fascia.
8. **Extract:** clip ureter, bag the specimen, remove via an extended port; close.

- **Problems:** peritoneal tear (space collapses), emphysema, pneumothorax, vascular injury, flank hernia (intramuscular balloon), rhabdomyolysis (about 1%). **Bail-out:** open the peritoneum and convert.

Diagram to draw: transverse section showing the three spaces, Gerota's laminae and the balloon anterior to psoas.

Key points to remember:

- Three spaces: anterior pararenal, perirenal, posterior pararenal; perirenal cone open below.
- Entry at the 12th rib tip, where the lumbodorsal fascia fuses.
- Psoas is the landmark; artery met first.
- Main advantage: shorter operative time, peritoneum untouched.

Section 10 · Testis, Penis & Urethra — Short answer questions (SAQ)

Surgical management of primary scrotal cancer

Type: SAQ (10 marks)

Source: 2023-Winter Paper 1, Q3e (also answered in 2021-Summer Paper 4, Q3c)

Introduction

- Rare; **SCC (squamous cell carcinoma)** commonest. **Percivall Pott (1775)**: chimney sweeps, the first occupational cancer.
- **Risks:** soot, tar, mineral oils, arsenic, PUVA (psoralen + ultraviolet A), HPV (human papillomavirus).
- **Others:** EMPD (extramammary Paget disease), BCC (basal cell carcinoma), melanoma, sarcoma.
- Drains to **superficial inguinal nodes, crossing the midline**.

Evaluation

- Painless nodule or ulcer on the anterolateral scrotum, often presenting late; **biopsy** confirms.
- Examine both groins (lymphatics cross the midline); FNAC (fine-needle aspiration cytology) of nodes; CT (computed tomography) of abdomen and pelvis for staging.

Staging (Lowe)

Stage	Extent
A1	Confined to scrotum
A2	Invades testis, cord, penis, pubis or perineum
B	Inguinal nodes
C	Pelvic nodes
D	Distant metastases

Primary tumour

1. **Wide local excision, 2 cm margin**, full thickness with dartos; frozen sections.
2. Testes spared unless invaded; hemi- or total scrotectomy for large lesions.
3. A2: en bloc removal of involved testis, penis, perineum or pubis.
4. Positive margin: re-excise.

Reconstruction (Chapter 87)

Defect	Management
<50%, no tension	Primary closure
<50%, tension	Scrotal advancement flap or secondary intention
>50%	Testes in thigh pockets; later STSG (split-thickness skin graft) or flap

Nodes (as in penile cancer)

- **Palpable, cytology positive:** ILND (inguinal lymph node dissection) ± pelvic dissection. Nodal status is the main prognostic factor.

- **Impalpable:** surveillance; sentinel node biopsy in high-grade tumours.
- **Bulky or fixed:** neoadjuvant cisplatin-based chemotherapy, then surgery; RT (radiotherapy) for palliation.
- Routine prophylactic dissection is not advised.

Other types

- **EMPD:** wide excision guided by mapping biopsies; screen for other cancers.
- **Also know:** BCC, excision or Mohs surgery; melanoma, excision by Breslow depth ± sentinel node; sarcoma, wide excision.

Diagram to draw: elliptical excision with 2 cm margin and bilateral inguinal drainage.

> **Key points to remember:**
> - Pott 1775: chimney sweeps.
> - 2 cm margin; testes spared.
> - Nodes decide prognosis.

Varicocele – Pathophysiology

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 1, Q3c

Definition and epidemiology

- Abnormal dilatation and tortuosity of the **pampiniform plexus** of the internal spermatic veins.
- About 15% of men, 35% with primary and 75–81% with secondary infertility. Appears after age 10, peaks at Tanner 3. About 90% left-sided.
- **Venous anatomy:** anterior (internal spermatic), middle (deferential) and posterior (cremasteric) groups; the last two give return after repair and cause recurrence.

Why it develops

1. **Left-sided anatomy:** the left vein enters the left renal vein at a right angle, 8–10 cm higher than the right enters the IVC (inferior vena cava). The taller column overloads its valves. The right vein enters obliquely, which acts as a valve.
2. **Valve incompetence** or absence; continuous reflux predicts progression.
3. **Nutcracker:** left renal vein compressed between aorta and SMA (superior mesenteric artery).
4. **Collateral pathways**; **genetic** (4–8 times commoner in first-degree relatives); tall, thin habitus.

- **Secondary:** a **right-sided or non-decompressing** varicocele suggests a retroperitoneal or renal mass, so image the abdomen.

How it damages the testis

Mechanism	Explanation
Heat (primary)	Lost countercurrent heat exchange; the testis normally runs 2–4 °C below core
Oxidative stress	Raised reactive oxygen species → sperm DNA fragmentation
Hypoperfusion and stasis	Hypoxia, toxin accumulation
Apoptosis	Germ cell loss
Leydig dysfunction	Lower testosterone than controls

- The injury is **progressive and duration-dependent**, and may affect **both** testes.

Consequences

- **Testicular hypotrophy** (asymmetry above about 20% guides adolescent repair); poor semen; hypogonadism; ache on standing (2–11%).
- Repair improves semen, DNA fragmentation, testosterone and pregnancy (**Madgar RCT [randomised controlled trial], 1995:** 44% versus 10% at 1 year).

Grading

Grade	Finding
Subclinical	Doppler only (not treated)
I	Palpable with Valsalva only
II	Palpable, not visible
III	Visible

Diagram to draw: left and right testicular veins draining into renal vein and IVC.

Key points to remember:

- Left vein: right-angled entry, 8–10 cm higher column; 90% left.
- Heat (lost countercurrent exchange) is the main injury; also oxidative DNA damage.
- Right or non-decompressing varicocele: image for a renal mass.
- Injury is progressive; subclinical varicocele is not treated.

Treatment of submeatal stricture urethra in males

Type: SAQ (10 marks)

Source: Practice Set 11 Paper 1, Q3c (meatal stenosis also answered in 2019-Summer Paper 2, Q3a)

"Submeatal stricture" is read as a stricture of the meatus and fossa navicularis (glanular urethra).

Causes

- **LS (lichen sclerosus, formerly BXO [balanitis xerotica obliterans]):** main inflammatory cause today; begins at the glans and meatus and progresses proximally.
- **Iatrogenic:** catheter or instrument trauma, TURP (transurethral resection of prostate).
- **Failed hypospadias repair**; post-circumcision ammoniacal meatitis (children).

Evaluation

- Spraying stream, straining; uroflowmetry.
- **RUG (retrograde urethrogram) before any meatal surgery**: a proximal stricture often coexists, especially in LS.
- Biopsy suspicious lesions: LS turns malignant (squamous cell carcinoma) in up to 8%.

Treatment

Option	Method	Comment
Topical steroid	Clobetasol 0.05% once or twice daily for 1–2 months, then alternate days	Early LS
Dilation or self-calibration	Meatal dilator lubricated with clobetasol	Elderly, unfit; seldom durable
Ventral meatotomy	Crush and incise ventrally; **three sutures** (apex and each side), mucosa to glans skin; dilate afterwards	Simple stenosis; in LS extend it with an inverted-V relaxing incision
Dorsal meatotomy (Y-V plasty)	Excise the scar ridge, advance a flap	Effective; cosmesis poorer
Fossa navicularis reconstruction (flaps)	**Jordan flap:** ventral urethrotomy through the stenosis; **transverse ventral penile skin island on dartos** transposed and inverted into the defect. **De Sy:** ventral longitudinal island advanced	Non-LS; excellent function and cosmesis
Graft	**BMG (buccal mucosal graft)** inlay or onlay, one or two stages; glans wings developed	**LS: oral mucosa, never genital skin**
Perineal urethrostomy	Bypass	Pan-anterior LS, failed repairs, elderly

- LS affects genital skin, so skin flaps recur; it spares the proximal urethra, so perineal urethrostomy works.
- Children: ventral meatotomy plus parental dilation for 7–10 days.

Complications

- Restenosis (commoner in LS), bleeding, spraying, poor cosmesis.

Diagram to draw: Jordan flap: transverse ventral skin island on dartos inverted into the ventral meatotomy.

Key points to remember:

- RUG first; adult meatal stenosis is usually the tip of a fossa stricture.
- Non-LS: meatotomy or Jordan/De Sy skin island flap.
- LS: clobetasol, buccal mucosa (never genital skin), perineal urethrostomy if extensive.

Perineal urethrostomy.

Type: SAQ (10 marks)

Source: Practice Set 12 Paper 1, Q3c (also answered in 2021-Summer Paper 2, Q3c)

Definition

- PU (perineal urethrostomy): the proximal bulbar urethra is opened onto perineal skin as a permanent or staged new meatus; the patient voids sitting. It **bypasses**, rather than reconstructs, a diseased anterior urethra.

Indications

- **Complex anterior strictures:** LS (lichen sclerosus) panurethral disease, multiple failed urethroplasties, failed hypospadias repair.
- **Elderly or high-risk men** unwilling or unfit for staged reconstruction.
- **First stage** of staged (Johanson-type) urethroplasty.
- After **penectomy** or distal urethrectomy for cancer.
- Temporary: after phalloplasty strictures; Fournier gangrene; resectoscope access in morbid obesity.
- AUA (American Urological Association) guideline (2016): PU is an acceptable long-term alternative to urethroplasty.
- LS spares the proximal urethra, so PU suits LS.

Contraindications and limits

- Disease reaching the proximal bulbar or membranous urethra; prior pelvic radiation and LS raise restricture.

Techniques

1. Classical (Blandy) PU: inverted-U perineal skin flap advanced into a ventral bulbar urethrotomy.

2. BMG (buccal mucosal graft) PU, steps:

1. High lithotomy; inverted-U incision; split bulbospongiosus in the midline.
2. Free and **transect** the spongiosum; **dorsal urethrotomy** into healthy mucosa.
3. Bring healthy ventral urethra to skin; calibrate proximally to 30 Fr; close distal urethra.
4. Quilt a BMG on the corpora to complete the dorsal wall up to the skin.

- The graft prevents the circumferential scar that narrows a deep PU.

Complications and outcomes

- **Stenosis of the urethrostomy** (main failure): revision or dilation.
- Spraying, dribbling, perineal wetness, wound infection, body-image concerns.
- Compared with staged reconstruction it is a shorter, single operation with less morbidity.
- High mid-term success and satisfaction in selected patients.

Diagram to draw: sagittal perineum with the inverted-U flap sutured into the opened bulbar urethra.

> **Key points to remember:**
> - PU bypasses the diseased anterior urethra; patient voids sitting.
> - Ideal for LS, failed repairs and elderly men.
> - BMG PU prevents circumferential scarring; calibrate to 30 Fr.

Testicular microlithiasis

Type: SAQ (10 marks)

Source: Practice Set 14 Paper 1, Q3a

Definition

- **TM (testicular microlithiasis):** multiple small (<3 mm) bright echogenic foci **without acoustic shadowing** scattered in the testicular parenchyma on ultrasound.
- **Classic TM:** ≥5 microliths per testis (or per image); **limited TM:** <5, of no clinical significance.

Pathology

- Laminated calcified bodies **inside seminiferous tubules**, formed on degenerated tubular cells that Sertoli cells fail to clear. Affected tubules are often obliterated.

Prevalence

Population	Prevalence
General population	5–10% (population-based 4.2%, rising with age)
Multicentre review, >37,000 males	2.9%
Boys with cryptorchidism	3.5%
Previously cryptorchid testis vs contralateral	14% vs 19% (no difference)

Associations

- Cryptorchidism, subfertility, **GCNIS (germ cell neoplasia in situ)** and TGCT (testicular germ cell tumour), Klinefelter syndrome, Down syndrome.

Cancer significance

- Tumours were found in 4.64% with TM vs 0.33% without, but cryptorchidism was not accounted for.
- **TM is not clearly an independent risk factor** for TGCT. An EAU (European Association of Urology) paediatric panel review of several hundred children found only one cancer through puberty.
- **Exception:** contralateral TM in a man with previous GCT (germ cell tumour) signals a higher GCNIS risk.

Management

- **Isolated TM, no risk factors:** reassure; **no further evaluation, no routine ultrasound follow-up**.
- **TM with risk factors** (previous GCT, cryptorchidism, infertility, atrophic testis): teach **monthly self-examination from puberty**; ultrasound only for a palpable change.
- **Biopsy:** not for TM alone. Contralateral biopsy at orchidectomy is considered when GCNIS risk is high: atrophic testis (<12 mL), cryptorchidism, age <40.
- Counsel carefully: incidental TM causes anxiety and unnecessary tests.

Differential diagnosis on ultrasound

- Coarse calcification, "burned-out" tumour scar, stellate scar (these need orchidectomy if intratesticular), scrotal pearls, rete testis ectasia.

Also know: routine screening ultrasound for infertility is discouraged because of incidental TM; follow-up schedules remain undefined.

Diagram to draw: ultrasound sketch of a testis with scattered non-shadowing echogenic dots.

Key points to remember:

- Classic TM ≥5 microliths; limited <5.
- Not an independent cancer risk factor in isolation.
- Self-examination only with risk factors; no routine imaging.

Growing teratoma syndrome

Type: SAQ (10 marks)

Source: Practice Set 20 SAQ Sheet, Q3a

Definition

- GTS (growing teratoma syndrome; Logothetis, 1982): metastatic masses of an NSGCT (non-seminomatous germ cell tumour) **enlarge during or after chemotherapy while AFP (alpha-fetoprotein) and hCG (human chorionic gonadotrophin) are normal or falling**. Resection shows **mature teratoma only**.
- Uncommon: about 2–8% of metastatic NSGCT.

Pathogenesis

- Teratoma is **histologically benign but genetically malignant** (i(12p), aneuploidy) and **chemoresistant**.
- Chemotherapy kills the sensitive elements (embryonal, yolk sac), so markers fall; the teratoma keeps growing, mostly by enlarging cysts. Chemotherapy-induced differentiation is a second theory.
- Main risk factor: teratoma in the orchidectomy specimen.

Clinical features

- Retroperitoneum commonest; also lung, mediastinum, supraclavicular nodes.
- Often silent; or compression: ureteric obstruction, IVC (inferior vena cava) compression or thrombosis, bowel obstruction, back pain.

Evaluation

- Markers before every cycle; **rising markers mean viable cancer, not GTS**.
- CT (computed tomography): enlarging, well-defined, **cystic** mass, sometimes with fat or calcification.
- **FDG-PET (fluorodeoxyglucose positron emission tomography) has no role**: teratoma is not FDG-avid.
- **Differential:** progressing viable GCT (germ cell tumour; markers rise); malignant transformation (somatic sarcoma or carcinoma); necrosis or fibrosis (mass shrinks).

Management

- **Stop chemotherapy**; do not escalate to salvage chemotherapy (ineffective).
- **Prompt complete surgical resection**: PC-RPLND (post-chemotherapy retroperitoneal lymph node dissection), with resection of every other site; adjacent vessel, kidney or bowel resection may be needed. Delay lets the mass become unresectable.
- Radiotherapy is ineffective.
- **Outcome:** favourable after complete resection (long-term survival 75–90% after resection of residual teratoma). Incomplete resection leads to regrowth, **malignant transformation** (rhabdomyosarcoma commonest) or **late relapse**.
- **Also know:** CDK4/6 (cyclin-dependent kinase 4/6) inhibitors studied for unresectable teratoma; lifelong CT and marker follow-up.

Diagram to draw: graph of mass size rising while marker curve falls during chemotherapy cycles.

Key points to remember:

- Enlarging mass plus normal or falling markers equals GTS.
- Teratoma is chemoresistant and not FDG-avid.
- Treatment is complete surgical resection, not more chemotherapy.

Rapid Revision Sheet – Section 10 · Testis, Penis & Urethra

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias (LAQ)

- Double zipper: canalisation, then fold fusion; glanular urethra endodermal.
- Bulbar urethra eccentric (dorsal); Cowper ducts 3 and 9 o'clock; dual retrograde glanular blood supply.
- DVIU: bulbar, <1.5–2 cm, first attempt; Santucci about 8% long-term.
- Optilume ROBUST III 74.6% vs 26.8%.
- Megalourethra: fusiform worse than scaphoid; duplication: ventral channel functional.
- About 70% of male epispadias are penopubic; diastasis >3 cm → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL.

Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications (LAQ)

- Right testis to (→) interaortocaval; left to left para-aortic; crossover only right to left.
- Limits/boundaries: renal vessels (hila), ureters, ureteric crossing of iliacs (ureter–iliac crossing); omit suprahilar and interiliac.
- Nerve-sparing of L1–L4 fibres gives antegrade ejaculation >90%; the fibres meet at the IMA: unilateral templates spare ejaculation.
- PC-RPLND (post-chemotherapy): full bilateral template remains standard unless mass ≤5 cm in landing zone.
- Scrotum and penis → inguinal nodes; never operate trans-scrotally.
- Penile drainage is bilateral and stepwise without skips: stage both groins, PLND only if inguinal nodes positive.
- pT1b or higher: DSNB or ILND within 3 months; Kroon 84% vs 35%.

Enlist various grafts used in urology and role of BMG in urology (LAQ)

- Take: imbibition 48 h, then inosculation 48 h; graft lives on its bed.
- BMG: non-keratinised, vascular lamina propria, wet-adapted.
- Avoid Stensen's duct; buccal equals lingual in success.
- Staged BMG, not skin, in lichen sclerosus.

Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy. (LAQ)

- Three spaces: anterior pararenal, perirenal, posterior pararenal; perirenal cone open below.
- Entry at the 12th rib tip, where the lumbodorsal fascia fuses.
- Psoas is the landmark; artery met first.
- Main advantage: shorter operative time, peritoneum untouched.

Surgical management of primary scrotal cancer (SAQ)

- Pott 1775: chimney sweeps.
- 2 cm margin; testes spared.
- Nodes decide prognosis.

Varicocele – Pathophysiology (SAQ)

- Left vein: right-angled entry, 8–10 cm higher column; 90% left.
- Heat (lost countercurrent exchange) is the main injury; also oxidative DNA damage.

- Right or non-decompressing varicocele: image for a renal mass.
- Injury is progressive; subclinical varicocele is not treated.

Treatment of submeatal stricture urethra in males (SAQ)

- RUG first; adult meatal stenosis is usually the tip of a fossa stricture.
- Non-LS: meatotomy or Jordan/De Sy skin island flap.
- LS: clobetasol, buccal mucosa (never genital skin), perineal urethrostomy if extensive.

Perineal urethrostomy. (SAQ)

- PU bypasses the diseased anterior urethra; patient voids sitting.
- Ideal for LS, failed repairs and elderly men.
- BMG PU prevents circumferential scarring; calibrate to 30 Fr.

Testicular microlithiasis (SAQ)

- Classic TM ≥5 microliths; limited <5.
- Not an independent cancer risk factor in isolation.
- Self-examination only with risk factors; no routine imaging.

Growing teratoma syndrome (SAQ)

- Enlarging mass plus normal or falling markers equals GTS.
- Teratoma is chemoresistant and not FDG-avid.
- Treatment is complete surgical resection, not more chemotherapy.

Section 11 · Stones, Obstruction & Trauma — Long answer questions (LAQ)

Describe Pathogenesis of Congenital Pelvi Ureteric Junction Obstruction its diagnosis and Management.

Type: LAQ (25 marks)

Source: 2025-Winter Paper 1, Q1 (related: 2021-Summer Paper 1, Q3e; 2016-Summer Paper 3, Q2)

Definition

- PUJO (pelviureteric junction obstruction) is impaired flow of urine from the pelvis into the ureter. **Koff:** obstruction is "any restriction to urinary outflow that, if left untreated, will lead to progressive kidney damage". **Dilatation is not obstruction.**
- About 1 in 1500 live births; commonest cause of antenatal hydronephrosis. Male and left predominance; **bilateral in 10–40%**.

Pathogenesis

Type	Lesion	Typical patient
Intrinsic (commonest)	Short adynamic segment: interrupted circular muscle, excess collagen, fewer pacemaker (interstitial Cajal-like) cells. The peristaltic wave cannot cross, so the pelvis empties poorly	Neonate or infant, antenatal finding
Extrinsic	**Lower-pole crossing vessel** anterior to the PUJ; the ureter kinks over it	Older child or adult with intermittent pain
Secondary	Kinking by a tortuous ureter in high-grade VUR (vesicoureteral reflux); fibroepithelial polyp; high insertion	Any age

- **Crossing vessels** are persistent embryonic renal arteries. They occur at a fifth to a quarter of **normal** PUJs, so a vessel alone does not prove cause.
- **Also know:** proposed incomplete ureteric recanalisation; horseshoe kidney and duplex lower moiety associations.

Effects of obstruction (pathophysiology)

- Pelvic pressure rises; filtration and perfusion fall.
- **Immature kidney:** apoptosis and impaired nephrogenesis. **Mature kidney:** hypertrophy, then fibrosis (TGF-β [transforming growth factor-beta] signalling). Fibrosis, not dilatation, causes loss of function.
- **Also know:** concentrating defect, salt wasting; rarely renin-mediated hypertension.

Clinical features

- **Antenatal ultrasound** (commonest); infants: UTI (urinary tract infection), flank mass, failure to thrive.
- Older children and adults: **intermittent loin pain with vomiting (Dietl crisis)**, haematuria after minor trauma, stones, infection.
- **Associated anomalies (up to 50%):** contralateral PUJO, dysplasia or MCDK (multicystic dysplastic kidney), VUR (up to 40%).

Diagnosis

1. Antenatal grading (APD, anteroposterior renal pelvic diameter)

	Second trimester	Third trimester
Mild	4 to <7 mm	4 to <9 mm
Moderate	7–10 mm	9–15 mm
Severe	>10 mm	>15 mm

- **UTD (urinary tract dilation) 2014 consensus:** six parameters (APD, central/peripheral calyces, parenchymal thickness and appearance, ureter, bladder). Prenatal A1 (low) or A2–3; postnatal **P1** (APD 10–<15 mm, central calyces), **P2** (≥15 mm or peripheral calyces or ureter), **P3** (abnormal parenchyma).
- **SFU (Society for Fetal Urology) grades 0–4:** 1 pelvis just seen; 2 pelvis plus major calyces; 3 all calyces, normal parenchyma; **4 parenchymal thinning**.

2. Postnatal ultrasound after **48 hours** (neonatal dehydration under-reads dilatation): large pelvis, rounded calyces, ureter not seen.

3. MAG3 (technetium-99m mercaptoacetyltriglycine) diuretic renogram ("well-tempered": hydrated, catheterised, furosemide)

- **DRF (differential renal function)** (normal 45–55%); drainage half-time <10 min normal, 10–20 equivocal, **>20 min obstructed** (adult chapter: >40 min).

4. Others: VCUG (voiding cystourethrogram) by UTI risk; MRU (magnetic resonance urography) or CT (computed tomography) angiography for vessels; **Whitaker test** (10 mL/min; >15–20 cm H_2O obstructed) if equivocal.

Management

Conservative: most resolve, about **one-third need surgery**; serial ultrasound and renogram.

Indications for pyeloplasty

- **DRF <40%** or a **fall of >10%** on serial scans.
- **Increasing APD** or SFU 4 / P3 with poor drainage.
- **Breakthrough UTI**, symptoms, stones, hypertension.
- Bilateral or solitary kidney.

Anderson–Hynes dismembered pyeloplasty: gold standard (~95% success)

1. Expose the PUJ; free the lower pole and look for a crossing vessel.
2. Stay sutures; transect the pelvis; reduce a redundant pelvis.
3. **Spatulate the ureter laterally** into healthy ureter; keep the narrow segment as a "no-touch" handle.
4. **Transpose** the PUJ anterior to any vessel (never divide it).
5. Watertight, tension-free, dependent anastomosis (5-0/6-0 absorbable); excise the segment.
6. Double-J or externalised stent (1–4 weeks equivalent); Foley 24–48 h with a double-J.

- **Route:** open preferred in infants; laparoscopic equals open with less pain; **robotic** (easier suturing, shorter learning curve) is now primary at most major centres.
- **Alternatives:** Foley Y-V plasty (high insertion); Culp spiral or vertical flap (long segment); **vascular hitch** (crossing vessel with normal PUJ); **endopyelotomy** (73% overall; **42% with vs 86% without a crossing vessel**), mainly for secondary PUJO.
- **Nephrectomy:** DRF **<15% in adults** (children: no agreed cut-off).

Complications and follow-up

- Urine leak, stent problems, UTI; early failure from oedema or clot (temporary nephrostomy).

- **Late failure:** redo pyeloplasty (best), endopyelotomy or ureterocalicostomy. Renogram and ultrasound after stent removal.

Recent advances

- Robotic redo with buccal graft and ICG (indocyanine green) fluorescence; pyramidal thickness ≤3 mm predicts surgery; urinary biomarkers.

Diagram to draw: Anderson–Hynes steps: transection, lateral spatulation, transposition anterior to the crossing vessel, dependent anastomosis.

> **Key points to remember:**
> - Intrinsic adynamic segment (infants) vs crossing vessel (older children, intermittent pain).
> - SFU 4 = parenchymal thinning; UTD P3 = abnormal parenchyma.
> - Operate for DRF <40%, fall >10%, rising APD, UTI or symptoms.
> - Anderson–Hynes ~95%; never divide a crossing vessel.

Discuss pathophysiology of renal stones and also describe how to prevent its recurrence

Type: LAQ (25 marks)

Source: Practice Set 4 Paper 1, Q1 (also answered in 2020-Summer Paper 4, Q2)

Introduction

- Stone disease is a **chronic metabolic disease** with recurrence up to **50% within 5 years**. Removal is surgical; recurrence is metabolic.
- Calcium oxalate (with or without apatite) forms about 80%; uric acid and struvite about 7% each; cystine 1–3%.

Risk factors

- **Epidemiological:** male sex, hot climate (low urine volume), animal protein and salt, family history.
- **Systemic:** metabolic syndrome and diabetes (insulin resistance lowers urine pH), gout, hyperparathyroidism, RTA (renal tubular acidosis), bowel disease or bypass, sarcoidosis.
- **Stasis:** PUJ (pelviureteric junction) obstruction, horseshoe kidney, medullary sponge kidney. **Drugs:** indinavir, topiramate, vitamin C and D excess.

Physicochemistry: state of saturation

State	Concentration product (CP)	What happens
Undersaturated	Below Ksp (solubility product)	No crystals; stones may dissolve
Metastable	Between Ksp and Kf (formation product)	No spontaneous nucleation, but growth on existing crystals; inhibitors act here. **Zone of therapy**
Unstable	Above Kf	Spontaneous nucleation, growth, aggregation

Steps of stone formation

1. **Supersaturation:** low volume; high calcium, oxalate or uric acid; low citrate.
2. **Nucleation:** homogeneous, or **heterogeneous** on debris or other crystals (needs less energy).
3. **Growth and aggregation:** aggregates, not single crystals, can block a tubule within the 5–7 minute transit (**free-particle theory**).
4. **Retention (fixed-particle theory): Randall plaque** (calcium apatite) starts in the basement membrane of the thin loop of Henle, spreads through the interstitium, erodes the papillary urothelium and anchors calcium oxalate growth (idiopathic formers). **Ductal plugs** (crystals in collecting ducts) point to a secondary cause: enteric hyperoxaluria, distal RTA, hyperparathyroidism, cystinuria.

Inhibitors and matrix

- **Inhibitors: citrate** (complexes calcium), **magnesium** (complexes oxalate), pyrophosphate, glycosaminoglycans, nephrocalcin, **Tamm–Horsfall protein**, osteopontin. **No inhibitor of uric acid** exists. Matrix is about 2.5% of stone weight.

Pathophysiology by urinary abnormality

Abnormality (definition)	Mechanism
Hypercalciuria (>200 mg/day, or >4 mg/kg/day)	Absorptive (gut), renal leak (secondary high PTH [parathyroid hormone]), resorptive (primary hyperparathyroidism): Pak classification
Hyperoxaluria (>40 mg/day)	**Enteric:** malabsorbed fat binds calcium, freeing oxalate; **primary:** hepatic enzyme defect (PH1, AGXT gene)
Hyperuricosuria (>600 mg/day)	Urate promotes calcium oxalate nucleation
Hypocitraturia (<320 mg/day)	Distal RTA, diarrhoea, hypokalaemia, animal protein (acid load)
Uric acid stones	**Low urine pH (<5.5)** more than excess uric acid; pKa 5.5
Cystine	Autosomal recessive defect of tubular cystine reabsorption
Struvite	**Urease** (Proteus) splits urea → ammonia, pH >7.2 → magnesium ammonium phosphate and carbonate apatite

Prevention of recurrence

1. Screen every first-time former (AUA [American Urological Association], EAU [European Association of Urology])

- History, serum electrolytes, creatinine, calcium, uric acid; urine pH and culture; **stone analysis**; imaging. This sorts patients into high and low risk.

2. General measures (all formers)

- **Fluid for urine ≥2.5 L/day** (drink about 3 L). Borghi 1996: 5-year recurrence 12.1% vs 27%.
- **Normal calcium**, low sodium, less animal protein, more fruit and vegetables (Borghi 2002: halved recurrence versus a low-calcium diet, which raises oxalate absorption).
- Limit high-oxalate foods (spinach, nuts, chocolate); weight loss.

3. Metabolic evaluation (high-risk or recurrent formers, children)

- One or two **24-hour urines**: volume, pH, calcium, oxalate, uric acid, citrate, sodium, potassium, creatinine. Therapy targeted to these results reduces stone events.

4. Selective drug therapy

Abnormality	Treatment	Rationale
Hypercalciuria	**Thiazide:** hydrochlorothiazide 25 mg twice daily, chlorthalidone 25–50 mg, indapamide 2.5 mg	Distal calcium reabsorption; add potassium citrate (hypokalaemia causes hypocitraturia)
Hypocitraturia, distal RTA	**Potassium citrate 30–60 mEq/day**	Alkali load reduces tubular citrate reabsorption
Hyperuricosuria with calcium stones	Allopurinol 300 mg/day	Watch hypersensitivity
Uric acid stones	Potassium citrate to **pH 6.0–6.5** (avoid >7.0)	pH disease; allopurinol only for hyperuricosuria
Enteric hyperoxaluria	**Calcium with meals**, low-oxalate, low-fat diet, magnesium, liquid citrate	Binds oxalate in gut
Primary hyperoxaluria type 1	Pyridoxine; **lumasiran** (RNA [ribonucleic acid] interference)	ILLUMINATE-A
Cystinuria	Urine ≥3 L/day, pH **7.0–7.5**; **tiopronin**	Cystine <250 mg/L; thiol forms soluble mixed disulfide

Abnormality	Treatment	Rationale
Struvite	**Complete clearance**, antibiotics; AHA (acetohydroxamic acid) 250 mg three times daily if unfit for surgery	Urease inhibitor; thrombosis

5. Monitoring

- 24-hour urine within **6 months** of starting therapy, then yearly; reanalyse any new stone.

Recent advances

- **ROKS (Recurrence of Kidney Stone) nomogram** predicts individual risk. **Also know:** NOSTONE (2023): hydrochlorothiazide did not reduce recurrence; citrus juices are adjuncts; *Oxalobacter formigenes* loss.

Diagram to draw: saturation zones (Ksp, Kf) and Randall plaque on a renal papilla.

Key points to remember:

- Metastable zone between Ksp and Kf: where inhibitors and therapy act.
- Randall plaque anchors idiopathic calcium oxalate stones; ductal plugs suggest a secondary cause.
- Urine ≥2.5 L/day; normal calcium, low salt and animal protein (Borghi).
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.

Section 11 · Stones, Obstruction & Trauma — Short answer questions (SAQ)

Flexible Ureterorenoscopy

Type: SAQ (10 marks)

Source: 2017-Summer Paper 1, Q3b

Introduction

- Flexible ureterorenoscopy (fURS) or RIRS (retrograde intrarenal surgery) inspects and treats the whole upper tract retrogradely. Marshall reported it in 1964; active deflection and a working channel made it practical.

Instrument

Feature	Details
Tip	4.5–7.5 Fr, tapered (digital tips up to about 8.5 Fr)
Working channel	3.6 Fr (laser fibre, basket)
Deflection	≥270° both ways; lower pole needs about 140° (up to 175°)
Types	**Fibreoptic** (smallest, image degrades); **digital** (chip on tip, better image, not more durable); **single-use** (LithoVue 2016, 7.7 Fr; cheaper below about 99 cases a year)

- Repair is needed about every 15 cases.

Indications

- **Stones:** renal <2 cm; lower pole <1 cm (or >1 cm as an alternative to PCNL, percutaneous nephrolithotomy); patients who cannot stop anticoagulants; failed SWL (shock wave lithotripsy); obesity.
- **Others:** diagnosis and ablation of low-grade UTUC (upper tract urothelial carcinoma); filling defects, positive cytology; endopyelotomy, short strictures; calyceal diverticulum; essential haematuria; foreign bodies.

Technique

1. Sterile urine; single-dose antibiotic.
2. Cystoscopy; **safety guidewire**; retrograde pyelogram.
3. **Access sheath** (usually 12/14 Fr) over a stiff wire under fluoroscopy: lowers intrarenal pressure, allows repeated passes; severe ureteric injury in about 1 in 8.
4. Low-pressure irrigation; **Ho:YAG (holmium:yttrium-aluminium-garnet) or thulium fibre laser**: dusting or fragment-and-basket. Keep power ≤10 W in ureter, ≤25 W in kidney (tissue damage above 43 °C).
5. Inspect ureter on withdrawal; stent unless uncomplicated.

- **Difficult lower pole:** steep infundibulopelvic angle, long (>10 mm) or narrow (<5 mm) infundibulum (EAU (European Association of Urology) factors against SWL; they also hinder fURS). Relocate the stone to an upper calyx with a basket.

Complications

- Sepsis about 5%; stricture about 3%; bleeding or subcapsular haematoma <1%; perforation; avulsion <1%; stent symptoms; scope damage.

Recent advances

- Suction access sheaths, intrarenal pressure sensors, thulium laser, robotic fURS.

Diagram to draw: flexible scope deflected into the lower pole calyx through an access sheath.

> **Key points to remember:**
> - 3.6 Fr channel; ≥270° deflection.
> - Access sheath lowers pressure but injures ureter in about 1 in 8.
> - Laser power ≤10 W ureter, ≤25 W kidney.

Distal Renal tubular acidosis.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 1, Q3d

Definition

- **Distal (type 1) RTA (renal tubular acidosis):** the collecting duct cannot secrete H^+ despite systemic acidosis. Result: **hypokalaemic, hyperchloraemic, normal anion gap metabolic acidosis** with **urine pH >6**. The only acidosis with alkaline urine.

Normal acidification and the defect

- **α-intercalated cells** secrete H^+ by apical H^+-ATPase; bicarbonate exits by basolateral Cl^-/HCO_3^- exchanger (AE1). Normally urine pH can fall to 4.5.
- In distal RTA, potassium is secreted in place of H^+ → hypokalaemia.

Causes

Inherited	Acquired (secondary)
SLC4A1 (AE1): mostly dominant	Sjögren syndrome, SLE (systemic lupus erythematosus)
ATP6V1B1, ATP6V0A4 (H^+-ATPase): recessive, **sensorineural deafness**	Obstructive uropathy, pyelonephritis, medullary sponge kidney
CA2: mixed proximal–distal	Hypercalciuria, hyperparathyroidism, amphotericin B

Why stones form (urological importance)

- **Stones in up to 70%**: **calcium phosphate** stones and **nephrocalcinosis**. Stones lead to diagnosis in up to half.
- **Profound hypocitraturia** (most important): acidosis increases proximal citrate reabsorption.
- **Hypercalciuria:** bone buffering releases calcium. **High urine pH** precipitates calcium phosphate.

Clinical features

- Stones, nephrocalcinosis, weakness (hypokalaemia), osteomalacia or rickets; children fail to thrive.
- **Incomplete distal RTA:** normal bicarbonate, but urine pH cannot fall below 5.5 after an acid load.

Diagnosis

- Low bicarbonate, high chloride, low potassium; urine pH >6 on a fasting morning sample.
- **UAG (urine anion gap) = $(Na^+ + K^+) - Cl^-$: positive** (low ammonium excretion) → renal cause; negative in diarrhoea.
- **Ammonium chloride loading** (0.1 g/kg orally; hourly urine pH): **failure to fall below 5.5** confirms. Furosemide–fludrocortisone test is better tolerated.
- 24-hour urine: low citrate, high calcium; ultrasound for nephrocalcinosis.

Treatment

- **Potassium citrate** (about 1–2 mEq/kg/day in divided doses): corrects acidosis, hypokalaemia and hypocitraturia, reduces calcium excretion and stone formation. Sodium alkali is less suitable because sodium raises urinary calcium.

- Treat the cause; manage stones endourologically.
- **Also know:** proximal (type 2) RTA (bicarbonate wasting, Fanconi, few stones); type 4 (hyperkalaemic, low urine pH; diabetes, obstruction); topiramate and acetazolamide mimic distal RTA.

Diagram to draw: α-intercalated cell with H^+-ATPase, AE1 and the stone mechanism.

Key points to remember:

- Urine pH >6 with acidosis; hypokalaemia; positive UAG.
- Calcium phosphate stones in up to 70%; hypocitraturia key.
- NH_4Cl test: pH fails to fall below 5.5.
- Potassium citrate is the treatment.

Radionuclide studies and diuretic renography in PUJ obstruction

Asked as:

- Radionuclide studies for PUJ obstruction. — *Summer 2021 Paper I Q3e (SAQ)*
- Radionuclear studies in PUJ obstruction — *Practice Set 4 Paper I Q3e (SAQ)*
- Diureteric renography — *Practice Set 2 Paper I Q3f (SAQ)*

Type: SAQ (10 marks)

(Practice Set 4: also answered in 2021-Summer Paper 1, Q3e. Practice Set 2: PUJ obstruction renography also in 2021-Summer Paper 1, Q3e; tracers: see Practice Set 2 Paper 1, Q3e)

"Diureteric" is read as diuretic renography.

Introduction / principle

- **Dilatation is not obstruction.** PUJ (pelviureteric junction) obstruction is impaired drainage that damages the kidney. **Diuretic renography** is the functional standard (functional gold standard for PUJ obstruction) and best follow-up test (best follow-up study).
- A capacious pelvis retains tracer passively and furosemide washes it out; an obstructed system cannot empty even under diuresis.

Tracers

Tracer	Handling	Use
^{99m}Tc-**MAG3** (technetium-99m mercaptoacetyltriglycine)	Tubular secretion	**Agent of choice** (tracer of choice): split function and drainage; good images (good curves) even with poor function
^{99m}Tc-**DTPA** (diethylenetriaminepentaacetic acid)	Glomerular filtration	GFR (glomerular filtration rate); poor in renal impairment; an alternative to MAG3
^{99m}Tc-**DMSA** (dimercaptosuccinic acid)	Cortical retention	Static split function and scars; no drainage data

"Well-tempered" diuretic renogram (standard protocol)

Element	How	Why
Hydration	Oral, or IV (intravenous) fluid in children (about 15 mL/kg)	Dehydration mimics obstruction
Bladder catheter	Urethral catheter	Removes full-bladder (back-pressure) and reflux (VUR, vesicoureteral reflux) effects
Furosemide	0.5 mg/kg (adult maximum 40 mg) in adults; 1 mg/kg in children	Maximal diuresis
Timing	**F+20** (classic; 20 min after tracer, when the pelvis is full; Campbell authors use F+15), **F0** (with tracer) or **F−15** (15 min before, so the system is already in diuresis)	Diuretic before filling (early diuretic into an unfilled system) misleads; always state timing
Regions of interest	Regions of interest over cortex (for split) and pelvis (dilated pelvis for drainage); then upright images	Wrong placement gives spurious values

- Well-tempered: hydrate, catheter, standard diuretic dose and timing.
- First study after about 4–6 weeks of age (immature function).

Outputs

- **DRF (differential renal function)** from counts at 1–3 min (1–3 minute counts); normal 45–55%.
- **Drainage half-time (T½, half-clearance time)** after furosemide, from maximal filling, not from injection:

T½ after furosemide	Interpretation
<10 min (minutes)	Not obstructed
10–20 min (minutes)	Equivocal (indeterminate)
>20 min (minutes)	Obstructed

- **O'Reilly curves:** I normal; II obstructed (keeps rising); IIIa dilated, non-obstructed (falls after furosemide); IIIb equivocal.
- **Also know:** P40 (activity remaining / left at 40 min) raises sensitivity; output efficiency; NORA (normalised residual activity); a **biphasic curve** and poor upright (post-void) drainage predict progression.

Indications

- PUJ (pelviureteric junction) obstruction, antenatal hydronephrosis, megaureter, after pyeloplasty, transplant ureter.

Indications for pyeloplasty from renography

- Consider surgery for:
- **DRF <40%**, **fall in DRF** on serial comparable scans (**>10%** on serial scans; ≥5% in children; 10% in older series), **T½ >20 min** (minutes) or upright retention; also (poor drainage with) symptoms, infection, increasing dilatation.
- Adult DRF <15%: likely non-salvageable; nephrostomy and reassess, or nephrectomy.

Pitfalls

- Dehydration, early diuretic (mistimed diuretic), full bladder, wrong ROI (region of interest), very poor function, very capacious pelvis.
- One Campbell chapter calls T½ >40 min obstructed; children have no agreed (paediatric T½) cut-off (false positives up to 44%; neonatal false-positive rates reach 44%); one Campbell chapter uses **T½ >40 min** in adults, so **serial trends and curve shape** matter / decide.

Follow-up and alternatives

- After pyeloplasty: renogram 1 month after stent removal; spontaneous drainage means success.
- Equivocal (cases) → **Whitaker test** (>15–20 cm H_2O obstructed) or MRU (magnetic resonance urography).

Diagram to draw: normal, obstructed and dilated non-obstructed renogram curves (O'Reilly curves I, II, IIIa and IIIb) with furosemide arrow.

Key points to remember:

- MAG3 is the tracer of choice; hydrate and catheterise.
- Well-tempered: hydrate, catheter, standard diuretic dose and timing.
- Furosemide 0.5 mg/kg (adult maximum 40 mg); 1 mg/kg in children.
- DRF normal 45–55%; T½ <10 normal (not obstructed); 10–20 indeterminate; >20 min obstructed.
- Operate (surgery) for DRF <40%, fall >10% (falling DRF), or poor drainage with symptoms.

Prone vs Supine PCNL: Indications, positioning, technique, specific complications, advantages and disadvantages of each.

Type: SAQ (10 marks)

Source: 2024-Winter Paper 1, Q3f (mini-PCNL: see 2022-Summer Paper 1, Q1)

Indications for PCNL (percutaneous nephrolithotomy)

- Renal stones **>2 cm**, staghorn, lower pole >1 cm, calyceal diverticulum, failed retrograde route or urinary diversion, UPJ (ureteropelvic junction) obstruction.

Comparison

Feature	Prone (Goodwin, 1955)	Supine (Valdivia-Uría, 1987)
Positioning	Lithotomy for ureteric catheter, then turned prone; chest and pelvic bolsters (horizontal bring kidney down more); arms ≤90°, face and knees padded	Ipsilateral flank at table edge on a bolster, arm across chest; **Galdakao-modified (Scoffone)**: legs in lithotomy for simultaneous retrograde access
Best suited	Standard anatomy; **upper pole and posterior calyx** access; multiple tracts	Obese, cardiopulmonary disease, kyphosis; ECIRS (endoscopic combined intrarenal surgery); ileal conduit or continent channel
Technique	Posterior calyx puncture under fluoroscopy or ultrasound; tract more vertical; large working area	Puncture in posterior axillary line; tract more horizontal and longer; kidney more mobile; air outlines anterior calyces
Drainage	Against gravity; higher pelvic pressure	**Gravity drains fragments**, low intrarenal pressure
Anaesthesia	Airway access difficult; turning needed	**Airway retained**; no repositioning

Specific complications

- **Prone:** ocular injury, neuropraxia (brachial plexus), pressure injury, airway tube dislodgement while turning.
- **Supine: colon injury** (if puncture too anterior), access failure to upper pole, kidney displacement during dilatation.
- **Both:** bleeding (transfusion now about 1%), pleural injury above the 12th rib, sepsis, bowel, spleen or liver injury.

Advantages and disadvantages

	Prone	Supine
Advantages	Wide working area, easy upper pole access, familiar	Single position, shorter time, ECIRS, gravity clearance, safer anaesthesia
Disadvantages	Repositioning, positioning injuries, needs split-leg table for ECIRS	Smaller working area, collapsed system, longer tract, harder upper pole

- **Evidence:** stone-free rates and complications similar; supine often shortens operating time; choice is mainly surgeon preference.

Diagram to draw: cross-section showing prone (vertical) and supine (horizontal) tracts.

> **Key points to remember:**
> - Prone: posterior and upper pole access, big working space.
> - Supine: airway, ECIRS, gravity drainage, no turning.
> - Similar outcomes; choose by anatomy and experience.

Metabolic Evaluation in Stone Disease

Type: SAQ (10 marks)

Source: 2025-Winter Paper 1, Q3b (also answered in 2015-Summer Paper 2, Q2)

Rationale

- Stones are a chronic metabolic disease: about **50% recur within 10 years**. Evaluation finds a treatable cause and directs targeted therapy, which reduces stone events.

Step 1: Screening evaluation (every first-time former; AUA [American Urological Association], EAU [European Association of Urology])

Component	What is assessed
History	GI (gastrointestinal) disease, gout, sarcoidosis, recurrent UTI (urinary tract infection), family history; diet; drugs (calcium, vitamins C and D, acetazolamide, steroids)
Blood	Creatinine, sodium, potassium, calcium, uric acid, bicarbonate; PTH (parathyroid hormone) if calcium is high
Urine	pH, crystals (hexagonal = cystine), culture
Stone analysis	Infrared spectroscopy or X-ray diffraction; uric acid, cystine and struvite point to a cause
Imaging	NCCT (non-contrast computed tomography): burden

Step 2: Extended evaluation (high-risk or recurrent formers)

- **High risk:** children, recurrent stones, solitary kidney, GI disease or bariatric surgery, hyperparathyroidism, RTA (renal tubular acidosis), metabolic syndrome, gout, family history.
- **One or two 24-hour urines** on a usual diet, ≥4 weeks after an episode, stone- and infection-free; creatinine confirms completeness.

Parameter	Target per day	Abnormality
Volume	**>2.5 L**	Low volume (commonest)
Calcium	<4 mg/kg (about 250 mg women, 300 mg men)	Hypercalciuria
Oxalate	**<40 mg**	Dietary, enteric, primary
Citrate	**>320 mg**	Hypocitraturia (RTA, diarrhoea)
Uric acid	<600 mg	Hyperuricosuria
pH	5.8–6.2	<5.5 uric acid; >6.5 calcium phosphate or infection
Sodium	<100 mmol	Raises urine calcium
Cystine	Negative screen	Cystinuria

- **Special tests:** ammonium chloride load (urine pH stays >5.5 in distal RTA); quantitative cystine.
- Fast and calcium-load subtyping of hypercalciuria is no longer advised (does not change treatment).

Follow-up

- Repeat 24-hour urine **within 6 months** of starting therapy, then yearly; blood tests for drug side effects.
- Pregnancy: defer 24-hour urine.

Diagram to draw: flowchart: first stone → screening → low risk (general advice) or high risk (24-hour urine → targeted therapy).

Key points to remember:

- Screening for all first-time formers; 24-hour urine for high-risk.
- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg.
- Repeat 24-hour urine within 6 months of therapy.

What are nanobacteria? Describe their role in urolithiasis.

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 1, Q3e

Definition

- **Nanobacteria**, now called **CNPs (calcifying nanoparticles)**, are spherical particles far smaller than bacteria with a **calcium apatite shell**, described in bovine and human serum by **Kajander and Çiftçioglu (1998)** as *Nanobacterium sanguineum*.
- **Their living nature is disputed.** Proposed as the smallest self-replicating bacteria, but later work showed "replication" is mineral nucleation, and that they are **mineralo-protein complexes** (apatite with serum proteins such as fetuin-A). Their bacterial DNA proved to be contamination. Most now regard them as **non-living**.

Claimed properties

Property	Claim
Size	Pass through bacteria-retaining filters
"Growth"	Very slow, in serum-containing medium
Shell	Apatite; resistant to heat and gamma irradiation
Detection	Electron microscopy, von Kossa and Hoechst staining, specific antibodies, culture
Susceptibility	Tetracycline and EDTA (ethylenediaminetetraacetic acid) inhibit them

Proposed role in urolithiasis

- **Nidus theory:** CNPs in blood or urine are filtered or excreted, deposit apatite in renal tubular cells and interstitium, and act as a **nucleating nidus** for crystal growth.
- **Link to Randall plaque:** idiopathic calcium oxalate stones grow on **Randall plaque**, an interstitial **apatite** deposit that starts in the basement membrane of the thin loop of Henle and erodes through the papilla. CNPs are **associated with**, but **not proven to cause**, Randall plaque (Campbell, Chapter 95).
- Early studies found apatite nanoparticles in a large majority of stones; rat studies reported renal calcification after injection.
- Also implicated in atherosclerotic calcification, polycystic kidney cysts and chronic prostatitis.

Therapeutic implications (unproven)

- Tetracycline-based regimens with EDTA were tried in stones and chronic prostatitis/chronic pelvic pain syndrome; no RCT (randomised controlled trial) evidence supports routine use.

Current position

- Not in stone guidelines; pathogenesis is explained by supersaturation, inhibitors, Randall plaque and ductal plugs.

Diagram to draw: renal papilla with interstitial apatite (Randall plaque) eroding through urothelium, CNPs as nidus, and calcium oxalate stone attached.

Key points to remember:

- Nanobacteria = calcifying nanoparticles; apatite shell; living nature disputed.
- Proposed nidus for apatite and Randall plaque; association, not causation.
- Tetracycline/EDTA therapy unproven.

Tear drop bladder

Type: SAQ (10 marks)

Source: Practice Set 5 Paper 1, Q3a

Definition

- A **tear-drop (pear-shaped or inverted-pear) bladder** is a bladder that is symmetrically **compressed from both sides** and elongated upwards on cystogram, IVU (intravenous urogram) or CT (computed tomography).
- It is a radiological sign of a **bilateral perivesical mass or collection**, not a disease itself.

Causes

Group	Causes
Collections	**Pelvic haematoma** (pelvic fracture: commonest in trauma); urinoma (extraperitoneal leak); bilateral lymphocele (after lymphadenectomy or transplant); pelvic abscess
Fat	**Pelvic lipomatosis**
Muscle	**Iliopsoas hypertrophy** (muscular young men)
Vascular	Dilated pelvic venous collaterals (IVC [inferior vena cava] occlusion); bilateral iliac artery aneurysms
Nodes and tumour	Pelvic lymphadenopathy (lymphoma, metastases); pelvic retroperitoneal fibrosis; pelvic tumours

Pelvic lipomatosis (the classic cause)

- Overgrowth of benign, non-encapsulated fat in the perivesical and perirectal spaces; mainly in men, more in Black men.
- **Presentation:** LUTS (lower urinary tract symptoms), suprapubic pain, constipation; often incidental.
- **Imaging:**
  - Plain film: pelvic lucency. IVU: elevated, elongated pear-shaped bladder; **medial deviation of both ureters**; hydronephrosis.
  - Barium enema: straightened, elongated rectosigmoid.
  - **CT or MRI (magnetic resonance imaging):** fat density is diagnostic.
- **Cystoscopy:** high bladder neck, difficult passage; often **cystitis glandularis**, which can progress to adenocarcinoma. Biopsy and surveillance are needed.
- **Management:** observe if no obstruction. Upper-tract obstruction → stent or nephrostomy, ureteric reimplantation or urinary diversion. Weight loss rarely helps; surgical excision of fat is difficult.

Evaluation of a tear-drop bladder

- History: trauma, pelvic surgery or transplant, malignancy.
- **CT** shows the cause (blood, fat, fluid, muscle, nodes, vessels). Ultrasound for collections; aspiration (creatinine for urinoma, lymph for lymphocele).

Management

- Treat the cause: pelvic haematoma usually observed; lymphocele drained or marsupialised (laparoscopic fenestration); urinoma drained and leak repaired.

Diagram to draw: cystogram outline of a bladder squeezed into a tear drop by bilateral pelvic haematoma.

Key points to remember:

- Tear-drop bladder = bilateral extrinsic compression.
- Trauma: pelvic haematoma; non-trauma: pelvic lipomatosis.
- Lipomatosis: fat on CT, medially deviated ureters, cystitis glandularis.

Retrograde endopyelotomy

Type: SAQ (10 marks)

Source: Practice Set 9 Paper 1, Q3f

Definition and principle

- Endoscopic **full-thickness incision of the UPJ (ureteropelvic junction)** from inside the ureter, through to peripelvic fat, followed by stenting so the ureter heals with a wider lumen (Davis "intubated ureterotomy" concept, 1943).
- **Retrograde ureteroscopic holmium laser endopyelotomy** is now the standard endourological route: direct vision without a percutaneous tract.

Indications and contraindications

Favourable	Unfavourable / contraindicated
Functionally significant UPJO (UPJ obstruction); short segment	**Long stenotic segment**
Secondary UPJO (failed pyeloplasty)	**Upper-tract stones** (treat percutaneously instead)
Moderate hydronephrosis, fair function	**Crossing vessel**, **gross hydronephrosis**, poor function

- Pre-operative CT (computed tomography) angiography or Doppler to look for crossing vessels.

Technique

1. General anaesthesia; retrograde pyelogram; hydrophilic safety wire into the pelvis.
2. Access sheath; flexible ureteroscope to the UPJ (semirigid in some women). Dilate a narrow intramural ureter or pre-stent.
3. Holmium laser, 200-µm fibre, 0.8–1.2 J, 10–15 Hz.
4. **Lateral incision** (no crossing vessels laterally), deepened into fat and extended into normal ureter and pelvis.
5. Optional balloon dilation to 24 Fr; DJ (double-J) stent and Foley catheter.
6. Stent out after some weeks; **diuretic renogram 4 weeks later**; follow up for 24–36 months.

Results

- Holmium series 70–80% at 5 years (range 44–89%). Excluding crossing vessels >4 mm raises success to about 90%.
- Crossing vessel present versus absent: 42% versus 86% (antegrade series).
- Still below pyeloplasty (about 95%), but most cost-effective endourological option.

Complications

- Urine leak, stent migration, infection, bleeding (lateral cut spares vessels), stricture (rare with small scopes); ureteroarterial fistula reported.
- **Also know:** antegrade (percutaneous) endopyelotomy for UPJO with stones; cautery wire balloon (more haemorrhage); balloon dilation alone (about 42% success); failure then needs pyeloplasty.

Diagram to draw: UPJ with lateral full-thickness incision and crossing vessel lying anterior/posterior.

Key points to remember:

- Lateral full-thickness incision over a stent.
- Crossing vessel and gross hydronephrosis predict failure.
- Success 70–80%, below pyeloplasty (about 95%).

Autotransplantation

Type: SAQ (10 marks)

Source: Practice Set 10 Paper 1, Q3c (also answered in 2020-Summer Paper 3, Q3c and 2025-Winter Paper 3, Q3c)

Taken as renal autotransplantation.

Definition and principle

- Removing a kidney and re-implanting it, usually in the **ipsilateral iliac fossa** on the iliac vessels, with **bench (ex vivo) surgery** if needed. First done in 1963 for proximal ureteric injury.
- Bringing the kidney to the bladder replaces the ureter without bowel.

Indications

Group	Examples
Ureteric loss (commonest)	Long iatrogenic or traumatic loss; failed repairs (**delayed setting, never acute**)
RPF (retroperitoneal fibrosis)	Ureterolysis impossible
Tumour	UTUC (upper tract urothelial carcinoma) in a solitary kidney; bench partial nephrectomy
Vascular	Complex renal artery aneurysm or AVM (arteriovenous malformation) unsuitable for endovascular repair; nutcracker
Pain	Selected loin pain–haematuria syndrome

- **Preferred over ileal ureter** when the other kidney is poor or bowel unsuitable (creatinine >2 mg/dL, bowel disease, radiation).
- **Contraindications:** poor renal function, severe iliac atherosclerosis, unfit patient.

Procedure

```
CT (computed tomography) angiography, renogram
→ harvest as for live donation: maximal vessel length,
  preserve ureteric blood supply
→ cold perfusion ± bench repair
→ Gibson incision, extraperitoneal iliac fossa
→ vein to external iliac vein (end-to-side);
  artery to external iliac (end-to-side) or internal iliac (end-to-end)
→ pyelovesicostomy (± Boari flap), ureteroneocystostomy,
  or native ureter left in continuity
→ stent, catheter, drain
```

- Laparoscopic harvest speeds recovery; robotic autotransplantation emerging.

Complications and outcome

- Arterial or venous thrombosis, bleeding, urine leak, anastomotic stricture, lymphocele, infection.
- **Follow-up:** Doppler ultrasound early; renogram and blood pressure later.
- **Renal unit loss 5–10%**; major (Clavien–Dindo ≥III) complications about 15–20%; **success about 90%**.

Also know: alternatives: Boari flap, TUU (transureteroureterostomy), ileal ureter, Yang–Monti.

Diagram to draw: kidney in the iliac fossa with vascular anastomoses and pyelovesicostomy.

Key points to remember:

- For long ureteric loss when the other kidney is poor or bowel unsuitable.
- Delayed setting only; bench surgery for vascular or tumour cases.
- Unit loss 5–10%; success about 90%.

Retroperitoneal fibrosis

Type: SAQ (10 marks)

Source: Practice Set 12 Paper 1, Q3a (also answered as an LAQ in 2023-Winter Paper 3, Q1)

Definition

- RPF (retroperitoneal fibrosis): inflammatory fibrous plaque encasing the aorta, IVC (inferior vena cava) and ureters; centred on the **distal aorta at L4–L5**. Age 40–60; males 2–3 times commoner.

Aetiology

Idiopathic (~70%)	Secondary (~30%)
Chronic periaortitis (large-vessel vasculitis); **IgG4 (immunoglobulin G4)-related disease**; autoimmune; ceroid antigen	**Drugs** (methysergide, ergots, β-blockers); **malignancy** (lymphoma commonest); radiotherapy; infection (tuberculosis)

Pathology

- Early: lymphocytes, plasma cells, capillaries (steroid-responsive). Late: acellular collagen.
- Ureteric obstruction is by compression and **loss of peristalsis**; lumen stays patent. Malignant RPF shows tumour islands: take deep biopsies.

Clinical features

- **Dull back or flank pain**, weight loss, malaise, hypertension; about half present with **uraemia**; leg oedema or DVT (deep venous thrombosis) from IVC compression.

Evaluation

- Raised ESR (erythrocyte sedimentation rate) and CRP (C-reactive protein), creatinine, IgG4.
- **CT (computed tomography):** periaortic mass with **medial deviation of mid-ureters**.
- **MRI:** high T2 = active. **PET-CT (positron emission tomography):** best marker of activity.
- **Biopsy** if atypical or malignancy suspected.

Management

```
Drain (stents or nephrostomy) → stop drug, exclude cancer
  → prednisone → failure / fibrotic disease → ureterolysis
```

- **Prednisone** 60 mg daily (~1 mg/kg), tapered over up to 2 years: ~80% respond; relapse 25–50%.
- **Vaglio RCT (randomised controlled trial; Lancet 2011):** relapse 17% prednisone vs 50% tamoxifen.
- Steroid-sparing: azathioprine, mycophenolate; **rituximab** for refractory or IgG4 disease.
- **Ureterolysis** (open, laparoscopic or robotic; ~15% need it): **bilateral**, biopsies, then **intraperitonealisation**, lateralisation or **omental wrap**.
- Others: long-term metallic stents; autotransplantation; nephrectomy for a dead kidney.

Follow-up

- ESR/CRP, creatinine and ultrasound; MRI or PET-CT for activity; remove stents once the ureters drain; watch for relapse and steroid side effects.

Diagram to draw: periaortic plaque pulling both ureters medially.

Key points to remember:

- 70% idiopathic; methysergide and lymphoma classic.
- Medially deviated ureters; PET-CT shows activity.
- Prednisone first (Vaglio 2011 beat tamoxifen).
- Bilateral ureterolysis with omental wrap.

Management of failed pyeloplasty

Type: SAQ (10 marks)

Source: Practice Set 13 Paper 1, Q3c

Definition of failure

- Persistent pain, UTI (urinary tract infection) or stones; **obstructive diuretic renogram** (T½ >20 minutes); increasing hydronephrosis; falling split function.
- Dilatation settles slowly; judge by renography.
- About 1 in 9 children need a secondary procedure. Most failures occur within 1–2 years, but up to 30% occur later.

Causes

- **Ischaemia** (over-stripping) and **tension** at the anastomosis.
- **Missed crossing vessel**; inadequate excision of the narrow segment; non-dependent anastomosis.
- **Urine leak** or infection causing periureteric fibrosis.
- Missed distal stricture or high-grade VUR (vesicoureteral reflux).

Early failure (first days or weeks)

- Pain, fever, increasing dilatation: usually **oedema or clot**; temporary nephrostomy or stent, and most settle.

Evaluation of late failure

- **MAG3 (mercaptoacetyltriglycine) diuretic renogram:** obstruction and split function.
- **CT (computed tomography) angiography:** crossing vessel.
- Antegrade or retrograde pyelography (stricture length); **Whitaker test** if equivocal. Nephrostomy first if infected.

Treatment options

Option	When	Results
Endopyelotomy (retrograde or antegrade)	Short stricture, no crossing vessel; first choice after failed open repair	About 70%
Redo pyeloplasty (open, laparoscopic, robotic)	Long or dense stricture, crossing vessel	About 86–89%; best choice in most children
Ureterocalicostomy	Scarred, small intrarenal pelvis; dilated lower calyx	About 70% (redo series)
Buccal mucosa onlay (robotic, omental wrap)	Long scarred UPJ (ureteropelvic junction)	Failure about 15–25% (CORRUS)
Nephrectomy	Split function <15%, normal other kidney; multiple failures	Definitive

- **Also know:** ileal ureter or autotransplantation for long ureteric loss.

Principles of redo surgery

- Transperitoneal route in children (retroperitoneum is scarred).
- Ureteric catheter; wide mobilisation of kidney and ureter; **excise all scar**; spatulate; **tension-free, watertight** anastomosis over a stent; wrap with omentum or perirenal fat.
- **Also know:** near-infrared fluorescence shows perfusion; downward nephropexy.

Follow-up

- Stent out at 4–6 weeks; renograms for 2–3 years (late failures).

Diagram to draw: ureterocalicostomy (lower calyx to spatulated ureter).

> **Key points to remember:**
>
> - Causes: ischaemia, tension, missed crossing vessel, leak.
> - Failed open repair: endourology first; redo pyeloplasty most reliable.
> - Ureterocalicostomy for small scarred pelvis; buccal onlay emerging.
> - Nephrectomy if split function <15%.

Post-obstructive diuresis

Type: SAQ (10 marks)

Source: Practice Set 19 SAQ Sheet, Q3b

Definition

- POD (post-obstructive diuresis): marked polyuria after relief of obstruction, as the kidney sheds retained salt, water and urea.
- No agreed rate; a common working definition is **>200 mL/h for 2 consecutive hours or >3 L in 24 hours**. Up to 300 mL/h is common.
- **Physiological** (appropriate, self-limited) versus **pathological** (continues after euvolaemia, causing hypovolaemia and electrolyte loss).

Incidence and setting

- Unilateral obstruction: **about 2%** (the other kidney compensates). Bilateral or solitary kidney: **about 50%**.
- Commonest after catheterising **chronic retention with renal impairment**; also bilateral ureteric obstruction and PUV (posterior urethral valves).

Mechanisms

1. **Transporter dysregulation:** down-regulation of Na^+/K^+-ATPase and aquaporin-2 → concentrating defect (nephrogenic DI (diabetes insipidus)); loss of medullary tonicity.
2. **Volume expansion:** on relief, raised ANP (atrial natriuretic peptide), low aldosterone and sympathetic tone → salt and water loss.
3. **Osmotic diuresis** from retained urea.

Risk factors

- Bilateral obstruction, raised creatinine, oedema, heart failure, hypertension, confusion.
- After valve ablation: oligohydramnios, preterm birth, raised creatinine.

Clinical features

- Starts within hours of relief; most settle within 24–48 hours.
- Danger signs: postural hypotension, tachycardia, weight loss, hypernatraemia (free-water loss), low potassium, magnesium, phosphate.

Management

```
Relieve obstruction (catheter / nephrostomy)
        |
Monitor hourly urine output, vitals, weight
        |
Alert, can drink -> free oral fluids (most resolve)
        |
Output persists / cannot drink / unstable
  -> admit; electrolytes every 4-6 h
  -> intravenous 0.45% saline: about 75% of urine output
  -> replace K, Mg, phosphate; adjust to urine sodium/osmolality
```

- **Do not overcorrect:** replacing every millilitre sustains the diuresis.
- **Gradual bladder decompression has no role**: it does not prevent haematuria or diuresis.

- PCN (percutaneous nephrostomy) for upper-tract obstruction; hydration alone for physiological POD.
- **Also know:** decompression haematuria; neonatal PUV concentrating defect; no approved drug therapy.

Diagram to draw: graph of urine output against time after catheterisation.

Key points to remember:

- Unilateral about 2%, bilateral about 50%.
- Mechanisms: aquaporin-2 loss, volume expansion, urea osmotic diuresis.
- Replace about 75% of output with 0.45% saline; do not overcorrect.
- Gradual decompression has no role.

Stone complexity scores

Type: SAQ (10 marks)

Source: Practice Set 20 SAQ Sheet, Q3b

Purpose

- Grade stone and kidney complexity, chiefly to **predict stone-free rate (SFR) after PCNL (percutaneous nephrolithotomy)**. Used for counselling, choosing the surgeon, auditing and comparing series.

Main scores

Score	Variables and scoring	Predicts
Guy's Stone Score (Thomas, 2011)	**Grade I:** single stone in mid or lower pole, or pelvis, simple anatomy. **II:** single upper-pole stone, or multiple stones with simple anatomy, or single stone with abnormal anatomy. **III:** multiple stones with abnormal anatomy, calyceal diverticulum stone, or partial staghorn. **IV:** complete staghorn, or any stone in spina bifida or spinal injury	SFR on KUB (kidney, ureter, bladder radiograph); not complications
S.T.O.N.E. nephrolithometry	**S**ize (area), **T**ract length (≤100 or >100 mm), **O**bstruction (hydronephrosis), **N**umber of involved calyces, **E**ssence (density, ≤950 or >950 HU [Hounsfield units]); total 5–13	SFR, blood loss, operative time, stay; needs CT (computed tomography)
CROES nomogram (Clinical Research Office of the Endourological Society)	Stone burden, location, staghorn, number, prior treatment, **annual centre volume**	SFR; only multicentre derivation, but SFR on KUB
S-ReSC (Seoul National University Renal Stone Complexity)	Number of **9 sites** involved on CT: pelvis, upper and lower major calyces, anterior and posterior minor calyces of the three groups; score 1–9	SFR; not complications
RUSS (Resorlu–Unsal Stone Score)	1 point each: stone >20 mm; lower-pole stone with infundibulopelvic angle <45°; stones in more than one calyx; abnormal anatomy; score 0–4	SFR after **RIRS** (retrograde intrarenal surgery)

Limitations

- Each score added what the previous lacked: anatomy, then CT depth and density, then centre factors, then calyceal distribution.
- Mostly predict clearance, **not complications**. Meta-analysis: heterogeneous; no single generalisable score.
- **Also know:** Triple D score predicts SWL (shock wave lithotripsy) success, not PCNL.

Diagram to draw: kidney outline showing the 9 S-ReSC sites.

> **Key points to remember:**
> - Guy's: grades I–IV; complete staghorn or spina bifida is grade IV.
> - S.T.O.N.E.: five variables, score 5–13.
> - CROES includes centre volume; RUSS is for RIRS.

Calyceal diverticulum – classification and management

Type: SAQ (10 marks)

Source: Practice Set 20 SAQ Sheet, Q3c

Definition

- Cystic cavity in the renal cortex, **lined by transitional epithelium**, connected to a calyx (or pelvis) by a **narrow neck**. Fills passively with urine.
- About 4.5 per 1000; equal in children and adults; no side or sex preference.

Classification

	Type I	Type II
Frequency	Commoner	Less common
Connects to	Minor calyx (usually upper pole)	Renal pelvis or major calyx
Size, symptoms	Small, usually silent	Larger, often symptomatic

Aetiology

- **Congenital** (favoured): persistent third- and fourth-generation ureteric bud branches.
- **Acquired:** cortical abscess draining into a calyx, infundibular stenosis, trauma, reflux.

Clinical features

- Stasis: **stones in up to 40%**, milk of calcium, infection, pain, haematuria.

Evaluation

- Ultrasound mimics a simple cyst. **CT (computed tomography) urography with delayed images shows contrast pooling**; milk of calcium layers and shifts with position.
- Retrograde pyelography maps the neck before intervention.
- **Differential:** hydrocalycosis (dilated major calyx from infundibular obstruction), megacalycosis, simple cyst.

Management

- **Asymptomatic:** observe with ultrasound.
- **Treat** pain, recurrent infection, haematuria, abscess, symptomatic stones.

Approach	Technique	Best for
Ureteroscopy	Find the ostium, laser-incise or balloon-dilate the neck, remove stones	Upper or mid-pole, small stones
PCNL (percutaneous nephrolithotomy)	Direct puncture onto stone, clear stones, fulgurate lining, dilate or create neck	Posterior, lower-pole, larger stones; long-term results good
Laparoscopic or robotic	Unroof, remove stones, close neck, obliterate cavity, pack with fat	Anterior, thin-parenchyma, failed endoscopy
Partial nephrectomy	Excision	Rare; destroyed segment

- Endoscopy keeps the cavity draining by widening the neck; laparoscopy abolishes it by closing the neck.
- **Also know:** SWL (shock wave lithotripsy) gives symptom relief but low stone clearance; fulguration of lining is debated.

Diagram to draw: kidney with upper-pole diverticulum, narrow neck to calyx, stone inside.

Key points to remember:

- Transitional lined, narrow neck; type I calyceal, type II pelvic.
- Stones in up to 40%; delayed CT shows pooling.
- Ureteroscopy or PCNL first; laparoscopy for anterior or failed cases.

Pelvic lipomatosis

Type: SAQ (10 marks)

Source: Practice Set 20 SAQ Sheet, Q3e

Definition

- Rare benign **overgrowth of non-encapsulated mature fat** in the perivesical and perirectal spaces, compressing the bladder, ureters, rectum and pelvic veins.
- Mostly men aged 20–60; associated with obesity and hypertension. Cause unknown; obesity alone does not explain it (weight loss rarely shrinks the fat). Chronic inflammation and venous obstruction are proposed.

Clinical features

- LUTS (lower urinary tract symptoms), haematuria, recurrent UTI (urinary tract infection), suprapubic or perineal pain, constipation.
- Leg oedema or venous thrombosis from venous compression.
- **Hydronephrosis and renal failure** from ureteric compression in a substantial minority.
- **Proliferative cystitis (cystitis glandularis)** is common; it can transform to **adenocarcinoma**.

Evaluation

- **IVU (intravenous urography) or cystogram:** elevated, elongated **"pear-" or "teardrop-shaped" bladder**; medially displaced pelvic ureters; high prostate.
- **Barium enema:** elongated, narrowed rectosigmoid.
- **Ultrasound:** echogenic pelvic fat; hydronephrosis.
- Medial deviation of **both** pelvic ureters also occurs in retroperitoneal fibrosis and after abdominoperineal surgery.
- **CT (computed tomography) or MRI (magnetic resonance imaging): diagnostic**; homogeneous fat density surrounding pelvic organs, separates it from tumour.
- **Cystoscopy:** elevated trigone, long prostatic urethra (rigid scope may be difficult); biopsy for cystitis glandularis.
- **Differential of pear-shaped bladder:** pelvic haematoma, lymphadenopathy, liposarcoma, iliopsoas hypertrophy, venous collaterals, pelvic fibrosis.

Management

- **Surveillance** (often progressive): ultrasound for hydronephrosis, renal function, periodic cystoscopy for cystitis glandularis.
- Weight loss, antibiotics and steroids give limited or no benefit; COX-2 (cyclo-oxygenase-2) inhibitors for cystitis glandularis are case reports only.
- **Ureteric obstruction:** stents or nephrostomy; ureteric reimplantation (difficult, dense fat).
- **Advanced or failed cases:** urinary diversion; cystectomy if adenocarcinoma or unmanageable bladder.
- Surgical excision of fat is difficult: no tissue planes, heavy bleeding.
- Resect cystitis glandularis transurethrally and treat obstruction or inflammation.

Diagram to draw: cystogram outline of pear-shaped bladder with medially deviated ureters.

Key points to remember:

- Pear-shaped bladder with medial ureters; CT shows fat.
- Cystitis glandularis common; adenocarcinoma risk.
- Watch for hydronephrosis; diversion if obstruction progresses.

Cystinuria

Type: SAQ (10 marks)

Source: Practice Set 21 SAQ Sheet, Q3a

Definition and genetics

- **Autosomal recessive** defect of proximal tubular (and jejunal) reabsorption of the dibasic amino acids **COLA: cystine, ornithine, lysine, arginine**. Only cystine is insoluble, so only cystine forms stones.
- Transporter subunits and ICC (International Cystinuria Consortium) types:

Gene	Subunit	Chromosome	Type
SLC3A1	rBAT (heavy)	2p	A
SLC7A9	$b^{0,+}$AT (light)	19q	B
Both	—	2 and 19	AB

- Fractional excretion of cystine rises from about 0.4% to up to 100%. Incidence 1 in 1000 to 1 in 17,000; **1–3% of adult stones, up to 10% of paediatric stones**; males more severe.

Pathophysiology

- **Pure supersaturation disease**: no specific inhibitor. Solubility is pH dependent: about 300 mg/L at pH 5, 400 mg/L at pH 7, 1000 mg/L at pH 9.
- Coexisting hypocitraturia (commonest), hyperuricosuria or hypercalciuria cause mixed calcium–cystine stones.

Clinical features

- First stone at mean about 12 years; recurrent, bilateral, staghorn stones; CKD (chronic kidney disease) from repeated surgery (about a quarter of adults have eGFR [estimated glomerular filtration rate] <60).

Diagnosis

- **Hexagonal crystals** in urine (always abnormal); cyanide–nitroprusside screen; 24-hour urinary cystine (normal <60 mg/day/1.73 m^2; homozygotes often >400).
- Stone: yellow, waxy, faintly radiopaque (sulphur); **resists SWL (shock wave lithotripsy)**.

Management

- **Fluids:** urine output **≥3 L/day**, including at night.
- **Diet:** sodium ≤2300 mg/day; limit animal protein (not in children).
- **Alkalinise to pH 7.0–7.5** with potassium citrate 30–60 mEq/day; avoid >7.5 (calcium phosphate stones) and sodium alkali (raises cystine excretion).
- **Thiols** if cystine stays >250 mg/L: thiol–cystine mixed disulfide is soluble.
  - **Tiopronin** (choice): 300 mg twice daily, up to 1200 mg/day.
  - **D-penicillamine:** 250 mg/day, titrated; rash, proteinuria, nephrotic syndrome, pancytopenia.
  - Captopril: weak evidence.
- Monitor blood count, liver tests, urine protein.
- **Surgery:** ureteroscopy or PCNL (percutaneous nephrolithotomy) with laser; nephron-sparing approach.

Diagram to draw: proximal tubule cell with rBAT/$b^{0,+}$AT transporter and hexagonal crystal.

Key points to remember:

- Autosomal recessive; SLC3A1 (type A), SLC7A9 (type B).
- Hexagonal crystals; target cystine <250 mg/L.
- Urine ≥3 L, pH 7.0–7.5, then tiopronin.

Rapid Revision Sheet – Section 11 · Stones, Obstruction & Trauma

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Describe Pathogenesis of Congenital Pelvi Ureteric Junction Obstruction its diagnosis and Management. (LAQ)

- Intrinsic adynamic segment (infants) vs crossing vessel (older children, intermittent pain).
- SFU 4 = parenchymal thinning; UTD P3 = abnormal parenchyma.
- Operate for DRF <40%, fall >10%, rising APD, UTI or symptoms.
- Anderson–Hynes ~95%; never divide a crossing vessel.

Discuss pathophysiology of renal stones and also describe how to prevent its recurrence (LAQ)

- Metastable zone between Ksp and Kf: where inhibitors and therapy act.
- Randall plaque anchors idiopathic calcium oxalate stones; ductal plugs suggest a secondary cause.
- Urine ≥2.5 L/day; normal calcium, low salt and animal protein (Borghi).
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.

Flexible Ureterorenoscopy (SAQ)

- 3.6 Fr channel; ≥270° deflection.
- Access sheath lowers pressure but injures ureter in about 1 in 8.
- Laser power ≤10 W ureter, ≤25 W kidney.

Distal Renal tubular acidosis. (SAQ)

- Urine pH >6 with acidosis; hypokalaemia; positive UAG.
- Calcium phosphate stones in up to 70%; hypocitraturia key.
- NH_4Cl test: pH fails to fall below 5.5.
- Potassium citrate is the treatment.

Radionuclide studies and diuretic renography in PUJ obstruction (SAQ)

- MAG3 is the tracer of choice; hydrate and catheterise.
- Well-tempered: hydrate, catheter, standard diuretic dose and timing.
- Furosemide 0.5 mg/kg (adult maximum 40 mg); 1 mg/kg in children.
- DRF normal 45–55%; T½ <10 normal (not obstructed); 10–20 indeterminate; >20 min obstructed.
- Operate (surgery) for DRF <40%, fall >10% (falling DRF), or poor drainage with symptoms.

Prone vs Supine PCNL: Indications, positioning, technique, specific complications, advantages and disadvantages of each. (SAQ)

- Prone: posterior and upper pole access, big working space.
- Supine: airway, ECIRS, gravity drainage, no turning.
- Similar outcomes; choose by anatomy and experience.

Metabolic Evaluation in Stone Disease (SAQ)

- Screening for all first-time formers; 24-hour urine for high-risk.
- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg.
- Repeat 24-hour urine within 6 months of therapy.

What are nanobacteria? Describe their role in urolithiasis. (SAQ)

- Nanobacteria = calcifying nanoparticles; apatite shell; living nature disputed.
- Proposed nidus for apatite and Randall plaque; association, not causation.

- Tetracycline/EDTA therapy unproven.

Tear drop bladder (SAQ)

- Tear-drop bladder = bilateral extrinsic compression.
- Trauma: pelvic haematoma; non-trauma: pelvic lipomatosis.
- Lipomatosis: fat on CT, medially deviated ureters, cystitis glandularis.

Retrograde endopyelotomy (SAQ)

- Lateral full-thickness incision over a stent.
- Crossing vessel and gross hydronephrosis predict failure.
- Success 70–80%, below pyeloplasty (about 95%).

Autotransplantation (SAQ)

- For long ureteric loss when the other kidney is poor or bowel unsuitable.
- Delayed setting only; bench surgery for vascular or tumour cases.
- Unit loss 5–10%; success about 90%.

Retroperitoneal fibrosis (SAQ)

- 70% idiopathic; methysergide and lymphoma classic.
- Medially deviated ureters; PET-CT shows activity.
- Prednisone first (Vaglio 2011 beat tamoxifen).
- Bilateral ureterolysis with omental wrap.

Management of failed pyeloplasty (SAQ)

- Causes: ischaemia, tension, missed crossing vessel, leak.
- Failed open repair: endourology first; redo pyeloplasty most reliable.
- Ureterocalicostomy for small scarred pelvis; buccal onlay emerging.
- Nephrectomy if split function <15%.

Post-obstructive diuresis (SAQ)

- Unilateral about 2%, bilateral about 50%.
- Mechanisms: aquaporin-2 loss, volume expansion, urea osmotic diuresis.
- Replace about 75% of output with 0.45% saline; do not overcorrect.
- Gradual decompression has no role.

Stone complexity scores (SAQ)

- Guy's: grades I–IV; complete staghorn or spina bifida is grade IV.
- S.T.O.N.E.: five variables, score 5–13.
- CROES includes centre volume; RUSS is for RIRS.

Calyceal diverticulum – classification and management (SAQ)

- Transitional lined, narrow neck; type I calyceal, type II pelvic.
- Stones in up to 40%; delayed CT shows pooling.
- Ureteroscopy or PCNL first; laparoscopy for anterior or failed cases.

Pelvic lipomatosis (SAQ)

- Pear-shaped bladder with medial ureters; CT shows fat.
- Cystitis glandularis common; adenocarcinoma risk.
- Watch for hydronephrosis; diversion if obstruction progresses.

Cystinuria (SAQ)

- Autosomal recessive; SLC3A1 (type A), SLC7A9 (type B).
- Hexagonal crystals; target cystine <250 mg/L.
- Urine ≥3 L, pH 7.0–7.5, then tiopronin.

Section 12 · Bladder Cancer & Diversion — Short answer questions (SAQ)

Instestinal segments in urology

Asked as:

- Instestinal segments in urology — *Summer 2022 Paper I Q3a (SAQ)*
- Instestinal segments in urology — *Practice Set 22 SAQ Q3f (SAQ)*

Type: SAQ (10 marks)
(also answered in 2022-Summer Paper 1, Q3a)

Assumed meaning: intestinal segments used in urinary diversion and bladder reconstruction.

Introduction

- Bowel is used for conduits, continent pouches, neobladders, augmentation and ureteric replacement. Choice depends on renal function, irradiation, previous surgery, mesenteric length and the reconstruction needed.

Segments compared

Segment	Electrolyte disturbance	Advantage	Drawback
Stomach	Hypochloraemic, hypokalaemic metabolic alkalosis	Secretes acid, less mucus; for renal failure, short gut	Haematuria–dysuria syndrome, ulcers
Jejunum	Hyponatraemic, hypochloraemic, **hyperkalaemic** acidosis	Outside radiation field	Severe salt loss; avoid
Ileum	**Hyperchloraemic metabolic acidosis**	Mobile, constant supply; default conduit	B12 and bile-salt loss; obstruction ~10%
Colon	Hyperchloraemic metabolic acidosis	Easy antireflux tunnel; transverse colon spared by pelvic radiation	Diarrhoea if ileocaecal valve used

- **Mechanism (ileum, colon):** NH_4^+ replaces Na^+ on the Na^+/H^+ exchanger, coupled with Cl^-/HCO_3^- exchange, so ammonium chloride is absorbed. Treat with sodium bicarbonate or citrate (potassium citrate if hypokalaemic); chlorpromazine or nicotinic acid limit chloride transport.

Other complications

- **Hypokalaemia** (worse with colon); correct potassium before acidosis.
- **Altered sensorium:** hyperammonaemia (drain diversion, lactulose), hypomagnesaemia, drug reabsorption (phenytoin, methotrexate).
- **Osteomalacia:** acidosis, vitamin D resistance, calcium loss; growth retardation in children.
- **Infection, stones** (ileal conduit 10–12%, continent caecal reservoir ~20%), short-gut effects.
- **Cancer:** ureterosigmoidostomy 6–29% after 10–20 years; endoscopy from 5 years.

Renal function needed for continent diversion

- Creatinine <2 mg/dL; GFR (glomerular filtration rate) >35 mL/min; urine pH ≤5.8 after ammonium chloride load; osmolality ≥600 mOsm/kg.

Technical principles

- Clear ≤8 cm of mesentery from a segment end; avoid watershed areas (Griffiths, Sudeck points).
- **Also know:** ERAS (enhanced recovery after surgery) with no routine mechanical bowel preparation; Wallace (stricture ~3%) and Bricker anastomoses; nipple stoma.

Diagram to draw: ileal loop with mesenteric arcade and Wallace anastomosis.

> **Key points to remember:**
> - Ileum and colon: hyperchloraemic acidosis (NH_4Cl absorption).
> - Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
> - Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.

Diagnosis and management of hemorrhagic cystitis

Type: SAQ (10 marks)

Source: 2023-Winter Paper 1, Q3d (also answered in 2022-Summer Paper 3, Q3b)

Definition

- Diffuse inflammation and bleeding of the bladder mucosa, after other sources are excluded.

Causes

Cause	Mechanism
Cyclophosphamide, ifosfamide	Liver forms **acrolein**, which sloughs urothelium; onset within 48 h
Pelvic radiation (about 5%; 6 months to 10 years)	**Obliterative endarteritis**, ischaemia, fibrosis
Viral	BK polyomavirus; adenovirus 11, 35; after transplant
Others	Bacteria; busulfan, thiotepa; penicillins; amyloidosis

Diagnosis

- History: drugs, radiotherapy, transplant, anticoagulants. Severity: microscopic, gross, clots, clot retention.
- Blood count, coagulation, renal function, urine culture, viral PCR (polymerase chain reaction) if transplanted.
- **Exclude cancer:** cytology, cystoscopy with biopsy, upper tract imaging (CT urography).

Prevention

- **Mesna** inactivates acrolein; hyperhydration, forced diuresis.

Management (escalate by severity)

```
Resuscitate, transfuse, correct coagulopathy, treat infection
→ 3-way catheter + CBI (continuous bladder irrigation)
→ clot evacuation, fulguration, biopsy under anaesthesia
→ intravesical alum 1%
→ bleeding after 48 h: hyperbaric oxygen (+ nephrostomies)
→ formalin 1%, then 4% (cystogram first)
→ cystectomy + conduit diversion
(Unstable at any stage: internal iliac embolisation)
```

Agent	Key point
Alum 1%	First line; no anaesthesia; avoid in renal failure (aluminium toxicity)
Aminocaproic acid	Antifibrinolytic CBI; clot-free bladder only
Prostaglandins, silver nitrate, hyaluronate	Second-line options
Formalin 1–4%	80–90% success; under anaesthesia, instilled by gravity for 10–15 min; **cystogram to exclude reflux** (occlude ureters if present); risks: contracted bladder, ureteric stricture

- **HBOT** (hyperbaric oxygen therapy): 100% oxygen at 2–3 atmospheres, 30–40 sessions; raises tissue oxygen and drives neovascularisation; 80–90% response, but many relapse; **RICH-ART** trial improved urinary scores in radiation cystitis. Nephrostomies divert urokinase-rich urine if bleeding continues.

- **Embolisation:** selective anterior division of internal iliac (posterior division causes gluteal pain); initial control about 90%.
- **Cystectomy** last; remove bladder (pyocystis).
- **Viral:** supportive (hydration, irrigation); cidofovir anecdotal.
- **Also know:** tranexamic acid and laser ablation are unproven.

Diagram to draw: escalation ladder.

Key points to remember:

- Acrolein (mesna); radiation endarteritis; BK virus.
- Exclude bladder cancer.
- Alum first; formalin after cystogram; HBOT for radiation.

BCG therapy in Bladder Cancer

Type: SAQ (10 marks)

Source: 2025-Winter Paper 1, Q3f (intravesical therapy also answered in 2017-Summer Paper 4, Q1)

Introduction

- BCG (bacille Calmette-Guérin), a live attenuated mycobacterium (Morales, 1976), is the only intravesical agent that reduces both **recurrence and progression** of NMIBC (non-muscle-invasive bladder cancer).

Mechanism

- BCG binds **fibronectin** on the urothelium; cytokines (IFN-γ [interferon-gamma], IL-2 [interleukin-2] and TNF-α [tumour necrosis factor-alpha]) and immune cells follow.
- A **Th1 (T-helper 1)** response gives cell-mediated tumour kill. It needs repeated stimulation, which is why **maintenance** is essential.

Indications

- **High-risk NMIBC:** HG (high-grade) T1, large/multifocal/recurrent HG Ta, **CIS (carcinoma in situ)** (initial complete response up to 84%).
- **Intermediate risk:** option. Not for low risk.

Technique and schedule

- Start **2–4 weeks after TURBT (transurethral resection of bladder tumour)**; urinalysis first; atraumatic catheter; retain **1–2 hours**.
- **Induction:** weekly × 6.
- **Maintenance (SWOG [Southwest Oncology Group], Lamm 2000):** weekly × 3 at 3 and 6 months, then every 6 months to **3 years** (recurrence-free survival 76.8 vs 35.7 months).
- **EORTC (European Organisation for Research and Treatment of Cancer) 30962:** intermediate risk, full dose for 1 year; high risk, full dose for 3 years. **NIMBUS:** reduced frequency was inferior.

Contraindications

- **Absolute:** immediately after TURBT, gross haematuria, traumatic catheterisation (all risk **intravasation**); previous BCG sepsis; total incontinence; immunosuppression (now debated).
- **Relative:** UTI (urinary tract infection), liver disease, prior TB (tuberculosis).

Toxicity

Grade	Features	Management
1	Cystitis, fever <38.5 °C, <48 h	Symptomatic
2	Severe or >48 h	Isoniazid + rifampicin; resume at ½–⅓ dose
3	Sepsis, organ involvement	Stop BCG; add ethambutol; steroids only with cover

- BCG is resistant to **pyrazinamide**.

BCG failure

- **BCG-unresponsive:** persistent or recurrent CIS or HG Ta/T1 despite **adequate BCG** (≥5 of 6 induction + ≥2 of 3 maintenance), or HG relapse within 6 months (Ta/T1) or 12 months

(CIS). **Timely radical cystectomy** is standard.

- **If unfit or unwilling:** pembrolizumab, nadofaragene firadenovec, N-803 + BCG, gemcitabine–docetaxel.
- **Recent:** PD-1/PD-L1 (programmed death-1/ligand-1) inhibitor + BCG in BCG-naive high risk improved event-free survival (CREST, POTOMAC); TAR-200 gemcitabine-releasing device.

Diagram to draw: SWOG schedule timeline.

Key points to remember:

- Only BCG reduces progression; maintenance is essential.
- Never immediately after TURBT; start at 2–4 weeks.
- Unresponsive: timely cystectomy.

Cancer in augmented bladder

Type: SAQ (10 marks)

Source: Practice Set 20 SAQ Sheet, Q3f

Introduction

- Bladder augmentation (enterocystoplasty) is mainly for neurogenic bladder. **Earliest reported tumour 4 years after surgery**, usually after decades; tumours are often **aggressive and advanced at diagnosis**.
- German registry (Kälble, 2011): secondary malignancy after cystoplasty **1.6%**, versus ureterosigmoidostomy 2.6% and ileal conduit 0.02%.

Pathology

- **Adenocarcinoma** commonest, usually near the **enterovesical anastomosis**; also urothelial carcinoma (aggressive, often metastatic), signet-ring and other types.
- **Nephrogenic adenoma**: benign, probably the commonest tumour after cystoplasty.

Pathogenesis

- **N-nitroso compounds** from bacteria in stagnant, infected urine (probably enhancers).
- Chronic inflammation at the anastomosis releases growth factors that drive proliferation; perianastomotic urothelium is genetically unstable.
- The underlying disease may itself carry risk: bladder cancer in spina bifida is about 2–4%, mostly stage III or IV, and non-gastric augmentation did not appear to increase that risk.
- Ureterosigmoidostomy is the extreme model: urothelium beside colonic mucosa bathed in faeces; mean latency about 26 years.

Risk factors

- **Gastrocystoplasty** (highest among augments), **exstrophy**, immunosuppression after transplant, smoking, chronic infection, long latency.

Clinical features

- **Gross haematuria**, recurrent UTI (urinary tract infection), pelvic pain, new incontinence, mass. Neurogenic patients may notice symptoms late.

Surveillance

- Chapters differ: Campbell bladder-surgery chapter advises **endoscopy yearly from 10 years**; paediatric chapter: no consensus, not shown cost-effective (Mayo group: unnecessary). Higher-risk groups benefit most.
- Investigate any haematuria promptly: cystoscopy with biopsy, cytology, CT (computed tomography) urography.
- **Prevention:** regular clean intermittent catheterisation, daily bladder irrigation (less stasis, mucus and stones), treat urea-splitting infection, stop smoking, lifelong transitional follow-up.

Management

- **Radical cystectomy with excision of the augment** and urinary diversion (conduit or continent pouch), with lymphadenectomy.
- Chemotherapy as for histology; prognosis poor because of late stage.
- **Also know:** stones, perforation, metabolic acidosis also occur after augmentation.

Diagram to draw: clam ileocystoplasty with tumour at the enterovesical suture line.

Key points to remember:

- Adenocarcinoma at the anastomosis; earliest at 4 years.
- Gastric segment, exstrophy, immunosuppression: highest risk.
- Haematuria needs cystoscopy; treatment is cystectomy.

Rapid Revision Sheet – Section 12 · Bladder Cancer & Diversion

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Instestinal segments in urology (SAQ)

- Ileum and colon: hyperchloraemic acidosis (NH_4Cl absorption).
- Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
- Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.

Diagnosis and management of hemorrhagic cystitis (SAQ)

- Acrolein (mesna); radiation endarteritis; BK virus.
- Exclude bladder cancer.
- Alum first; formalin after cystogram; HBOT for radiation.

BCG therapy in Bladder Cancer (SAQ)

- Only BCG reduces progression; maintenance is essential.
- Never immediately after TURBT; start at 2–4 weeks.
- Unresponsive: timely cystectomy.

Cancer in augmented bladder (SAQ)

- Adenocarcinoma at the anastomosis; earliest at 4 years.
- Gastric segment, exstrophy, immunosuppression: highest risk.
- Haematuria needs cystoscopy; treatment is cystectomy.

Section 13 · Kidney & UTUC — Long answer questions (LAQ)

Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal malignancy

Asked as:

- Discuss anatomy of kidney. Describe role of nephron sparing surgery in small renal mass. — *Summer 2017 Paper I Q1 (LAQ)*
- Discuss anatomy of kidney. Describe role of nephron sparing nephrectomy in small renal mass. — *Practice Set 1 Paper I Q1 (LAQ)*
- Describe embryology and anatomy of kidney.. Describe role of nephron sparing surgery in a small renal mass. — *Practice Set 13 Paper I Q2 (LAQ)*
- Nepron Sparing Surgery in renal malignancy. — *Winter 2025 Paper I Q2 (LAQ)*

Type: LAQ (25 marks)
(also answered in 2017-Summer Paper 1, Q1; NSS also in 2022-Summer Paper 3, Q1 and 2025-Winter Paper 1, Q2) (also answered in 2017-Summer Paper 1, Q1 and Practice Set 1 Paper 1, Q1; NSS techniques: see Practice Set 13 Paper 2, Q1) (also answered in 2022-Summer Paper 3, Q1; 2017-Summer Paper 1, Q1)

"Nephron sparing nephrectomy" is read as nephron-sparing surgery (NSS): partial nephrectomy and ablation. "Nepron" is read as nephron.

Embryology

- Three successive kidneys form from **intermediate mesoderm**, in cranial-to-caudal order:

Kidney	Time and site	Fate
Pronephros	Week 3–4, cervical	Non-functional; regresses
Mesonephros	Week 4, thoracolumbar	Functions briefly; tubules become efferent ductules; **mesonephric (Wolffian) duct** persists
Metanephros	Week 5, sacral	**Definitive kidney**

- **Reciprocal induction:** the **ureteric bud (UB)** grows from the caudal Wolffian duct (about week 5) into the **metanephric blastema**.
  - UB branches into ureter, pelvis, calyces and collecting ducts.
  - Blastema, induced by UB tips, forms nephrons (glomerulus to distal tubule).
- **Molecular control:** GDNF (glial cell line-derived neurotrophic factor) from the blastema acts on RET on the UB; SLIT2/ROBO2 restricts GDNF to one bud. Also WT1, PAX2.
- Urine production is established by 14 weeks; nephrogenesis ends before birth.
- **Ascent and rotation (weeks 6–9):** the kidney rises from the pelvis to the lumbar region and rotates 90° so the hilum faces medially. It takes successively higher arteries; persistent lower ones become accessory arteries.
- **Clinical correlation:** no UB gives agenesis; abnormal induction gives MCDK (multicystic dysplastic kidney); failed ascent gives ectopia; fused lower poles trapped by the IMA (inferior mesenteric artery) give horseshoe kidney; also malrotation, duplication.

Surgical anatomy of the kidney

- **Size and position:** about 11 cm, 125–170 g; T12–L3, lies on (the cone-shaped) psoas, so the upper pole is more medial and posterior, and the kidney is angled 30–50° behind the

coronal plane. The right kidney is 1–2 cm lower (liver).

- **Relations:**

Aspect	Right	Left
Superior	Liver, adrenal, 12th rib	Spleen, stomach, adrenal, 11th–12th ribs
Medial/anterior	Duodenum (2nd part), hepatic flexure	Pancreatic tail with splenic vessels, splenic flexure, jejunum
Posterior (both)	Diaphragm, psoas, quadratus lumborum, subcostal nerve	Same

- **Coverings (inside out):** true capsule, perirenal fat, **Gerota fascia**, pararenal fat. Gerota fascia is closed above and laterally but open below. So it contains tumour and abscess (collections), but (yet) fluid can track (tracks) into the pelvis.
- **Hilum (anterior to posterior, front to back):** renal vein, artery, pelvis, then the posterior segmental artery.
- **Vascular anatomy (the basis of partial nephrectomy)**
  - **Arteries:** renal artery from the aorta at L1–L2; the right passes behind the IVC (inferior vena cava).
  - **Five segmental end arteries**: posterior (first branch, behind the pelvis) plus four from the anterior division (apical, upper, middle, lower). With no anastomoses, occlusion causes segmental infarction, but each segment can be clamped or resected separately. Then interlobar (columns of Bertin), arcuate, interlobular arteries.
  - **Accessory arteries** in 25–28%, end arteries too (see Practice Set 1 Paper 2, Q3c).
  - **Brödel line:** an avascular plane between anterior (two-thirds) and posterior (one-third) territories, just behind the lateral border. Incision here bleeds least (anatrophic nephrolithotomy).
  - **Veins anastomose freely**, so a segmental vein can be ligated safely. The long left renal vein receives adrenal, gonadal and lumbar veins.
- **Parenchyma / internal structure:** cortex and 8–18 pyramids; 0.4–1.2 million nephrons per kidney.
- **Collecting system:** papilla, minor calyces (7–9), major calyces, pelvis. Puncture through the **calyceal fornix**, not the infundibulum (Sampaio). (Collecting system: see Q3d.)
- **Also know:** lymph drains to para-aortic (left) or paracaval/interaortocaval nodes (right); sympathetic T8–L1.

Diagram to draw: segmental arteries with the Brödel line on a transverse kidney section.

Small renal mass (SRM)

- **Definition:** an enhancing renal mass **≤4 cm (cT1a)**, usually (mostly) incidental.
- **Biology:** 20–30% of masses ≤4 cm and about 40% ≤2 cm are benign (oncocytoma, fat-poor angiomyolipoma); most cancers are indolent (under/<2% of the cancers metastasise). About 20% benign, 60% indolent, 20% potentially aggressive.

Definition and rationale of NSS / why spare nephrons

- NSS (nephron-sparing surgery) is complete local removal of the tumour with **maximal preservation of functioning parenchyma**. PN (partial nephrectomy) is the standard form; thermal ablation is the non-excisional form.
- **Why:** RN (radical nephrectomy) causes CKD (chronic kidney disease), linked to (which raises) cardiovascular events and death. PN gives equal cancer control for small masses (cT1a) with better function and keeps about **90%** of global function (RN 65–70%).

Work-up / preoperative evaluation

- Multiphase CT (computed tomography) or MRI (magnetic resonance imaging) with 3D (three-dimensional) reconstruction; chest imaging; eGFR (estimated glomerular filtration rate) and proteinuria (CKD [chronic kidney disease] stage).
- **Renal mass biopsy (RMB)** if (when) it may change management (non-diagnostic about 14%). Biopsy when it changes management; genetic counselling if age ≤46, multifocal or bilateral.
- **Nephrometry scores** predict complexity and complications:

R.E.N.A.L. component	Scored as
Radius	≤4, 4–7, ≥7 cm
Exophytic/endophytic	≥50% exophytic, <50%, entirely endophytic
Nearness to collecting system	≥7, 4–7, ≤4 mm
Anterior/posterior	Descriptor
Location to polar lines	Polar, crossing, central

- Total: low 4–6, moderate 7–9, high 10–12 complexity. **PADUA** and C-index are alternatives.

Indications for nephron-sparing surgery

Category	Situations
Imperative	Solitary (functioning) kidney, bilateral tumours (bilateral RCC (renal cell carcinoma)), RN would mean dialysis
Relative	Contralateral kidney threatened: stone, stones, (renal artery) stenosis, hydronephrosis, diabetes, severe hypertension; hereditary RCC
Elective	**cT1a (≤4 cm)** with a normal contralateral kidney: **standard of care**

- Also used for selected cT1b–T2, complex or hilar tumours when function matters.
- **Relative contraindications: <20% of nephron mass retained**; pedicle encasement; collecting-system invasion; **venous tumour thrombus**; cT4; cN1.

Choosing the option (AUA (American Urological Association) 2021)

Option	Role
Active surveillance	<2 cm or predominantly cystic; elderly, comorbid (frail patients), life expectancy <5 years. Imaging every 3–6 months. Intervene if >3 cm or growth >5 mm a year (DISSRM registry data: almost no metastasis/metastases)
Thermal ablation (cryoablation, radiofrequency)	cT1a **<3 cm** (masses <3 cm), percutaneous, biopsy first. Higher local recurrence (than PN), salvaged by repeat ablation
Partial nephrectomy	**Preferred / prioritised (prioritise PN) for cT1a** needing treatment. Imperative: solitary kidney, bilateral tumours, CKD or proteinuria; also familial RCC (renal cell carcinoma), young patients
Radical nephrectomy	RN is preferred only if **all three** apply: high complexity; no CKD or proteinuria; normal contralateral kidney with new (post-RN) eGFR/GFR likely **>45** after RN. Otherwise PN

- **PN principles:** negative margin (width does not matter); maximise preserved parenchyma; avoid long warm ischaemia; enucleation for familial or multifocal tumours. Approaches, steps and complications: see Practice Set 13 Paper 2, Q1.

Techniques of PN

Approach: open (flank extraperitoneal or transperitoneal), laparoscopic, or robotic **RAPN (robot-assisted PN)**: transperitoneal for anterior and retroperitoneal for posterior tumours; equal oncology.

Steps

1. **Early hilar control** of artery, vein and accessory vessels, even if off-clamp is planned (selective or off-clamp possible).
2. Expose / defat the kidney; **intraoperative ultrasound** maps (marks) the tumour borders and depth.
3. Clamp the artery (± vein). Use **ice slush** if ischaemia is expected to be **≥25–30 minutes** (solitary kidney, CKD).
4. Excise with a thin rim of normal parenchyma (or enucleate). **A negative margin matters (is the priority); its width does not.**
5. **Two-layer renorrhaphy** (sliding clip): inner layer closes vessels and collecting system; outer capsular layer over a haemostatic bolster.
6. Unclamp; check haemostasis; close Gerota fascia; drain.

Excision options

	Standard (wedge) resection	Enucleation
Plane	Through normal parenchyma, thin rim	Along the pseudocapsule
Advantage	Fewer positive margins	Less bleeding, maximal parenchyma
Use	Default	Hereditary RCC (VHL [von Hippel–Lindau]), familial, multifocal (disease), severe CKD

- Polar resection or heminephrectomy for large polar tumours.

Vascular control (ischaemia)

- Traditional warm ischaemia (traditionally) **<25–30 minutes** (min); ice slush if longer is expected. Function depends mainly on the **quantity of parenchyma preserved**; ischaemia is secondary within accepted limits.
- **Off-clamp, selective or early unclamping:** randomised trials show (gave) no long-term (lasting) eGFR advantage / gain. Hypothermia and mannitol give no benefit with a normal contralateral kidney.
- **Trifecta:** negative margin, no major complication, warm ischaemia ≤25 min (minutes). RAPN: shorter ischaemia than laparoscopy, less blood loss than open.

Ablative NSS / other nephron-sparing options

- **Cryoablation** (double freeze–thaw), **RFA (radiofrequency ablation)**, **MWA (microwave ablation)**; mostly percutaneous.
- AUA: alternative for **cT1a <3 cm**, especially elderly or comorbid; biopsy before or at ablation. Recurrence is higher than PN but treatable (salvaged) by repeat ablation.
- **Also know:** SBRT (stereotactic body radiotherapy) for non-surgical candidates; active surveillance for frail patients.

Special situations

- **Hereditary RCC:** observe (treat) until the largest tumour is **3 cm**, then enucleate (except HLRCC [hereditary leiomyomatosis and RCC]: wide excision). (von Hippel–Lindau: treat at 3 cm, enucleation.)
- **Bilateral tumours:** bilateral PN, usually staged; if one side needs RN, do the PN first.

- **Solitary kidney:** temporary dialysis about 3.5%; local recurrence 3–5%.
- **Also know:** neoadjuvant TKI (tyrosine kinase inhibitor) downsizing in imperative cases; bilateral Wilms tumour; kidney-sparing upper-tract urothelial cancer surgery.

Complications of PN

- **Urine leak** (drain, stent); **haemorrhage / bleeding, pseudoaneurysm or AV (arteriovenous) fistula** (**delayed haematuria**: angioembolisation); acute kidney injury.
- **Positive margin 0–7%:** surveillance, not reoperation.
- **Hyperfiltration injury** if >50% of a solitary kidney is removed (lost): (hyperfiltration) proteinuria, FSGS (focal segmental glomerulosclerosis); ACE (angiotensin-converting enzyme) inhibitor.
- **Follow-up:** AUA risk-based imaging; eGFR and urine protein at every visit.

Outcomes and evidence

- Elective PN: **local recurrence 1–2%**, cancer-free survival >90%; imperative 3–5%.
- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** RN vs PN (PN versus RN) for ≤5 cm tumours: PN better function, similar oncology (cancer control); overall survival slightly better with (favoured) RN — 10-year overall survival 81% RN vs 76% PN, but the trial is flawed.
- **DISSRM (Delayed Intervention and Surveillance for Small Renal Masses) registry:** surveillance almost never led to metastasis; about 24% had delayed intervention.

Recent advances

- 3D (three-dimensional) models (randomised: shorter ischaemia, less blood loss); segmental clamping; ICG (indocyanine green) fluorescence for selective clamping; single-port RAPN.

Diagram to draw: clamped renal artery, tumour excised with a rim, two-layer renorrhaphy.

Key points to remember:

- Ureteric bud gives collecting system; blastema gives nephrons (GDNF–RET).
- Five segmental end arteries; veins anastomose; Brödel line avascular; puncture at the fornix.
- SRM ≤4 cm: 20–30% benign; PN is standard for cT1a, ablation <3 cm, surveillance <2 cm.
- Imperative, relative, elective.
- R.E.N.A.L. 4–6 / 7–9 / 10–12; trifecta includes warm ischaemia ≤25 min.
- AUA: RN only if complex + no CKD + eGFR/GFR likely >45.
- Preserved parenchyma, not clamp time, governs function.
- EORTC 30904: PN better function, similar cancer control.

Section 13 · Kidney & UTUC — Short answer questions (SAQ)

Role of renal nephrometry in management of RCC.

Type: SAQ (10 marks)

Source: 2020-Summer Paper 1, Q3f

Definition

- **Nephrometry scores** grade the anatomical complexity of a renal mass on CT (computed tomography) or MRI (magnetic resonance imaging), to standardise RCC (renal cell carcinoma) reporting and plan treatment.

R.E.N.A.L. score (Kutikov and Uzzo, 2009)

Component	1 point	2 points	3 points
Radius (maximum diameter)	≤4 cm	>4 to <7 cm	≥7 cm
Exophytic/endophytic	≥50% exophytic	<50% exophytic	Entirely endophytic
Nearness to collecting system or sinus	≥7 mm	>4 to <7 mm	≤4 mm
Anterior/posterior	Suffix a, p or x (no points)		
Location to polar lines	Entirely above or below	Crosses a polar line	>50% across, crosses midline, or entirely between

- Suffix "h": touches main renal vessels. **Total 4–12:** low 4–6, moderate 7–9, high 10–12.

PADUA (Preoperative Aspects and Dimensions Used for an Anatomical classification) score (Ficarra, 2009)

- Polar location, exophytic rate, renal rim (lateral or medial), renal sinus, collecting system and size, plus anterior/posterior face. **Total 6–14:** 6–7 low, 8–9 intermediate, ≥10 high risk.

C-index (centrality index)

- Distance from tumour centre to kidney centre ÷ tumour radius. **Lower value = more central, more complex.**

Role in management

Decision	Use
Partial vs radical nephrectomy	Feasibility and risk of PN (partial nephrectomy) vs RN (radical nephrectomy)
Approach	Open vs minimally invasive; hilar clamping
Predict outcomes	Warm ischaemia, blood loss, complications, conversion, collecting-system entry; urine leak (R and N components)
Function	Parenchymal loss, eGFR (estimated glomerular filtration rate) decline
Ablation	Cryoablation complications rise with complexity
Pathology	Higher scores: malignancy, higher grade

- **AUA (American Urological Association) 2021:** with high oncological potential, prefer RN only if PN would be challenging even for experts, there is no CKD (chronic kidney disease) or proteinuria, and new baseline eGFR will likely exceed 45.

Limitations

- Interobserver variation; ignores patient factors and surgeon experience.

Also know: ABC (arterial based complexity), DAP (diameter–axial–polar) scores; **MAP (Mayo Adhesive Probability) score** for adherent perinephric fat.

Diagram to draw: kidney with polar lines and the five R.E.N.A.L. components.

> **Key points to remember:**
>
> - RENAL 4–12: low 4–6, moderate 7–9, high 10–12.
> - PADUA 6–14; C-index lower = more central.
> - Predicts PN feasibility, ischaemia time and complications.
> - MAP score: adherent perinephric fat.

Rapid Revision Sheet – Section 13 · Kidney & UTUC

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal malignancy (LAQ)

- Ureteric bud gives collecting system; blastema gives nephrons (GDNF–RET).
- Five segmental end arteries; veins anastomose; Brödel line avascular; puncture at the fornix.
- SRM ≤4 cm: 20–30% benign; PN is standard for cT1a, ablation <3 cm, surveillance <2 cm.
- Imperative, relative, elective.
- R.E.N.A.L. 4–6 / 7–9 / 10–12; trifecta includes warm ischaemia ≤25 min.
- AUA: RN only if complex + no CKD + eGFR/GFR likely >45.
- Preserved parenchyma, not clamp time, governs function.
- EORTC 30904: PN better function, similar cancer control.

Role of renal nephrometry in management of RCC. (SAQ)

- RENAL 4–12: low 4–6, moderate 7–9, high 10–12.
- PADUA 6–14; C-index lower = more central.
- Predicts PN feasibility, ischaemia time and complications.
- MAP score: adherent perinephric fat.

Section 14 · Adrenal & Retroperitoneum — Long answer questions (LAQ)

Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma

Type: LAQ (25 marks)

Source: Practice Set 6 Paper 1, Q1 (pre-operative management also answered in 2020-Summer Paper 2, Q3c)

Assumed meaning: "arenal" = adrenal glands.

A. Anatomy

Gross

- Paired retroperitoneal glands on the upper renal poles (11th–12th ribs), inside Gerota fascia, separated from the kidney by thin connective tissue. Yellow-orange, 4–6 × 2–3 cm, about 5 g each.

Feature	Right	Left
Shape, level	Triangular; higher	Crescentic; lower (to L1)
Relations	Liver anteriorly, duodenum anteromedially, IVC (inferior vena cava) medially, diaphragmatic crus behind	Pancreas body and splenic vessels anteriorly, aorta medially, crus behind
Vein	**Short (<1 cm), into posterior IVC**: tears easily	**Longer, into left renal vein**, often joined by inferior phrenic vein

Blood supply

- **"Many arteries in, one vein out."** Superior adrenal (from inferior phrenic), middle (aorta), inferior (renal artery) arteries break into 10–50 small branches.
- Cortical sinusoids drain into the medulla, so cortisol-rich blood reaches the chromaffin cells. **Cortisol induces PNMT (phenylethanolamine-N-methyltransferase)**, which is why the medulla makes adrenaline.

Nerves: preganglionic sympathetic fibres (T11–L2, via splanchnic nerves, no synapse in coeliac ganglion) end directly on chromaffin cells. The medulla is a modified sympathetic ganglion.

Embryology: cortex from coelomic mesoderm (adrenogonadal ridge, weeks 3–4); medulla from neural crest. Hence adrenals stay orthotopic in renal agenesis, and adrenal rests occur along the path of gonadal descent.

Histology (zones from outside in: glomerulosa, fasciculata, reticularis = "salt, sugar, sex")

Zone	Share	Product	Main control
Glomerulosa	~15% of cortex	Aldosterone	Angiotensin II, potassium
Fasciculata	~80%	Cortisol (up to 20 mg/day)	ACTH (adrenocorticotropic hormone)

Zone	Share	Product	Main control
Reticularis	5–7%	DHEA (dehydroepiandrosterone), DHEA-S, androstenedione	ACTH
Medulla	~10% of gland	Adrenaline 80%, noradrenaline 19%, dopamine 1%	Sympathetic nerves

B. Steps in hormone production

1. Steroids (cortex)

- Cholesterol (mainly from LDL [low-density lipoprotein]) is carried into mitochondria by **StAR (steroidogenic acute regulatory protein)**: the rate-limiting step, driven by ACTH.
- **Side-chain cleavage enzyme (CYP11A1)** converts it to **pregnenolone**. The enzymes present in each zone then decide the product:

```
Pregnenolone -3βHSD-> Progesterone -21OH-> 11-deoxycorticosterone -11βOH-> Corticosterone -
aldosterone synthase (CYP11B2)-> ALDOSTERONE (glomerulosa only)
Pregnenolone -17αOH-> 17-OH pregnenolone/progesterone -21OH-> 11-deoxycortisol -11βOH->
CORTISOL (fasciculata)
17-OH pregnenolone -17,20-lyase-> DHEA -> androstenedione (reticularis)
```

- 3βHSD (3β-hydroxysteroid dehydrogenase). Steroids are not stored; they diffuse out and act on intracellular receptors.
- **Regulation:** CRH (corticotropin-releasing hormone) → ACTH → cortisol, with negative feedback and morning peak; RAAS (renin–angiotensin–aldosterone system) and potassium for aldosterone.
- **Clinical link:** 21-hydroxylase deficiency is the commonest cause of CAH (congenital adrenal hyperplasia).

2. Catecholamines (medulla)

```
Tyrosine -tyrosine hydroxylase (rate-limiting)-> L-DOPA -decarboxylase-> Dopamine
-dopamine β-hydroxylase-> Noradrenaline -PNMT (cortisol-induced)-> Adrenaline
```

- Stored in vesicles, released by exocytosis. **COMT (catechol-O-methyltransferase)** inside chromaffin cells forms **metanephrine and normetanephrine** continuously, which is why metanephrines are the best test. MAO (monoamine oxidase) and liver form VMA (vanillylmandelic acid).

C. Perioperative management of phaeochromocytoma

Why: handling the tumour causes catecholamine surges (crisis, arrhythmia); vein ligation causes sudden withdrawal (hypotension, hypoglycaemia). Mortality fell from up to 50% to <3% with blockade.

Pre-operative (7–14 days)

- Confirm with plasma free or urinary fractionated metanephrines; CT (computed tomography) or MRI; functional imaging (^{68}Ga-DOTATATE PET [positron emission tomography], MIBG [metaiodobenzylguanidine]) if extra-adrenal or hereditary; **never biopsy**; genetic testing (about one-third hereditary).
- ECG (electrocardiogram) and echocardiography for catecholamine cardiomyopathy.

Drug	Use
Phenoxybenzamine (irreversible non-selective α)	First line: 10 mg twice daily, increased every 2–3 days; about 1 mg/kg total
Doxazosin, prazosin (selective α1)	Fewer side effects

Drug	Use
Calcium channel blockers	Alternative strategy or add-on
β1-blocker (atenolol, metoprolol)	**Only after α-blockade**, for tachycardia
Metyrosine	Blocks tyrosine hydroxylase; add-on; sedation, extrapyramidal effects

- **Never β before α:** losing β2 vasodilatation leaves α-vasoconstriction unopposed: crisis.
- **Target:** seated BP 120–130/80 mmHg with mild postural drop.
- **Volume expansion:** high-salt diet and fluids; 1–2 L IV (intravenous) fluid the night before. Last phenoxybenzamine dose the night before surgery.

Intra-operative

- Arterial and central lines; laparoscopic adrenalectomy (open for large or invasive tumours); minimal handling; **early adrenal vein ligation** after warning the anaesthetist.
- Surges: nitroprusside, nicardipine, phentolamine. Tachyarrhythmia: esmolol, lidocaine. Post-ligation hypotension: fluids ± vasopressors.
- Cortical-sparing adrenalectomy in MEN2 (multiple endocrine neoplasia type 2) and VHL (von Hippel–Lindau).

Post-operative

- HDU (high dependency unit): hypotension, **rebound hypoglycaemia** (insulin no longer suppressed), steroid cover if bilateral.
- Metanephrines after surgery, then **yearly for life** (recurrence up to 16%).

Diagram to draw: posterior view of both adrenals with three arteries and the two veins.

> **Key points to remember:**
> - Right vein short into IVC; left vein into left renal vein.
> - StAR is rate-limiting for steroids; tyrosine hydroxylase for catecholamines.
> - Cortisol induces PNMT; COMT forms metanephrines inside the tumour.
> - α-blockade 7–14 days, β only after α, volume expansion; watch glucose.

Rapid Revision Sheet – Section 14 · Adrenal & Retroperitoneum

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma (LAQ)

- Right vein short into IVC; left vein into left renal vein.
- StAR is rate-limiting for steroids; tyrosine hydroxylase for catecholamines.
- Cortisol induces PNMT; COMT forms metanephrines inside the tumour.
- α-blockade 7–14 days, β only after α, volume expansion; watch glucose.

Section 15 · Renal, Nephrology & Transplant — Long answer questions (LAQ)

Describe in details basic patho-physiology of urine production and transport

Type: LAQ (25 marks)

Source: 2015-Summer Paper 1, Q1

Introduction

- Urine is formed by filtration, reabsorption, secretion and concentration, then propelled to a low-pressure bladder.

A. Functional anatomy

- Glomerular barrier (fenestrated endothelium, basement membrane, podocytes) restricts by size and negative charge, so albumin is excluded.
- **Juxtamedullary nephrons** (long loops) build the medullary gradient.

B. Renal blood flow (RBF) and autoregulation

- RBF ~25% of cardiac output. The medulla is hypoxic (10–20 mmHg), endangering the thick ascending limb and S3 segment.
- **Autoregulation** keeps RBF and GFR (glomerular filtration rate) constant at mean arterial pressure 80–180 mmHg:
  - **Myogenic:** the afferent arteriole constricts when stretched and dilates (prostaglandins) when pressure falls; NSAIDs (non-steroidal anti-inflammatory drugs) abolish this.
  - **Tubuloglomerular feedback:** low NaCl at the macula densa releases renin; angiotensin II constricts the efferent arteriole. Angiotensin blockade abolishes this.

C. Glomerular filtration

- GFR ~125 mL/min (180 L/day); filtration fraction ~20%.
- **Net filtration pressure** ≈ glomerular hydrostatic 60 – Bowman's space 18 – oncotic 32 = **~10 mmHg**.
- Efferent constriction lowers RBF but **raises** GFR; afferent constriction lowers both. Obstruction raises Bowman's space pressure, so GFR falls.
- **Also know:** GFR estimated from creatinine or cystatin C (race-free 2021 equations).

D. Tubular handling

Segment	Key transporter	Handling	Drug/disease link
Proximal tubule	Na^+/H^+ exchanger, SGLT2 (sodium–glucose cotransporter 2)	~**65%** Na^+ and water (iso-osmotic); all glucose; most bicarbonate and phosphate; secretes drugs	SGLT2 inhibitors
Thin descending limb	Aquaporin-1	Water out, solute stays: fluid concentrates	–
Thick ascending limb	NKCC2 (Na^+-K^+-$2Cl^-$)	~**25%** NaCl; water-impermeable "diluting segment"	Loop diuretics

Segment	Key transporter	Handling	Drug/disease link
Distal convoluted tubule	NCC (Na^+-Cl^- cotransporter)	5–7% Na^+; Ca^{2+} (parathyroid hormone)	Thiazides lower urinary Ca^{2+}
Collecting duct	ENaC (epithelial Na^+ channel), AQP2	Principal cells: Na^+/K^+ (aldosterone), water (ADH); intercalated cells: acid	Amiloride, spironolactone; distal RTA (renal tubular acidosis)

E. Concentration

- **Countercurrent multiplication:** ascending limbs absorb NaCl without water, raising medullary osmolality from 300 to ~**1200 mOsm/kg**; urea recycling adds about half. Vasa recta exchange without washing out the gradient.
- **Antidiuretic hormone (ADH)** is released by high osmolality or low volume. Signal chain: V2 receptor → cAMP (cyclic adenosine monophosphate) → protein kinase A → aquaporin-2 (AQP2) inserted into the apical membrane → water reabsorbed. Concentration needs both a hypertonic medulla and AQP2; urine osmolality ranges ~50–1200 mOsm/kg.

Hormone	Site	Effect
Angiotensin II, aldosterone	Efferent arteriole, proximal tubule; collecting duct	Na^+ retention, K^+/H^+ loss, GFR maintained
Atrial natriuretic peptide	Afferent arteriole, collecting duct	Natriuresis, lowers renin
Parathyroid hormone	Proximal and distal tubule	Phosphaturia, Ca^{2+} retention

F. Ureteric transport

- **Pacemaker:** atypical smooth muscle cells at the minor calyces/pelvicalyceal border.
- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves (T11–L1, S2–4) carry only pain.
- Calcium entry triggers action potentials spread by gap junctions; calcium–calmodulin then activates myosin light-chain kinase.
- Walls coapt behind each bolus: 2–6 waves/min at 20–80 cmH_2O. Very high flow makes boluses coalesce, dilating the ureter **without obstruction**.
- Ureterovesical tunnel length:diameter 5:1 prevents reflux; bladder pressure >40 cmH_2O stops efflux.

Factor	Effect on transport
Obstruction	Acute: stronger contractions; chronic: dilatation, lost coaptation
Infection	Gram-negative endotoxin inhibits peristalsis (dilatation without obstruction)
Pregnancy	Right > left dilatation: compression at pelvic brim, progesterone
Drugs	α1-blockers and calcium-channel blockers relax (medical expulsive therapy); NSAIDs lower pelvic pressure in colic

G. Bladder storage and voiding

- **Storage** below ~10 cmH_2O: compliance, sympathetic β3 detrusor relaxation, α1 bladder-neck and pudendal sphincter tone.
- **Voiding:** pelvic parasympathetic (S2–4) acetylcholine on M3 receptors contracts the detrusor; nitric oxide relaxes the urethra.

H. Pathophysiology of obstruction (unilateral: triphasic)

```
Phase 1 (0–1.5 h): RBF up,   pressure up    (afferent dilatation:
                                             prostaglandins, nitric oxide)
Phase 2 (1.5–5 h): RBF down, pressure up    (efferent constriction)
Phase 3 (>5 h):    RBF down, pressure down  (afferent constriction: angiotensin II,
                                             thromboxane A2, endothelin)
```

- Ischaemic injury starts **within 2 hours** of complete block.
- **Tubules:** down-regulated aquaporins and Na^+ transporters cause concentrating defect, salt wasting and distal RTA.
- **Fibrosis** (final path to failure): stretch, inflammation, TGF-β (transforming growth factor-β), angiotensin II, microRNAs.
- **Bilateral obstruction or solitary kidney:** pressure stays high longer and natriuretic peptides and urea accumulate, so **post-obstructive diuresis** follows relief in ~50% (vs ~2% unilateral).
- **Clinical:** hypertension (~77% bilateral, ~20% unilateral); contralateral hypertrophy; pyelorenal backflow acts as a pop-off; forniceal rupture causes urinoma.
- **Also know:** immature kidneys fail to develop; outlet obstruction causes detrusor hypertrophy then decompensation; diuretic MAG3 (mercaptoacetyltriglycine) renogram (half-time >40 min = obstructed); Whitaker test (>15–20 cmH_2O = obstructed); tolvaptan in polycystic disease.

Diagram to draw: Nephron with transporters; triphasic graph.

Key points to remember:

- RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
- Proximal 65%, loop 25%, distal 5–7% of Na^+; ADH → V2 → AQP2.
- Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL

Asked as:

- Describe anatomy of pelvicalyceal system in detail and its application in endourological surgeries — *Winter 2017 Paper I Q1 (LAQ)*
- Anatomy of renal collecting system and its relevance to PCNL — *Practice Set 1 Paper I Q3d (SAQ)* (also answered in 2022-Summer Paper 1, Q1 and 2017-Winter Paper 1, Q1)
- Describe anatomy of kidney in relation to PCNL and describe indication, contraindication, procedure and complications of mini-PCNL. — *Summer 2022 Paper I Q1 (LAQ)* (also answered in 2017-Winter Paper 1, Q1 (pelvicalyceal anatomy) and 2017-Winter Paper 4, Q3d (Miniperc))

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction

- Papilla → minor calyx → infundibulum → major calyx → renal pelvis → UPJ (ureteropelvic junction); lined by urothelium. Anatomy is very variable, so map it (CT (computed tomography) urogram or pyelogram) before every endourological procedure.
- Safe PCNL (percutaneous nephrolithotomy) access depends on the kidney's position, neighbours, vessels and calyces.

Surgical anatomy relevant to PCNL: position and orientation

- Lies T12–L3 (right 1–2 cm lower); descends about 3 cm on inspiration.
- It lies on the cone-shaped psoas, so the upper pole is medial and posterior and the lower pole lateral and anterior.
- Tract: skin, flank muscles, thoracolumbar fascia, paranephric fat, Gerota fascia, perirenal fat, cortex, then **pyramid and papilla** into the calyx.

Anatomy of the calyces (collecting system)

- **Minor calyx:** a cup around one papilla; its two side projections are the **fornices**; it drains by an infundibulum to a major calyx and pelvis. **Compound calyces** (2–3 papillae) are commonest at (cluster at) the poles. Average 7–9 calyces per kidney (range 4–19 or more).
- **Distribution:** upper pole usually 3 calyces, interpolar 3–4, lower pole 2–3.
- **Drainage pattern:** upper pole by a single midline infundibulum; lower pole by a single midline infundibulum or paired calyces; hilar region (hilar ones) by **anterior and posterior rows** of paired calyces.
- **Orientation:** the kidney lies on psoas, rotated 30–50° behind the coronal plane (posteriorly), so **posterior calyces point posterolaterally** (towards the prone surgeon). On a pyelogram anterior and posterior calyces (rows) overlap.
- **Brödel vs Hodson pattern (Kaye and Reinke):** in a Brödel-type kidney the posterior calyces lie lateral and the anterior calyces medial; in a Hodson-type kidney the reverse. Brödel type is commoner on the right, Hodson type on the left, so confirm calyx orientation with contrast/air or ultrasound before puncture.

Renal pelvis

- Pyramidal, apex to the ureter; capacity 3–10 mL. **Intrarenal** (in the sinus), **extrarenal** (larger, longer infundibula, mimics dilatation) or mixed.
- **Duplex (divided) pelvis:** upper and lower parts join at the hilum, with no direct connection between upper and lower calyces (does not link upper and lower calyces). Suspect it when one pole cannot be reached from a puncture.

Sampaio classification (Sampaio types; 140 resin endocasts)

Type	Frequency	Midzone (hilar) drainage
A-I	45%	Midzone calyces drain into the superior or inferior calyceal group (via a polar group)
A-II	17.2%	**Crossed calyces**: one to the superior and one to the inferior group (to both groups)
B-I	21.4%	Independent major calyceal group (major calyx) from the midzone
B-II	16.4%	1–4 minor calyces enter the pelvis directly (straight into the pelvis)

- **Perpendicular minor calyx (11.4%):** drains straight into the pelvis, hidden on a pyelogram (pyelography); narrow neck (≤4 mm) hinders clearance.
- **Crossed calyces:** the limb to the inferior group is **ventral in 87.5%**.

Vascular relations (the basis of safe access)

- Hilum, anterior to posterior: vein, artery, pelvis, **posterior segmental artery** (the only branch passing behind the pelvis).
- Five segmental **end arteries**; interlobar arteries run in the **columns of Bertin**, hugging the anterior and posterior faces of the infundibula (beside the infundibula).
- **Brödel line:** avascular plane between anterior and posterior arterial territories, just posterior to (behind) the lateral border, along the posterior calyces. Posterior calyces along it are the target.

Puncture route / site (Sampaio)	Vascular injury
Upper-pole infundibulum	67% (arterial 26%); posterior segmental artery
Mid-pole infundibulum	Arterial 23%
Lower-pole infundibulum	Infundibular artery (present) in 38%
Calyceal fornix	Venous <8%, **no arterial injury**

- So the needle passes **through the papilla along the calyx axis**, never into an infundibulum or the pelvis.

Relations (the hazards of access)

- **Relations:** pleura reaches the 12th rib posteriorly, lung the 10th intercostal space; liver and spleen descend in inspiration; **retrorenal colon** is commoner at the lower pole and when prone.

Structure	Relation	Practical point
Pleura, lung	Pleura reaches the 12th rib; lung the 10th intercostal space	Prefer subcostal or 11th-space access; risk rises above the 12th rib, highest above the 11th; puncture in expiration
Liver, spleen	Over the upper poles; descend in inspiration	Risk with supracostal access, inspiration, organomegaly
Colon	Retrorenal colon, mainly at the lower pole, commoner prone and on the left	Review CT; ultrasound shows bowel

Applications in endourology

1. PCNL (percutaneous nephrolithotomy) access

- Ideal target: a **posterior calyx with a short, wide infundibulum**, punctured (entered) **through the papilla and fornix** along the calyx axis, never into (through) the infundibulum or pelvis; identified by injected air, C-arm rotation or ultrasound. This passes through a pyramid and avoids interlobar arteries.

- **Upper-pole posterior access** gives the straightest route (best reach) to the pelvis, UPJ and most calyces (staghorn; staghorn branches). Chest risk (pneumothorax, hydrothorax) rises above the 12th rib, highest above the 11th. Puncture in expiration.
- **Lower-pole access** is chest-safe but risks (retrorenal) colon (left) and reaches the UPJ poorly (poor UPJ reach).
- Identifying (identify) posterior calyces: rainbowing the C-arm towards the surgeon shortens them (they shorten when the C-arm rotates towards the surgeon); injected air fills non-dependent (posterior, when prone) calyces (rises posteriorly when prone); ultrasound also shows bowel and pleura.
- **Also know:** traps: perpendicular or crossed calyces, duplex pelvis.

2. RIRS (retrograde intrarenal surgery)

- Entering the lower-pole infundibulum needs about **140° deflection** (104–175°); ≥270° scopes see it, but working there is harder.
- **Unfavourable lower-pole anatomy:** steep infundibulopelvic angle, long calyx (>10 mm), narrow infundibulum (<5 mm). Relocate stones to an upper calyx with a basket.
- Lower-pole stone >1 cm: PCNL preferred (**Lower Pole I, 2001:** stone-free 91% PCNL vs 21% SWL (shock wave lithotripsy)).

3. Endopyelotomy

- Incise the UPJ **laterally**, where crossing vessels are absent. A crossing vessel lowers success (42% vs 86%).
- Antegrade route: mid-posterior or superolateral calyx; not lower pole (acute angle to the UPJ).

4. Infundibulotomy and calyceal diverticulum

- Incise infundibular stenosis **superiorly or inferiorly**; never anteriorly.
- **Calyceal diverticulum:** cortical cavity joined to the fornix by a narrow neck; treat if symptomatic. Upper or interpolar: ureteroscopy, incise or dilate the neck. Lower pole: direct puncture onto the stone, mini-PCNL; fulguration optional.

5. Open/anatrophic nephrolithotomy: incision along the Brödel line.

- **Also know:** horseshoe kidney (medial access, upper pole usually subcostal); pelvic kidney (ultrasound or laparoscopic guidance); transplant kidneys; megacalycosis mimics obstruction; AV (arteriovenous) fistula or pseudoaneurysm → angioembolisation.

Mini-PCNL: definition

Term	Tract size
Standard	24–34 Fr
Mini	15–22 Fr
Ultra-mini	11–14 Fr
Micro-perc	<10 Fr

- First mini-perc: Jackman, 1998 (13 Fr).

Indications

- Renal stones 1–2 cm; lower-pole stones >1 cm; failed SWL (shock wave lithotripsy) or RIRS (retrograde intrarenal surgery).
- Calyceal diverticulum (lower pole), children, poor retrograde access (urinary diversion, ureteric stricture), selected stones >2 cm or extra tracts for staghorns.

Contraindications

- Untreated UTI (urinary tract infection), uncorrected coagulopathy, pregnancy (EAU, European Association of Urology).
- Tumour in the access tract or suspected malignant renal tumour; no safe window; unfit for anaesthesia.

Procedure

1. Urine culture, blood count, coagulation, creatinine; **non-contrast CT (computed tomography)**; antibiotic prophylaxis.
2. Lithotomy: ureteric catheter and retrograde pyelogram.
3. Prone, or supine (Galdakao-modified) for single-position retrograde access.
4. 18 G puncture of a posterior calyx under fluoroscopy (bull's-eye or triangulation) or ultrasound; urine return; guidewire and safety wire.
5. One-step, serial or balloon dilatation to 15–22 Fr; place the sheath.
6. Mini-nephroscope; holmium/thulium laser or pneumatic lithotripsy. Fragments wash out by the irrigation **"vacuum-cleaner" effect** or a **suction sheath**.
7. Flexible nephroscopy to clear calyces.
8. Exit: tubeless, small nephrostomy (≤10 Fr: less pain) or double-J stent.

Advantages and limitations

- Meta-analysis: **non-inferior stone-free rate**, smaller haemoglobin drop, **fewer transfusions**, shorter stay.
- Longer operation, poorer vision. A tight sheath raises **intrarenal pressure**, causing pyelovenous backflow and **fever or sepsis**. Advantages disappear against 24 Fr tracts.

Complications (grade by Clavien–Dindo)

Type	Points
Bleeding	Acute tract bleeding: tamponade sheath, balloon or Kaye catheter. Delayed: pseudoaneurysm or AV (arteriovenous) fistula, treated by angioembolisation
Infection	Fever, SIRS (systemic inflammatory response syndrome), urosepsis
Thoracic	Pneumothorax, hydrothorax: chest drain
Visceral	Colon (stent plus nephrostomy withdrawn into the colon), spleen, liver, duodenum
Collecting system	Perforation, extravasation, later infundibular stenosis; residual stones
Others	Irrigation absorption, venous gas embolism, DVT (deep vein thrombosis), neuropraxia, renal function loss, death

Diagram to draw: transverse section of kidney showing anterior and posterior calyces (posterior calyx), Brödel line and the needle through the posterior calyceal fornix.

Key points to remember:

- Sampaio: A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
- Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
- Target a posterior calyx with a short, wide infundibulum; upper pole reaches most.
- Lower pole: steep angle, long calyx >10 mm, infundibulum <5 mm unfavourable.
- Endopyelotomy lateral; infundibulotomy superior or inferior, never anterior.
- Pleura to 12th rib; retrorenal colon at the lower pole, commoner prone.
- Mini 15–22 Fr (Jackman 1998): fewer transfusions, longer operation.
- High intrarenal pressure in mini-PCNL causes fever and sepsis.

Lymphatic drainage of the genitourinary system and chyluria (etiopathogenesis and management)

Asked as:

- Describe anatomy of lymphatic drainage of genitourinary system, etiopathogenesis and management of Chyluria — *Summer 2018 Paper I Q1 (LAQ)*
- Describe lymphatic system of urinary tract with reference to etiopathogenesis of chyluria. — *Practice Set 8 Paper I Q1 (LAQ)*
- Management of chyluria — *Practice Set 10 Paper I Q1 (LAQ)*

Type: LAQ (25 marks)
(also answered in 2018-Summer Paper 1, Q1 and 2024-Winter Paper 3, Q2)

General plan of lymphatic drainage

- Lymph flows **cephalad and right → left** and returns to the left innominate (brachiocephalic) vein.
- Pelvic nodes (external iliac, internal iliac, obturator, presacral) → common iliac → lumbar (para-aortic, interaortocaval, paracaval) nodes → **lumbar trunks**. These join the intestinal trunk at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta). The **thoracic duct** rises behind the aorta to the left innominate vein.
- Pelvic lymph crosses freely between right and left (the two sides). Organs supplied by lateral aortic branches (kidney, adrenal, ureter, gonad) drain to lumbar nodes.
- **Lumbar groups:** left lumbar (preaortic, left para-aortic, retroaortic), interaortocaval, and right lumbar (precaval, paracaval, retrocaval). They drain organs supplied by lateral aortic branches: kidneys, adrenals, ureters and gonads.
- Preaortic nodes (coeliac, superior and inferior mesenteric) form the **intestinal trunk**, which carries **chyle** from the intestinal lacteals.
- Lumbar and intestinal trunks join at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta). It continues as the **thoracic duct**, which rises behind the aorta to the left innominate vein.
- **Lymphatic anatomy (summary):** intestinal lacteals → intestinal trunk → **cisterna chyli** (L1–L2) → **thoracic duct** → left subclavian–jugular junction. Renal lumbar lymphatics join the cisterna; valves normally prevent backflow.

Drainage by organ

Organ	Primary nodes	Notes
Left kidney	Left lateral para-aortic	Capsular and deep hilar networks drain to renal vein nodes; may go retrocrural
Right kidney	Right paracaval, interaortocaval	May cross to left para-aortic
Ureter	Abdominal: para-aortic (left), paracaval/interaortocaval (right); middle: common iliac; pelvic: common, external, internal iliac	Plexus in muscle and adventitia
Bladder	Mainly **external iliac**; also internal iliac, obturator	Begins in lamina propria; bilateral cross-drainage
Prostate	**Obturator and internal iliac**; presacral	Three routes: superolateral (side wall), posterior (presacral), apical (internal pudendal)

Organ	Primary nodes	Notes
Testis	Left: left para-aortic; right: interaortocaval, then right paracaval	Follows gonadal vessels; crossover only right → left. Epididymal cauda and vas → external iliac
Penis	Skin → superficial inguinal (both groins); glans and corpora → superficial and deep inguinal, then pelvic	Shaft lymphatics converge dorsally
Scrotum	Ipsilateral superficial inguinal	Does not cross the raphe
Urethra (male)	Penile → inguinal; bulbomembranous and prostatic → internal/external iliac, obturator	Female: proximal → pelvic; distal → inguinal

Lymphatic drainage of the urinary tract (intrinsic lymphatics and regional nodes)

Organ	Intrinsic lymphatics	Regional nodes
Kidney	Superficial capsular and deep hilar networks along the interlobular, arcuate and interlobar arteries, to nodes at the renal vein	**Left:** left lateral para-aortic (sometimes retrocrural, or thoracic duct above the diaphragm). **Right:** interaortocaval and right paracaval (may cross to left para-aortic)
Ureter	Plexuses in the muscle and adventitia	Abdominal: left para-aortic (left); paracaval and interaortocaval (right). Middle: common iliac. Pelvic: common, external and internal iliac
Bladder	Begins in the lamina propria; a few perivesical nodes	Mainly **external iliac**; also internal iliac and obturator; cross-drainage to both sides
Prostate	Periprostatic plexus	Obturator and internal iliac; presacral; external iliac
Urethra	Submucosal plexus	Male prostatic and membranous: internal iliac, obturator; penile: inguinal, then external iliac. Female: proximal to pelvic, distal to inguinal nodes

- **Applied:** upper tract cancers spread to lumbar nodes; bladder and prostate cancers to pelvic nodes. Renal drainage is asymmetric: right renal lymph can cross left; left stays left.
- Valves in the renal hilar lymphatics normally stop backflow from the cisterna chyli. **Failure of these valves is the basis of chyluria.**

Chyluria: definition

- Passage of **chyle** (intestinal lymph rich in chylomicrons, triglyceride and protein) in urine, giving milky urine (making it milky; urine looks milky). Caused by an abnormal **lymphourinary (chylourinary) fistula**, usually at the renal fornices / a renal calyceal fornix (**pyelolymphatic fistula**).

Etiology

Parasitic (commonest; India, China)	Non-parasitic
Wuchereria bancrofti (lymphatic filariasis; *Culex* mosquito)	Congenital lymphangiectasia, renal lymphangioma
Brugia (lymphatic filariasis that spares the genitals; rarely urological)	Trauma, surgery (e.g. partial nephrectomy, RPLND (retroperitoneal lymph node dissection))
	Retroperitoneal tumour or fibrosis, tuberculosis, pregnancy

Pathogenesis (obstructive theory)

- **Life cycle:** mosquito-borne larvae mature into adult worms that live for years in retroperitoneal and genital lymphatics. Females release microfilariae with **nocturnal periodicity**.

- Adult worms live in retroperitoneal lymphatics and cause lymphangitis and fibrosis. This **obstructs** flow between the intestinal lymph trunks and the thoracic duct.
- Proximal lymphatics dilate and their valves fail, so chyle **refluxes** into renal hilar lymphatics.
- Raised pressure ruptures them into a calyceal fornix → **pyelolymphatic fistula**. Rarely the fistula is in the ureter or bladder.
- **Obstructive (reflux) theory:**

```
Adult worms + host immune reaction -> lymphangitis and fibrosis
 -> obstruction between intestinal trunks/cisterna chyli and thoracic duct
 -> dilatation of retroperitoneal lymphatics, valve incompetence
 -> reflux of chyle into renal hilar and peripelvic lymphatics
 -> raised pressure ruptures thin-walled lymphatics into a calyceal fornix
 -> pyelolymphatic fistula -> chyluria
```

- **Pathogenesis (filarial):** adult *Wuchereria bancrofti* worms and the host reaction cause **lymphangitis and fibrosis** between the intestinal trunks, cisterna chyli and thoracic duct. Lymphatics below the block dilate and their valves fail. Chyle refluxes into renal hilar lymphatics, which rupture into a calyx.
- The fistula is usually renal (hilar lymphatics lie against thin fornices); rarely ureteric or vesical.
- Chyluria appears **earlier than genital elephantiasis**, is usually **intermittent** and may remit spontaneously. It worsens after fatty meals, when lacteal flow is highest. So every treatment is judged against natural remission.
- **Non-parasitic causes** work the same way: lymphatic obstruction (tumour, fibrosis), direct disruption (surgery, trauma) or congenital lymphangiectasia.
- **Effects:** loss of fat, protein and lymphocytes causes weight loss, hypoproteinaemia and lowered immunity. Fibrin in chyle forms clots.

Clinical features

- Milky urine, worse after fatty meals; **chylous clots** (retention, colic); **haematochyluria**.
- Loss of fat and protein → weight loss, **hypoproteinaemia**, oedema, anaemia, fatigue, lowered immunity.
- **Severity:** mild (intermittent, no systemic effect); moderate (persistent, clots); severe (haematochyluria, clot retention, weight loss, hypoalbuminaemia).

Grade	Features
Mild	Intermittent; no systemic effect
Moderate	Persistent; clots
Severe	Haematochyluria, clot retention, weight loss, hypoalbuminaemia

Evaluation (before treatment)

1. Confirm chyle

- **Ether (chloroform) test**: milky urine shaken with ether clears because fat dissolves. **Urine triglycerides** (and protein) positive/raised; Sudan III stains fat globules.
- Distinguish from / exclude **phosphaturia** (clears with acetic acid) and **pyuria** (leucocytes, no clearing).

2. Filarial status (filarial tests)

- Night blood smear (often negative at this stage / late), **circulating filarial antigen** (card) test, eosinophilia; PCR (polymerase chain reaction).

3. Lateralise the side and localise the fistula

- **Cystoscopy after a fatty meal** shows milky efflux from the ureteric orifice (one, sometimes both).
- **Retrograde pyelography** shows **pyelolymphatic backflow**.
- **Define fistula: lymphoscintigraphy** (non-invasive, shows tracer in kidney and bladder), pedal lymphangiography (historical gold standard), MR (magnetic resonance) lymphangiography or CT.

4. Baseline: serum albumin, haemoglobin, renal function; urine culture before instillation.

Management (outline / algorithm)

```
Mild → diet + DEC
Persistent / moderate → add sclerotherapy (can repeat)
Severe or failed sclerotherapy → renal hilar lymphatic disconnection
```

```
Mild               -> low-fat, high-protein diet + MCT oil + DEC
Moderate/persistent -> endoscopic sclerotherapy (silver nitrate 0.5-1%, povidone-iodine)
Severe or failed   -> renal hilar lymphatic disconnection (open, laparoscopic, robotic)
```

```
Confirm chyle -> grade severity
   Mild             -> diet + antifilarial drugs; observe
   Moderate/persist -> add endoscopic sclerotherapy (repeatable)
   Severe or failed -> renal hilar lymphatic disconnection
```

1. Conservative (all patients)

- **Low-fat, high-protein diet with MCT (medium-chain triglyceride) oil**, plenty of fluids.
- **Why MCT:** MCTs (medium-chain fats) are absorbed straight/directly into the portal blood, not into lacteals, so chyle formation falls (less chyle is formed).
- **DEC (diethylcarbamazine)** 6 mg/kg/day for 12 days kills microfilariae (another source: 2 mg/kg orally three times daily, traditionally 12 days; a 1-day course is as effective). It has only modest effect on adults.
- **Doxycycline** 200 mg daily for 4–8 weeks kills *Wolbachia* (endosymbiont; anti-*Wolbachia*) and so sterilises or shortens the life of adult worms. Albendazole and ivermectin are alternatives.
- Correct anaemia and protein deficit. Most mild cases settle on this, and many remit spontaneously.

2. Endoscopic sclerotherapy (renal pelvic instillation sclerotherapy)

- Ureteric catheter on the affected (leaking) side, head-down position. Instil **silver nitrate** (0.5–1%) or **povidone-iodine**, repeated over several sessions.
- **Mechanism:** chemical lymphangitis and fibrosis seal the fistula.
- **Contraindications:** active UTI (urinary tract infection), bilateral simultaneous instillation (risk of anuria).
- Good initial success/response; recurrence is common, but it is repeatable and avoids surgery.
- **Complications:** flank pain, haematuria, papillary necrosis, ureteric stricture, anuria; rarely renal failure or death (silver nitrate).

3. Surgery: renal hilar lymphatic disconnection (chylolymphatic disconnection)

- **Indications:** severe disease, failed sclerotherapy, haematochyluria with clot retention, marked weight loss.
- **Steps:**

1. Full renal mobilisation (nephrolysis).
2. **Skeletonise the renal artery and vein**, ligating, clipping or sealing every/all hilar lymphatics.
3. Strip the upper ureter; remove perinephric fat.
4. Nephropexy ± optional **omental wrap** around the hilum as a barrier.
5. **On-table cystoscopy** confirms clear urine/efflux.

- Reflux occurs only through **hilar lymphatics**, so upper-pole dissection can be limited.
- **Approach:** open, laparoscopic (transperitoneal or retroperitoneal) or robotic. **Retroperitoneoscopic** ligation beats open in meta-analysis: shorter operation and stay, less blood loss, faster bowel recovery, fewer complications.
- **Complications:** lymphocele, renal vein injury, recurrence.
- **Also know:** lymphovenous anastomosis; autotransplantation (rare); nephrectomy (now) obsolete; prevent by clipping lymphatics during renal and nodal surgery.

4. Follow-up and prevention

- Urine triglycerides and weight; recurrence needs repeat sclerotherapy or surgery.
- Mass drug administration (albendazole with DEC or ivermectin), bednets; clip lymphatics during renal and nodal surgery.

Diagram to draw: Retroperitoneal lymphatics with cisterna chyli, obstruction (a block above it / obstructed thoracic duct), dilated hilar lymphatics, and reflux to a pyelolymphatic fistula at a calyceal fornix.

Key points to remember:

- Lymph flows right → left; cisterna chyli at L1–L2, behind the right side of the aorta.
- Bladder drains mainly to external iliac nodes; left kidney to left para-aortic nodes.
- Chyluria = pyelolymphatic fistula from lymphatic obstruction and valve failure, usually filarial (*W. bancrofti*); often intermittent.
- Ether test and (urine) triglycerides confirm (ether test clears milky urine); cystoscopy after (a) fatty meal lateralises.
- Diet (MCT) + DEC (diet with MCT plus DEC) → (silver nitrate) sclerotherapy → laparoscopic hilar lymphatic disconnection; disconnection surgery for severe or refractory cases.
- Retroperitoneoscopic disconnection beats open.

Pathophysiology of Renovascular hypertension

Type: LAQ (25 marks)

Source: Practice Set 8 Paper 1, Q2 (also answered in 2022-Summer Paper 2, Q2, with treatment in detail)

Definitions

- **RVH (renovascular hypertension):** hypertension caused by RAS (renal artery stenosis) and relieved by revascularisation or nephrectomy. RAS is common in older vascular patients; RVH is proved only by cure.
- **Ischaemic nephropathy:** loss of GFR (glomerular filtration rate) from RAS affecting the **whole renal mass** (bilateral RAS or stenosis to a solitary kidney).

Causes

	Atherosclerotic RAS	FMD (fibromuscular dysplasia)
Share	Most (about 90%)	Most of the rest
Patient	Older, smokers, diabetic, vascular disease	Young women
Site	**Ostium and proximal third**	**Mid and distal artery, branches**; "string of beads" (medial fibroplasia)
Course	Progresses to occlusion and renal loss	Rarely occludes

Also know: Takayasu arteritis, neurofibromatosis, midaortic syndrome (children), aneurysm, dissection, embolism, transplant RAS, **Page kidney** (perinephric compression, e.g. subcapsular haematoma).

The RAAS (renin–angiotensin–aldosterone system)

- **Renin** is released by juxtaglomerular cells of the afferent arteriole in response to:
  1. low afferent arteriolar pressure (intrarenal baroreceptor);
  2. low sodium chloride delivery to the **macula densa**;
  3. sympathetic **β1** stimulation.
- Renin cleaves hepatic **angiotensinogen** to angiotensin I. **ACE (angiotensin-converting enzyme)**, mainly in lung endothelium, converts it to **angiotensin II**.
- **Angiotensin II (AT1 receptor) actions:** systemic arteriolar vasoconstriction; **aldosterone** release (sodium retention, potassium loss); proximal tubular sodium reabsorption; thirst and ADH (antidiuretic hormone); sympathetic facilitation; **efferent arteriolar constriction**, which maintains GFR; cardiac and vascular hypertrophy and fibrosis.
- Normally, rising blood pressure (BP) and sodium suppress renin (negative feedback). In RVH the stenosed kidney "sees" low pressure, so this feedback fails.

Goldblatt experiments

- **Goldblatt (1934)** clipped the canine renal artery and produced sustained hypertension, reversed by early clip removal.

	Two-kidney, one-clip (2K1C)	One-kidney, one-clip (1K1C)
Human equivalent	Unilateral RAS, normal opposite kidney	Bilateral RAS or stenosed solitary kidney

	Two-kidney, one-clip (2K1C)	One-kidney, one-clip (1K1C)
Sodium	Normal kidney excretes extra sodium (**pressure natriuresis**)	No normal kidney: sodium and water retained
Renin	**High** from stenotic side; contralateral suppressed	Rises, then normalises as volume expands
Type	**Renin (angiotensin)-dependent**	**Volume-dependent**
ACE inhibitor effect	BP falls; captopril tests positive	Less BP effect; may cause **acute kidney injury**

Phases of 2K1C hypertension

1. **Acute phase:** renin-dependent; ACE inhibitor or clip removal reverses it.
2. **Transition:** sodium retention and secondary hyperaldosteronism; renin falls towards normal.
3. **Chronic phase:** hypertension damages the contralateral kidney (nephrosclerosis) and resistance vessels. BP no longer depends on the stenosis, so **revascularisation cures less**. This explains poorer results in long-standing hypertension.

Other contributing mechanisms

- Renal afferent nerve signals raise **sympathetic activity**.
- Angiotensin II generates **oxidative stress** (reactive oxygen species), lowers **nitric oxide** and raises **endothelin**, worsening vasoconstriction.
- Inflammation and fibrosis; left ventricular hypertrophy and diastolic dysfunction.

Pathophysiology of ischaemic nephropathy

- The kidney receives about **20–25% of cardiac output**, far more than its metabolic need. **Autoregulation** holds GFR until stenosis is severe (about 70% or more, or a significant translesional pressure gradient).
- Beyond this, GFR depends on angiotensin II-mediated efferent constriction. Blocking it (ACE inhibitor or ARB [angiotensin receptor blocker]) drops GFR: hence **creatinine rise >30%** after these drugs.
- Chronic hypoperfusion causes hypoxia, inflammation, microvascular rarefaction, **tubulointerstitial fibrosis** and atrophy. Atheroemboli and hypertensive injury add damage.
- Early disease is reversible ("hibernating" kidney); established fibrosis is not.
- **Flash pulmonary oedema:** bilateral RAS causes volume retention plus angiotensin II-driven diastolic dysfunction.

Clinical clues

- Onset <30 years (FMD) or severe hypertension after 55; **resistant** (≥3 drugs) or accelerated hypertension; sudden loss of control.
- Flash pulmonary oedema; kidneys differing by >1.5 cm; abdominal bruit; **hypokalaemia** (secondary hyperaldosteronism).

Investigations based on pathophysiology

- **Duplex Doppler:** high peak systolic velocity, renal/aortic ratio >3.5. **RI (resistive index) >0.8** means parenchymal disease and poor response.
- **CT (computed tomography) or MR (magnetic resonance) angiography** for anatomy; **catheter angiography** with translesional gradient is the gold standard.
- **Captopril renography:** removing angiotensin II drops GFR in the stenosed kidney (delayed uptake), proving renin-dependence.

- **Renal vein renin ratio >1.5** lateralises.

Treatment implications (brief)

- **ACE inhibitor or ARB** treats renin-dependent RVH (monitor creatinine and potassium), plus statin, antiplatelet and smoking cessation.
- **ASTRAL (2009)** and **CORAL (2014):** stenting added no benefit over medical therapy in atherosclerotic RAS.
- **FMD:** balloon angioplasty without stent. Revascularise atherosclerotic RAS for flash pulmonary oedema, rapidly falling function or truly resistant hypertension.

Diagram to draw: RAAS cascade from renin release to angiotensin II actions, beside the 2K1C and 1K1C models.

Key points to remember:

- Goldblatt 1934: 2K1C = renin-dependent; 1K1C = volume-dependent.
- Chronic phase: contralateral nephrosclerosis makes RVH stenosis-independent.
- Angiotensin II keeps GFR by efferent constriction; ACE inhibitor causes acute kidney injury in bilateral RAS.
- ASTRAL and CORAL: no routine stenting; FMD = angioplasty alone.

Section 15 · Renal, Nephrology & Transplant — Short answer questions (SAQ)

Renin-angiotensin system

Type: SAQ (10 marks)

Source: 2015-Summer Paper 1, Q3c

Definition

- Hormonal cascade controlling **blood pressure, extracellular volume, Na^+/K^+ balance and GFR (glomerular filtration rate)**.

Cascade

```
Angiotensinogen (liver)
   ↓ renin (juxtaglomerular cells) — rate-limiting
Angiotensin I
   ↓ ACE (angiotensin-converting enzyme), mainly lung
Angiotensin II → AT1 / AT2 receptors
   ↓ ACE2
Angiotensin (1-7) → Mas receptor (protective)
```

Juxtaglomerular apparatus and renin release

- Granular cells in the afferent arteriole secrete renin and act as baroreceptors; the **macula densa** senses tubular NaCl.

Renin stimulated by	Renin inhibited by
Low renal perfusion pressure (baroreceptor)	High perfusion pressure
Low NaCl at macula densa	High NaCl at macula densa
Sympathetic β1 activity	Angiotensin II (negative feedback)
Prostaglandins, hypovolaemia	Atrial natriuretic peptide, β-blockers, NSAIDs (non-steroidal anti-inflammatory drugs)

Actions of angiotensin II

- **AT1 receptor:** vasoconstriction; **aldosterone** (distal Na^+ retention, K^+ loss); direct proximal Na^+ reabsorption; thirst and ADH (antidiuretic hormone); **fibrosis**.
- **AT2 receptor:** vasodilatation; fetal urinary tract development.
- **Intrarenal:** efferent constriction maintains GFR when perfusion falls, so ACE inhibitors and ARBs (angiotensin receptor blockers) drop GFR in bilateral or solitary-kidney renal artery stenosis.

Urological relevance

- **Renovascular hypertension (Goldblatt kidney):** atherosclerosis (older men, ostial) or fibromuscular dysplasia (young women). Diagnose by duplex ultrasound, computed tomography or magnetic resonance angiography, captopril renography. **Renal vein renin ratio >1.5** with contralateral suppression predicts cure by revascularisation or nephrectomy.
- **Page kidney:** subcapsular haematoma compresses parenchyma; ACE inhibitor/ARB, then drainage.

- **Reninoma (juxtaglomerular cell tumour):** young women, ≤3 cm mass, refractory hypertension, **hypokalaemia**; excision cures most.
- **Obstruction:** hypertension (~77% bilateral, ~20% unilateral); angiotensin II drives fibrosis.
- **Also know:** post-traumatic hypertension (arteriovenous fistula, scarring); reflux nephropathy (renin-guided segmental resection); hyperkalaemia with angiotensin blockade in CKD (chronic kidney disease).
- **Drugs:** ACE inhibitors (cough, angio-oedema from bradykinin), ARBs, aliskiren (direct renin inhibitor), aldosterone antagonists; all contraindicated in pregnancy.

Diagram to draw: Juxtaglomerular apparatus and cascade.

Recent advances

- Renal denervation (US approval 2023) for resistant hypertension.

Key points to remember:

- Renin is rate-limiting; released by low pressure, low NaCl, β1.
- Efferent constriction maintains GFR, so ACE inhibitors risk renal failure in bilateral stenosis.
- Renin ratio >1.5 lateralises; reninoma excision cures most.

Patho-physiology of Brain stem death

Type: SAQ (10 marks)

Source: 2015-Summer Paper 1, Q3d

Definition

- **Irreversible loss of the capacity for consciousness and to breathe** from permanent brainstem failure; legally death. Donation after brain death (DBD) keeps circulation until procurement, minimising warm ischaemia.

Pathophysiology

```
Brain injury → oedema → raised intracranial pressure (ICP)
 → falling cerebral perfusion → herniation ("coning")
 → Pons: Cushing reflex (hypertension, bradycardia)
 → Medulla: catecholamine storm (hypertension, arrhythmia)
 → Loss of sympathetic tone → vasodilatation, hypotension
```

- The storm injures myocardium and causes pulmonary oedema. Loss of antidiuretic hormone causes **diabetes insipidus (DI)**: polyuria, hypernatraemia; thyroid hormone and cortisol fall.
- **Hypothermia** (hypothalamic loss) and **disseminated intravascular coagulation (DIC)** from brain thromboplastin.

Diagnosis

- **Preconditions:** apnoeic coma of known irreversible cause; exclude sedatives, relaxants, hypothermia (35–36 °C by protocol), metabolic causes.
- **Brainstem reflexes, all must be absent:**

Test	How it is done	Cranial nerves
Pupillary light reflex	Pupils fixed, no response to bright light	II, III
Corneal reflex	Touch cornea; no blink	V, VII
Oculovestibular (caloric)	~50 mL ice-cold water into each ear (drum visible); no eye movement	VIII, III, VI
Oculocephalic (doll's eye)	Turn head; eyes move with the head	VIII, III, VI
Motor response	Central painful stimulus (supraorbital pressure); no facial grimace	V, VII
Gag and cough	Stimulate pharynx; pass suction catheter down trachea; no response	IX, X

- **Apnoea test:** pre-oxygenate with 100% oxygen, disconnect the ventilator with oxygen insufflation, and watch for breathing effort until **arterial CO_2 ($PaCO_2$) ≥60 mmHg**. No effort = positive. Abort if hypotension, desaturation or arrhythmia.
- **Ancillary tests** (electroencephalography (EEG), cerebral angiography, perfusion scan, transcranial Doppler) are not mandatory; used when clinical testing cannot be completed.

Indian law

- **THOA (Transplantation of Human Organs Act) 1994**, amended 2011, Rules 2014: **4-member board** (hospital in-charge, independent specialist, neurologist/neurosurgeon, treating doctor; no transplant team member); **two tests at least 6 hours apart**.

Donor management

- **Rule of 100s:** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Vasopressin** preferred (also treats DI); thyroid hormone, methylprednisolone, insulin.
- **Also know:** cold storage at 4 °C (University of Wisconsin solution); donor exclusions (metastatic cancer, melanoma, untreatable viral infection).

Recent advances

- Hypothermic machine perfusion reduces delayed graft function.

Diagram to draw: Herniation sequence leading to storm, DI, hypothermia, DIC.

Key points to remember:

- Raised ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- THOA: 4-member board, two tests ≥6 h apart; Rule of 100s.

Drugs associated with Acute Renal Failure.

Type: SAQ (10 marks)

Source: 2016-Summer Paper 1, Q3c

Introduction

- AKI (acute kidney injury), KDIGO (Kidney Disease: Improving Global Outcomes): creatinine rise ≥0.3 mg/dL in 48 hours or ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for 6 hours.

Classification by mechanism

Mechanism	Drugs	How
Haemodynamic (prerenal)	**NSAIDs** (non-steroidal anti-inflammatory drugs)	Block prostaglandin afferent dilatation
	ACE (angiotensin-converting enzyme) inhibitors, ARBs (angiotensin receptor blockers)	Remove efferent constriction that holds GFR (glomerular filtration rate)
	Ciclosporin, tacrolimus, contrast; diuretics	Afferent vasoconstriction; volume loss
ATN (acute tubular necrosis)	**Aminoglycosides**, amphotericin B, vancomycin, cisplatin, ifosfamide, contrast	Direct tubular toxicity
AIN (acute interstitial nephritis)	**β-lactams**, sulfonamides, quinolones, rifampicin, NSAIDs, **PPIs (proton pump inhibitors)**, allopurinol, **checkpoint inhibitors**	Cell-mediated hypersensitivity
Crystals	Aciclovir, methotrexate, indinavir, sulfadiazine; tumour lysis	Intratubular obstruction
Microangiopathy	Gemcitabine, VEGF (vascular endothelial growth factor) inhibitors	Endothelial injury
Rhabdomyolysis	Statins (with fibrates)	Myoglobin toxicity, precipitation

- **"Triple whammy":** NSAID + ACE inhibitor/ARB + diuretic in a dehydrated patient.
- **False creatinine rise:** trimethoprim and cimetidine block creatinine secretion.

Recognition

Type	Urine
Prerenal	FeNa (fractional excretion of sodium) <1%, bland
ATN	FeNa >1%, muddy-brown casts
AIN	Sterile pyuria, white cell casts, ± rash, fever, eosinophilia

- AIN appears days to weeks after the drug starts; it is the commonest biopsy lesion with checkpoint inhibitors. Urine eosinophils are unreliable.

Prevention and management

- Identify risk (CKD (chronic kidney disease), age, diabetes). Dose by eGFR (estimated GFR); monitor drug levels; once-daily aminoglycoside.
- Hydrate before contrast or cisplatin; alkalinise for methotrexate.
- **Stop the culprit**, withhold NSAIDs and ACE inhibitors, restore volume. **Early steroids for AIN** (only about half of antibiotic AIN recovers fully).

- Dialysis for refractory AEIOU: acidosis, electrolytes, intoxication, overload, uraemia.

Diagram to draw: glomerulus with afferent (prostaglandin) and efferent (angiotensin II) control.

> **Key points to remember:**
> - NSAIDs block afferent dilatation; ACE inhibitors/ARBs block efferent constriction.
> - ATN: aminoglycosides, amphotericin, cisplatin, contrast.
> - AIN: β-lactams, PPIs, NSAIDs, checkpoint inhibitors; stop and give steroids.
> - Crystals: aciclovir, methotrexate, indinavir.

GFR estimation in a prospective renal donor and creatinine clearance

Asked as:

- Describe methods to estimate GFR in a prospective renal donor — *Summer 2019 Paper I Q3a (SAQ)*
- Creatinine clearance — *Practice Set 4 Paper I Q3d (SAQ)*

Type: SAQ (10 marks)

(Practice Set 4: GFR assessment also answered in 2023-Winter Paper 2, Q3a)

Why it matters

- Donor loses about half the nephron mass. **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** sets GFR (glomerular filtration rate) thresholds and asks for **measurement, not just estimation**, at a decision point.

Methods (and comparison with other GFR measures)

1. Serum creatinine and eGFR (estimated GFR)

- Creatinine is freely filtered (and not reabsorbed), but generation depends on **muscle mass**, diet (cooked meat, creatine supplements) and age; it is also secreted, so it overestimates GFR.
- **CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) 2021 creatinine equation:** age and sex, **no race term** (no race); currently recommended; used for staging. MDRD (Modification of Diet in Renal Disease) and Cockcroft–Gault are older.
- Less accurate at GFR >60, where donors lie: a screening test only.

2. Cystatin C

- 13-kDa protein made by all nucleated cells; freely filtered, fully reabsorbed and broken down. Independent of muscle mass; affected by thyroid state, steroids, inflammation.
- **Combined creatinine–cystatin C equation** is more accurate than either alone (combined equation most accurate).

3. Measured 24-hour creatinine clearance

- **Definition:** creatinine clearance (CrCl) is the volume of plasma cleared of creatinine per minute. It estimates **GFR (glomerular filtration rate)** because creatinine is freely filtered and not reabsorbed.
- Normal about 120 mL/min/1.73 m² in young men, slightly lower in women; falls with age.
- **CrCl = (U × V) / P** ((urine creatinine × volume) ÷ serum creatinine): U = urine creatinine, V = urine flow (mL/min), P = plasma creatinine.
- **Method:** discard the first morning void, collect all urine for 24 hours including the next morning's first void; take serum creatinine during the collection.
- **Completeness check:** creatinine excretion should be **20–25 mg/kg/day in men, 15–20 mg/kg/day in women**.

Limitations of creatinine clearance

- **Tubular secretion** of creatinine: small in health but rising as function falls, so CrCl (secretion) **overestimates GFR** in CKD (chronic kidney disease).
- Prone to collection errors (collection errors are common). Creatinine generation depends on **muscle mass**, diet (cooked meat, creatine supplements) and age.
- Drugs blocking secretion (**trimethoprim, cimetidine**) raise serum creatinine without real injury.

- Invalid in AKI (acute kidney injury). **Also know:** cimetidine pretreatment blocks secretion to improve accuracy; averaging urea and creatinine clearance helps in advanced CKD.

Estimated clearance: Cockcroft–Gault

- **CrCl = (140 − age) × weight (kg) / (72 × serum creatinine mg/dL)**, × 0.85 in women.
- Used for **drug dosing**; not normalised to body surface area; inaccurate in obesity.

4. mGFR (measured GFR): reference (standard)

Marker	Method / comment
Inulin	Urinary clearance; gold standard, research only
Iohexol	Plasma clearance; safe, accurate
Iothalamate	Urinary or plasma clearance
^{51}Cr-EDTA (ethylenediaminetetraacetic acid)	Plasma clearance
^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid)	Plasma sampling, or gamma camera (Gates method); DTPA also gives split function

5. Split function

- DTPA renogram gives differential function; MAG3 or DMSA alternatives. A difference >10% (beyond 45:55) matters: **leave the better kidney with the donor**.

KDIGO 2017 approach

- Initial **eGFRcr**; confirm with **mGFR, measured creatinine clearance or eGFRcr-cys**.

GFR (mL/min/1.73 m²)	Decision
≥90	Accept
60–89	Individualise (age, lifetime risk)
<60	Decline

- Indian practice: **DTPA GFR** with split function plus CT (computed tomography) angiography is standard.

Clinical uses of creatinine clearance

- Drug dosing; **cisplatin eligibility** (CrCl <60 mL/min usually unfit).
- Extremes of body size, amputees and pregnancy, where eGFR is unreliable.
- Living donor evaluation (confirm with measured GFR).

Diagram to draw: plasma tracer clearance curve after a single injection; nephron showing filtration plus a small secreted fraction of creatinine.

Key points to remember:

- CKD-EPI 2021 race-free; cystatin C adds accuracy.
- Inulin is gold standard; iohexol or DTPA in practice.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.
- Better kidney stays with the donor.
- CrCl = UV/P; check completeness by creatinine excreted per kg.
- Secretion makes CrCl overestimate GFR as function falls.
- Cockcroft–Gault: (140 − age) × weight / (72 × SCr), × 0.85 women.

Berger's Disease.

Type: SAQ (10 marks)

Source: 2020-Summer Paper 1, Q3a

Berger's disease is IgA (immunoglobulin A) nephropathy (IgAN).

Definition

- **Commonest primary glomerulonephritis (GN) worldwide**, in children too. Dominant **mesangial IgA deposits** on immunofluorescence.
- Commonest cause of **glomerular haematuria** (about one-third). Second–third decades; male predominance.
- **IgA vasculitis** (Henoch–Schönlein purpura) is the systemic form.

Pathogenesis

- "Multi-hit" model: abnormal (galactose-deficient) IgA1 → autoantibodies → immune complexes → mesangial deposition, complement activation, proliferation. Mucosal (respiratory) infection triggers flares.

Clinical features

- **Recurrent painless macroscopic haematuria during or within days of an upper respiratory infection** ("synpharyngitic"), or after exercise. Post-streptococcal GN follows infection by weeks.
- Persistent microscopic haematuria, proteinuria, hypertension; occasional loin pain; rarely nephrotic syndrome or RPGN (rapidly progressive GN).

Evaluation (urologist's role)

- **Dysmorphic RBCs (red blood cells), RBC casts, proteinuria** → glomerular source. **Normal C3 and C4.**
- Without casts, complete the urological evaluation (imaging ± cystoscopy) to exclude a urinary tract cause.
- **Renal biopsy** confirms. Once diagnosed, repeat haematuria work-ups are not needed.

Oxford MEST-C score (biopsy)

Item	Grades
M: mesangial hypercellularity	M0, M1
E: endocapillary hypercellularity	E0, E1
S: segmental sclerosis	S0, S1
T: tubular atrophy/interstitial fibrosis	T0 ≤25%, T1 26–50%, T2 >50%
C: crescents	C0 none, C1 <25%, C2 ≥25% of glomeruli

Prognosis

- Often benign, but **20–30% reach ESKD (end-stage kidney disease)**.
- Poor: persistent proteinuria, hypertension, raised creatinine, older onset, high MEST-C.

Management

- No cure. **Supportive:** ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker), BP (blood pressure) control, low salt, stop smoking.
- **Newer drugs:** SGLT2 (sodium–glucose cotransporter 2) inhibitors; sparsentan (dual endothelin–angiotensin receptor blocker); targeted-release budesonide acting on gut

mucosal lymphoid tissue.

- Persistent proteinuria above about 0.75–1 g/day despite supportive care: consider a 6-month steroid course (KDIGO (Kidney Disease: Improving Global Outcomes) 2021); weigh infection risk.
- Crescentic RPGN: steroids plus cyclophosphamide.
- **Transplant:** recurs histologically often, clinically rarely.

Diagram to draw: glomerulus with mesangial IgA deposits.

> **Key points to remember:**
>
> - Commonest primary GN; haematuria with upper respiratory infection.
> - Normal complement; biopsy shows mesangial IgA.
> - Oxford MEST-C; 20–30% ESKD.
> - ACE inhibitor or ARB is the mainstay.

Renal allograft rejection.

Type: SAQ (10 marks)

Source: 2020-Summer Paper 1, Q3d

Definition

- Recipient immune attack on donor (allo) antigens: **HLA (human leucocyte antigen)** class I (A, B, C) and II (DR, DQ, DP), ABO and non-HLA antigens.
- Acute rejection has fallen from over 60% (pre-ciclosporin) to **below 10%**. **Chronic antibody injury** now limits graft survival.

Mechanism

- **T-cell**: direct (donor antigen-presenting cells) and indirect pathways → cytotoxic T cells → tubulitis, arteritis.
- **Antibody**: DSA (donor-specific antibody) binds endothelium → complement, leaving **C4d** → microvascular inflammation.

Types

Type	Timing	Features
Hyperacute	Minutes–hours	Preformed ABO/HLA antibody; thrombosis; graft turns blue. **Prevented by crossmatch**
Accelerated	2–5 days	Memory response in sensitised recipient
Acute TCMR (T-cell-mediated rejection)	Mostly first 6 months	Interstitial infiltrate, tubulitis, arteritis
Acute AMR (antibody-mediated rejection)	Any time	DSA, C4d, capillaritis
Chronic active AMR	Months–years	Transplant glomerulopathy, arterial intimal fibrosis; main cause of late loss
Subclinical	—	Histological rejection, stable function

- **Risk factors:** HLA mismatch, DSA, ABO incompatibility, long ischaemia and DGF (delayed graft function), **non-adherence**, retransplant, previous rejection.

Diagnosis

- Rising creatinine, oliguria; fever and graft tenderness now rare.
- **Exclude:** obstruction, urine leak, lymphocele, renal artery stenosis or thrombosis (ultrasound with Doppler), CNI (calcineurin inhibitor) toxicity, BK virus nephropathy, recurrent disease.
- **Biopsy is the gold standard (Banff):**

Banff category	Criteria
Borderline	Foci of tubulitis with mild inflammation
TCMR IA/IB	Interstitial inflammation >25% with moderate/severe tubulitis
TCMR IIA/IIB	Mild-moderate/severe intimal arteritis
TCMR III	Transmural arteritis or fibrinoid necrosis
Active AMR	Tissue injury + antibody–endothelium interaction (C4d) + circulating DSA

- **Monitoring:** de novo DSA predicts loss; dd-cfDNA (donor-derived cell-free DNA) has negative predictive value >90%; gene-expression panels.

Treatment

- **TCMR:** IV (intravenous) pulse methylprednisolone; **ATG (antithymocyte globulin)** for Banff II–III or steroid-resistant.
- **AMR:** not standardised: steroids, **plasmapheresis, IVIG (intravenous immunoglobulin)**, rituximab; eculizumab in selected patients; anti-IL-6 (interleukin-6) for chronic active AMR.
- Optimise maintenance (tacrolimus, mycophenolate, steroids); address adherence.

Diagram to draw: Banff biopsy features: tubulitis, arteritis, peritubular capillary C4d.

> **Key points to remember:**
>
> - Hyperacute rejection prevented by crossmatch.
> - Banff AMR = injury + C4d + DSA.
> - TCMR: pulse steroids, ATG; AMR: plasmapheresis, IVIG.
> - Chronic active AMR is the main cause of late loss.

Immunosuppressants in renal transplantation (classification, protocols, ciclosporin)

Asked as:

- Immunosuppressants. — *Summer 2021 Paper I Q3f (SAQ)* (also answered in 2020-Summer Paper 4, Q1)
- Classify immunosuppresants. Describe few common protocols. — *Practice Set 3 Paper I Q3b (SAQ)* (also answered in 2021-Summer Paper 1, Q3f and 2020-Summer Paper 4, Q1)
- Cyclosporins — *Practice Set 4 Paper I Q3f (SAQ)* (immunosuppressants also answered in 2021-Summer Paper 1, Q3f; tacrolimus in 2017-Winter Paper 3, Q3c)

Type: SAQ (10 marks)

Taken as immunosuppressant drugs (immunosuppressants) in renal transplantation; "Cyclosporins" taken as ciclosporin (cyclosporin A).

Principle

- T-cell activation needs three signals: **signal 1** (TCR [T-cell receptor] binding donor antigen → calcineurin → IL-2 [interleukin-2] transcription), **signal 2** (costimulation, CD28–CD80/86) and **signal 3** (IL-2 receptor → mTOR [mammalian target of rapamycin] → proliferation). Each drug blocks a step (drugs block these steps).

Phases

- **Induction** (peri-transplant), **maintenance** (lifelong), **rescue** (**treatment of rejection**).

Classification

Class	Drug(s)	Mechanism / action	Main toxicity
CNI (calcineurin inhibitor(s))	**Tacrolimus**, ciclosporin	Block calcineurin (signal 1)	**Nephrotoxicity**, diabetes, tremor, hypertension; ciclosporin: gum hypertrophy, hirsutism
Antimetabolites (antiproliferatives)	**Mycophenolate**, azathioprine	Block purine synthesis (IMPDH [inosine monophosphate dehydrogenase])	Diarrhoea, leucopenia
mTOR inhibitors	Sirolimus, everolimus	Block signal 3	**Lymphocele, poor wound healing**, proteinuria, dyslipidaemia; not nephrotoxic, antitumour
Corticosteroids	Prednisolone, methylprednisolone	Block cytokine transcription	Diabetes, bone loss, cataract
Costimulation blocker	**Belatacept**	CTLA4-Ig binds CD80/86 (signal 2)	**PTLD** (post-transplant lymphoproliferative disorder); avoid if EBV (Epstein–Barr virus)-negative
Depleting antibodies	rATG (rabbit antithymocyte globulin), alemtuzumab	T-cell depletion	Infection, PTLD
Non-depleting (antibody)	**Basiliximab**	Anti-CD25 (IL-2 receptor)	Few

Class	Drug(s)	Mechanism / action	Main toxicity
B-cell, antibody (antibody/B-cell agents)	Rituximab, IVIG (intravenous immunoglobulin), plasmapheresis, bortezomib, eculizumab, imlifidase	Antibody-mediated rejection, desensitisation; antibody removal or blockade	—

Standard regimen and common protocols

- **Over 90%** receive induction and **tacrolimus plus mycophenolate** (tacrolimus–mycophenolate); steroids in more than two-thirds. Acute rejection has fallen from ≥60% to <10%.

1. **Standard-risk (most recipients):** basiliximab 20 mg IV (intravenous) on days 0 and 4 + intraoperative methylprednisolone; then **tacrolimus + mycophenolate + tapering prednisolone**.
2. **High-immunological-risk** (sensitised, retransplant, DGF [delayed graft function] expected): **rATG** induction, then triple therapy.
3. **Steroid-avoidance or early withdrawal:** low-risk recipients, usually after depleting induction.
4. **CNI-sparing:** belatacept- or mTOR-based regimens to spare nephrotoxicity; TRANSFORM: everolimus with reduced CNI non-inferior.
5. **ABO-incompatible or HLA (human leucocyte antigen)-sensitised:** desensitisation with rituximab, plasmapheresis and IVIG; imlifidase for crossmatch-positive.
6. **Rejection:** T-cell-mediated → (pulse) IV (intravenous) steroids / methylprednisolone, then rATG (for Banff grade II–III); antibody-mediated → steroids, plasmapheresis, IVIG, rituximab.

- **Monitoring: tacrolimus trough monitoring** (trough levels: higher early, lower later); CYP3A4 (cytochrome P450 3A4) interactions (azoles, macrolides raise levels; rifampicin lowers).

Ciclosporin (cyclosporin A)

Introduction

- Ciclosporin is a cyclic peptide of 11 amino acids from the fungus *Tolypocladium inflatum*. It was the first **CNI (calcineurin inhibitor)**.
- Its introduction in the early 1980s cut acute rejection from at least 60% (pre-ciclosporin era) towards today's <10%. Tacrolimus has now largely replaced it.

Mechanism

```
Ciclosporin + cyclophilin (cytoplasm)
   → complex inhibits calcineurin
   → NFAT (nuclear factor of activated T cells) stays phosphorylated
   → no IL-2 (interleukin-2) transcription → T-cell activation blocked (signal 1)
```

Pharmacokinetics and monitoring

- Variable oral absorption; the **microemulsion form** absorbs more predictably.
- Metabolised by **CYP3A4** (cytochrome P450 3A4) and P-glycoprotein; narrow therapeutic index, so whole-blood **trough (C0) or 2-hour (C2) levels** guide dosing.

Uses

- **Transplantation:** maintenance with an antimetabolite and steroids, when tacrolimus is not tolerated.

- **IC/BPS (interstitial cystitis/bladder pain syndrome):** refractory **Hunner lesions**, 1–3 mg/kg/day; better than pentosan polysulfate.
- **Others:** retroperitoneal fibrosis after steroid failure; frequently relapsing nephrotic syndrome; severe psoriasis and other dermatoses.

Toxicity

System	Effect
Kidney	**Nephrotoxicity**: acute afferent arteriolar vasoconstriction (reversible); chronic striped fibrosis, arteriolar hyalinosis; TMA (thrombotic microangiopathy)
Metabolic	Hypertension, hyperkalaemia, hypomagnesaemia, **hyperuricaemia** (gout), dyslipidaemia
Cosmetic	**Gingival hyperplasia, hirsutism**
Neurological	Tremor, PRES (posterior reversible encephalopathy syndrome)
Immune	Infection, skin cancer, PTLD (post-transplant lymphoproliferative disorder)

Interactions

- **Levels rise:** azoles, erythromycin and clarithromycin, diltiazem, verapamil, grapefruit.
- **Levels fall:** rifampicin, phenytoin, carbamazepine, St John's wort.
- More nephrotoxicity with NSAIDs (non-steroidal anti-inflammatory drugs) and aminoglycosides.

Versus tacrolimus

- Ciclosporin: more hirsutism, gum hypertrophy, hypertension and hyperlipidaemia; less diabetes and tremor.
- **ELITE-Symphony (2007):** low-dose tacrolimus gave less rejection and better GFR (glomerular filtration rate) than ciclosporin regimens.

Urological relevance

- **Infection:** UTI (urinary tract infection), **BK virus** (ureteric stricture, nephropathy), CMV (cytomegalovirus).
- **Malignancy:** skin cancer, PTLD, RCC (renal cell carcinoma, mostly native kidneys), **bladder cancer about 3-fold** and more aggressive.
- **Surgery:** lymphocele and wound problems (mTOR; lymphocele with mTOR inhibitors); stones (CNI, steroids).
- **Urological note:** CNIs promote transplant stones (hyperuricaemia, hyperoxaluria, hypocitraturia).
- **Also know:** belatacept (BENEFIT trial: better glomerular filtration rate), extended-release tacrolimus, TRANSFORM (everolimus with reduced CNI non-inferior).

Diagram to draw: antigen-presenting cell (APC)–T cell (synapse) with signals 1–3 and the drug acting at each; T cell with the cyclophilin–calcineurin–NFAT–IL-2 pathway.

Key points to remember:

- Tacrolimus + mycophenolate ± steroids (maintenance) after induction (basiliximab or ATG / rATG).
- CNI nephrotoxic; mTOR inhibitors cause lymphocele and poor healing.
- Belatacept blocks signal 2; avoid if EBV-negative.
- Rejection: steroids for T-cell-mediated; plasmapheresis, IVIG, rituximab for antibody-mediated.
- Ciclosporin: cyclophilin → calcineurin → NFAT → IL-2 blocked.
- Nephrotoxic; gum hyperplasia and hirsutism; CYP3A4 interactions.
- Urology uses: transplant, refractory Hunner lesions, retroperitoneal fibrosis.

Kidney retrieval techniques for organ donation

Type: SAQ (10 marks)

Source: 2023-Winter Paper 1, Q3c (also answered in 2021-Summer Paper 2, Q2, with donor selection)

Types of donor

- **Deceased:** DBD (donation after brain death; circulation maintained) or DCD (donation after circulatory death; warm ischaemia).
- **Living:** related or unrelated (THOA, Transplantation of Human Organs Act, in India).

A. Deceased multi-organ retrieval (DBD)

1. Midline laparotomy with sternotomy.
2. Mobilise bowel off retroperitoneum (Cattell–Braasch); expose aorta and IVC (inferior vena cava); **look for and preserve accessory renal arteries**.
3. Heparin; cannulate distal aorta; clamp supracoeliac aorta; **vent IVC**; flush cold UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution; surface ice; open Gerota's fascia laterally.
4. Heart, lungs, liver and pancreas removed first.
5. Kidneys removed **en bloc** with aorta, IVC, perinephric fat and ureters, keeping periureteric tissue.

- **DCD:** super-rapid laparotomy and direct aortic cannulation to cut warm ischaemia; usually machine perfusion.
- **Back table:** separate kidneys; aortic **Carrel patch**; IVC lengthens the short right renal vein; ligate lymphatics; biopsy marginal kidneys; package at 4 °C.
- Paediatric donors <15 kg: en bloc kidneys.

B. Living donor nephrectomy

- **Principles:** donor safety first; maximal vessel length; keep periureteric tissue and the lower pole–ureter triangle; minimal handling; short warm ischaemia.
- **Left kidney preferred** (longer vein).

Technique	Features
Open (flank or mini-incision)	Complex anatomy, more often right side
Laparoscopic (Ratner, 1995)	Preferred; less pain, faster recovery
Hand-assisted	About two-thirds of US cases; tactile control
Retroperitoneoscopic, LESS (laparoendoscopic single-site), robotic	Selected centres

- Sequence: hilar dissection → ureter divided → artery then vein stapled or clipped → rapid extraction → cold flush.
- **Risks:** complications about 16%; mortality 3 per 10,000; conversion 0.26–3.34%.

Preservation

- Cooling slows metabolism; hypothermic machine perfusion reduces DGF (delayed graft function), not long-term loss.

Diagram to draw: aortic cannula, supracoeliac clamp, IVC vent, en bloc kidneys.

Key points to remember:

- Cannulate, clamp, vent, cool; kidneys removed last, en bloc.
- Left kidney preferred for living donation.
- Machine perfusion reduces DGF.

Extended Criteria for Selection of Live Renal Donor

Type: SAQ (10 marks)

Source: 2025-Winter Paper 1, Q3a (donor assessment also answered in 2022-Summer Paper 3, Q3a; marginal donor in 2024-Winter Paper 4, Q3e)

Read as the expanded (marginal, "medically complex") living donor.

Definition and principle

- A living donor with a risk factor that would once have excluded donation; used because of organ shortage.
- **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** replaced fixed cut-offs with a **risk-based approach**: the donor's estimated lifetime ESKD (end-stage kidney disease) risk is compared with the centre's acceptable threshold. Donor safety comes first.
- Counsel with figures: donor lifetime ESKD 90 vs 14 per 10,000 matched non-donors (Muzaale 2014); mortality 3 per 10,000.

Extended criteria

Factor	Standard	Extended (individualised)	Exclude
Age	Adult	Older donors (often >60) acceptable; no upper limit	Under legal age
GFR (glomerular filtration rate), mL/min/1.73 m²	≥90	**60–89**	<60
Albuminuria, mg/day	<30	**30–100**	>100
Hypertension	None	Controlled on 1–2 drugs, no target-organ damage	Uncontrolled, organ damage
Obesity	BMI (body mass index) <30	**30–35**	Many centres: >35
Glucose	Normal	Prediabetes; selected type 2 diabetes	Type 1 diabetes
Stones	None	Single past stone, low recurrence risk; small stone removed ex vivo	Bilateral or recurrent; cystinuria, primary hyperoxaluria
Anatomy	Single vessels	Multiple arteries, simple cyst, mild FMD (fibromuscular dysplasia)	Major anomaly

- **Why older donors are acceptable:** their remaining lifespan is short, so lifetime ESKD risk is low. Young donors carry the highest lifetime risk.
- **Also know:** isolated microscopic haematuria after negative work-up; treated low-risk cancer; APOL1 (apolipoprotein L1) high-risk genotype; future pregnancy (pre-eclampsia risk).

Safeguards

- Measured GFR and split function; **the better kidney stays with the donor**.
- CT (computed tomography) angiography; laparoscopic retrieval.
- India: **THOA (Transplantation of Human Organs Act)** authorisation; independent donor advocate.

- **Outcome:** grafts from older living donors do about as well as standard deceased-donor kidneys.

Diagram to draw: risk-threshold scale (accept, individualise, decline).

> **Key points to remember:**
> - KDIGO 2017: risk-based lifetime ESKD threshold.
> - GFR 60–89 and albuminuria 30–100 are individualised.
> - Controlled hypertension on 1–2 drugs may donate; type 1 diabetes cannot.

Vascular complications after renal transplantation, including transplant renal artery stenosis

Asked as:

- Vascular Complications after Renal Transplantation in Recipient — *Winter 2025 Paper I Q3d (SAQ)*
- Transplant renal artery stenosis — *Practice Set 20 SAQ Q3d (SAQ)*

Type: SAQ (10 marks)

(vascular complications of transplantation also answered in 2025-Winter Paper 1, Q3d)

Introduction

- Uncommon but a major cause of **early graft loss**. Doppler ultrasound is the first test.
- **Concept:** no inflow gives anuria; blocked outflow gives a swollen, tender graft with haematuria; reduced inflow gives hypertension and fluid retention.

Main complications

	Arterial thrombosis	Venous thrombosis	TRAS (transplant renal artery stenosis)
Incidence	0.2–5%	0.1–8.2%	1–23% (commonest)
Timing	First days	First week	Months (usually 3 months–2 years)
Presentation	Sudden anuria	Painful swollen graft, haematuria, oliguria	Refractory hypertension, rising creatinine, flash pulmonary oedema
Doppler	No arterial flow	**Reversed diastolic flow**, clot in vein	High velocity at stenosis; **parvus-tardus** waveform distally
Treatment	Immediate exploration; thrombectomy (catheter thrombolysis described); nephrectomy if non-viable	Immediate exploration, thrombectomy; nephrectomy if necrotic	**Angioplasty or stenting** (first choice); surgical revision if that fails

- **Thrombosis risks:** extreme donor or recipient age, **cold ischaemia >24 hours**, multiple arteries, technical error (kink, intimal flap), hypovolaemia, rejection, thrombophilia (factor V Leiden, antiphospholipid antibodies).
- **TRAS causes:** anastomotic technique, clamp injury, recipient atherosclerosis (iliac inflow stenosis mimics it). Contrast angiography confirms.

Transplant renal artery stenosis (TRAS) in detail

Definition and incidence

- TRAS (transplant renal artery stenosis): narrowing of the graft artery or its inflow; **commonest vascular complication (1–23%)**, usually 3 months to 2 years after transplant.

Types and causes

- **Anastomotic:** suture technique, end-to-end anastomosis to the internal iliac artery, no aortic (Carrel) patch.
- **Post-anastomotic (distal):** clamp or perfusion-cannula intimal injury, kinking, rejection.
- **Pre-anastomotic ("pseudo-TRAS"):** recipient iliac atherosclerosis.
- **Also know:** CMV (cytomegalovirus) infection, delayed graft function, deceased donor.

Pathophysiology and clinical features

- Reduced graft perfusion activates the **renin–angiotensin–aldosterone system**.
- **Refractory hypertension**, rising creatinine, fluid retention, **flash pulmonary oedema**; graft bruit.
- Clue: creatinine rises sharply after an ACE (angiotensin-converting enzyme) inhibitor.

Evaluation

- **Doppler ultrasound first:** high peak velocity at stenosis (about >200–250 cm/s), and distal **parvus-tardus** waveform with low resistive index.
- MRA (magnetic resonance angiography) or CO_2 angiography to spare contrast.
- **Contrast angiography is the gold standard** and allows treatment at once. Iodinated contrast risks nephrotoxicity; gadolinium is avoided at low GFR (glomerular filtration rate).
- **Differential of graft dysfunction with hypertension:** rejection, calcineurin-inhibitor toxicity, ureteric obstruction.

Management

- **Medical:** for mild stenosis with stable function: blood pressure control (calcium-channel blockers), statin, antiplatelet; avoid or monitor ACE inhibitors.
- **Percutaneous angioplasty with or without stent: treatment of choice** for significant stenosis (usually >50–60%). High technical success; reverses hypertension and graft dysfunction. Stents reduce restenosis.
- **Surgical revision** (resection and re-anastomosis, interposition graft, bypass) if angioplasty fails or anatomy unsuitable; risks graft loss.
- Follow-up Doppler after angioplasty detects restenosis; untreated TRAS ends in graft loss.

Other vascular complications

- **Haemorrhage:** hilar vessels or anastomosis; re-explore if unstable.
- **Pseudoaneurysm (<1%):** technical or **mycotic**; rupture risk; endovascular or open repair, often graft nephrectomy, prolonged antibiotics.
- **Post-biopsy AV (arteriovenous) fistula:** persistent haematuria; usually closes; large or with steal: **coil embolisation**. Subcapsular haematoma can cause a **Page kidney**.
- **Graft torsion:** intraperitoneal grafts; prevent with nephropexy.
- **DVT (deep vein thrombosis)** of the iliac vein; **lymphocele** compressing the iliac or renal vein.
- **Limb ischaemia** from iliac dissection; internal iliac use: gluteal claudication, erectile dysfunction.

Prevention

- Anastomose away from atheroma; check the lie for kinking; avoid clamp injury and kinking; use a Carrel patch where possible.
- Extend a short right renal vein with donor IVC (inferior vena cava).
- Short cold ischaemia; anticoagulate thrombophilic patients; avoid hypovolaemia.
- At reperfusion, a soft kidney suggests poor inflow; a tense, engorged kidney suggests outflow block.

Diagram to draw: Doppler waveforms: normal, parvus-tardus (TRAS; distal to stenosis) and reversed diastolic flow (venous thrombosis).

Key points to remember:

- TRAS is the commonest vascular complication; presents months later; angioplasty or stent first (first-line); angiography is gold standard.
- Refractory hypertension plus rising creatinine: Doppler first.
- Venous thrombosis: reversed diastolic flow, swollen tender graft.
- Arterial thrombosis: sudden anuria; explore immediately.

Management of acute hyperkalaemia

Asked as:

- Management of Acute Hyperkalemia — *Winter 2025 Paper I Q3e (SAQ)*
- Mgt. of acute hyperkalemia. — *Practice Set 6 Paper I Q3e (SAQ)*

Type: SAQ (10 marks)

(also answered in 2025-Winter Paper 1, Q3e)

Definition and physiology

- Serum potassium **>5.5 mmol/L**; mild 5.5–5.9, moderate 6.0–6.4, **severe ≥6.5** (UK Kidney Association). Above 6.0 is generally urgent.
- About 98% of potassium is intracellular; the ratio sets the resting membrane potential, so a high level (high levels) causes **arrhythmia and cardiac arrest**.
- Defence: rapid shift into cells (insulin, β_2-agonists via Na^+/K^+-ATPase), then renal excretion (needs distal sodium delivery, urine flow, aldosterone).

Causes

Mechanism	Examples
Pseudohyperkalaemia	Haemolysis, tourniquet, fist clenching, thrombocytosis
Reduced excretion	AKI (acute kidney injury), CKD (chronic kidney disease), **obstructive uropathy** (type 4 RTA [renal tubular acidosis]), adrenal insufficiency; drugs: ACE (angiotensin-converting enzyme) inhibitors, ARBs (angiotensin receptor blockers), spironolactone, NSAIDs (non-steroidal anti-inflammatory drugs), tacrolimus
Shift out of cells	Acidosis, insulin deficiency, β-blockers, suxamethonium
Cell release	**Rhabdomyolysis** (prolonged lithotomy or flank position), tumour lysis, crush
Urological	**Jejunal conduit** (reabsorbs urinary potassium)

Assessment

- Repeat the sample to **exclude pseudohyperkalaemia** if the ECG (electrocardiogram) is normal.
- **ECG changes:** peaked T (earliest) → prolonged PR, wide QRS → loss of P → sine wave → ventricular fibrillation (VF). Ascending weakness.
- Creatinine, glucose, blood gas, CK (creatine kinase); stop potassium sources.

Emergency treatment (in order)

Step	Drug and dose	Onset and effect
1. Protect the heart	**Calcium gluconate 10%, 10 mL IV (intravenous)** over 5–10 min; repeat if ECG unchanged	1–3 min; lasts 30–60 min; does not lower potassium
2. Shift into cells	**Insulin 10 units regular IV with 50 mL of 50% dextrose**	15–30 min; lowers about 0.6–1 mmol/L; check (monitor) glucose (hypoglycaemia)
	Salbutamol 20 mg nebulised	Additive to insulin
	Sodium bicarbonate	Only with metabolic acidosis
3. Remove	Loop diuretic if passing urine; **binders**: sodium zirconium cyclosilicate (about 1 h), patiromer (about 7 h), sodium polystyrene sulfonate (slow; colonic necrosis)	Hours

Step	Drug and dose	Onset and effect
	Haemodialysis	Refractory or oliguric, ESKD (end-stage kidney disease)

- **Relieve obstruction** (catheter, nephrostomy); watch for post-obstructive diuresis.
- Monitor ECG, potassium and glucose; treat the cause.

Diagram to draw: flowchart: ECG change or K^+ ≥6.5 → calcium → shift → remove.

Key points to remember:

- Calcium stabilises the membrane but does not lower potassium.
- Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
- Binders, diuretics and dialysis remove potassium; relieve obstruction.

Vascular grafts in urology

Type: SAQ (10 marks)

Source: Practice Set 5 Paper 1, Q3e (also answered in 2021-Summer Paper 3, Q2 and 2015-Summer Paper 2, Q3e)

Introduction

- A vascular graft **replaces, bypasses, extends or widens** a vessel when primary repair would leave tension or narrowing. Patch widens a wall; interposition graft replaces a segment; bypass routes flow around a lesion.

Types

Type	Examples	Advantages	Disadvantages
Autologous vein	Saphenous, gonadal, donor external iliac vein	Resists infection; small vessels	Limited length
Autologous artery	Internal iliac, **Carrel aortic patch**	Best for renal artery	Limited supply
Xenograft	Bovine pericardium	Pliable patch	Patch only
Synthetic	**PTFE (polytetrafluoroethylene)**, Dacron	Large calibre, any length	Infection; venous grafts thrombose

Applications

1. **Renal transplantation:** multiple arteries (18–43%) joined by Carrel patch or bench reconstruction; **preserve the lower-pole artery** (supplies the ureter). Short right renal vein extended with donor IVC (inferior vena cava), external iliac or spiral gonadal vein.
2. **RCC (renal cell carcinoma) with IVC thrombus:** residual lumen <50% → **patch cavoplasty**; circumferential wall invasion → **PTFE replacement (16–20 mm)**, then lifelong aspirin.
3. **PC-RPLND (post-chemotherapy retroperitoneal lymph node dissection):** IVC narrowing >25% → patch or graft; IVC ligation tolerated. **Aorta never ligated**: PTFE or Dacron replacement (about 1%).
4. **Retroperitoneal sarcoma:** caval resection and graft; thrombosis 10–40%.
5. **Renovascular repair and autotransplantation:** ex vivo bench repair with internal iliac artery or saphenous vein.
6. **Dialysis access:** PTFE AV (arteriovenous) graft when no native fistula is possible.
7. **Penile:** Grayhack saphenous–cavernosal shunt for priapism.

Principles

- Autologous in contaminated fields and small vessels; synthetic for large, high-flow vessels.
- Venous grafts: use the widest graft. Heparin before clamping; tension-free spatulated anastomosis with monofilament polypropylene; **omental wrap** keeps synthetic grafts away from bowel and urine.

Complications

- Thrombosis, stenosis, infection (synthetic usually removed), graft–enteric fistula, **ureteroarterial fistula**.

Diagram to draw: IVC patch cavoplasty and PTFE caval graft.

Key points to remember:

- Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
- IVC: lumen <50% → patch; invaded wall → PTFE 16–20 mm.
- PC-RPLND: IVC may be ligated; aorta must be grafted.

Hypokalemia

Type: SAQ (10 marks)

Source: Practice Set 5 Paper 1, Q3f

Definition and physiology

- **Hypokalaemia:** serum potassium **<3.5 mEq/L**; severe <2.5 mEq/L.
- About **98% of body potassium is intracellular**; the ratio sets the resting membrane potential.
- Two defences: **shift into cells** (insulin, β2-adrenergic stimulation, alkalosis) and **renal excretion** (distal nephron). Distal secretion rises with high urine flow, distal sodium delivery, **aldosterone** and alkalosis.

Causes

Mechanism	Causes
Low intake	Malnutrition
Shift into cells	Insulin, catecholamines (stress, β-agonists), alkalosis, refeeding, periodic paralysis
GI (gastrointestinal) loss	Diarrhoea, villous adenoma, **ureterosigmoidostomy and colonic diversions** (colon secretes potassium, ileum absorbs it)
Renal loss, high BP (blood pressure)	**Primary hyperaldosteronism (Conn)**, renin-secreting juxtaglomerular tumour, Cushing, Liddle, **abiraterone** (mineralocorticoid excess)
Renal loss, normal BP	Loop and thiazide diuretics, **postobstructive diuresis**, osmotic diuresis, **RTA (renal tubular acidosis) types 1 and 2**, Bartter, Gitelman, hypomagnesaemia, amphotericin, cisplatin

- **Vomiting:** potassium is lost mainly in urine (alkalosis → kaliuresis), not in gastric juice.

Clinical features

- Muscle weakness, cramps, ileus, **flaccid paralysis**, rhabdomyolysis.
- Arrhythmias; potentiates digoxin toxicity.
- Polyuria (nephrogenic diabetes insipidus); **hypocitraturia** → calcium stones.
- **ECG (electrocardiogram):** flat T wave, **prominent U wave**, ST depression, prolonged QU; ventricular arrhythmias.

Evaluation

- Repeat potassium; magnesium, bicarbonate, glucose.
- **24-h urine potassium:** <20 mmol/day suggests extrarenal loss; higher suggests renal loss.
- Renal loss: check blood pressure and acid–base; **aldosterone–renin ratio** if hypertensive (only a minority of Conn patients are hypokalaemic).

Management

- Treat the cause; stop offending drugs.
- **Potassium chloride** preferred (potassium citrate if acidotic, phosphate if hypophosphataemic). **Oral** 20–40 mEq repeated when possible.
- **Intravenous** for severe or symptomatic cases: usually ≤10 mEq/h peripherally; faster only centrally with ECG monitoring. Never as a bolus.
- **Correct magnesium**, or potassium will not correct.

- **Urinary diversion:** replace potassium **before or with bicarbonate**; correcting acidosis alone drives potassium into cells → paralysis.
- Correct before abiraterone and adrenalectomy for Conn syndrome (spironolactone preoperatively).

Diagram to draw: ECG strip showing flat T and U wave.

Key points to remember:

- Shift or loss; urine potassium separates renal from extrarenal.
- ECG: flat T, U wave.
- Diversion acidosis: give potassium with or before bicarbonate.
- Correct magnesium first.

Lymphocele and postoperative lymphatic leak (chylous ascites)

Asked as:

- Lymphocele — *Practice Set 8 Paper I Q3a (SAQ)*
- Postoperative lymphatic leak — *Practice Set 14 Paper I Q3b (SAQ)*

Type: SAQ (10 marks)

Postoperative lymphatic leak taken as lymphocele and chylous ascites after RPLND (retroperitoneal lymph node dissection), nephrectomy, pelvic node dissection and transplantation.

Definition and types

- **Lymphocele:** a collection of lymph **without an epithelial lining** (a pseudocyst with a fibrous wall), caused by divided lymphatics that keep leaking. It is the **commonest complication of PLND (pelvic lymph node dissection)**.
- **Chylous ascites:** lipid-rich lymph in the peritoneum from the cisterna chyli or major trunks.

Settings and incidence

- **PLND** (radical prostatectomy, cystectomy): radiological in up to 60%, clinically significant 2–5%. An extended template roughly doubles the rate.
- **Renal transplantation:** from lymphatics around the recipient iliac vessels or donor hilum. Radiological 0.6–33.9% (symptomatic 0.03–26%); most appear within 6 months, peaking at 6 weeks.
- Also after RPLND (retroperitoneal lymph node dissection; lymphocele ≤1.7%), inguinal dissection, nephrectomy.
- **Chylous ascites:** RPLND: 0.2–2.1% primary, **2–12% post-chemotherapy**; left nephrectomy with hilar dissection.

Type	Definition	Setting and frequency
Lymphocele	Lymph collection **without epithelial lining**	**PLND (pelvic lymph node dissection):** radiological up to 60%, significant 2–5%. **Transplant:** symptomatic 0.03–26%, peak 6 weeks. **RPLND:** ≤1.7%
Chylous ascites	Lipid-rich lymph in the peritoneum from the cisterna chyli or major trunks	RPLND: 0.2–2.1% primary, **2–12% post-chemotherapy**; left nephrectomy with hilar dissection

- **Risk factors:** extensive dissection, extended templates, suprahilar dissection near the cisterna chyli (L1–L2), post-chemotherapy surgery, obesity, heparin prophylaxis, mTOR (mammalian target of rapamycin) inhibitors, acute rejection, retransplantation (re-transplant).

Clinical features

- **Lymphocele:** most are silent (usually silent). Large ones compress:
  - iliac vein: leg oedema, **DVT (deep vein thrombosis)**;
  - bladder: frequency, urgency, pelvic pressure;
  - ureter or graft: hydronephrosis, rising creatinine (graft dysfunction).
- Infection (may become infected): fever, rigors, sepsis.
- **Chylous ascites:** rising girth, early satiety, dyspnoea; milky drain fluid.

Diagnosis

- **Ultrasound:** anechoic, often septated, thin-walled collection. **CT (computed tomography)** maps it before intervention. Air or rim enhancement means infection.
- **Aspirate:** lymph has creatinine equal to (equals) serum (urinoma: much higher), lymphocytes; culture excludes abscess. Chyle: **triglyceride 2–8 × serum**, protein >3 g/dL.

Management

Lymphocele (treat by symptoms, not size alone)

Situation	Treatment
Asymptomatic, <5 cm	Observe; most resolve
Symptomatic	**Percutaneous drain** ± sclerotherapy (povidone-iodine, doxycycline)
Recurrent or thick-walled	**Laparoscopic fenestration (marsupialisation)** into a peritoneal window (best)
Infected	Antibiotics + percutaneous drainage; exclude urine leak
Laparoscopy unsafe	Open drainage

```
Asymptomatic, <5 cm → observe
Symptomatic → percutaneous drain (± sclerotherapy)
Recurrent / thick-walled → laparoscopic peritoneal fenestration (best)
Infected → drainage + antibiotics; exclude urine leak
```

- **Success:** laparoscopic fenestration best, open next, drain or sclerotherapy intermediate, **aspiration alone worst** (reaccumulates, recurs, risks infection).
- Sclerosants (povidone-iodine, doxycycline) may hinder later fenestration.
- Fenestration lets the peritoneum absorb lymph (internal drainage): the definitive cure.
- **Lymphangiography with embolisation:** high success; long-term data awaited.

Chylous ascites (stepwise)

1. Paracentesis; drain if it reaccumulates.
2. **Very low-fat, high-protein diet with MCT (medium-chain triglycerides)**, which are absorbed directly into portal blood, bypassing lacteals; plus **octreotide**.
3. Nil by mouth with **TPN (total parenteral nutrition)**.
4. Refractory: lymphangiography with embolisation, peritoneovenous shunt or surgical ligation.

- Continue a low-fat diet for **≥4 weeks** after resolution.

Prevention

- **Clip or ligate visible lymphatics** (clips often preferred; no robust data) (cephalad at the renal hilum in RPLND; perivascular at transplant). Spare lymphatics over the external iliac artery (leg lymph). Avoid unneeded suprahilar dissection.
- **Peritoneal interposition flap** fixed to the bladder (peritoneal flap after robotic PLND) reduces clinically significant lymphocele. Transperitoneal access is **not** protective (fluid loculates within 48–72 h).
- Drain after transplantation and major nodal dissections.

Diagram to draw: pelvic lymphocele compressing the external iliac vein and bladder, with a laparoscopic peritoneal window.

Key points to remember:

- Lymphocele: commonest PLND complication; radiological up to 60%, significant 2–5%.
- Lymph creatinine = serum; urinoma creatinine higher.
- Laparoscopic fenestration best (is best); aspiration alone worst.
- Chylous ascites: 2–12% after post-chemotherapy RPLND; triglyceride 2–8 × serum.
- Ladder: diet with MCT and octreotide → TPN → embolisation or shunt.

Hyperfiltration injury in AKI and CKD

Type: SAQ (10 marks)

Source: Practice Set 14 Paper 1, Q3c

Definition

- **Hyperfiltration:** a rise in **SNGFR (single-nephron glomerular filtration rate)** in surviving nephrons when nephron mass falls; whole-kidney GFR (glomerular filtration rate) may look normal.
- **Brenner hypothesis (1982):** the adaptation itself damages glomeruli.

Physiology

- Net filtration pressure ≈ 10 mmHg (hydrostatic 60 − Bowman's 18 − oncotic 32).
- Afferent dilatation raises flow and filtration; **efferent constriction (angiotensin II) raises filtration pressure**.

Mechanism

```
Nephron loss → afferent dilatation + efferent constriction
→ raised glomerular capillary pressure and flow (hyperfiltration)
→ glomerular hypertrophy; podocytes cannot divide → stretch, detachment
→ proteinuria → FSGS (focal segmental glomerulosclerosis)
→ tubulointerstitial inflammation, fibrosis → more nephron loss (vicious cycle)
```

In AKI (acute kidney injury)

- Incomplete repair leaves fewer nephrons, which hyperfilter; creatinine may normalise while **renal reserve is lost**.
- This drives the **AKI-to-CKD (chronic kidney disease) transition**; AKI persisting ≥90 days becomes CKD.
- Each perioperative AKI episode is a **modifiable risk factor** for CKD progression.

In CKD

- Final common pathway of progression; marked by **albuminuria (KDIGO (Kidney Disease: Improving Global Outcomes) A stages)**, hypertension, falling eGFR (estimated GFR).

Urological settings

Setting	Point
Nephrectomy or donation	Compensatory hypertrophy 20–30%; donor function 60–80% of pre-donation
Partial nephrectomy, solitary kidney	>50% parenchymal loss → proteinuria, FSGS; marked when total nephron mass falls >80%; ESRD (end-stage renal disease) if <20–30% of one kidney remains
Congenital solitary kidney, MCDK (multicystic dysplastic kidney)	Monitor BP (blood pressure), proteinuria, yearly ultrasound

- **Also know:** reflux nephropathy, infundibulopelvic stenosis, diabetes, obesity, pregnancy, low birth weight (low nephron endowment).

Prevention and treatment

- Nephron sparing; avoid NSAIDs (non-steroidal anti-inflammatory drugs) and nephrotoxins.

- **BP <130/80 mmHg**; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)** in proteinuric disease: efferent dilatation lowers glomerular pressure.
- **SGLT2 (sodium–glucose cotransporter-2) inhibitors:** restore tubuloglomerular feedback, causing afferent constriction. **DAPA-CKD** and **EMPA-KIDNEY** slowed progression with or without diabetes.
- Finerenone (type 2 diabetic CKD); dietary protein restriction.

Diagram to draw: glomerulus with dilated afferent and constricted efferent arterioles.

> **Key points to remember:**
> - Hyperfiltration = raised SNGFR; Brenner 1982.
> - Leads to FSGS, proteinuria and the AKI-to-CKD transition.
> - ACE inhibitors and SGLT2 inhibitors lower glomerular pressure.

Management of CAPD peritonitis

Type: SAQ (10 marks)

Source: Practice Set 21 SAQ Sheet, Q3d (CAPD as a whole answered in 2018-Summer Paper 2, Q3e and Practice Set 1 Paper 2, Q3a)

Introduction

- CAPD (continuous ambulatory peritoneal dialysis) peritonitis is the leading cause of technique failure and transfer to haemodialysis. Commonest organism: **coagulase-negative *Staphylococcus*** (touch contamination).
- ISPD (International Society for Peritoneal Dialysis) 2022 target: centre rate **≤0.40 episodes per patient-year**.

Diagnosis (any 2 of 3)

1. Abdominal pain and/or **cloudy effluent**.
2. Effluent **WCC (white cell count) >100/μL with >50% neutrophils** (dwell ≥2 hours).
3. Positive effluent culture.

Initial steps

- Effluent cell count, Gram stain, **culture in blood-culture bottles** (culture-negative rate should be <15%).
- Examine exit site and tunnel; analgesia; heparin in bags if fibrin.

Empirical antibiotics

- **Intraperitoneal route preferred** (intravenous if septic); start immediately after samples.
- Cover **Gram-positives** (vancomycin or first-generation cephalosporin, e.g. cefazolin) **and Gram-negatives** (third-generation cephalosporin, e.g. ceftazidime, or aminoglycoside). Choice follows centre sensitivities.
- Intermittent (once daily in long dwell) or continuous dosing; avoid prolonged aminoglycoside (residual function, ototoxicity).
- **Antifungal prophylaxis** (nystatin or fluconazole) during antibiotics.

Organism-specific duration

Organism	Duration
Coagulase-negative *Staphylococcus*, streptococci, culture-negative	2 weeks
S. aureus, enterococcus, Gram-negatives	3 weeks
Pseudomonas	3 weeks, two antipseudomonal drugs
Polymicrobial enteric	Add metronidazole; surgical review (bowel perforation)
Fungal	**Immediate catheter removal** + antifungal

Catheter removal

- **Refractory** (effluent not clear after 5 days of right antibiotics), relapsing, fungal, refractory exit-site or tunnel infection.
- Haemodialysis meanwhile; new catheter after at least 2 weeks.

Definitions

- **Relapse:** same organism (or sterile) within 4 weeks of finishing therapy. **Recurrent:** different organism within 4 weeks. **Repeat:** same organism after 4 weeks.

Prevention

- Training, exit-site mupirocin or gentamicin, prophylaxis before colonoscopy or gynaecological procedures.

Diagram to draw: management flowchart from cloudy bag to catheter removal.

> **Key points to remember:**
> - Two of: cloudy fluid or pain, WCC >100/µL with >50% neutrophils, positive culture.
> - Intraperitoneal Gram-positive plus Gram-negative cover.
> - Remove catheter for fungal or refractory (5 days) peritonitis.

Rapid Revision Sheet – Section 15 · Renal, Nephrology & Transplant

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Describe in details basic patho-physiology of urine production and transport (LAQ)

- RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
- Proximal 65%, loop 25%, distal 5–7% of Na^+; ADH → V2 → AQP2.
- Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL (LAQ)

- Sampaio: A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
- Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
- Target a posterior calyx with a short, wide infundibulum; upper pole reaches most.
- Lower pole: steep angle, long calyx >10 mm, infundibulum <5 mm unfavourable.
- Endopyelotomy lateral; infundibulotomy superior or inferior, never anterior.
- Pleura to 12th rib; retrorenal colon at the lower pole, commoner prone.
- Mini 15–22 Fr (Jackman 1998): fewer transfusions, longer operation.
- High intrarenal pressure in mini-PCNL causes fever and sepsis.

Lymphatic drainage of the genitourinary system and chyluria (etiopathogenesis and management) (LAQ)

- Lymph flows right → left; cisterna chyli at L1–L2, behind the right side of the aorta.
- Bladder drains mainly to external iliac nodes; left kidney to left para-aortic nodes.
- Chyluria = pyelolymphatic fistula from lymphatic obstruction and valve failure, usually filarial (*W. bancrofti*); often intermittent.
- Ether test and (urine) triglycerides confirm (ether test clears milky urine); cystoscopy after (a) fatty meal lateralises.
- Diet (MCT) + DEC (diet with MCT plus DEC) → (silver nitrate) sclerotherapy → laparoscopic hilar lymphatic disconnection; disconnection surgery for severe or refractory cases.
- Retroperitoneoscopic disconnection beats open.

Pathophysiology of Renovascular hypertension (LAQ)

- Goldblatt 1934: 2K1C = renin-dependent; 1K1C = volume-dependent.
- Chronic phase: contralateral nephrosclerosis makes RVH stenosis-independent.
- Angiotensin II keeps GFR by efferent constriction; ACE inhibitor causes acute kidney injury in bilateral RAS.
- ASTRAL and CORAL: no routine stenting; FMD = angioplasty alone.

Renin-angiotensin system (SAQ)

- Renin is rate-limiting; released by low pressure, low NaCl, β1.
- Efferent constriction maintains GFR, so ACE inhibitors risk renal failure in bilateral stenosis.
- Renin ratio >1.5 lateralises; reninoma excision cures most.

Patho-physiology of Brain stem death (SAQ)

- Raised ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- THOA: 4-member board, two tests ≥6 h apart; Rule of 100s.

Drugs associated with Acute Renal Failure. (SAQ)

- NSAIDs block afferent dilatation; ACE inhibitors/ARBs block efferent constriction.
- ATN: aminoglycosides, amphotericin, cisplatin, contrast.
- AIN: β-lactams, PPIs, NSAIDs, checkpoint inhibitors; stop and give steroids.
- Crystals: aciclovir, methotrexate, indinavir.

GFR estimation in a prospective renal donor and creatinine clearance (SAQ)

- CKD-EPI 2021 race-free; cystatin C adds accuracy.
- Inulin is gold standard; iohexol or DTPA in practice.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.
- Better kidney stays with the donor.
- CrCl = UV/P; check completeness by creatinine excreted per kg.
- Secretion makes CrCl overestimate GFR as function falls.
- Cockcroft–Gault: (140 − age) × weight / (72 × SCr), × 0.85 women.

Berger's Disease. (SAQ)

- Commonest primary GN; haematuria with upper respiratory infection.
- Normal complement; biopsy shows mesangial IgA.
- Oxford MEST-C; 20–30% ESKD.
- ACE inhibitor or ARB is the mainstay.

Renal allograft rejection. (SAQ)

- Hyperacute rejection prevented by crossmatch.
- Banff AMR = injury + C4d + DSA.
- TCMR: pulse steroids, ATG; AMR: plasmapheresis, IVIG.
- Chronic active AMR is the main cause of late loss.

Immunosuppressants in renal transplantation (classification, protocols, ciclosporin) (SAQ)

- Tacrolimus + mycophenolate ± steroids (maintenance) after induction (basiliximab or ATG / rATG).
- CNI nephrotoxic; mTOR inhibitors cause lymphocele and poor healing.
- Belatacept blocks signal 2; avoid if EBV-negative.
- Rejection: steroids for T-cell-mediated; plasmapheresis, IVIG, rituximab for antibody-mediated.
- Ciclosporin: cyclophilin → calcineurin → NFAT → IL-2 blocked.
- Nephrotoxic; gum hyperplasia and hirsutism; CYP3A4 interactions.
- Urology uses: transplant, refractory Hunner lesions, retroperitoneal fibrosis.

Kidney retrieval techniques for organ donation (SAQ)

- Cannulate, clamp, vent, cool; kidneys removed last, en bloc.
- Left kidney preferred for living donation.
- Machine perfusion reduces DGF.

Extended Criteria for Selection of Live Renal Donor (SAQ)

- KDIGO 2017: risk-based lifetime ESKD threshold.
- GFR 60–89 and albuminuria 30–100 are individualised.
- Controlled hypertension on 1–2 drugs may donate; type 1 diabetes cannot.

Vascular complications after renal transplantation, including transplant renal artery stenosis (SAQ)

- TRAS is the commonest vascular complication; presents months later; angioplasty or stent first (first-line); angiography is gold standard.

- Refractory hypertension plus rising creatinine: Doppler first.
- Venous thrombosis: reversed diastolic flow, swollen tender graft.
- Arterial thrombosis: sudden anuria; explore immediately.

Management of acute hyperkalaemia (SAQ)

- Calcium stabilises the membrane but does not lower potassium.
- Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
- Binders, diuretics and dialysis remove potassium; relieve obstruction.

Vascular grafts in urology (SAQ)

- Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
- IVC: lumen <50% → patch; invaded wall → PTFE 16–20 mm.
- PC-RPLND: IVC may be ligated; aorta must be grafted.

Hypokalemia (SAQ)

- Shift or loss; urine potassium separates renal from extrarenal.
- ECG: flat T, U wave.
- Diversion acidosis: give potassium with or before bicarbonate.
- Correct magnesium first.

Lymphocele and postoperative lymphatic leak (chylous ascites) (SAQ)

- Lymphocele: commonest PLND complication; radiological up to 60%, significant 2–5%.
- Lymph creatinine = serum; urinoma creatinine higher.
- Laparoscopic fenestration best (is best); aspiration alone worst.
- Chylous ascites: 2–12% after post-chemotherapy RPLND; triglyceride 2–8 × serum.
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Hyperfiltration injury in AKI and CKD (SAQ)

- Hyperfiltration = raised SNGFR; Brenner 1982.
- Leads to FSGS, proteinuria and the AKI-to-CKD transition.
- ACE inhibitors and SGLT2 inhibitors lower glomerular pressure.

Management of CAPD peritonitis (SAQ)

- Two of: cloudy fluid or pain, WCC >100/µL with >50% neutrophils, positive culture.
- Intraperitoneal Gram-positive plus Gram-negative cover.
- Remove catheter for fungal or refractory (5 days) peritonitis.

Section 16 · BPH — Long answer questions (LAQ)

Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer

Asked as:

- Anatomy of prostate and its neurovascular bundles with its clinical details. — *Practice Set 1 Paper I Q2 (LAQ)*
- Describe in detail, anatomy of prostate with special reference to concepts in surgical management if localised CA prostate. — *Practice Set 9 Paper I Q1 (LAQ)*

Type: LAQ (25 marks)

(Practice Set 1: nerve-sparing techniques also in 2023-Winter Paper 4, Q3a. Practice Set 9: also answered in Practice Set 2 Paper 4, Q1)

"if localised" is read as "of localised" prostate cancer.

Introduction

- A (20-g) fibromuscular, glandular organ around the proximal urethra, from bladder neck (**base**) to urogenital diaphragm (**apex**), in front of the rectum; about 20 g.
- Buds from the urogenital sinus at week 10, driven by androgen acting on the **mesenchymal** AR (androgen receptor).
- RP (radical prostatectomy) aims at the "trifecta" of cancer control, continence and potency. Each anatomical step either secures a clear margin or saves a named structure (nerves, sphincter, anterior support).

Relations

- **Anterior:** retropubic space (of Retzius), puboprostatic ligaments, DVC (dorsal venous complex).
- **Posterior:** Denonvilliers fascia, then rectal ampulla, hence DRE (digital rectal examination) and TRUS (transrectal ultrasound).
- **Lateral:** levator ani with its fascia; NVB (neurovascular bundle) posterolaterally.
- **Base:** bladder neck; seminal vesicles behind. **Inside:** the urethra bends about 35° at the **verumontanum**, where the ejaculatory ducts open beside the utricle.

Zonal anatomy (McNeal 1981)

Zone	Glandular share (share of glands)	Position	Clinical / surgical relevance
Peripheral (PZ)	About 70%	Posterior and lateral; longest ducts	**70% of cancers**; felt on DRE; biopsy target; posterolateral, so ECE (extracapsular extension) runs along the nerves
Central (CZ)	About 25%	Cone around the ejaculatory ducts at the base	Few cancers; mimics HGPIN (high-grade prostatic intraepithelial neoplasia)
Transition (TZ)	About 5%	Two lobes beside the preprostatic sphincter; shortest ducts	**BPH (benign prostatic hyperplasia)**; anterior cancers (about 15%) missed by DRE (digital rectal examination)

Zone	Glandular share (share of glands)	Position	Clinical / surgical relevance
Periurethral glands	Small	Within the preprostatic sphincter	Middle-lobe BPH
Anterior fibromuscular stroma (AFMS)	No glands	Anterior	Anterior tumours reach it; margin risk in posterior approaches

- **Capsule:** no true histological capsule (there is **no true capsule**), so cancer (breaching the gland) shows "extraprostatic extension". In BPH the compressed PZ forms the **surgical capsule**, the plane of enucleation. The capsule transmits growth pressure to the urethra, so TUIP (transurethral incision) relieves obstruction.

Fascial investments

Layer	Position	Surgical rule
Prostatic fascia	On the gland; carries DVC tributaries anteriorly	Stays on the prostate in nerve sparing
Levator fascia	Covers levator; fuses with prostatic fascia as the lateral pelvic fascia	Incised to reach the NVB
Denonvilliers fascia	Between prostate and rectum; thickest at base and vesicles	Excised completely for the posterior margin

- **Denonvilliers fascia** stays on the rectum when ECE risk is low; it is excised when posterolateral ECE is suspected.

Blood supply and lymphatics

- **Inferior vesical artery** ends as: **urethral group** (enters at the vesicoprostatic junction, 5 and 7 o'clock; supplies the adenoma) and **capsular group** (posterolateral, in the lateral pelvic fascia, marking the cavernous nerves).
- **Accessory pudendal arteries** run over the anterolateral prostate; dividing them reduces penile inflow. Preserve them; they may be the main penile supply.
- **Veins:** the deep dorsal vein passes under the pubis into a superficial branch and lateral (**Santorini**) plexuses, draining to the internal iliac vein. They anastomose freely, so tears bleed heavily.
- DVC (dorsal venous complex): CO_2 pneumoperitoneum tamponades it, so robotic blood loss is low.
- **Lymph:** obturator, internal and external iliac, common iliac, presacral nodes.

Neurovascular bundle

- **Pelvic plexus:** parasympathetic S2–S4 plus sympathetic (T10–L2) via the hypogastric nerves (hypogastric sympathetic); beside the rectum, centred on the seminal vesicle tip. It gives the **cavernous nerves**.
- **NVB of Walsh (Walsh and Donker, 1982):** capsular vessels plus **cavernous nerves**. The nerves are invisible; the vessels mark them (marked by capsular vessels).
- It runs posterolaterally **between prostatic and levator fascia**, **3–5 mm from the capsule**, outside Denonvilliers fascia.
- At the apex it (they) lie(s) at **5 and 7 o'clock**, posterolateral to the sphincter. The nerves then swing to about **3 and 9 o'clock** at the mid-urethra and enter the corpora behind the dorsal penile artery. The apex is where they are most often injured (most nerve injuries occur here).

- Distribution varies (is diffuse): posterolateral bundle, posterior "hammock", anterolateral fibres (which is the basis of high anterior release).
- **Striated sphincter:** a vertical tube over the apex (Oelrich), pudendal nerve; passive continence.

Diagram to draw: axial prostate showing the three fasciae, NVBs at 5 and 7 o'clock and the intra-, inter- and extrafascial planes.

Continence anatomy

Passive	Active
Smooth muscle of bladder neck and urethra; anterior support (endopelvic fascia, detrusor apron, puboprostatic ligaments, DVC)	Horseshoe striated sphincter at the apex, supplied by pudendal nerve branches

Clinical details

- **BPH surgery:** TURP (transurethral resection of the prostate) stays proximal to the verumontanum (sphincter lies just distal); urethral-group vessels at 5 and 7 o'clock bleed.
- **Cancer:** PZ cancer exits posterolaterally along nerves; anterior TZ tumours are missed by DRE and biopsy, so MRI (magnetic resonance imaging) helps.
- **Radical prostatectomy (RP):**
  1. Incise endopelvic fascia; spare the pubourethral ligament.
  2. Ligate the DVC over the urethra before division.
  3. Preserve accessory pudendal arteries.
  4. Excise Denonvilliers fascia; the apex is the commonest positive-margin site.

Nerve-sparing plane	Course (lies)	Verdict (trade-off)
Interfascial	Between prostatic and levator fascia	**Standard plane** for organ-confined disease
Intrafascial	Under the prostatic fascia (between capsule and prostatic fascia)	Most nerves (kept); more positive margins
Extrafascial	Outside levator (and Denonvilliers) fascia	Widest margin; ED (erectile dysfunction) likely

- **NVB injury** (injury comes from): incision, entrapment, **heat**, **traction**, so dissect athermally without traction. Excise an NVB for palpable extracapsular disease; margins are equal with or without nerve sparing.
- **Continence:** Retzius-sparing (Bocciardi 2010) and hood techniques preserve anterior support; continence returns earlier.
- **Also know:** rectal injury at the ampulla; PDE5 (phosphodiesterase type 5) inhibitor rehabilitation.

Concepts in surgical management of localised cancer

1. **Selection:** RP suits localised disease (mainly intermediate and selected high risk) with a life expectancy of more than 10 years; low risk is usually offered active surveillance.
2. **RARP (robot-assisted RP):** 3D, magnified view and wristed instruments removed the suturing barrier. **Yaxley 2016** RCT (randomised controlled trial): equal early function versus open; less blood loss and shorter stay. Surgeon matters more than platform.
3. **Risk-graded nerve sparing:** each side is graded from full (intrafascial) through partial to none, using grade, positive cores and mpMRI (multiparametric magnetic resonance imaging) capsular contact. **NeuroSAFE** (intraoperative frozen section of the posterolateral surface) aims to allow more nerve sparing without more positive margins.

4. **Anterior-support preservation:**
    - **Retzius-sparing** (Bocciardi, 2010): fully posterior through the pouch of Douglas. Earlier continence at 3 months, equal at 12; more positive margins, so avoid for anterior tumours.
    - **Hood technique** (Tewari, 2021): anterior approach keeping detrusor apron, puboprostatic ligaments and DVC.
5. **Apical dissection:** the apex is the commonest positive-margin site. Keep maximal urethral length and divide sharply without cautery. Posterior (Rocco) reconstruction may aid early continence.
6. **PLND (pelvic lymph node dissection) templates:**

Template	Nodes	Drainage captured
Limited	Obturator	Least
Standard	+ External iliac	About 38%
Extended	+ Internal iliac	About 63%
+ Common iliac	+ Common iliac	About 75%

- Extended PLND when nodal risk exceeds 5% (EAU [European Association of Urology]). It stages but does not improve survival; lymphocele is the commonest complication.

7. **Focal therapy anatomy:** ablate the **index lesion**. Transrectal HIFU (high-intensity focused ultrasound) suits posterior PZ lesions; anterior lesions need transperineal cryotherapy. Spare sphincter, NVB and rectum.

- **Also know:** single-port robot (extraperitoneal); PSMA PET (prostate-specific membrane antigen positron emission tomography) staging; ERAS (enhanced recovery after surgery).

Key points to remember:

- PZ (gives) 70% of cancers; TZ gives BPH; CZ around ejaculatory ducts; no true capsule.
- NVB between prostatic and levator fascia, 5 and 7 o'clock; apex most vulnerable (the commonest margin and nerve-injury site).
- Interfascial plane is (the) standard (nerve-sparing plane); avoid heat and traction.
- Ligate DVC; excise Denonvilliers fascia; spare accessory pudendals.
- Retzius-sparing: earlier continence, more positive margins.
- Extended PLND captures about 63% of drainage.

Section 16 · BPH — Short answer questions (SAQ)

HoLEP in BPH

Type: SAQ (10 marks)

Source: 2019-Summer Paper 1, Q3c

Principle

- **HoLEP (holmium laser enucleation of the prostate)** removes the whole adenoma along the **surgical capsule**, as in open simple prostatectomy, but endoscopically.
- **Ho:YAG (holmium:yttrium–aluminium–garnet), 2120 nm**: absorbed by water, pulsed, **0.4 mm penetration**, coagulates vessels. Normal saline irrigation (no TUR (transurethral resection) syndrome).
- Evolution: HoLAP → HoLRP → **HoLEP (1998)**, made practical by the **morcellator**.

Indications

- **Size-independent** (<30, 30–80, >80 mL); median lobe; **anticoagulated or high bleeding risk** (AUA (American Urological Association) recommendation); alternative to open prostatectomy for large glands.

Steps

- 60–150 W laser, 550 µm end-firing fibre, 24–28 Fr resectoscope; about 2 J × 40–50 Hz to enucleate.

1. Bladder-neck incisions at 5 and 7 o'clock to capsule (median lobe) or one at 6 o'clock.
2. Perimontanal incision; **early apical release** to spare the sphincter.
3. Develop the capsular plane round to 12 o'clock; detach lobes into the bladder (**en bloc, 2-lobe or 3-lobe**).
4. Haemostasis with a defocused beam.
5. **Morcellation** with a full bladder; 22 Fr three-way catheter.

Outcomes

- Versus **TURP (transurethral resection of the prostate)** (trials, meta-analyses): **better IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate)**, less bleeding, shorter catheter time and stay; longer operation. The **only endoscopic method superior to TURP**.
- Versus **open prostatectomy** for large glands (**Kuntz, 5-year**): equal function with less blood loss and shorter stay.
- **Durable**: whole transition zone removed; **retreatment rare**.

Complications

- **Retrograde ejaculation** about 75%; transient storage symptoms and early SUI (stress urinary incontinence).
- BNC (bladder neck contracture) 0–3.2%; stricture; capsular perforation; ureteric orifice injury; **morcellator bladder injury**; retained adenoma.
- Erectile function unchanged.

Limitation

- **Steep learning curve**: efficiency plateaus after about 50 cases.

Recent advances

- **Moses pulse modulation**; thulium enucleation; ejaculation-sparing HoLEP; same-day discharge. EAU (European Association of Urology): enucleation is an alternative to TURP, and to open surgery for >80 mL.

Diagram to draw: sagittal prostate showing enucleation along the capsule.

Key points to remember:

- Ho:YAG 2120 nm, water chromophore, 0.4 mm.
- Size-independent; best for anticoagulated men.
- Better IPSS and Qmax than TURP; durable; retreatment rare.
- Retrograde ejaculation about 75%; learning curve about 50 cases.

Millin's prostatectomy

Type: SAQ (10 marks)

Source: 2022-Summer Paper 1, Q3f (also answered in 2015-Summer Paper 2, Q3c)

Definition

- **Retropubic transcapsular simple prostatectomy** (Millin, 1945): adenoma enucleated through a transverse anterior capsulotomy **without opening the bladder**. Plane: the surgical capsule (compressed peripheral zone around the transition-zone adenoma).

Indications

- BPH (benign prostatic hyperplasia), gland **>80–100 g**, with refractory LUTS (lower urinary tract symptoms), retention, recurrent UTI (urinary tract infection), haematuria or upper-tract damage.
- Prefer suprapubic (Freyer) for a large median lobe, diverticulum or bladder stones.
- EAU (European Association of Urology) and AUA (American Urological Association): simple prostatectomy or HoLEP (holmium laser enucleation) for large glands.

Work-up

- PSA (prostate-specific antigen), MRI (magnetic resonance imaging) ± biopsy; TRUS (transrectal ultrasound) volume; cystoscopy; urodynamics if bladder function is doubtful.

Steps

1. Lower midline or Pfannenstiel; **space of Retzius**.
2. Optional prophylactic ligation of DVC (dorsal venous complex) and pedicles (Walsh–Oesterling).
3. **Transverse capsulotomy ~2 cm distal to the bladder neck** (a longitudinal cut may extend into the sphincter).
4. Find the adenoma–capsule plane with scissors; **finger enucleation**.
5. **Divide the apical urethra under vision** proximal to the sphincter.
6. **Figure-of-eight sutures at 5 and 7 o'clock** (prostatic arteries); advance bladder neck mucosa into the fossa.
7. 22 Fr three-way catheter, **watertight capsule closure**, drain.

Millin versus Freyer

Aspect	Millin (retropubic)	Freyer (suprapubic)
Bladder	Not opened	Opened
Apex	Divided under vision	Pinched by finger
Risk	Venous bleeding	Cystotomy leak

Complications

- Bleeding (transfusion <5%): catheter traction, Malament purse-string, O'Conor plication.
- Urine leak, UTI, bladder neck contracture 2–6%, stricture, DVT (deep vein thrombosis).
- **Retrograde ejaculation 80–90%**; ED (erectile dysfunction) 3–5%; incontinence very rare.

Advantages over TURP (transurethral resection of prostate)

- Complete removal, low retreatment, no TUR syndrome; but incision, longer stay.
- **Recent:** laparoscopic, robotic multiport and single-port transvesical simple prostatectomy: less bleeding, shorter stay.

Diagram to draw: sagittal view of transverse capsulotomy and enucleation plane.

Key points to remember:

- Millin 1945: capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%.

TUR Syndrome

Type: SAQ (10 marks)

Source: Practice Set 12 Paper 1, Q3f (also answered in 2017-Winter Paper 3, Q3f)

Definition

- **TUR (transurethral resection) syndrome:** neurological and cardiovascular disturbance from **absorbing hypo-osmolar, non-conducting irrigant** during monopolar TURP (transurethral resection of the prostate). Incidence low and falling.

Pathophysiology

- Prostatic venous pressure about **10 mmHg**; irrigant at higher pressure enters open sinuses or, after perforation, the periprostatic space.
- Monopolar needs ion-free fluid: **glycine 1.5%**, sorbitol, mannitol or water (all hypo-osmolar).
- **Fluid overload** → hypertension, pulmonary oedema.
- **Dilutional hyponatraemia and hypo-osmolality** → water enters brain → **cerebral oedema**.
- **Glycine toxicity:** ammonia → encephalopathy; retinal inhibitory transmitter → **transient blindness**.

Risk factors

- Gland **>45 mL**, resection **>90 minutes** (Veterans Affairs Cooperative Study), high irrigation pressure, capsular perforation, sterile water.

Clinical features

- Awake (spinal) patient: restlessness, confusion, **nausea, vomiting, visual disturbance**.
- **Hypertension with bradycardia**, tachypnoea, later hypotension.
- Severe: seizures, coma, herniation. **Na <120 mEq/L** is dangerous.

Prevention

- **Bipolar TURP or laser with saline** (NNT [number needed to treat] about 50).
- Bag **~60 cm** above patient (each extra 10 cm more than doubles absorption); continuous flow; low pressure.
- Limit resection time; stage huge glands; check sodium in long cases.
- Spinal anaesthesia for early detection; ethanol-tagged irrigant monitors absorption.

Treatment

1. Stop surgery, secure haemostasis, insert catheter.
2. ABC (airway, breathing, circulation), oxygen, monitoring; send **sodium**, osmolality, glucose.
3. Overload: **furosemide**; restrict fluids.
4. Severe neurological signs or Na <120: **3% saline 100 mL bolus** (raises Na 2–3 mEq/L), repeated at 10 minutes, maximum 300 mL. **Target rise 4–6 mEq/L.**
5. Limit: **≤12 mEq/L in 24 h** for acute post-TURP hyponatraemia (8–10 mEq/L if symptoms start >48 h after surgery), avoiding **osmotic demyelination**.
6. Seizures: benzodiazepines; ICU (intensive care unit).

- **Also know:** occurs in TURBT (transurethral resection of bladder tumour), PCNL (percutaneous nephrolithotomy), hysteroscopy.

Diagram to draw: flowchart: absorption → overload + hyponatraemia → cerebral oedema.

Key points to remember:

- Venous pressure 10 mmHg; gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.

TURP instruments and techniques

Asked as:

- TURP instruments — *Practice Set 14 Paper I Q3e (SAQ)*
- TURP techniques — *Practice Set 21 SAQ Q3b (SAQ)*

Type: SAQ (10 marks)
(irrigation fluids also answered in 2017-Summer Paper 2, Q3a) (irrigation fluids answered in 2017-Summer Paper 2, Q3a; complications in Practice Set 9 Paper 2, Q3b)

Introduction

- **TURP (transurethral resection of the prostate):** an electrified (wire) loop removes transition-zone adenoma **from bladder neck to verumontanum, down to the surgical capsule**. Gold standard for glands up to 80 mL (AUA [American Urological Association] 2023); larger glands need enucleation or simple prostatectomy.

Resectoscope

Part	Description
Outer sheath	24–28 Fr; **insulated beak** prevents arcing to the urethra
Inner sheath	Creates **continuous flow**: inflow through the inner, outflow between sheaths (passive or suction). Lower bladder pressure, clear vision, no repeated emptying
Obturator	Blunt or visual; atraumatic passage
Working element	Moves the loop. **Passive:** spring holds loop inside sheath; cutting stroke by spring recoil. **Active:** surgeon's thumb pulls the extended loop in
Telescope	**Hopkins rod-lens**, usually **30°** (12° alternative)

Electrodes

- **Cutting loop:** tungsten (monopolar); **platinum–iridium (bipolar plasma)**. Plasma loop (platinum–iridium) cuts at lower temperature and seals vessels.
- Roller ball or barrel (coagulation), vaporisation button, **Collins knife** (TUIP (transurethral incision of the prostate)).

Energy (energy variants)

- **Generator:** about 500 kHz, too fast to stimulate nerves.
- **Cut:** continuous waveform. **Coagulation:** interrupted, higher voltage. **Blend:** better haemostasis, cuts less well.

	Monopolar	Bipolar
Current	Loop → patient → **dispersive pad**	Active and return poles on the loop; stays at loop
Irrigant	**Non-conducting, hypo-osmolar irrigant:** glycine 1.5%, sorbitol, water	**Normal saline**
TUR (transurethral resection) syndrome	Risk; yes; keep resection <90 min	Virtually none; eliminated
Bleeding	More	Less transfusion, clot retention
Efficacy	Equal	Equal

Imaging and irrigation

- **Light source:** xenon or LED (light-emitting diode) with fibreoptic cable; camera and monitor.
- **Irrigant** at body temperature; bag about **60 cm** high (each extra 10 cm more than doubles absorption); lowest height giving vision.

Accessories

- **Ellik evacuator or Toomey syringe** for chips.
- Van Buren sounds or Otis urethrotome: dilate meatus 2 Fr above sheath size (prevents stricture).
- **Three-way Foley catheter** (22–24 Fr) with catheter guide; continuous irrigation set.

Preparation

- Sterile urine; spinal or general anaesthesia; lithotomy with peroneal padding.
- **Inspect bladder first** (tumour, diverticula); identify ureteric orifices, verumontanum, external sphincter.

Resection techniques

- **Median lobe first**, then lateral lobes.
- **Modified Nesbit "encirclement":** start at 11–9 and 1–3 o'clock; expose but do not resect bladder neck fibres.
- **Floor-first:** channel at 5 or 7 o'clock to the capsule (sets depth early); widen laterally; lateral lobes "fall" into the fossa.
- **Anterior start:** 12 o'clock bladder neck, then quadrants down to the apex.
- **Chips:** long "canoe-like" chips with synchronised rocking.
- **Anterior tissue and apex last** in a bloodless field; leaving a little tissue is better than incontinence.
- Haemostasis: arterial bleeders must be fulgurated; venous ooze settles with catheter traction.
- Evacuate chips (Ellik evacuator); three-way catheter.
- **Also know:** TUVP (transurethral vaporisation, button electrode; ≤80 mL); TUIP (transurethral incision, <30 mL, no median lobe); bipolar enucleation; high bleeding risk: AUA prefers HoLEP (holmium laser enucleation) or PVP (photoselective vaporisation).

Troubleshooting a loop that will not cut

```
Scope–generator connection → foot pedal →
irrigant matches generator (glycine for monopolar, saline for bipolar) →
patient grounding (pad contact)
```

Outcomes

- About 80% symptom improvement; low retreatment; retrograde ejaculation about two-thirds.

Recent advances

- Bipolar plasma resection; chip-on-tip digital optics; laser enucleation scopes; single-use resectoscopes.

Diagram to draw: assembled resectoscope labelling sheaths, working element, loop and flow channels.

Diagram to draw: resection field with clock positions, bladder neck, verumontanum and sphincter.

Key points to remember:

- Continuous-flow 24–28 Fr sheath, 30° telescope.
- Tungsten loop (monopolar) vs platinum–iridium (bipolar).
- Glycine 1.5% for monopolar, saline for bipolar; bag 60 cm.
- Bladder neck to verumontanum, depth to capsule.
- Median lobe first; apex last.
- Bipolar saline TURP: no TUR syndrome.

Methods to estimate prostate size

Type: SAQ (10 marks)

Source: Practice Set 14 Paper 1, Q3f (TRUS also answered in 2017-Winter Paper 1, Q3e)

Why measure

- Chooses treatment: 5ARI (5α-reductase inhibitor) if >30 mL (AUA (American Urological Association)) or >40 mL (EAU (European Association of Urology)); type of BPH (benign prostatic hyperplasia) surgery.
- **PSAD (PSA density)** = PSA (prostate-specific antigen) ÷ volume; >0.15 suggests cancer. Also AUR (acute urinary retention) risk, brachytherapy planning.
- Normal adult gland 20–25 g; about 48 g at 60–70 years. **1 cm^3 ≈ 1 g**.

Methods

Method	Technique	Accuracy and use
DRE (digital rectal examination)	Estimate size, consistency	**Underestimates** above 30 mL (lateral lobes impalpable); flat gland predicts <30 mL
PSA as surrogate	Age-specific cut-offs for **>40 mL:** >1.6 ng/mL (50s), >2.0 (60s), >2.3 (70s)	AUC (area under curve) 0.76–0.78; within ±20% of TRUS in most
TAUS (transabdominal ultrasound)	Full bladder; transverse width and height, sagittal length	Less precise than TRUS; also PVR (post-void residual) and **IPP (intravesical prostatic protrusion)**
TRUS (transrectal ultrasound)	Ellipsoid formula; planimetry	**Method of choice**
MRI (magnetic resonance imaging)	T2-weighted ellipsoid or segmentation (manual or automated)	Most accurate and reproducible
CT (computed tomography)	Axial images	Overestimates; poor capsule definition
Cystoscopy	Urethral length, lobes, median lobe	Rough; guides procedure choice

TRUS formulas

Shape assumed	Formula
Ellipsoid (standard)	**π/6 (0.52) × transverse × AP (anteroposterior) × length**
Sphere	π/6 × $transverse^3$
Prolate spheroid	π/6 × $transverse^2$ × AP
Planimetry	Sum of serial transverse areas

- Transverse and AP at the widest axial section; length sagittally. All formulas correlate >0.90 with specimen weight.
- **Planimetry:** probe on a stepper, images every 3–5 mm, areas summed; most precise; used for brachytherapy.

IPP grading (suprapubic, sagittal, bladder 100–200 mL)

- Grade I <5 mm, II 5–10 mm, III >10 mm; ≥10 mm predicts BOO (bladder outlet obstruction) and failed trial without catheter.

Diagram to draw: TRUS transverse and sagittal views with three measured diameters.

Key points to remember:

- Ellipsoid volume = 0.52 × T × AP × L; 1 mL ≈ 1 g.
- DRE underestimates above 30 mL; PSA >1.6/2.0/2.3 by decade predicts >40 mL.
- Planimetry most precise; MRI most reproducible.

Rapid Revision Sheet – Section 16 · BPH

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer (LAQ)

- PZ (gives) 70% of cancers; TZ gives BPH; CZ around ejaculatory ducts; no true capsule.
- NVB between prostatic and levator fascia, 5 and 7 o'clock; apex most vulnerable (the commonest margin and nerve-injury site).
- Interfascial plane is (the) standard (nerve-sparing plane); avoid heat and traction.
- Ligate DVC; excise Denonvilliers fascia; spare accessory pudendals.
- Retzius-sparing: earlier continence, more positive margins.
- Extended PLND captures about 63% of drainage.

HoLEP in BPH (SAQ)

- Ho:YAG 2120 nm, water chromophore, 0.4 mm.
- Size-independent; best for anticoagulated men.
- Better IPSS and Qmax than TURP; durable; retreatment rare.
- Retrograde ejaculation about 75%; learning curve about 50 cases.

Millin's prostatectomy (SAQ)

- Millin 1945: capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%.

TUR Syndrome (SAQ)

- Venous pressure 10 mmHg; gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
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TURP instruments and techniques (SAQ)

- Continuous-flow 24–28 Fr sheath, 30° telescope.
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Section 17 · Prostate Cancer — Long answer questions (LAQ)

Describe staging of CA prostate and management of nonorgan confined CA prostate

Type: LAQ (25 marks)

Source: Practice Set 14 Paper 1, Q2 (also answered in Practice Set 10 Paper 2, Q1)

Staging: TNM (AJCC (American Joint Committee on Cancer) 8th edition, 2017)

Stage	Definition
T1	Not palpable: T1a incidental in ≤5% of resected tissue; T1b >5%; T1c found on needle biopsy
T2	Palpable, confined: T2a ≤ half of one lobe; T2b > half of one lobe; T2c both lobes
T3	**T3a** extraprostatic extension (including microscopic bladder neck invasion); **T3b** seminal vesicle invasion
T4	Fixed or invading external sphincter, rectum, bladder, levators, pelvic wall
N	N0; **N1 regional (pelvic) nodes**
M	**M1a** non-regional nodes; **M1b** bone; **M1c** other sites

- **Traps:** seminal vesicle invasion is T3b, not T4; non-regional nodes are M1a.
- **ISUP (International Society of Urological Pathology) grade groups (GG):** GG1 ≤6; GG2 3+4; GG3 4+3; GG4 8; GG5 9–10.
- **Risk groups (EAU (European Association of Urology)):** high risk = PSA (prostate-specific antigen) >20 ng/mL, GG4–5 or cT2c; **locally advanced = cT3–4 or cN1**. AJCC prognostic groups combine T, N, M, PSA and grade.

Staging work-up

- DRE (digital rectal examination) sets clinical T stage but under- and over-stages. PSA predicts extent: organ-confined about 80% if PSA <4, under 50% if >10; nodes in about 75% if >50 ng/mL.
- **mpMRI (multiparametric magnetic resonance imaging):** extraprostatic extension and seminal vesicle invasion; best specificity in higher-risk men.
- **High risk:** bone scan plus CT (computed tomography) or MRI; **PSMA (prostate-specific membrane antigen) PET (positron emission tomography)**. **proPSMA:** accuracy 92% vs 65% for conventional imaging.
- Nomograms (Partin, Kattan, CAPRA (Cancer of the Prostate Risk Assessment)) and genomic classifiers (Decipher) refine risk.

Management principle

- Non-organ-confined disease (T3–4 and/or N1, M0) has a high chance of micrometastases. **Single-modality treatment usually fails**, so treatment is **multimodal**: local therapy plus systemic ADT (androgen deprivation therapy).

A. Locally advanced (cT3–4 N0 M0)

1. EBRT (external beam radiotherapy, RT) plus long-term ADT: standard

- Dose-escalated IMRT (intensity-modulated radiotherapy) with image guidance, about 76–78 Gy, with **ADT for 18–36 months** (NCCN (National Comprehensive Cancer Network); EAU 2–3 years). Neoadjuvant ADT shrinks the target and is synergistic with radiation.

- **EORTC 22863:** 3 years of ADT with RT raised 5-year overall survival (78% vs 62%). **RTOG 92-02:** 28 vs 4 months; survival gain in Gleason 8–10.
- **SPCG-7 and MRC PR07/NCIC PR.3:** adding RT to ADT reduced cancer death (RR (relative risk) 0.44; HR (hazard ratio) 0.46). ADT alone is inferior.
- **Brachytherapy boost** improves biochemical control (ASCENDE-RT).
- **Very high risk:** add **abiraterone for 2 years** (STAMPEDE M0: metastasis-free survival HR 0.53). Eligible: N1, or two of cT3–4, Gleason ≥8, PSA ≥40 ng/mL.

2. RP (radical prostatectomy) with ePLND (extended pelvic lymph node dissection): selected men

- Fit, life expectancy >10 years, resectable (mainly cT3a). Clinical overstaging occurs in 7–27%; 10-year disease-specific survival about 80–90%.
- Neoadjuvant ADT downstages clinically but gives **no survival gain**.
- **After surgery:** pT3 or positive margin: **observation with early salvage RT** (RADICALS-RT, RAVES, GETUG-AFU 17); adjuvant RT only for very high-risk pathology (GG4–5, pT3b, margins).

3. Other options

- **ADT alone:** only for men unfit or unwilling for local therapy; immediate if symptomatic, high PSA or rapid doubling.
- **Docetaxel with RT and ADT:** RTOG 0521 overall survival HR 0.69; not routine.
- Cryotherapy or HIFU (high-intensity focused ultrasound): investigational in high risk.

B. Node-positive (N1 M0)

- **cN1:** RT to prostate and pelvic nodes with long-term ADT plus 2 years of abiraterone; RP with ePLND in selected men.
- **pN1 after prostatectomy: ECOG 7887**: immediate ADT improved overall survival; ADT plus pelvic RT beats ADT alone. Observation is an option for ≤2 nodes with undetectable PSA.

C. Metastatic (M1), in brief

- ADT plus an **ARPI (androgen receptor pathway inhibitor)** (abiraterone, enzalutamide, apalutamide) or a triplet with docetaxel; prostate RT for low-volume disease (see 2025-Winter Paper 3, Q3b).

```
Non-organ-confined, M0
 ├─ Fit, cT3a, resectable → RP + ePLND → early salvage RT ± ADT
 ├─ cT3–4 or cN1 (most) → EBRT + ADT 18–36 mo
 │       very high risk / N1 → + abiraterone 2 years
 └─ Unfit for local therapy → ADT (immediate if symptomatic)
```

Local complications and follow-up

- Bladder outlet obstruction (channel TURP (transurethral resection of the prostate)), ureteric obstruction (stent or nephrostomy), haematuria (palliative RT).
- PSA every 3–6 months; bone health, metabolic and cardiovascular monitoring on ADT.

Diagram to draw: sagittal pelvis showing T3a, T3b and T4 spread.

Key points to remember:

- T3b = seminal vesicle; M1a = non-regional nodes.
- EBRT + 2–3 years ADT is standard; EORTC 22863 and SPCG-7.
- STAMPEDE: abiraterone for 2 years in very high-risk or N1 disease.
- After RP: early salvage RT preferred to routine adjuvant RT.

Section 17 · Prostate Cancer — Short answer questions (SAQ)

Transrectal ultrasound study of prostate

Type: SAQ (10 marks)

Source: 2017-Winter Paper 1, Q3e

Principle and technique

- **TRUS (transrectal ultrasound):** the prostate lies directly in front of the rectum. Probes: end-fire, side-fire or biplane, **6–10 MHz** (higher frequency: better resolution, less depth). Left lateral (or lithotomy for transperineal work), after DRE (digital rectal examination). Transverse and sagittal planes, base to apex.

Normal sonographic anatomy

Zone (McNeal)	Glandular share	TRUS appearance	Cancer share
Peripheral	~70%	Homogeneous, echogenic (posterior)	~70%
Central	~25%	Echogenic, at base	~5–10%
Transition	5–10% (BPH (benign prostatic hyperplasia))	Heterogeneous, anterior	~20–25%
Anterior fibromuscular stroma	Non-glandular	Hypoechoic	–

- Zones are not seen as such; TRUS separates echogenic posterior PZ (peripheral zone) from heterogeneous TZ (transition zone). **Corpora amylacea** mark the surgical capsule.
- Seminal vesicles 4.5–5.5 × 2 cm; absent seminal vesicle → ipsilateral renal agenesis in 79%.

Measurements

- **Volume (ellipsoid) = 0.52 (π/6) × transverse × AP (anteroposterior) × length**; 1 mL ≈ 1 g; normal 20–25 g. **Planimetry** (serial sections) for brachytherapy.
- **PSAD (prostate-specific antigen density)** = PSA/volume; **>0.15** suggests cancer.

Findings

- **Cancer:** classically a hypoechoic PZ lesion, but **39% are isoechoic**; false positives: BPH nodules, prostatitis, infarct.
- **Cysts:** Müllerian (midline), ejaculatory duct cyst (just off midline), seminal vesicle cyst (lateral, renal agenesis).
- Abscess, calculi, extraprostatic extension.
- After ADT (androgen deprivation therapy) or radiotherapy the gland shrinks and becomes hypoechoic.

Indications

- **Guided biopsy** (12-core systematic ± MRI (magnetic resonance imaging)-targeted fusion).
- Volume before BPH surgery, brachytherapy, cryotherapy; fiducial markers.
- **Azoospermia:** ejaculatory duct obstruction (seminal vesicle >1.5 cm wide, midline cyst).
- Drainage of prostatic abscess or cysts.

Biopsy notes

- Periprostatic block. Complications: bleeding, **sepsis** (fluoroquinolone-resistant *E. coli*); transperineal route lowers infection.

Advances

- Colour/power Doppler, contrast-enhanced TRUS, **elastography**, multiparametric ultrasound.
- **Micro-ultrasound (29 MHz;** 70 µm resolution) with PRI-MUS scoring, comparable to MRI.
- MRI-first pathway: PROMIS (2017) mpMRI (multiparametric MRI) sensitivity 93% vs 48% for TRUS biopsy.

Diagram to draw: transverse and sagittal TRUS views labelling PZ, TZ, seminal vesicles and urethra.

> **Key points to remember:**
> - 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15.
> - 39% of cancers isoechoic; hypoechoic PZ lesion classic.
> - Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.

Robotic radical prostatectomy

Type: SAQ (10 marks)

Source: 2023-Winter Paper 1, Q3b

Introduction

- RARP (robot-assisted radical prostatectomy): Walsh's anatomical nerve-sparing principles done with **3D, 10× magnified vision and wristed instruments**. The robot removed the laparoscopic suturing barrier; da Vinci from 2000 (S, Si, Xi, SP).

Indications and contraindications

- Localised cancer (cT1–T2, selected cT3); life expectancy over 10 years.
- **Absolute:** uncorrectable bleeding diathesis; unfit for general anaesthesia.
- **Relative:** morbid obesity, prostate >80 g, median lobe, prior TURP (transurethral resection of the prostate), salvage surgery.

Set-up

- Low lithotomy, arms tucked, **steep Trendelenburg (22–25°)**, padding; pressure 12–15 mmHg; camera port 15–17 cm above pubis, three robotic and two assistant ports.

Steps (transperitoneal anterior approach)

1. Drop bladder; enter space of Retzius.
2. Incise endopelvic fascia; ligate DVC (dorsal venous complex).
3. Divide bladder neck (fat line, balloon traction); check for median lobe.
4. Seminal vesicles and vasa; incise Denonvilliers fascia.
5. Clip pedicles athermally; **interfascial nerve sparing** (extrafascial if high risk).
6. Apical dissection preserving urethral length.
7. Extended PLND (pelvic lymph node dissection) if indicated.
8. Posterior (Rocco) reconstruction; **Van Velthoven running anastomosis**.

Postoperative care

- ERAS (enhanced recovery after surgery); discharge within 24 h; catheter about 1 week (not before day 5: retention); pelvic floor training; penile rehabilitation with PDE5 (phosphodiesterase type 5) inhibitors.

Outcomes

- **Yaxley 2016** randomised trial: equal function and early margins versus open; less blood loss (300 versus 1400 mL) and shorter stay.
- Margins about 15% (9% if pT2); 1-year continence mid-90s% in expert hands.

Complications

- Transfusion 1–2%; bladder neck stricture <2% (open up to 16%); rectal injury; ureteric injury; lymphocele; positioning neuropathy, corneal abrasion; robot failure <0.5%.

Recent advances

- **Retzius-sparing** (earlier continence, more positive margins); Hood technique; single-port robot; same-day discharge.

Diagram to draw: port layout and fascial planes (intra-, inter-, extrafascial).

Key points to remember:

- Surgeon, not platform, decides outcome (Yaxley 2016).
- Interfascial plane is the standard nerve-sparing plane.
- Bladder neck stricture <2%; transfusion 1–2%.

Prostate multiparametric MRI and endorectal coil MRI

Asked as:

- Multiparametric MRI : its components and interpretation for prostate cancer. — *Winter 2024 Paper I Q3d (SAQ)*
- Endorectal coil MRI — *Practice Set 7 Paper I Q3f (SAQ)*

Type: SAQ (10 marks)

Part A: Multiparametric MRI

Introduction

- **mpMRI (multiparametric magnetic resonance imaging):** anatomical plus functional sequences, usually at 3 T without endorectal coil. Reported by **PI-RADS v2.1 (Prostate Imaging Reporting and Data System, 2019)**.

Components

Sequence	Principle	Cancer appearance
T2W (T2-weighted), triplanar	Anatomy; normal PZ (peripheral zone) bright	Low-signal lesion; "erased charcoal" in TZ (transition zone); shows EPE (extraprostatic extension) and SVI (seminal vesicle invasion)
DWI (diffusion-weighted imaging) with **ADC (apparent diffusion coefficient)** map; high b-value (≥1400)	Cellular tumour restricts water movement	Bright on high-b DWI, **dark on ADC**; ADC falls as Gleason rises
DCE (dynamic contrast-enhanced) T1 with gadolinium	Angiogenesis: early enhancement and washout	Early focal enhancement (type 3 curve)

- **Biparametric MRI (T2W + DWI):** no contrast, cheaper, similar detection; poorer staging.
- **PI-QUAL (Prostate Imaging Quality):** quality score (1–5) before reporting.

Interpretation: PI-RADS v2.1

Score	Clinically significant cancer
1	Highly unlikely
2	Unlikely
3	Equivocal
4	Likely; lesion <1.5 cm
5	Highly likely; ≥1.5 cm or definite EPE

Zone	Dominant sequence	Tie-breaker
Peripheral	**DWI**	DWI 3 + DCE positive → 4
Transition	**T2W**	T2W 3 + DWI 5 → 4; T2W 2 + DWI ≥4 → 3

- **DCE only upgrades PZ 3 to 4**; it never changes 4–5 or TZ scores.
- Cancer in about one-third of PI-RADS 3, over half of 4, 90% of 5.
- **PRECISE** score (1–5) tracks change during active surveillance.

Clinical use and evidence

- **PROMIS (2017):** sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; 27% could avoid biopsy.

- **PRECISION (2018):** MRI pathway with targeted biopsy of PI-RADS ≥3 detected more significant cancer (38% vs 26%), 28% avoided biopsy.
- EAU (European Association of Urology) and AUA (American Urological Association): MRI **before biopsy**; also staging, active surveillance, focal therapy planning.
- Limitations: low specificity (Cochrane 0.37); reader-dependent.

Diagram to draw: prostate zones with a PZ lesion dark on ADC.

Part B: Endorectal coil MRI

Principle

- **ERC (endorectal coil):** a receive-only surface coil inside an inflatable balloon placed in the rectum against the prostate, used with an external pelvic phased-array coil.
- **"The currency of MRI is signal."** A coil close to the prostate gives higher **SNR (signal-to-noise ratio)**, allowing **higher spatial resolution**.

Technique

- Antispasmodic to reduce rectal motion artefact; balloon filled with air or perfluorocarbon (fluid reduces susceptibility artefact).
- Sequences as for mpMRI (Part A): **T2W** triplanar for anatomy, **DWI with ADC** map, **DCE**; ERC was traditionally needed for **MRSI (MR spectroscopic imaging)**.

Uses

- **Local staging:** extraprostatic extension and seminal vesicle invasion (both low T2 signal) before radical prostatectomy.
- **Local recurrence** after prostatectomy: recurrence isointense on T1, slightly hyperintense on T2; DCE raises accuracy at low PSA (prostate-specific antigen).
- **Also know:** focal therapy planning (ERC recommended at 1.5 T); seminal vesicles; female urethral diverticulum.

ERC versus no ERC

	With ERC	Without ERC (3 T, pelvic coil)
Signal and resolution	Higher	Adequate at 3 T
csPCa (clinically significant cancer) detection	GG (Grade Group) ≥3 equal; GG2 slightly better	GG ≥3 equal
Comfort, cost, time	Worse	Better
Gland	Deformed (fusion and radiotherapy planning harder)	Natural shape

Disadvantages

- Artefacts: **near-field signal flare** next to the coil (needs intensity correction), air susceptibility at the rectal wall, malposition.
- **Contraindications:** absent rectum (abdominoperineal resection), recent rectal surgery, anorectal fistula or inflammatory bowel disease, severe haemorrhoids or stricture, latex allergy.

Current position

- At **3 T the ERC is largely historical**; PI-RADS v2.1 does not require it. **Consider it at 1.5 T** or with older scanners, or when body-coil quality is poor.

Diagram to draw: sagittal pelvis with inflated endorectal coil against the posterior prostate.

Key points to remember:

- T2W anatomy, DWI/ADC cellularity, DCE vascularity.
- PZ ruled by DWI, TZ by T2W; DCE only lifts PZ 3 to 4.
- PRECISION and PROMIS: MRI before biopsy.
- Endorectal coil close to prostate: higher SNR and resolution.
- Useful at 1.5 T; largely unnecessary at 3 T (GG ≥3 detection equal).
- Drawbacks: discomfort, gland deformation, near-field flare.

Docetaxel

Type: SAQ (10 marks)

Source: Practice Set 8 Paper 1, Q3b

Introduction

- Semisynthetic **taxane** cytotoxic. Since **TAX 327 (2004)** it is the standard first-line chemotherapy for mCRPC (metastatic castration-resistant prostate cancer).

Mechanism of action

- Binds **β-tubulin** and **stabilises microtubules** (GTP [guanosine triphosphate]-independent polymerisation). Cells arrest in **G2/M**.
- Induces apoptosis independent of TP53, through **BCL2 phosphorylation** and p27 induction.
- Prostate-specific: blocks microtubule transport of ligand-bound **AR (androgen receptor)** into the nucleus.

Dose and administration

- **75 mg/m² IV (intravenous) every 3 weeks with prednisone 5 mg twice daily**; 6 cycles in mHSPC (metastatic hormone-sensitive prostate cancer). Steroid premedication limits fluid retention and hypersensitivity. Hepatic metabolism: caution in liver dysfunction.

Uses in urology

Setting	Evidence
mCRPC	**TAX 327:** overall survival (OS) 18.9 vs 16.4 months versus mitoxantrone (HR [hazard ratio] 0.76); better pain and PSA (prostate-specific antigen) response. SWOG 9916 (with estramustine) also positive
mHSPC	**CHAARTED:** OS 57.6 vs 44.0 months (HR 0.61), benefit in **high volume**. STAMPEDE: OS HR 0.78. GETUG-AFU 15 not significant
Triplet	ADT (androgen deprivation therapy) + docetaxel + abiraterone (**PEACE-1**) or darolutamide (**ARASENS**) improves OS
High-risk localised	RTOG 0521: added to RT (radiotherapy) + long-term ADT, OS HR 0.69
Bladder	Second-line single agent (response about 13%); **intravesical gemcitabine–docetaxel** for BCG (bacillus Calmette–Guérin)-unresponsive or BCG-naive high-grade NMIBC (non-muscle-invasive bladder cancer)
Penis	**TPF** (docetaxel, cisplatin, 5-fluorouracil)

CHAARTED high volume: ≥4 bone metastases with ≥1 outside the axial skeleton, or visceral metastasis.

Toxicity

- **Neutropenia** and febrile neutropenia (dose-limiting), fatigue, **fluid retention** (peripheral oedema), **peripheral neuropathy**, hyperlacrimation, **nail dystrophy**, alopecia, liver dysfunction, hypersensitivity.
- 3-weekly: more neutropenia; weekly: more fatigue, diarrhoea, tearing.

Resistance and sequencing

- P-glycoprotein efflux pump mediates resistance. **Cabazitaxel** has poor P-glycoprotein affinity, so it works after docetaxel (**TROPIC**; **CARD** beat a second ARPI [androgen receptor

pathway inhibitor]).
- **Also know:** failed combinations (bevacizumab, atrasentan).

Diagram to draw: microtubule stabilised by docetaxel causing G2/M arrest and blocking AR entry into the nucleus.

Key points to remember:

- 75 mg/m² 3-weekly + prednisone; TAX 327 HR 0.76.
- CHAARTED: benefit in high-volume mHSPC (HR 0.61).
- Neutropenia, oedema, neuropathy, nail change; cabazitaxel after failure.

Rapid Revision Sheet – Section 17 · Prostate Cancer

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Describe staging of CA prostate and management of nonorgan confined CA prostate (LAQ)

- T3b = seminal vesicle; M1a = non-regional nodes.
- EBRT + 2–3 years ADT is standard; EORTC 22863 and SPCG-7.
- STAMPEDE: abiraterone for 2 years in very high-risk or N1 disease.
- After RP: early salvage RT preferred to routine adjuvant RT.

Transrectal ultrasound study of prostate (SAQ)

- 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15.
- 39% of cancers isoechoic; hypoechoic PZ lesion classic.
- Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.

Robotic radical prostatectomy (SAQ)

- Surgeon, not platform, decides outcome (Yaxley 2016).
- Interfascial plane is the standard nerve-sparing plane.
- Bladder neck stricture <2%; transfusion 1–2%.

Prostate multiparametric MRI and endorectal coil MRI (SAQ)

- T2W anatomy, DWI/ADC cellularity, DCE vascularity.
- PZ ruled by DWI, TZ by T2W; DCE only lifts PZ 3 to 4.
- PRECISION and PROMIS: MRI before biopsy.
- Endorectal coil close to prostate: higher SNR and resolution.
- Useful at 1.5 T; largely unnecessary at 3 T (GG ≥3 detection equal).
- Drawbacks: discomfort, gland deformation, near-field flare.

Docetaxel (SAQ)

- 75 mg/m² 3-weekly + prednisone; TAX 327 HR 0.76.
- CHAARTED: benefit in high-volume mHSPC (HR 0.61).
- Neutropenia, oedema, neuropathy, nail change; cabazitaxel after failure.